FAST MOVING TECHNOLOGY

Arm - TX2 series 60 family

Instruction manual



D.280.942.04-L - 05/10/2026
TX2-60 - Master
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1 - INTRODUCTION

1.1 - FOREWORD

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Electrical drawings provided in this manual are for information only. The reference for robot electrical drawings is provided in a separate manual. The photos are used to make the document easier to understand, they cannot be construed as being of a contractual nature.
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- https://www.staubli.com/contacts

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1.1.1 - CONTACT

M0009594.1

Manufacturer:
Stäubli Faverges SAS
Place Robert Stäubli CS 30070
FR-74210 Faverges-Seythenex / FRANCE

Robotics Customer Support contact for your country:

- https://www.staubli.com/global/en/contacts.html

Please have the following information ready before you contact our Customer Support:

- Robot type
- Serial number of the machine (as shown on the identification plate)
- Software version: (Access this information on the teach pendant's display screen by selecting ROBOT / INFORMATION)
- Error logger of the robot: (Access this information on the teach pendant's display screen by selecting ROBOT / LOGGER)

1.1.2 - OBJECTIVE OF THIS MANUAL

M0006370.1

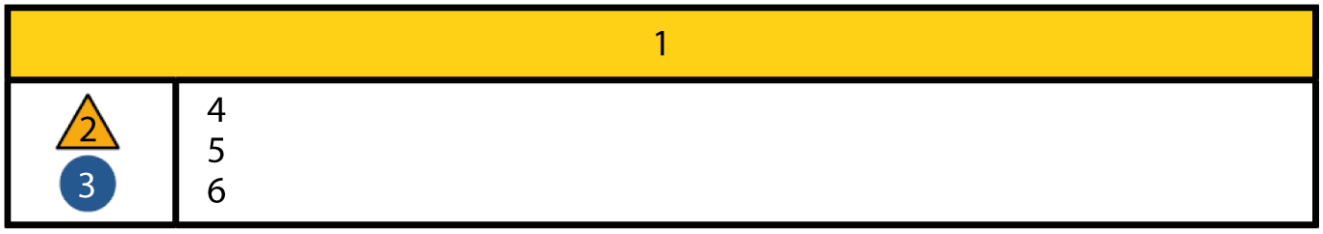
The objective of this manual is to provide information concerning the installation, operation, and maintenance of Stäubli TX2-60 STANDARD ACCESS STER STER+. It provides help for people working on the equipment, for reference purposes only. Stäubli recommends attending training courses for easier understanding of this manual, better use and robot maintenance.

The CS9 Instruction, CS9 Safety, Electrical diagrams and Spare parts manuals along with the arm manual, and software manuals (VAL 3, Stäubli Robotics Suite...) are needed to have a complete overview of the robot.

1.1.3 - SPECIAL MESSAGES AND HAZARD WARNINGS

M0009573.1

This document contains ISO-3864 hazard warnings which are structured as follows:



Item	Description
1	Hazard level
2	Triangular warning pictogram
3	Circular obligation or prohibition pictogram (optional)
4	Type of hazard
5	Consequence: specifies the consequence of failing to avoid a hazard
6	How to avoid: contains information about how a user can avoid a hazard

The warning levels are as follows, in descending order of importance:

DANGER

Type of hazard: Danger
Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.

WARNING

Type of hazard: Warning
Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION

Type of hazard: Caution
Indicates a potentially hazardous situation which, if not avoided, might result in minor or moderate injury.

NOTICE

Type of hazard: Notice
Indicates a potential situation which, if not avoided, might result in property damage or in an undesirable result or state.

There is always a triangular warning prictogram on the left side of the hazard warning:

Pictogram	Meaning
	Generic hazard
	Robot tilting hazard
	Lifting hazard
	Object fall hazard
	Electrical hazard

A circular obligation or prohibition pictogram may be added below the first pictogram:

Pictogram	Meaning
	Support contact is mandatory
	Stacking is prohibited
	Wear of protective gloves is mandatory
	Wearing safety goggles is mandatory
	Electrical grounding is mandatory
	Reading the manual is mandatory

This document may also contain the following special messages:



Supplies further information, or underlines a point or an important procedure. This information must be memorized to make it easier to apply and ensure correct sequencing of the operations described.



ESD
This symbol is a reminder of the possible consequences of touching electrostatically sensitive components (see chapter 7.3).

1.2 - DEFINITIONS AND GLOSSARY

M0006409.1

In this manual, a specific vocabulary is used according to our products. This chapter allows an easier general understanding.
People in contact with the product are designated by different names:

Person: General term identifying all individuals likely to come close to the Stäubli machine.
Staff: Identifies people specifically employed and trained to install, operate, and service the Stäubli machine.
User: Refers to people or the company responsible for operating the Stäubli machine.
Operator: Refers to the person who starts or stops the robot, or controls its operation.
Integrator: Refers to people or the company responsible for the design and manufacture of the robot cell.

Item	Definition
AIB	Arm interface PCB (arm base)
BACKPLANE	Link card between URPS 325 (or RPS 325) power supply and Drive Power Drawer
BRK	Brake
COD	Arm Encoder
CPT	Computer drawer
CW	Compact wrist
DP	Drive Power drawer
DPM 325	Dual Power Module 325 VDC (Dual drive)
DPMI-CAP	Drive Power Module Interface & Capacitors
DPMI-VBUS	Drive Power Module Interface VBUS distribution
DSI9	Digital Sensor Interface: Safety board at robot's base
EOAT	End Of Arm Tooling
EV	Solenoid valve
FDI	Fast Digital Input
FDO	Fast Digital Output
HW	Hollow wrist
IC	Interconnect Cable (between controller and arm)
MMI	Human-machine interface PCB
MOT	Motor
PCIe	Digital interface standard for optional peripheral boards
RBR	Remote Brake Release
RPS 325 (SP)	Robot Power supply 325 VDC single-phase
RSI9	Robot Safety Interface for CS9 controller
S1	Main switch
SBL/CPU	Stäubli Bus Link & Central Processing Unit
SDI	Safe Digital Input
SDO	Safe Digital Output
SP2/SP2+	Stäubli pendant
SPOC	Single Point of Control
SRC	The embedded software running on the CS9 Stäubli controllers
SRS	The PC software to configure, develop, tune and maintain a robotics application
STARC9	Stäubli Advanced Robot Control for CS9 controller
TDO	Digital Output Test signal for Safe Inputs

Item	Definition
Th	Thermal Sensor
UCB	User connection Board
uniVAL	A SRC control mode allowing to drive the robot from an external computer through an industrial fieldbus
URPS 325 (3P)	Universal Robot Power supply 325 VDC three phases
USI	User Safe Input
USO	User Safe Output
VAL 3	The Stäubli robotics programing language for CS9 Stäubli controllers
WMS9	Working Mode Selection control panel for CS9 controller
WMSES	Emergency stop button on the operating mode selection panel

2 - DESCRIPTION

2.1 - OVERVIEW

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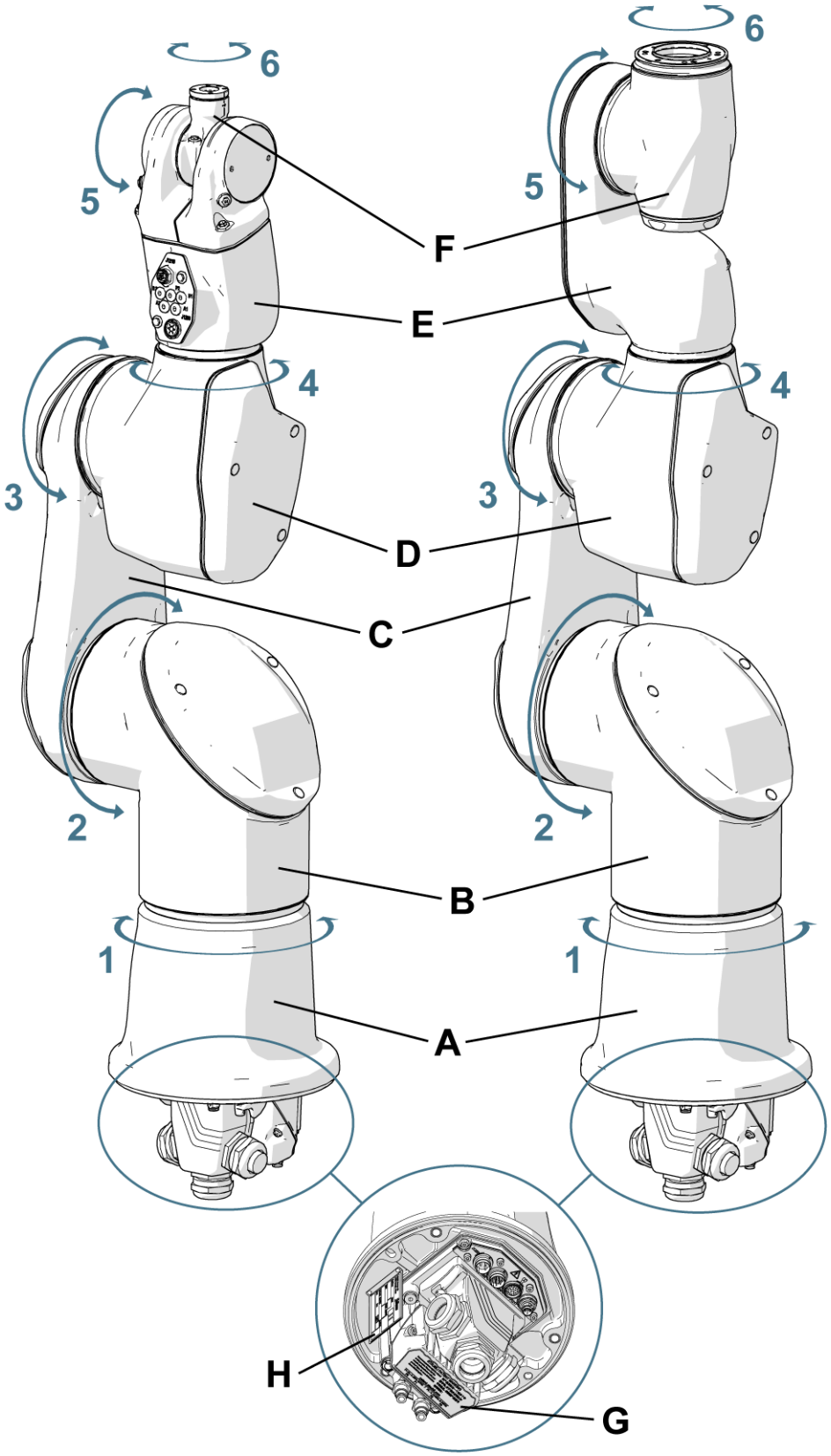


Figure 2.1: Pharma environment: Arm overview, CW (left) and HW (right)

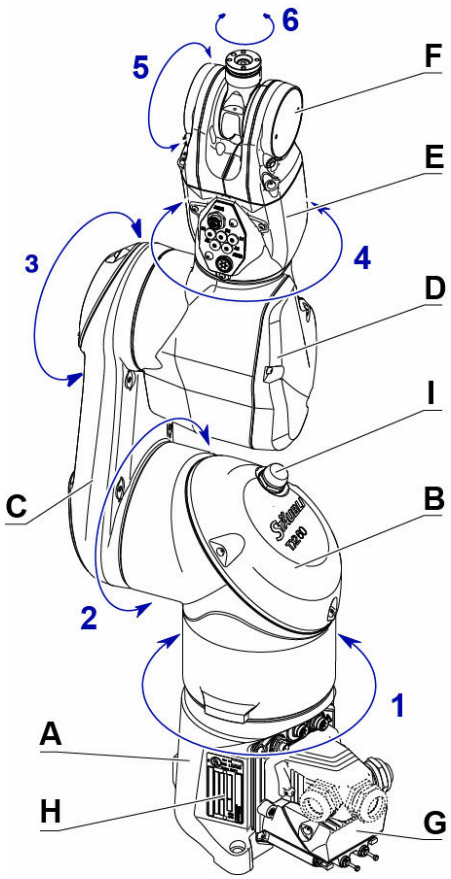


Figure 2.2: Standard environment: Arm Overview

The arm consists of links interconnected by joints. The different links are: The base **(A)**, the shoulder **(B)**, the inner link **(C)**, the elbow **(D)**, the forearm **(E)** and the wrist **(F)**.

The arm joints movements are generated by servomotors coupled to safe digital absolute encoders. Motor brakes maintain the arm at a fixed position when servomotors are disabled. A safe digital processor (DSI9) **(G)** serializes the encoders information to the controller, and controls the brakes and the electro-valves.

The arm includes electric wiring and pneumatic plumbing for customer use, linking the connectors at the base to the connectors on the forearm. The forearm also includes points to attach devices to be connected to the robot tooling.

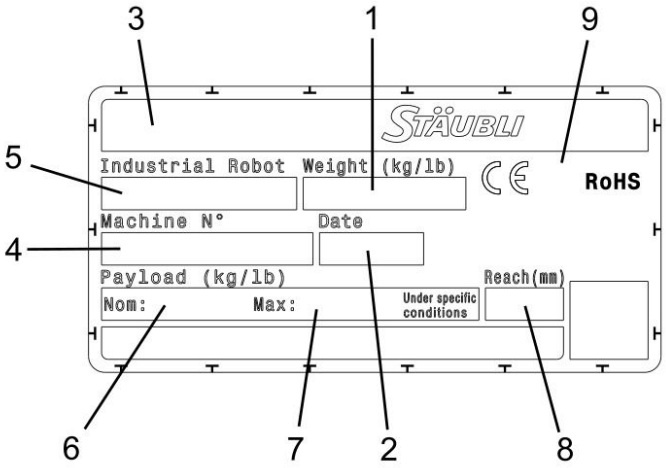
The arm can be mounted in any position, with a horizontal or a vertical cabling connection to match the specific machine constraints.

Different options for the arm are possible to match end-user appearance or environmental requirements, such as cleanroom (CR, SCR), food safe oil (H1), UL version, humid (HE) or bio-contaminated environments (Stericlean) and (Stericlean+) for aseptic environment.

Standard environment: For the UL version or the specific option, a yellow indicator light **(I)** is fitted on the robot shoulder to show that drive power is available, that motion is possible and that it represents a possible cause of risks for the operator (see chapter 4.6).

The speed, accuracy, and repeatability of the arm allow for a great variety of applications including automotive and equipment manufacturers, food, life sciences and pharma, machine tending, plastics, etc...

A robot is identified by a plate attached to the controller and to the arm.
For STERICLEAN + version the identification plate is located under the robot's base (vertical outlet).
Example: Identification plate **(H)** (see figure 2.1).



Item	Description
1	Arm mass
2	Manufacture date (MM/YY)
3	Manufacturer's address
4	Serial number
5	Machine type and model
6	Nominal payload
7	Max payload (under specific conditions)
8	Reach (operating reach)
9	Regulatory marking

Figure 2.3:

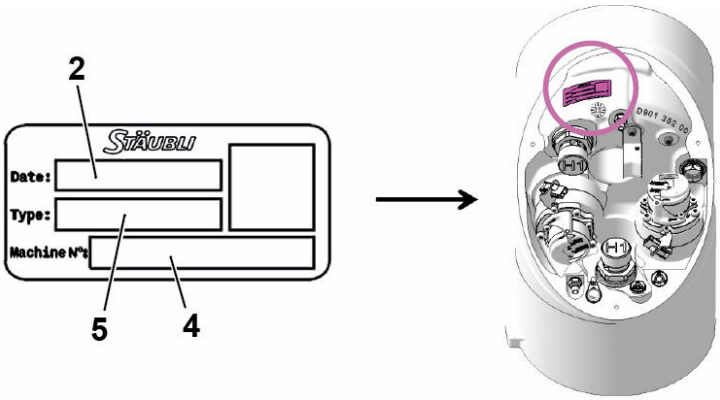


Figure 2.4: A second identification plate inside the robot shoulder summarizes the information

The CE mark placed on the identification plate refers to the "Electromagnetic Compatibility" directive and “low voltage” directive for CS9LP and CS9MP version. It refers only to the “low voltage” directive for CS9EP version, meaning the EMC compliance shall be handled by the integrator at machine level.

The manufacturer's identification plate must remain present and legible throughout the robot service life.

Model	TX2–60 CW	TX2–60L CW	TX2–60 HW	TX2–60L HW
Characteristics				
Limiting load (see chapter 3.3.4)	9 kg	5 kg	9 kg	5 kg
Nominal load	3.5 kg	2 kg	3.5 kg	2 kg
Reach (between axis 1 and flange) (see chapter 3.3.2)	670 mm	920 mm	710 mm	945 mm
Number of degrees of freedom	6	6	6	6
Repeatability - ISO 9283	± 0.02 mm	± 0.03 mm	± 0.02 mm	± 0.03 mm
Maximum speed at load center of gravity (using CS9LP controller version)	8.4 m/s	11.1 m/s	9.0 m/s	11.1 m/s
Maximum speed at load center of gravity (using CS9EP controller version)	0.8 m/s	1.1 m/s	0.9 m/s	1.1 m/s
Noise at nominal conditions	70 dBA			
Power consumption - VDMA 24608 (using CS9LP controller version (worst case))	0,405 kW	0,4 kW	0,405 kW	0,4 kW
Standard cleanliness - ISO 14644-1	Class ISO5 and cGMP Grade A			
Protection class - EN 60529	STD, HE, ACCESS, STER : IP65 (and IP67 with pressurized version). CR, SCR, STER+ : IP65		IP65 with sealed tool on flange	
Weight (see chapter 4.1)	52 kg	53 kg	52 kg	53 kg
Stäubli Controller	CS9LP and CS9EP			
Brakes	All axis			
RBR (option) (see chapter 2.6)	All models			
Universal attachment methods (see chapter 4.4)	Floor / Wall / Ceiling			
I/O	Electrical / Ethernet / Pneumatic (option)			
Specific versions (see chapter 3.3.1)				
Vertical cable outlet (option)	All models			
Pressurized version (option)	All models		No	
HE version (vertical cable outlet only)	All models		No	
SCR version (cleanliness class ISO 2)	All models		No	
CR version (cleanliness class ISO 4)	All models		No	
Stericlean (vertical cable outlet only)	All models			
Stericlean+ (vertical cable outlet only)	All models			
H1 oil	All models			
ESD	All models		No	
Euromap 12/67 interface Plastics	All models		No	
Spherical working space				
Working space can be restricted by different means (software, mechanical stops, …) to match the available or desired maximum space. See chapter 3.3.2 for detailed information.				

The arm type and model are defined as specified below:

TX2	6	0	STER
1	2	3	4

Item	Description
1	TX2 family arm
2	Maximum working radius between joint 1 and joint 5, expressed in decimeters and rounded off to one significant digit: Ⓐ+Ⓑ (see figure 2.1).
3	Number of active axis: ■ 0 = 6 active axis.
4	Letters to indicate an option: ■ L = For arms longer than the standard version These letters can be combined. Example: L HW STER = long arm with hollow wrist for cleanroom applications.

Figure 2.5:

In the manual, the following terminology is used:
- Standard arm: arm with standard geometry
- L arm: long arm with a different geometry where the forearm and the inner link are longer

2.2 - SAFETY LABELS

M0009960.1

The robot is fitted with safety labels intended to draw the user's attention to residual hazards and important information related to installation, handling and maintenance operations.

⚠ CAUTION

Illegible safety labels

Safety labels shall remain present and legible throughout the service life of the robot. Damaged, missing or illegible labels shall be replaced without delay.

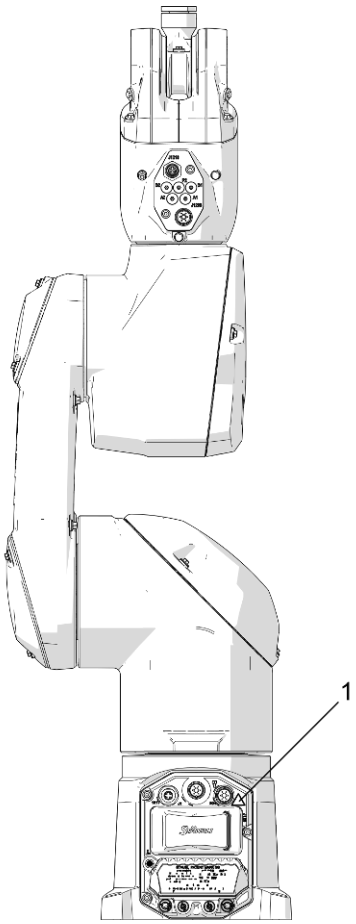


Figure 2.6:

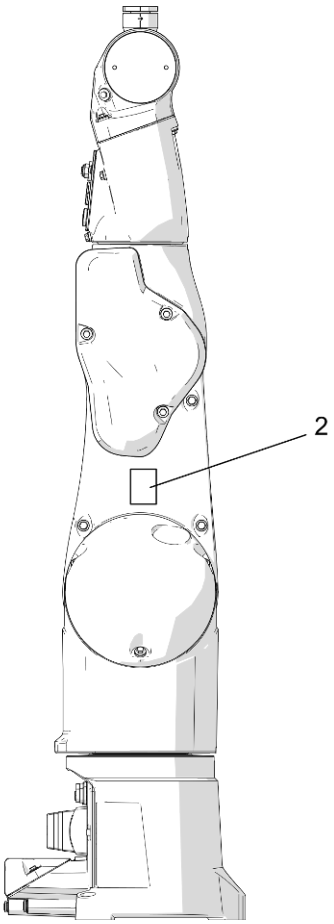


Figure 2.7:

Item	Safety symbol	Description
1		**Electrical hazard** Hazardous voltage may be present. Only qualified service personnel may install or service the robot system.
2		**Lifting hazard** Follow the indicated lifting instructions when handling, transporting or installing the robot. Failure to comply may result in equipment damage or injury.

The safety labels identify residual risks that cannot be completely eliminated by design measures. Operators, integrators and maintenance personnel shall understand and follow the instructions associated with each label before performing any work on the robot.

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2.3 - BEST PRACTICES

M0006728.1

Stäubli robots are designed to be compact, dynamic and accurate. To have the best performance on your application please have a look on the following integration best practices.

- The tool must be as stiff as possible to get the best accuracy performances.
- The robot support must be as stiff as possible to get the best accuracy performances.
- If your application needs absolute positioning, please assure a controlled temperature around the robot and make sure that the robot is at an established temperature.
- The maximum payload must be respected to prevent any safety issue (see chapter 3.3.4.1).
- Avoid singularities to get the best accuracy performances.
- Please take into account the maintenance operations in your design to enhance your robot lifetime with periodic interventions.
- Check your screw tightening torques when the temperature varies a lot.
- If your application needs the best duty cycle, it is possible to overpass the nominal speed up to maximum speed. Double check the support and tool and verify the temperature and lifetime of your robot. It will lead to a loss of accuracy along the path (vibrations, …).
- The path shape and type must be chosen with care when accurate movement is needed. A smooth path will lead to a better precision.
- It is also possible to optimize the duty cycle by optimizing the robot position with respect to the cycle path. A path close to the foot will allow having big accelerations while a path far from the foot will allow increasing the Cartesian speed.

If your application needs a specific evaluation Stäubli is at your disposal for a custom evaluation.

2.4 - BASE INTERFACE

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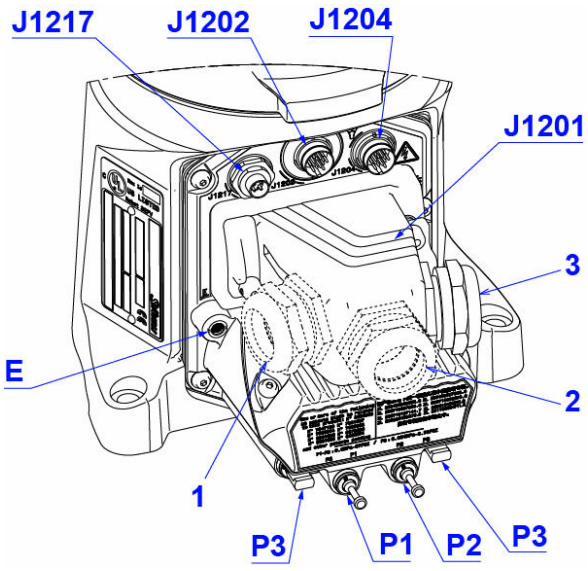


Figure 2.8: CW, horizontal outlet

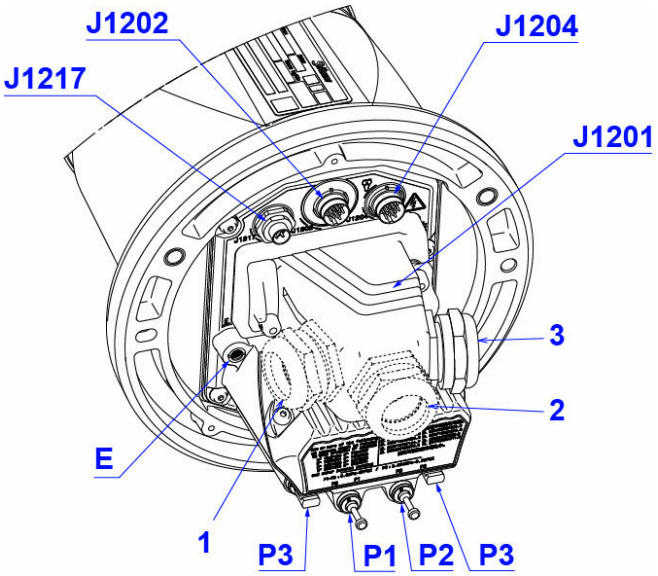


Figure 2.9: CW, vertical outlet

Item	Description
J1201	Arm/controller interconnect cable connection
J1202	Electrical cable connection to J1203
J1204	RBR interface connection
J1217	Ethernet Cat 5e or electrical cable connection to J1218
P1, P2	Pneumatic connections (6 mm)
P3	Pressurization unit connection (8 mm)
E	Pneumatic exhaust muffler
1	Interconnect cable, left outlet
2	Interconnect cable, straight outlet
3	Interconnect cable, right outlet

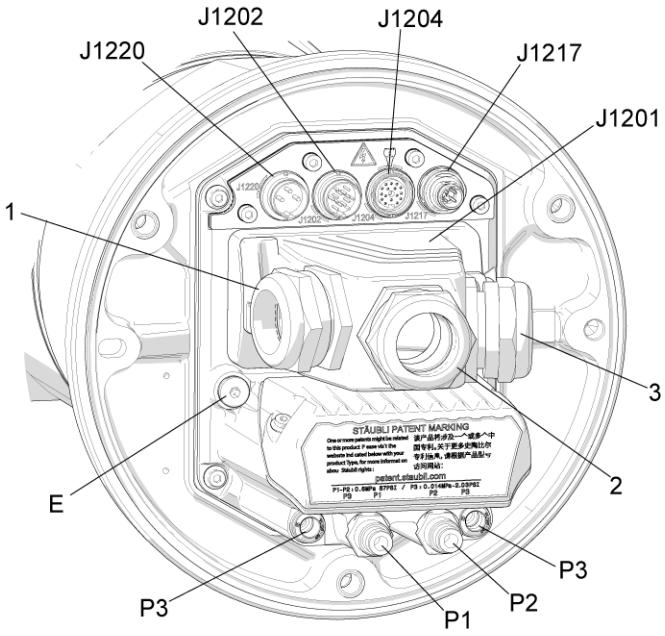


Figure 2.10: HW, vertical outlet

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Item	Description
J1201	Arm/controller interconnect cable connection
J1202	Electrical I/O or power cable connection to flange cable outlet
J1204	RBR interface connection
J1217	Ethernet Cat 5e or electrical cable connection to J1230
J1220	Electrical power or I/O cable connection to flange cable outlet
P1, P2	Pneumatic connections (6 mm)
P3	Pressurization unit connection (8 mm)
E	Pneumatic exhaust muffler
1	Interconnect cable, left outlet
2	Interconnect cable, straight outlet
3	Interconnect cable, right outlet

2.5 - FOREARM CONNECTIONS

2.5.1 - FOREARM CONNECTIONS

M0006716.1

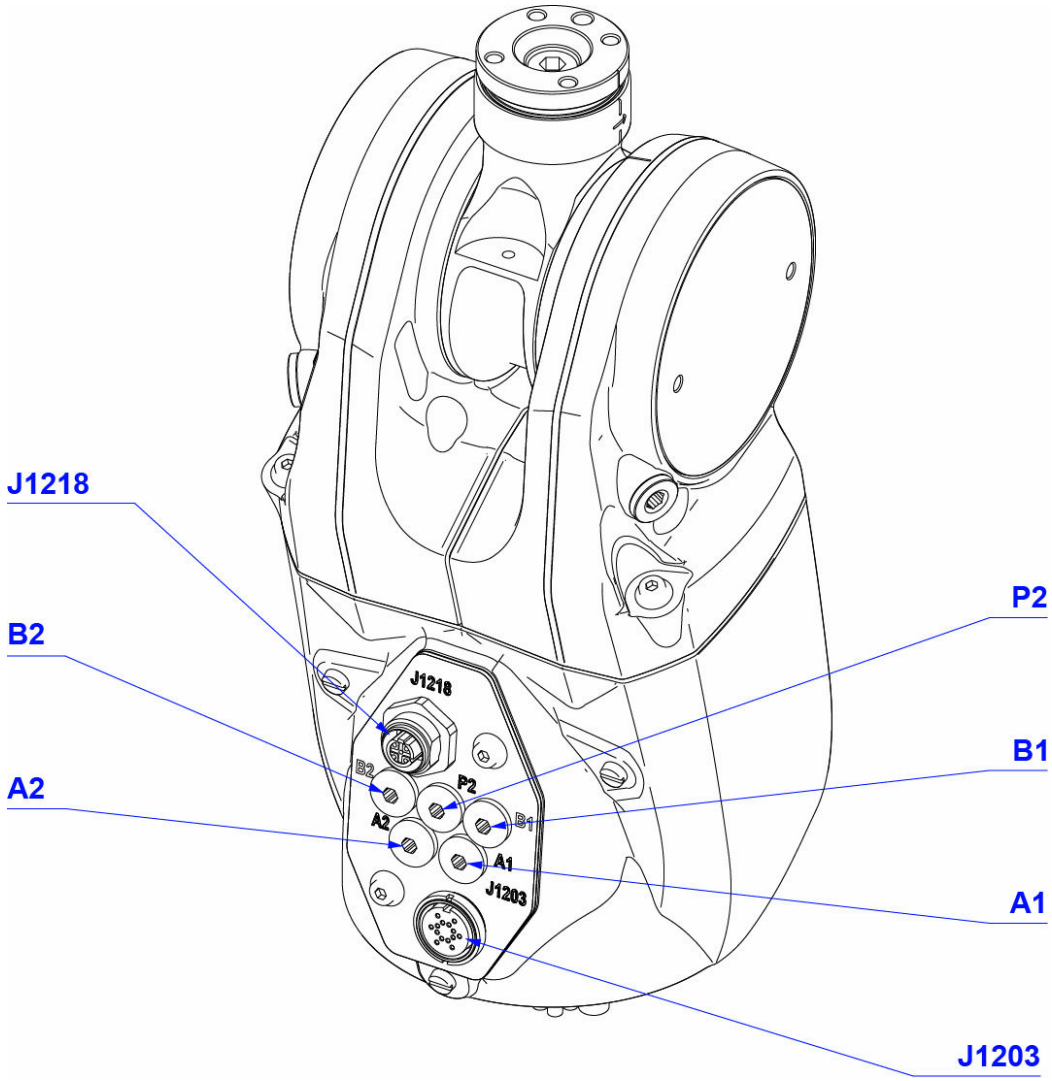


Figure 2.11:

Item	Description
J1203	Socket for electrical connection of the tool by the user where applicable
J1218	Socket for a Cat 5e Ethernet link or an electrical connection for the tooling
A1, A2, B1, B2, P2	Pneumatic connections

Tapped holes are available for the user to attach additional load, harness,... (see chapter 3.3.5).

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2.5.2 - FLANGE CONNECTIONS (HOLLOW WRIST ONLY)

M0009501.1

No valve 2 lines + Fieldbus

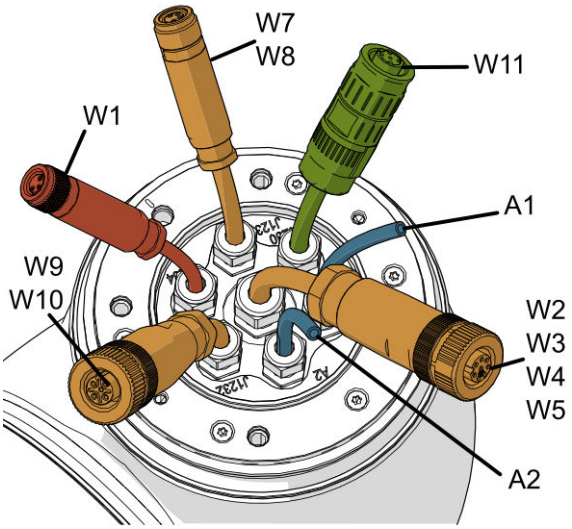


Figure 2.12: Flying connectors

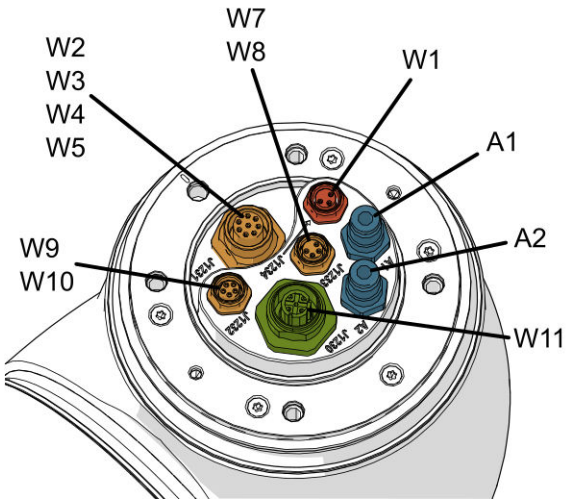


Figure 2.13: Flange-mounted connectors

Item	Description
W1	Power supply triplon (3 x AWG20, Black, Red, White)
W2 - W10	8 I/O pairs (2 x AWG24, Red, Blue) labelled W2 to W10
W11	Ethernet
A1, A2	Pneumatic connections

No valve 2 lines

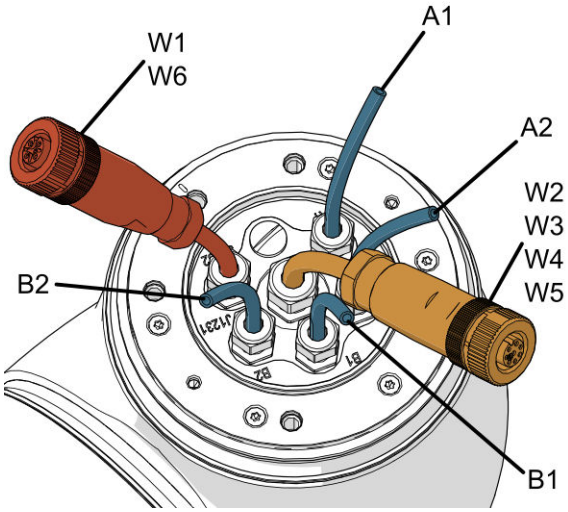


Figure 2.14: Flying connectors

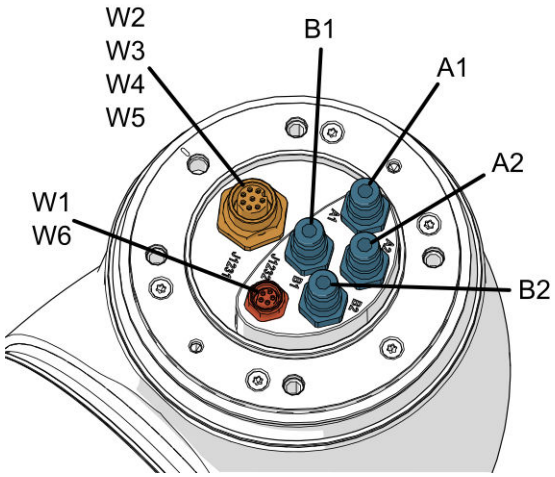


Figure 2.15: Flange-mounted connectors

Item	Description
W1, W6	Power supply triplon (3 x AWG20, Black, Red, White) (W1) and 1 I/O pair (2 x AWG24, Red, Blue) (W6)
W2 - W5	4 I/O pairs (2 x AWG24, Red, Blue) labelled W2 to W5
A1, A2, B1, B2	Pneumatic connections

2.6 - BRAKE RELEASE FUNCTION (OPTION)

M0006457.1

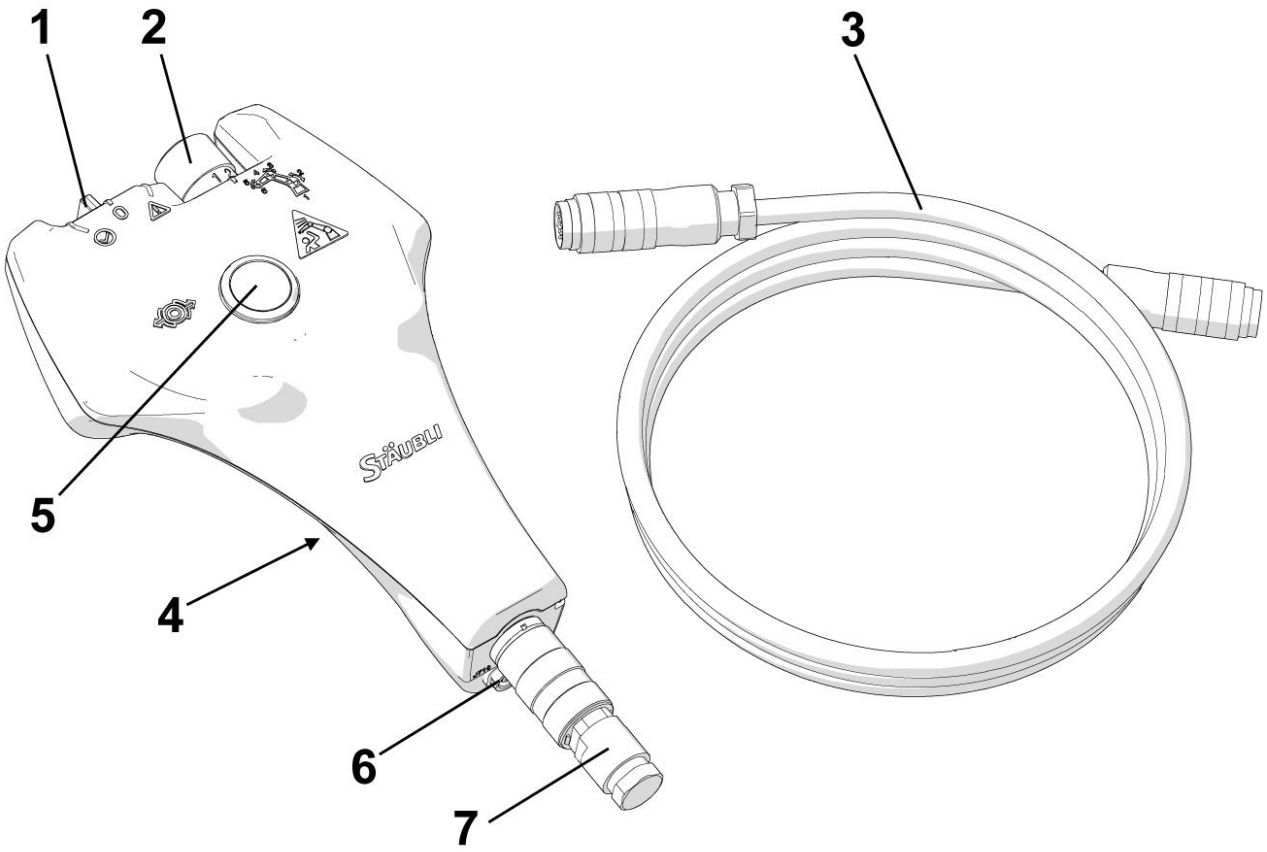
The brake release function from SP2/SP2+ is secured with the enabling device, an electrical braking of the motor, and a low safe speed limit.

- The electrical braking and the speed limit allows only slow movements; the RBR brake release function must be used when faster movements are required.

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2.6.1 - RBR INTERFACE DESCRIPTION

M0006489.1



Item	Description
1	RBR mode: Normal/OFF/Reduced safety.
2	6-position turn switch to select the axis to release.
3	Connection cable to the arm (J1204).
4	Enabling device to enable brake release opening.
5	Push button to release the selected brake.
6	Plug for external 24 V ±5% (1.5 A) PELV* power supply, not protected against overcurrent.
7	Connector for the arm connection cable.

Figure 2.16:

The RBR interface is designed to be usable with a single hand, left or right.

2.6.2 - RELEASING JOINT BRAKE

M0006898.1

The brake on a joint can be released to move the robot manually, for example to quickly clear the cell during an emergency stop, or following a failure affecting the controller (amplifiers) or the arm (motor, coder).

The SP2/SP2+ includes an interface to control each brake from the pendant. The use of the SP2/SP2+ interface for brake release is described in the chapter "Manual Brake Release" of the controller instruction manual.

* PELV according to EN 60204-1 6.4.2

The RBR interface is an alternate way to release a joint brake, when the SP2/SP2+ is not operational during installation or maintenance phases, for example.
The behavior when releasing a brake depends on the brake control mode:

- Using the SP2/SP2+ interface, an electrical braking is activated to limit the speed and acceleration of the axis falling due to gravity. Moreover, a safety function engages the brake when the speed exceeds a predefined threshold.
- Using the RBR normal mode, no electrical braking is activated, but the speed-limiting safety function remains active.
- Using the RBR reduced safety mode, neither electrical braking nor speed limit are activated.

⚠ WARNING

Residual risks
The operator shall be informed of the risks when operating near the arm before using the brake release interface. Residual risks are listed in chapter 3.2.5.

⚠ DANGER

Mandatory training
Use of the RBR by untrained personnel will result in serious injury or death.
• Wherever possible, operation near the arm shall be done out of the reach of a falling axis.
• The arm must be secured before any operation under its mechanical parts.
• The RBR interface shall be used only with an arm whose base has been fastened to the mounting surface.
• Depending on the arm and the axis, releasing the brake may lead to an unexpected move.

2.6.2.1 - Normal mode

M0006749.1

- Connect the RBR interface to the J1204 connector at the arm base.
- Select the normal mode position on the switch **(1)**.
- Wait 3 seconds.
- Select the brake to open with the switch **(2)**.

⚠ DANGER

Wrong selection
If the wrong brake is selected, an unexpected arm movement will occur.
Make sure the selection is correct.

- Press the enabling device **(4)** in its intermediate position.
- Press the push button **(5)** to open the brake, release it to close the brake.

There is a delay of 3 seconds until brakes can be opened when coming from mode "OFF" to "Normal mode".

- After use, select the OFF position on the switch **(1)**, or remove the RBR interface from connector J1204. Press the "Update" button on the controller, or reboot it, to resume operation.

DANGER

Safety function for RBR:
Using the RBR normal mode, no electrical braking is activated, but a safety function engages the brake when the speed exceeds a predefined threshold.
The axis must be supported before opening the brake.

- The brake is engaged automatically when the speed exceeds a predefined threshold. The enabling device must be released and pressed again to reopen the brake.
- If this safety function cannot be activated due to a fault, for example a defective encoder, the brake release function will not be possible on normal mode position. In that case, the switch **(1)** must be placed in reduced safety mode position to enable brake release.

2.6.2.2 - Reduced safety mode

M0006700.1

- Connect the RBR interface to the **(J1204)** connector at the arm base.
- Select the normal mode position on the switch **(1)**.
- Wait 3 seconds.
- Select the brake to open with the switch **(2)**.

DANGER

Wrong selection
If the wrong brake is selected, an unexpected arm movement will occur.
Make sure the selection is correct.

- Select the reduced function mode position on the switch **(1)**.
- Press the enabling device **(4)** in its intermediate position.
- Press the push button **(5)** to open the brake, release it to close the brake.

- There is a delay of 3 seconds until brakes can be opened when coming from mode "OFF" to "Normal mode".
- Before selecting the reduced function mode, it is necessary to stay first 3 seconds in the normal mode position with switch **(1)**.

- After use, select the OFF position on the switch **(1)**, or remove the RBR interface from connector J1204. Press the "Update" button on the controller, or reboot it, to resume operation.

DANGER

Safety function for RBR Reduced safety mode:
Using the RBR reduced safety mode, neither electrical braking nor speed limit are activated.
The axis must be supported before opening the brake.

3 - MACHINE DESIGN

3.1 - ROBOT USAGE

M0009592.1

Preamble

This documentation is aimed at industrial users and manufacturers only. Integration activities must be performed only by appropriately trained and skilled personnel with the following knowledge and skills:

- Advanced knowledge of mechanical engineering,
- Advanced knowledge of electrical engineering,
- Advanced knowledge of the Stäubli robot through the Stäubli course training program.

Intended use

As a partly completed machinery, the Stäubli robot is not a ready-to-use product and must not be used as such. The robot is intended to be undergo further construction, namely, to integrate a system, to perform the specific application for which the robot is designed.

The robot is intended to be integrated in industrial environments, in consistency with ISO 10218-1 perimeter, for a wide range of industrial applications including automotive and equipment manufacturing, machine tending, plastics, where the operator is safely separated from the moving robot with protective devices.

Counter-indications

⚠ WARNING

Counter-indications

With or without integration, use of the Stäubli Robot is expressly counter indicated in any of the following circumstances as well as for any of the following applications:

- Any outdoor operation,
- In explosive areas (ATEX),
- Airborne and space, including outer space,
- Military (defence, harmful or lethal to human) and law enforcement,
- As part of a medical or healthcare robot
- Operation in direct contact with food or drugs,
- As prosthetics and other aids for the physically impaired,
- As service robots, which provide a service to a person and as such where the public can have access.

⚠ WARNING

Robot modification

Modifying the robot voids the warranty and exposes the user to unpredictable risks.

Do not modify or alter the Stäubli robot without Stäubli's prior indication.

⚠ WARNING

Personnel lifting

Do not use the Stäubli robot as a personnel lift, footbridge or climbing aid.

Integration best practices

To have the best performance on your application please have a look on the following integration best practices:

- The tool must be as stiff as possible to get the best accuracy performances.
- The robot support must be as stiff as possible to get the best accuracy performances.
- If your application needs absolute positioning, please assure a controlled temperature around the robot and make sure that the robot is at an established temperature.
- The maximum payload must be respected to prevent any safety issue (see chapter 3.3.4.1).
- Please take into account the maintenance operations in your design to enhance your robot lifetime with periodic interventions.

If your application needs a specific evaluation Stäubli is at your disposal for a custom evaluation.

Use best practices

- Check your screw tightening torques when the temperature varies a lot.
- Avoid singularities to get the best accuracy performances.
- The path shape and type must be chosen with care when accurate movement is needed. A smooth path will lead to a better precision.
- If your application needs the best duty cycle, it is possible to overpass the nominal speed up to maximum speed. Double check the support and tool and verify the temperature and lifetime of your robot. It will lead to a loss of accuracy along the path (vibrations, …).
- It is also possible to optimize the duty cycle by optimizing the robot position with respect to the cycle path. A path close to the foot will allow having big accelerations while a path far from the foot will allow increasing the Cartesian speed.

3.2 - SAFETY REQUIREMENTS

M0006759.1

3.2.1 - SAFETY REGULATIONS

M0006877.1

European regulations

In the European Union, the applicable safety regulations for a machine are defined by the Machinery Directive 2006/42/EC. This directive specifies that a risk assessment shall be carried out in order to determine the health and safety requirements which apply to the machinery. The machinery must then be designed and constructed taking into account the results of the risk assessment. The directive defines essential health and safety requirements that are mandatory.

International standards have been developed to provide a framework for the correct application of the Machinery Directive. The Machinery Directive, and harmonized standards list, can be found at http://eur-lex.europa.eu. For robot integration, application of the specifications of the EN ISO 10218-2:2011 standard on the basis of the integrator risk assessment confers a presumption of conformity with the essential health and safety requirements of the Machinery Directive covered by the standard.

The Machinery Directive defines the procedures to evaluate the machinery conformity. For completed machinery, the CE mark accompanied by the EC conformity declaration confers presumption of compliance with the Directive. For partly completed machinery, the incorporation declaration defines the essential health and safety requirements and other applicable Directives that are satisfied.

The robot is delivered with its incorporation declaration. It can also be requested from the Stäubli Customer Support contact.

The robot incorporation declaration is based on the applicable requirements implementation. This declaration is as follows: The official declaration of incorporation is formalized at the end of robot production and is delivered with the robot; it can also be requested later from the Stäubli Customer Support contact. The declaration of incorporation below is an illustration. Text could differ from the one provided with the delivered robot.

Declaration of Incorporation in accordance with Machinery Directive 2006/42/EC, annex II B
Declaration of Conformity with EMC Directive 2014/30/EU and 2013/35/EU
Declaration of Conformity with Low Voltage Directive 2014/35/EU
Declaration of Conformity with RoHS Directive 2011/65/EU and 2015/863
Declaration of Conformity with REACH Regulation 1907/2006

The manufacturer: **STÄUBLI FAVERGES**
Address: Place Robert Stäubli, 74210 FAVERGES, France

We hereafter declare that:
The [1] made by **STÄUBLI**
Serial number: Arm [2] [3] F--/------/-/01
Controller [5] [4] F--/------/-/01

- constitute a [6] axis industrial robot,
- The unit is partly completed machinery designed to be incorporated in a machine or to be assembled with other machines, in accordance with the specifications set out in the documentation, to constitute a machine covered by the "Machines" Directive 2006/42/EC
    - the essential requirements in annex 1 have been considered and deliberately met within the limits of the contents of the delivery:
        - Chapter 1 except requirements 1.1.4; 1.1.7; 1.1.8; 1.3.5; 1.4.2.3; 1.5.12; 1.5.14; 1.6.5
        - Chapter 2 except requirements 2.1.1.a; 2.1.1.b; 2.2.1; 2.2.2; 2.3; 2.4;
        - Chapter 4 except requirements 4.1.2.2; 4.1.2.5; 4.1.2.6.e; 4.1.2.7; 4.1.2.8; 4.2.2; 4.2.3; 4.3.1; 4.3.2; 4.4.1
    - Deliberate compliance with these essential requirements at the level of the partly completed machinery does not imply that the corresponding requirements at the level of the machine are complied with
    - The following harmonized standards have been used:
        - EN ISO 12100: 2010
        - EN ISO 10218-1:2011
        - EN 60204-1:2018
        - EN IEC 62061:2021
        - EN ISO 13849-1: 2023
    - has a technical file drawn up in accordance with annex VII B, and that we undertake to forward to the competent authorities on receipt of a duly motivated request to that effect. The person authorized to draw up this file is the signatory of the present declaration.

The manufacturer also declares that under the responsibility of the end user, this partly completed machinery must only be put into production once the final machine in which it is to be incorporated, or of which it forms a part, has been considered and declared as being in conformity with the requirements set out in the "Machines" Directive and in the national legislations transposing it, and with the other legislation in effect in the country in which it is put into service.

- the unit complies with the requirements set out in the "Electromagnetic Compatibility" Directives 2014/30/EU and 2013/35/EU
    - the following European standards have been used:
        - EN 61000-6-2: 2019
        - EN 61000-6-4: 2019
        - EN 61000-6-7: 2015
        - EN 62311: 2020

- the unit complies with the safety objectives set out in the Low Voltage Directive 2014/35/EU.
- the unit complies with RoHS Directives 2011/65/EU and 2015/863
- the unit complies with REACH Regulation 1907/2006
- the CE marking placed on the robot only applies to Directives 2014/30/EU, 2014/35/EU, and 2011/65/EU, in accordance with European legislation.

Faverges, [7]
[8]

Item	Description
1	Commercial designation
2	Arm: Model
3	Arm: Serial number
4	Controller: Serial number
5	Controller: Model
6	Number of axes
7	Date
8	Signature

In the United States, there are no standardized Occupational Safety and Health Administration regulatory standards for robots. However, standard ANSI/RIA R15.06-2012 is referenced as the standard applicable to robot systems. That standard is harmonized to make it basically identical to standards ISO 10218-1:2011 and ISO 10218-2:2011. These ISO standards for industrial robots have also been adopted in Korea.
Each country can draw up further safety standards, and implement specific local rules. For example, the UL standard for robots and robot equipment UL 1740 sets out further requirements covering the robot and electrical safety.

For the UL version, the robot system has been evaluated and certified conform to the following standards:

- Standard UL 1740 Robots and Robotics Equipment.

UL version robot is suitable to be integrated in robotic systems:
- Which conform to Electrical Standards for Industrial Machinery NFPA 79.
- Which conform to National Electrical Code NFPA 70.
- Evaluated following applicable sections of American National Standard for Industrial Robots and Robot Systems, Safety Requirements ANSI/RIA R15.06.

For relative information about REACH 1907/2006 regulation (CE) consult the following internet address: https://www.staubli.com/en/robotics-reach/

3.2.2 - ESSENTIAL HEALTH AND SAFETY REQUIREMENTS

M0006891.1

When addressing the essential health and safety requirements defined by the applicable machinery regulations, the integrator shall address the requirements related to the robot application. To have a clear view of the essential requirement covered by staubli please refer to intended use in chapter 3.2.1 and compliance declaration.

General-purpose requirements related to the robot are highlighted hereafter:

Principles of safety integration

The completed machinery shall be designed so that the robot is used according to its specifications, and can be used, adjusted and maintained without risk.

Ergonomics

Robot tuning conditions (point teaching, motion tuning) shall be addressed when designing the robot end-effector and cell environment.

Control & information devices

The interaction means between the robot and the operator shall be compliant at least with requirements set out in applicable machinery regulations and with additional regulations that may apply locally.
When there is more than one robot control position in automatic mode, the control system shall be designed so that the use of one of them precludes the use of the others, except for stop controls and emergency stops.

Risks related to moving parts

Contacts between the robot and the operator shall be prevented by designing adequate guards or protective devices.

Instructions

The integrator may have to translate the robot instructions if the completed machinery is put into service in a country differing from the country where the robot was delivered.

The instructions for the completed machinery shall include the robot manuals, or manuals parts that are needed to use, adjust, and maintain the robot.

The sales literature of the completed machinery shall be compliant with the robot instructions.

3.2.3 - SAFETY FUNCTIONS

3.2.3.1 - Performance of safety functions

M0006660.1

Function	ISO 13849-1	IEC 62061	PFH (h^{-1})	Description
5/2-way solenoid valve control	61.6 million	-	-	B_{10d} (number of commutation cycles reached by 90% of the devices without dangerous failure). See also "Safe control of external actuators" chapter of the Safety manual.
5/3-way solenoid valve control	61.6 million	-	-	
3/2-way solenoid valve control	36 million	-	-	
Brakes For all axis of the CW, and for axis 1, 2, 3 of the HW	20 million	-	-	B_{10d} (number of commutation cycles reached by 90% of the devices without dangerous failure). See also "Brake testing" chapter of the Safety manual.
Brake (Axis 4) For HW only	6 million	-	-	
Brakes (Axis 5 and 6) For HW only	4 million	-	-	
RBR enabling device brake lock	Cat4, PLe	SIL3, HFT1	$<10^{-7}$	In remote brake release mode: ■ Maintain the brakes closed when not activated, ■ Close the brake immediately when released with an open brake. [1]
RBR limited speed	Cat3, PLd	SIL2, HFT1	$<10^{-7}$	In remote brake release mode, the brake is engaged immediately when the axis speed exceeds a predefined threshold. [1]
[1] The RBR enabling device and limited speed safety functions disable the safety brake power. The brakes must be tested regularly for safe operation.				

A SIL2/PLd Category 3 safety function is usually adequate for a temporary protection of the operator. A frequent protection of the operator may require a higher performance of the safety function.

DANGER

Safety functions performance and parameters definition

The required performance and parameters for the use of the robot safety functions shall be defined from the identified risks in the completed machinery.

DANGER

Safety functions for regular operator intervention near the robot
If regular operator intervention near the robot is planned, the safety functions of the control system shall be designed with an adequate performance level, and the residual risks near the robot listed in chapter 3.2.5.

3.2.3.2 - Characteristics of the solenoid valves

M0006863.1

Test conditions:

- No lubrication,
- Dry air,
- Cycle frequency: 3 Hz (5/3) - 5 Hz (3/2 and 5/2),
- Pressure: 7 bar (5/2 and 5/3) -0.53 bar (3/2),
- Ambient temperature.

3.2.3.3 - Diagnostic tests

M0006590.1

When safe position limits are used, diagnostic tests are required to ensure that the robot abide by the limits over time:

- The referencing test checks the robot calibration.
- The brake test checks the braking torque.

The criteria to select the period of the referencing and brake tests are defined in the Safety Programming manual.
The minimum acceptable brake level is 50%, if the brake test level is lower, please plan a maintenance. Additionally when the SafeCell/SafeCell+ option is enabled, a degrated mode will be forced to limit the arm speed (250 mm/s).
A regular brake test is also recommended when there is a risk that a person is exposed to the consequence of a brake failure. The brake test can be initiated from the pendant interface (see the brake test chapter of the controller manual), or with a VAL 3 instruction (see the brake test chapter of the VAL 3 reference manual).

3.2.4 - SAFETY EQUIPMENT

3.2.4.1 - Remote Brake Release interface (RBR)

M0006690.1

The RBR is equipped with 4 safety devices:

- A 6-position rotary switch to select the axis to be released,
- A push button to release the brake selected with the rotary switch,
- An enabling device to enable brake release opening,

The RBR is also equipped with a plug for an external 24 V PELV power supply.
The RBR is an optional equipment. A brake release interface is also provided on the SP2/SP2+.
The RBR is needed:

- During installation or maintenance phases, when the controller is switched off or not connected to the arm,
- After a fault in the controller or SP2/SP2+ hardware/software that prevents the use of the SP2/SP2+ brake release interface.

DANGER

Unstudied use of the RBR
The use of the RBR in a situation that does not require it is hazardous.
This risk shall be covered by the integrator's risk analysis of the cell.

DANGER

Misplaced RBR
A misplaced RBR can be inaccessible when needed or misused.
The integrator's risk analysis shall define the location of the RBR.
The RBR may remain connected to the arm, or it can be connected only when needed.

DANGER

Untested RBR interface
Using a defective RBR interface during an emergency situation will result in serious injury or death.
Test the RBR interface yearly.

3.2.4.2 - RBR external 24V PELV power supply (24V2 circuit)

M0006868.1

When the controller is switched off, the RBR interface requires a PELV external power supply to be activated.

DANGER

Non-PELV 24V power supply
Without a PELV 24V power supply, the system is exposed to overvoltages.
The external 24V power supply for the RBR shall be protected extra-low voltage (PELV) (according to EN 60204-1 6.4.2).

DANGER

Unstudied use of RBR external 24V PELV power supply
There is a risk of a trapped person or production stop if a failure in the controller power supply prevents the use of the RBR to release the arm.
The choice to use of an external 24V PELV power supply for the RBR shall be covered by the integrator's risk analysis of the cell.

3.2.5 - RESIDUAL RISKS

M0006600.1

WARNING

Residual risks
The identified residual risks and foreseeable misuse shall be handled by the integrator, and completed with specific risks and foreseeable misuse in the context of the completed machinery. See Safety Programming manual for more information.
The residual risks shall be eliminated or reduced as far as possible.
Protective measures shall be taken in relation to the risks that cannot be eliminated.
Users shall be informed of the residual risks.

3.2.5.1 - Absence of defects

M0006578.1

The robot must only be operated in faultless conditions. It is recommended to keep records of all safety related actions on the robot and on the cell in a "safety book" or the like. Special care must be taken during maintenance and restarting phases. See Safety Programming manual and maintenance manual.

- Risks of using the robot under degraded conditions (restarting under conditions of reduced safety, with damaged parts or safety equipment, etc.).
- Risks of using unsuitable spare parts (fuses, fixings, inappropriate second-hand parts, etc.).
- Risks of using the robot with a damage harness: Loss of the earth grounding of on segment of the arm and/or loss of air pressure in pneumatic hoses used by customer tool.

3.2.5.2 - Adequate conditions of use

M0006691.1

Risk-free operation of the robot in its cell does not depend solely on its safety functions.

- Risks of using the robot with protective equipment disabled (sensors, switches, validation devices, etc.).
- Risks of changes in the surrounding conditions (electrical, electromagnetic, temperature, dust, wet, etc.).
- Risks of changes in the operating conditions (load, speed, acceleration, etc.).
- Risks of using the robot as a tool for maintenance operations (part holder, support for personnel, etc.).
- Risks occurring when several people are working near the robot, with only one person being protected by safety equipment.
- Risks of inadequate earth grounding: If an external electrical power supply is brought to equipment added on the robot arm, an adequate earth grounding must be added on this equipment.
- Risks of wearing inadequate clothing near the robot: pinching loose clothing, loose long hair, jewelry, etc....

3.2.5.3 - Risk signaling

M0007145.1

ISO 10218-1, 5.7.3 recommends that "wherever possible, the manual mode of operation shall be performed with all persons outside the safeguarded space". It also reminds that prior to selecting automatic mode, any suspended safeguards shall be returned to their full functionality.

- Risks of uncontrolled movements, due to gravity (faulty brake) or an outside force (maintenance operation on the arm or its tool). It is necessary to ensure that the brakes are operational before working near the arm. The robot must be secured before working underneath the arm.
- Risk of being entrapped by a moving part, or crushed when caught between the moving robot and its environment in the cell.
- Risks of contact with sharp edges on the moving robot, its tool, or the part held, even at low speed.
- Risks of contact with sharp edges on the controller.
- Risks of contact with hot parts:
  - The arm surface temperature can reach 80°C (176°F) after intense use.
  - Any restriction of movement that could limit the user's ability to move away from the robot (direct contact) is prohibited.
- Risks stemming from activation of digital outputs (fall of a part on opening of a gripper, activation of movement for other parts in the cell, etc.).
- Risks stemming from reduced levels of attention to other hazards in the cell when concentrating on the robot (slipping, tripping, falls, contact with moving parts, sharp edges, etc.).
- Risks of sudden operator reaction to unexpected robot motion, even at low speed (inappropriate operator command, large joint move induced by a small TCP move, unexpected command from program...).
- Risk of oil leakage, despite the presence of sealing on all axes.

3.2.5.4 - Emergency rescue procedure

M0006954.1

If, after a hazardous event during a collaborative operation, a tuning or a maintenance task, the operator may possibly be trapped or injured by the robot or its environment, a rescue procedure shall be established, and the staff shall be trained to react promptly. The procedure and training shall identify the means to stop the hazards in the cell (such as emergency stops), and to unblock a possibly defective robot using the brake release safety function (see chapter 2.6).

3.3 - MACHINERY REQUIREMENTS

3.3.1 - WORK ENVIRONMENT

M0006663.1

3.3.1.1 - Standard recommendations

M0007033.1

NOTICE

Arm damage
Failure to observe the following environmental conditions will result in damage to the arm and void the warranty.
- Working temperature: + 5°C to + 40°C (+ 41°F to + 104°F)
- Temperature for storage and transport: -25°C (-13°F) to 55°C (131°F)
- Humidity: 30% to 90% (non-condensing)
- Altitude: 2000 m max.
- Vibrations during operation: Please consult us

NOTICE

Impact of temperature on robot performance
In hot conditions, the robot thermal protections may lead to robot cycle time limitations.
In cold conditions, it may be necessary to perform a warm-up cycle before nominal performances are obtained.

Cleaning and disinfecting recommendations (see chapter 7.7).

NOTICE

Contamination:
A possible leakage from the robot arm might contaminate the food or drug.
The integrator must install some system to prevent any contamination.

- Noise at nominal conditions: 70 dBA (measured at 1 m away from movement and at 1.6 m high).
- Impact protection: The mechanical protection index is IK08 for the arm, except IK07 for the external connectors on base interface and forearm.

NOTICE

Radiations
Exposing the equipment to unusual levels of radiation (e.g., microwave, ultraviolet, laser, x-ray) may result in undesirable effects.
Contact the Stäubli support team to determine what additional measures must be taken to avoid malfunction of the equipment and accelerated deterioration of its insulation.

3.3.1.2 - Additional information for cleanroom application

M0006943.1

- SCR: Cleanliness class ISO 2 in accordance with standard ISO 14644-1.
- CR: Cleanliness class ISO 4 in accordance with standard ISO 14644-1.
- Stericlean +: Cleanliness class ISO 5 in accordance with standard ISO 14644-1 and cGMP Grade A.

3.3.1.3 - Additional information for humid environment application (HE)

M0006528.1

- Limits of application:
  - 4,5 < pH < 8,5.
  - Resistance to salt spray: 300 hours in accordance with standard NS EN 60068-2-11.
  - Use of chlorinated products for cleaning is prohibited.
  - The arm must be pressurized with the Stäubli pressurization unit.
    Stäubli declines its warranty if the arm is pressurized by a pressurization unit which doesn't provide exactly the same characteristics.

⚠ CAUTION

Pressurization unit in HE environment
The pressurization unit must be protected from the humid environment.

NOTICE

Inadequate pressurization
Inadequate pressurization may damage the arm and void Stäubli's warranty.
Use a separate pressurization unit for each robot.
Use only the Stäubli pressurization unit or one with the same characteristics.

  - The following elements are not qualified for a humid environment (HE) and they must be protected from that environment:
    - The screw(s) on the flange mechanical interface.
    - The underside of the base.
    - The electrical connection on the forearm (CW robots) or on the flange of the robot (HW robots).
    - The interconnect plate.
    - The equipment on the plate.

    This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.
  - In the event of use of an optional mechanical hard stop system, depending on the environment, the system may show limited chemical resistance. Regularly inspect the system to check that it is in good condition. In the event of deterioration, the system can be replaced. It should be noted that because of its complex shape, the mechanical hard stop system makes the arm more difficult to clean and disinfect.
- Vertical cable outlet only.

3.3.1.4 - Additional information for Accesspharma version (ACCESS)

M0009531.1

TX2-60 Accesspharma robots are specifically designed for Pharma controlled area, and are suitable for Cleanliness class ISO 5 in accordance with standard ISO 14644-1.

- Limits of application:
  - Resistance to isopropanol

3.3.1.5 - Additional information for Stericlean version (STER)

M0007518.1

TX2-60 Stericlean robots are specifically designed for Pharma controlled area, and are suitable for Cleanliness class ISO 5 in accordance with standard ISO 14644-1.

- Limits of application:
  - 4,5 < pH < 8,5.
  - Resistance to salt spray: 300 hours in accordance with standard NF EN ISO 9227.
  - Bio-decontamination with hydrogen peroxide on a wipe (concentration 50%) or in vapor phase.
  - Use of chlorinated products for cleaning is prohibited.
  - The following elements are not qualified for a bio-contamination controlled environment (Stericlean) and they must be protected from that environment:
    - The screw(s) on the flange mechanical interface.
    - The underside of the base.
    - The electrical connection on the forearm (CW robots) or on the flange of the robot (HW robots).
    - The interconnect plate.
    - The equipment on the plate.

    This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.
  - In the event of use of an optional mechanical hard stop system, depending on the environment, the system may show limited chemical resistance. Regularly inspect the system to check that it is in good condition. In the event of deterioration, the system can be replaced. It should be noted that because of its complex shape, the mechanical hard stop system makes the arm more difficult to clean and disinfect.
- Vertical cable outlet only.

3.3.1.6 - Additional information for Stericlean+ version (STER+)

M0007340.1

TX2-60 Stericlean + robots are specifically designed for Pharma aseptic area, and are suitable for Cleanliness class ISO 5 in accordance with standard ISO 14644-1, and cGMP Grade A.
A pharma special coating allows high chemical resistance to decontamination process and global hygienic design fits with aseptic environments requirements.

- Limits of application:
  - 4,5 < pH < 8,5 (pH 2 to 12 under condition, please consult us).
  - Resistance to salt spray: 300 hours in accordance with standard NF EN ISO 9227.
  - Bio-decontamination with hydrogen peroxide on a wipe (concentration 50%) or in vapor phase.
  - Use of chlorinated products for cleaning is prohibited.
  - The following elements are not qualified for a bio-contamination controlled environment (Stericlean) and they must be protected from that environment:
    - The screw(s) on the flange mechanical interface.
    - The underside of the base.
    - The electrical connection on the forearm (CW robots) or on the flange of the robot (HW robots).
    - The interconnect plate.
    - The equipment on the plate.

    This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.
- Vertical cable outlet only.

3.3.2 - WORKING SPACE AND SPEED

M0009203.1

Environment	Wrist	Outlet	Covers	Screws or caps of the covers	TX2-60 Overall footprint	TX2-60L Overall footprint
Standard (STD) Clean Room (CR)	CW	Horizontal / vertical	Standard	Standard head screws	3.1	3.5
Super Clean Room (SCR) Acess Pharma				Hexagonal head screws		
Humid Environment (HE)		Vertical only				
Stericlean (STER)	CW					
	HW		Standard (shoulder, arm, elbow) and hygienic (forearm and wrist)	Hexagonal head screws (shoulder, arm, elbow) and hexagonal head caps (forearm and wrist)	3.2	3.6
Stericlean+ (STER+)	CW		Hygienic	Hexagonal head or flat caps (shoulder, arm, elbow) and hexagonal head screws (forearm and wrist)	3.3	3.7
	HW			Hexagonal head or flat caps	3.4	3.8

The main dimensions are summarized below:

Working space	TX2-60	TX2-60 HW	TX2-60L	TX2-60L HW
R.M max. reach between axis 1 and 5	600 mm	615 mm	850 mm	850 mm
R.m1 min. reach between axis 1 and 5	190 mm	201 mm	209 mm	209 mm
R.m2 min. reach between axis 2 and 5	189 mm	200 mm	208 mm	208 mm
R.b reach between axis 3 and 5	310 mm	325 mm	450 mm	450 mm

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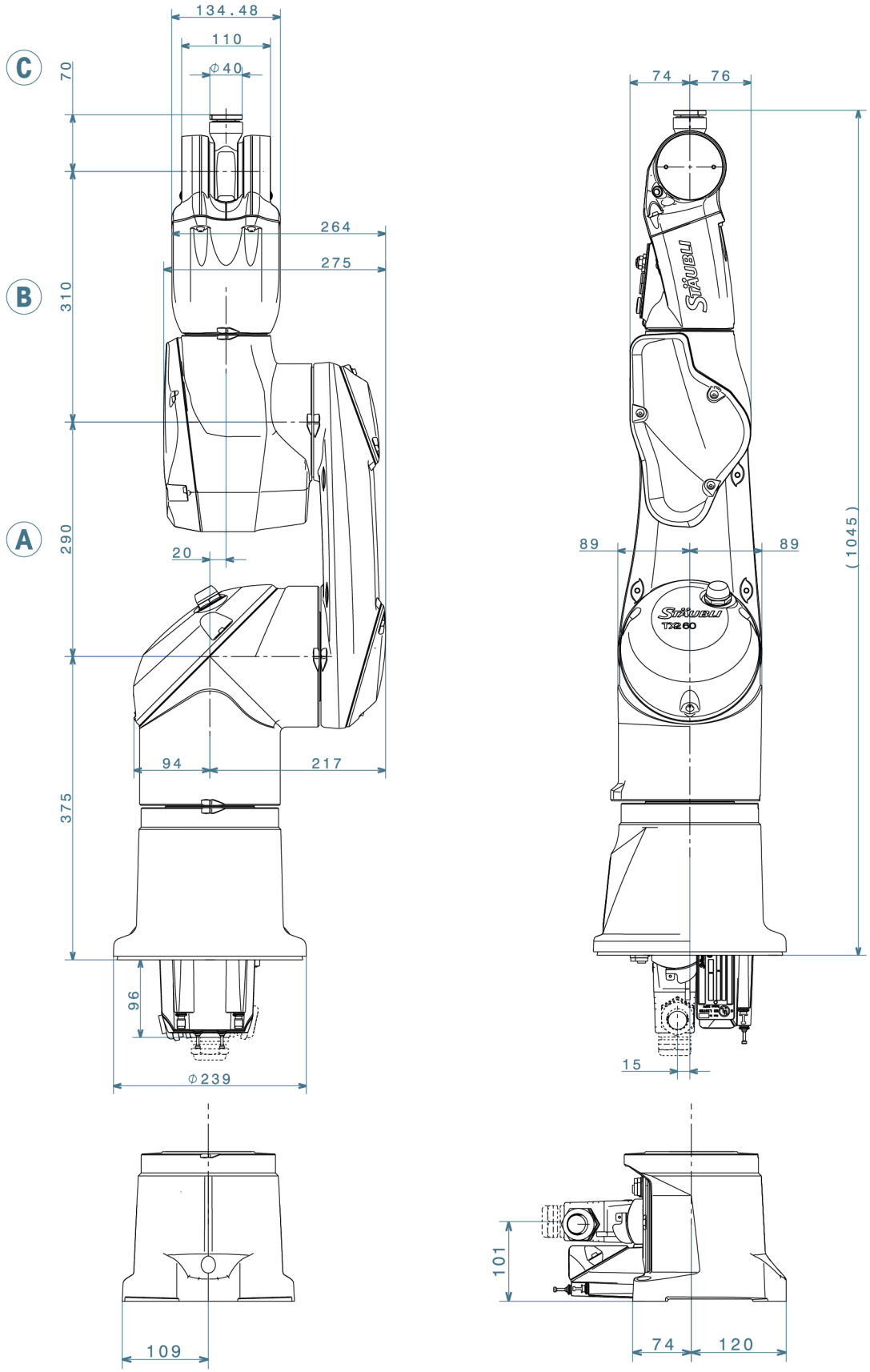


Figure 3.1: Overall footprint of TX2-60 CW with standard covers and standard or hexagonal screws

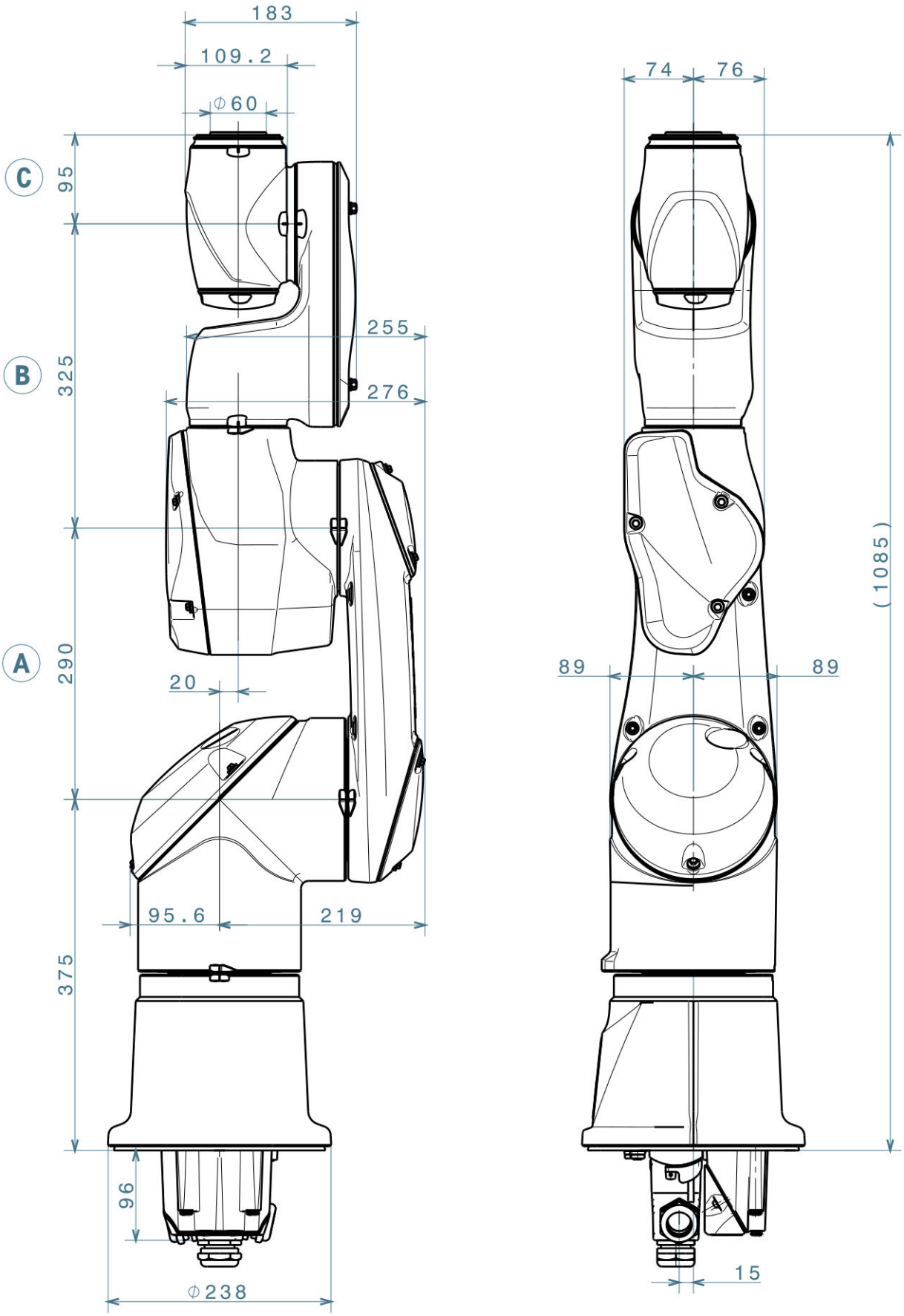


Figure 3.2: Overall footprint of TX2-60 HW with shoulder/arm/elbow with [standard covers and hexagonal screws], and forearm/wrist with [hygienic covers and and hexagonal head caps]

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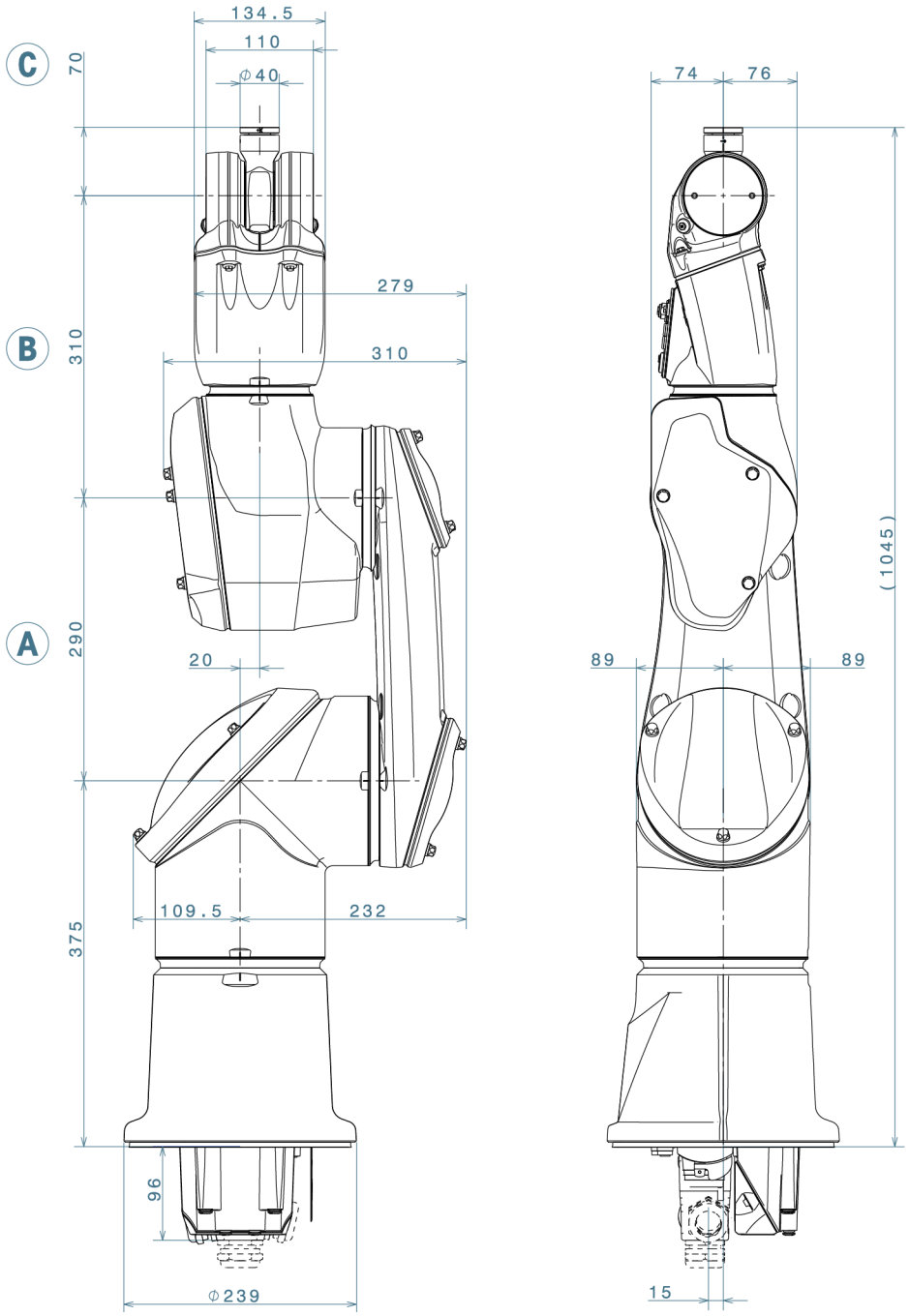


Figure 3.3: Overall footprint of TX2-60 CW with hygienic covers and flat or hexagonal screws

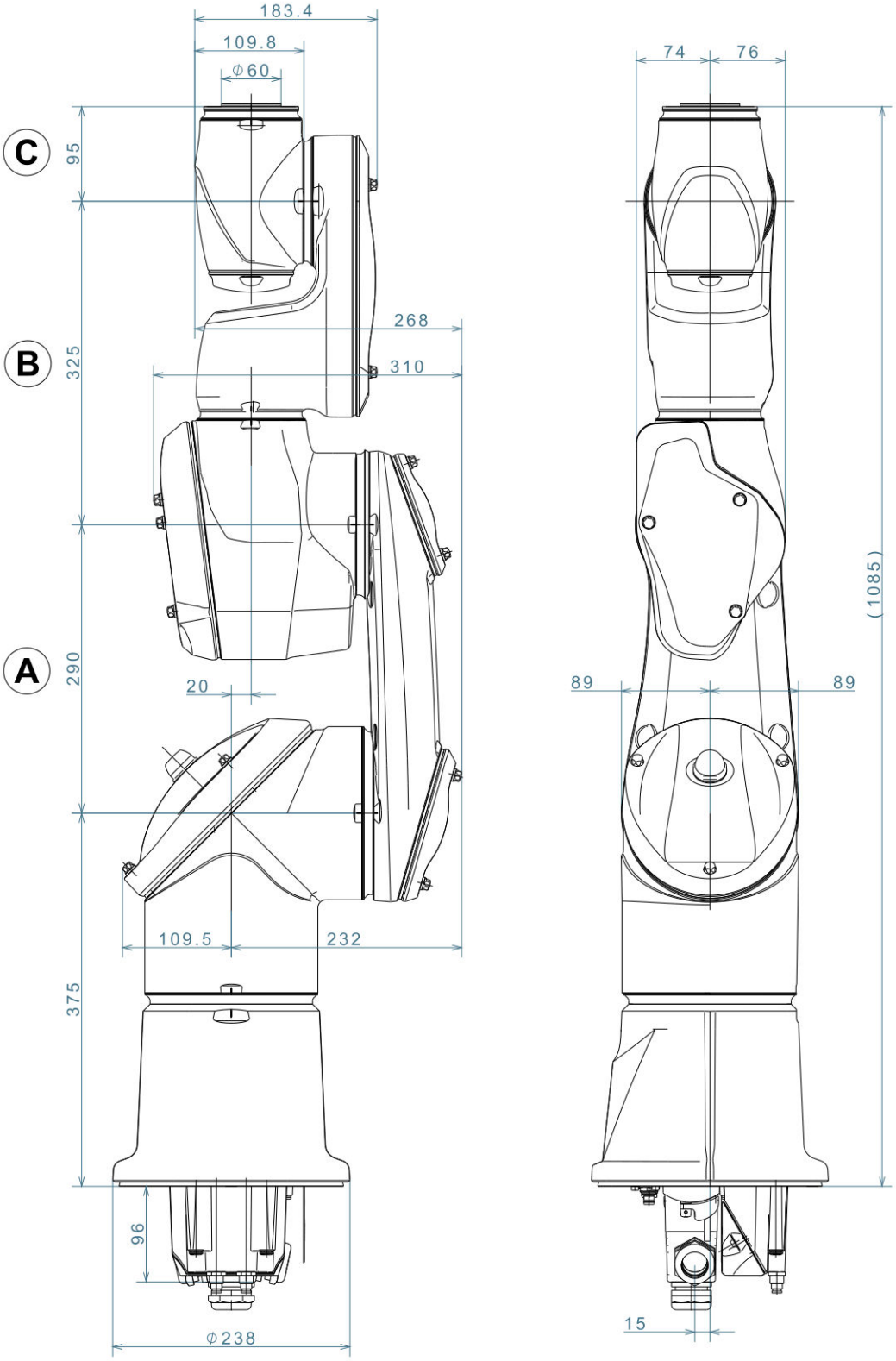


Figure 3.4: Overall footprint of TX2-60 HW with hygienic covers and flat or hexagonal screws

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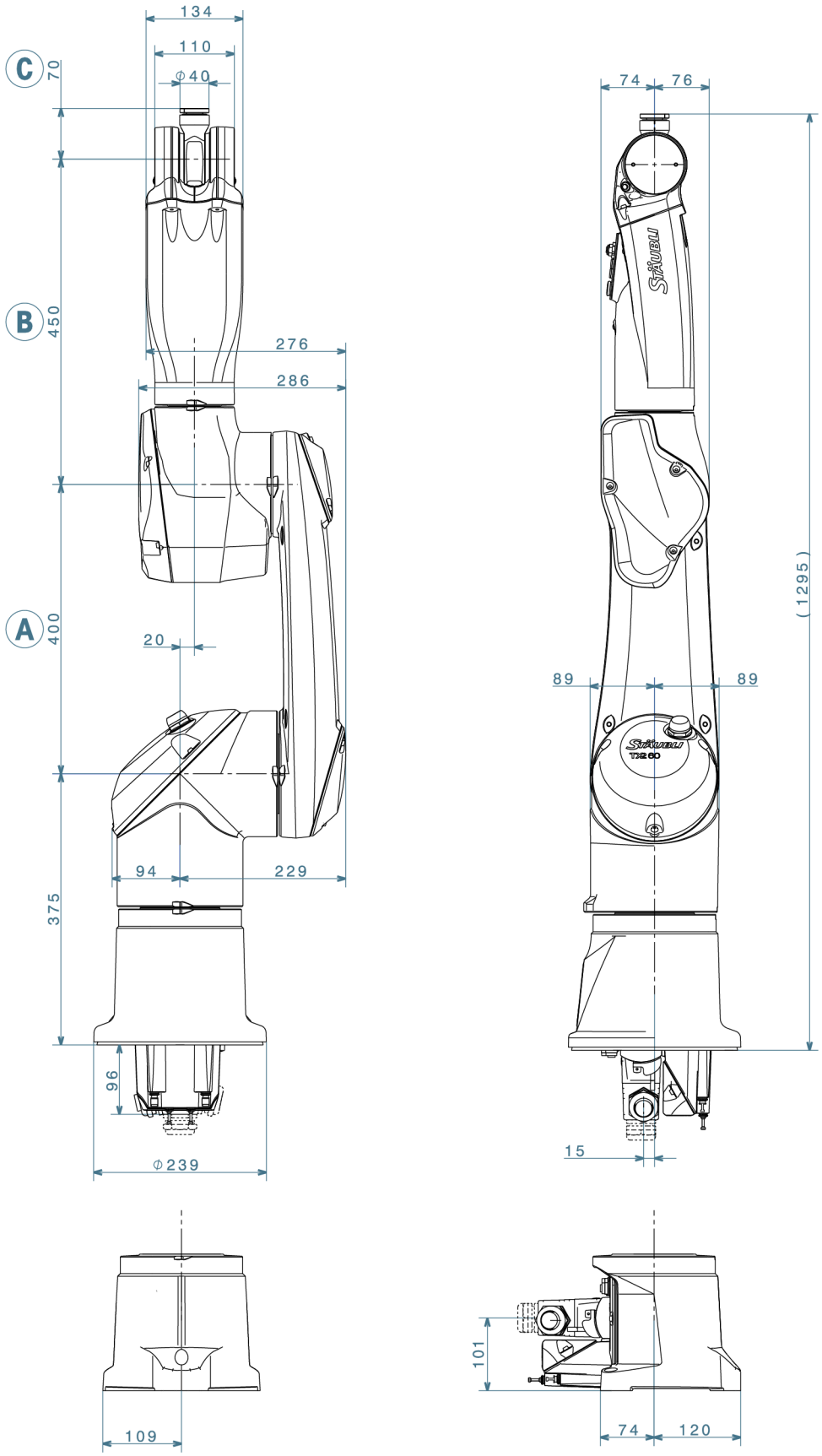


Figure 3.5: Overall footprint of TX2-60L CW with standard covers and standard or hexagonal screws

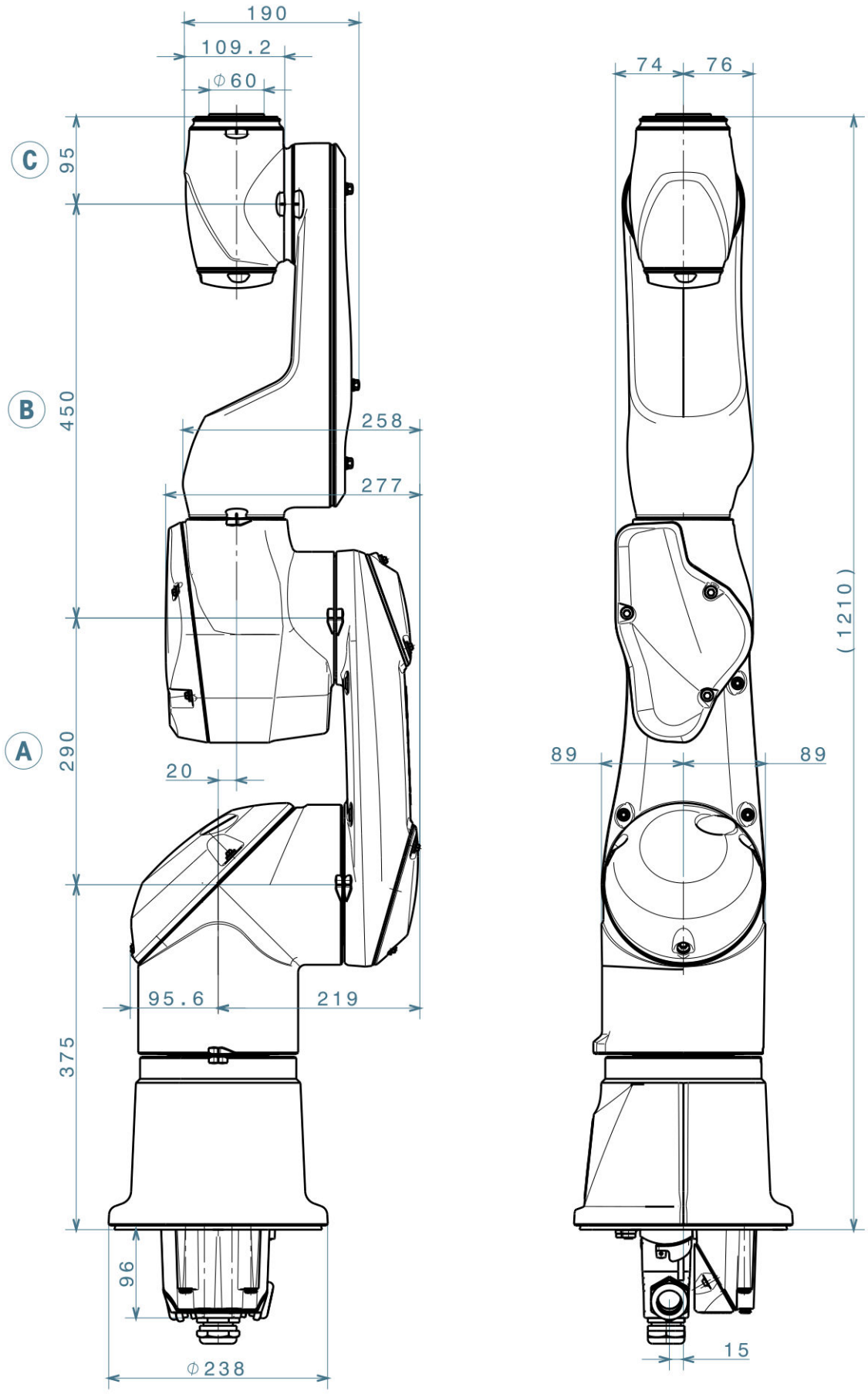


Figure 3.6: Overall footprint of TX2-60L HW with shoulder/arm/elbow with [standard covers and hexagonal screws], and forearm/wrist with [hygienic covers and and hexagonal head caps]

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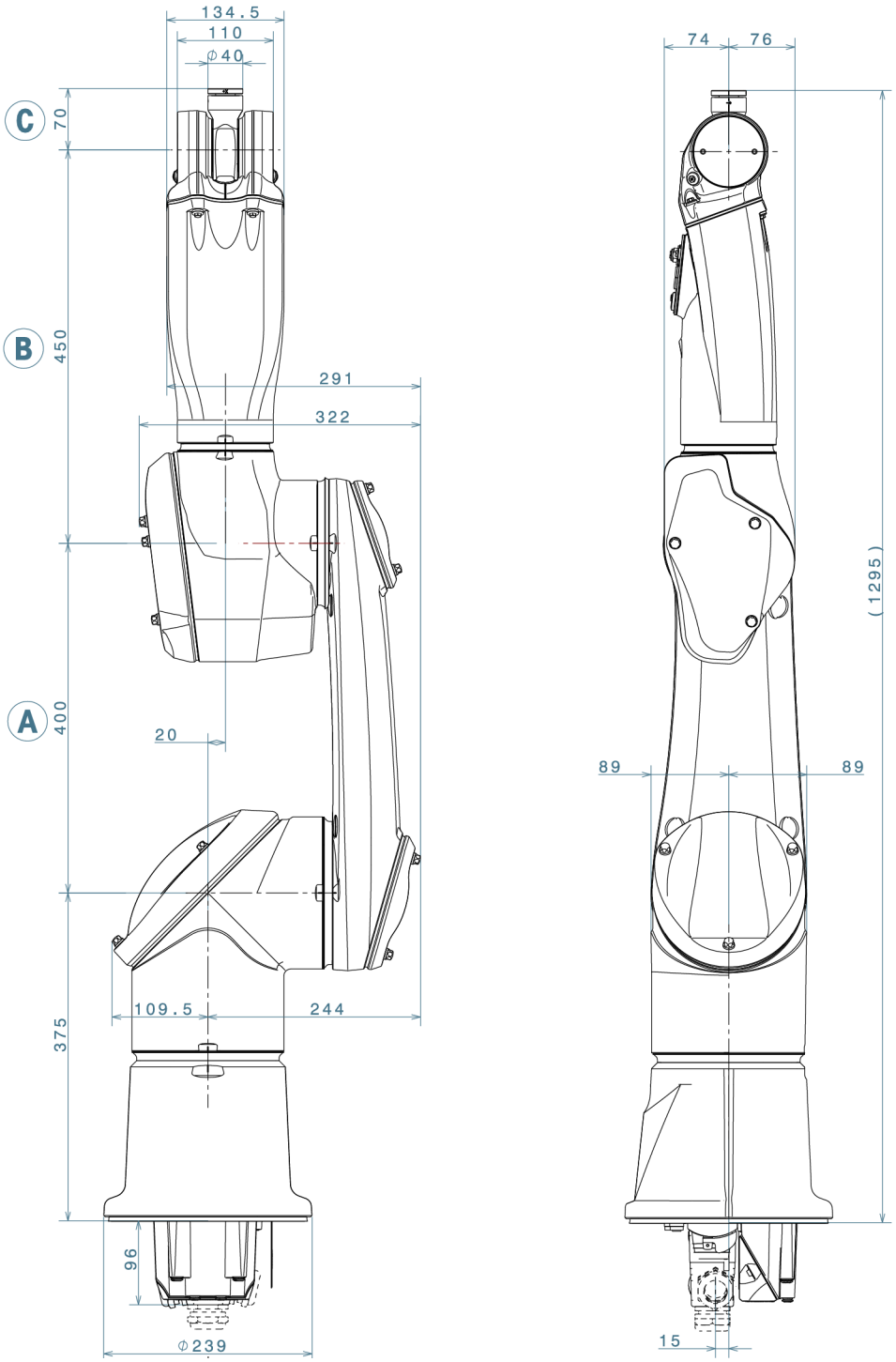


Figure 3.7: Overall footprint of TX2-60L CW with hygienic covers and flat or hexagonal screws

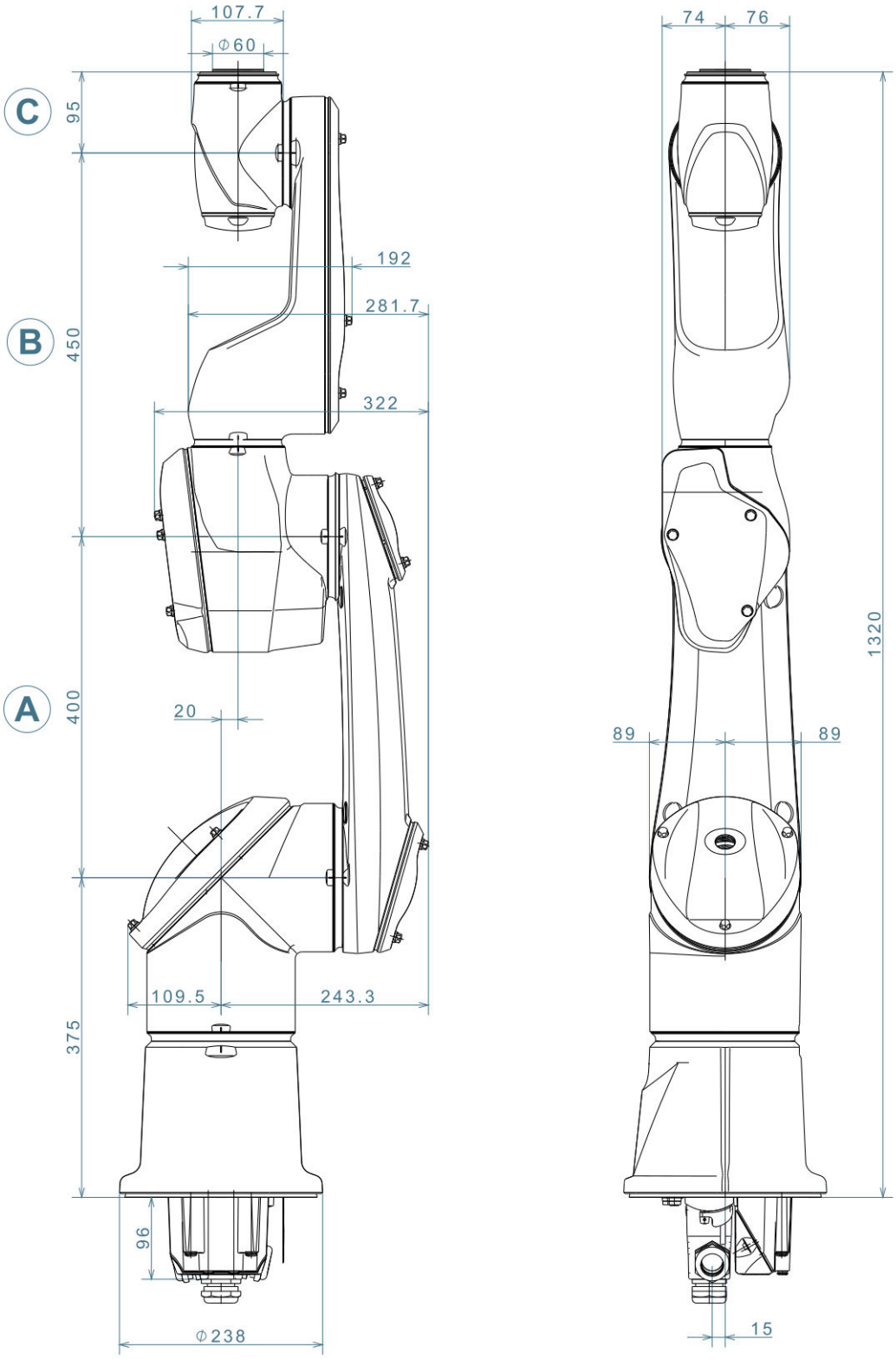


Figure 3.8: Overall footprint of TX2-60L HW with hygienic covers and flat or hexagonal screws

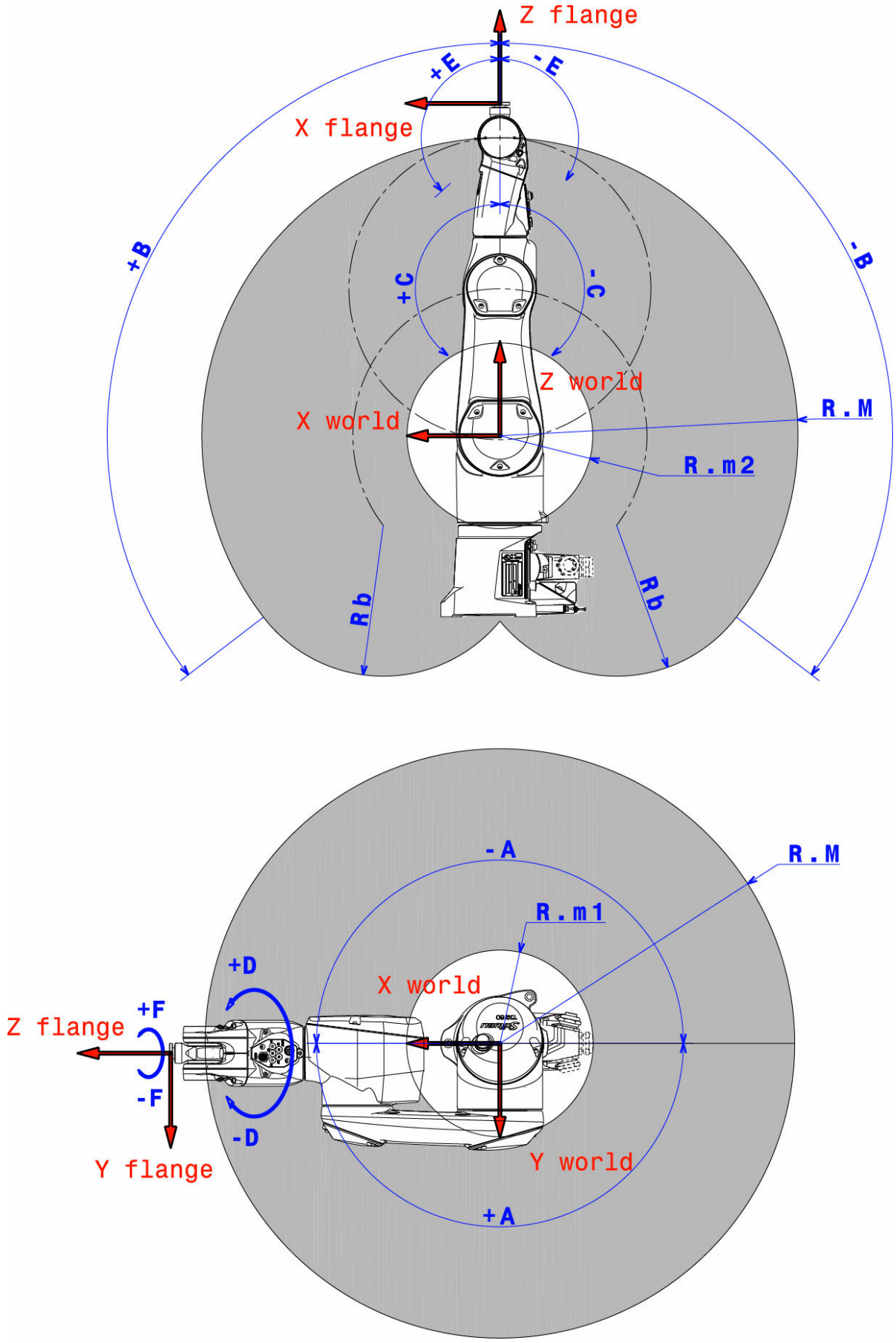


Figure 3.9: Working space, CW

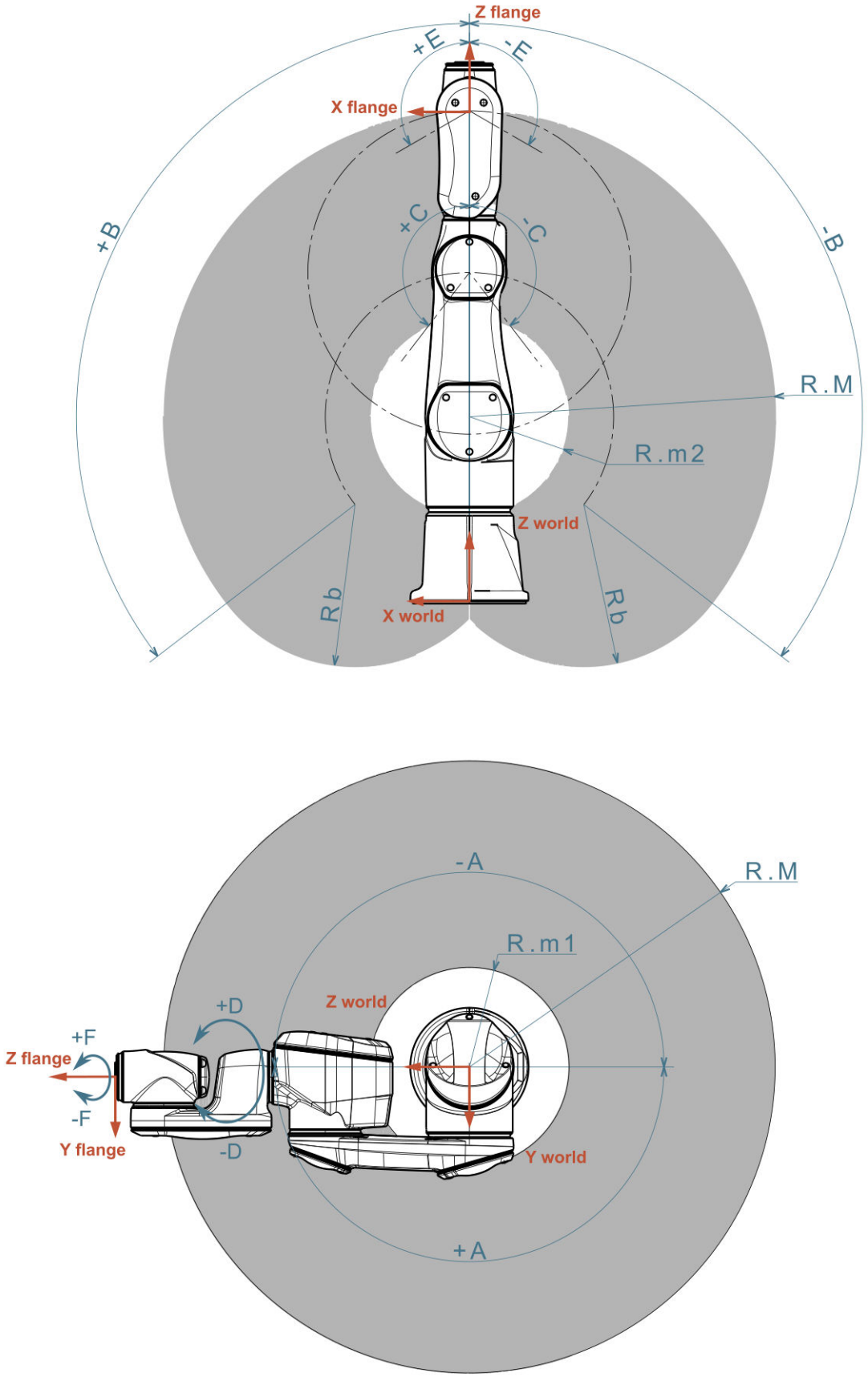


Figure 3.10: Working space, HW

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Configuration for compact wrist:

Axis		1	2	3	4	5	6
Range (°)	TX2–60 CW	360	255	285	540	253.5	540[1]
	TX2–60L CW	360	255	305	540	253.5	540[1]
Joint Limits (°)	TX2–60 CW	±180	±127.5	±142.5	±270	+132.5/-121	±270
	TX2–60L CW	±180	±127.5	±152.5	±270	+132.5/-121	±270
Nominal speed (°/s) (with CS9LP controller version)	TX2–60 CW	301	301	453	431	336	735
	TX2–60L CW	301	301	453	431	336	735
Nominal speed (°/s) (with CS9EP controller version)	TX2–60 CW	35	35	35	80	80	120
	TX2–60L CW	35	35	35	80	80	120
Maximum speed (°/s) (with CS9LP controller version)	TX2–60 CW[2]	435	410	540	995	1065	1445
	TX2–60L CW[2]	435	385	500	995	1065	1445
Maximum speed (°/s) (with CS9EP controller version)	TX2–60 CW[3]	120	120	120	220	220	400
	TX2–60L CW[3]	120	120	120	220	120	220
Angular resolution ($°.10^{-3}$)	CW	0.007	0.007	0.007	0.015	0.015	0.021

[1] Can be configured by software up to ± 11 250° (± 31.25 turns). To do so, see the "Software configuration" chapter in the controller instruction manual. Can be configured as a "continuous" axis under the continuousAxis license.

[2] Maximum speed for reduced conditions of load and inertia.

[3] Maximum speed for reduced conditions of load, inertia and for maximum motor drive voltage level.

Configuration for hollow wrist:

Axis		1	2	3	4	5	6
Range (°)	TX2–60 HW	360	255	285	540	210	360
	TX2–60L HW	360	255	305	540	240	360
Joint Limits (°)	TX2–60 HW	±180	±127.5	±142.5[1]	±270	+120/-90	±180
	TX2–60L HW	±180	±127.5	±152.5[1]	±270	±120	±180
Nominal speed (°/s) (with CS9LP controller version)	TX2–60 HW	271	271	408	388	302	662
	TX2–60L HW	271	271	408	388	302	662
Nominal speed (°/s) (with CS9EP controller version)	TX2–60 HW	35	35	35	80	35	80
	TX2–60L HW	35	35	35	80	35	80
Maximum speed (°/s) (with CS9LP controller version)	TX2–60 HW[2]	435	410	540	700	600	885
	TX2–60L HW[2]	435	385	500	700	600	885
Maximum speed (°/s) (with CS9EP controller version)	TX2–60 HW[3]	120	120	120	220	220	400
	TX2–60L HW[3]	120	120	120	220	120	220
Angular resolution ($°.10^{-3}$)	HW	0.007	0.007	0.007	0.015	0.007	0.008

[1] Those robots are delivered with a limitation of ±127° for TX2-60 HW and ±145° for TX2-60L HW in order to prevent collisions between the shoulder and the forearm. Those additionnal restrictions are removable.

[2] Maximum speed for reduced conditions of load and inertia.

[3] Maximum speed for reduced conditions of load, inertia and for maximum motor drive voltage level.

NOTICE

Collisions
The base of the robot is part of the working space; the working space with the end-effector may also include other parts of the robot. Therefore, there is a risk of collision with the robot or the environment, in case of misuse or programming error.
Reduce unused axes ranges to mitigate the risk.

The maximum space can be reduced by different means:

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- Fences,
- Mechanical hard stop on axis 1,
- Safe joint limits on all axes,
- Safe zones in the Cartesian space.

The configuration of safe joint limits and safe zones is described in the controller instruction manual. The arm-specific information required to define the controlled space around the robot are given in chapter 3.3.10.

A 3D CAD simulation is required for an accurate positioning of the arm in the cell, to check reachability and prevent collisions. Our Stäubli Robotics Suite PC software allows a complete 3D simulation, as described in chapter 3.5 (Robot simulation). CAD files are available in our CAD library at http://www.staubli.com/ for other simulation tools.

3.3.3 - ARM MOUNTING

M0006865.1

3.3.3.1 - Working space

M0006996.1

The user or integrator is responsible for performing all preparatory work required to complete the on-site installation of the robot. Working space must be sufficient, installation surface appropriate; the power supplies must be available (for the electric power supplies, see the controller characteristics).

DANGER

Hazardous deformation of the fences or perimeter guards

If the arm can cause dangerous deformation of the cell fences or perimeter guards, the user is at risk of serious injury or death.

The robot shall not be able to cause a hazardous deformation of the fences or perimeter guards (ISO 10218-2, 5.4.3).

- The cell must be designed in such a way that the robot is used in accordance with its specifications, and can be used, adjusted, and maintained without risk.
- The adjustment conditions (teaching points, movement adjustment) must be taken into account when designing the cell.
- All contact between the robot and the operator must be prevented by designing suitable guards or protective devices.

To enable all maintenance operations to be carried out, it is essential to provide easy access to the robot (e.g. removable safety barriers, etc.) and anchoring points for easy handling of the arms, especially those whose weight exceeds 70 kg / 154 lb.

3.3.3.2 - Mounting floor quality

M0006713.1

DANGER

Ground strength

The arm will fall by breaking its attachment if the ground is not strong enough.

The user has to make sure that the mechanical characteristics of the floor and the means of attachment will hold up to the maximum forces caused by the moving arm.

3.3.3.3 - Attachment

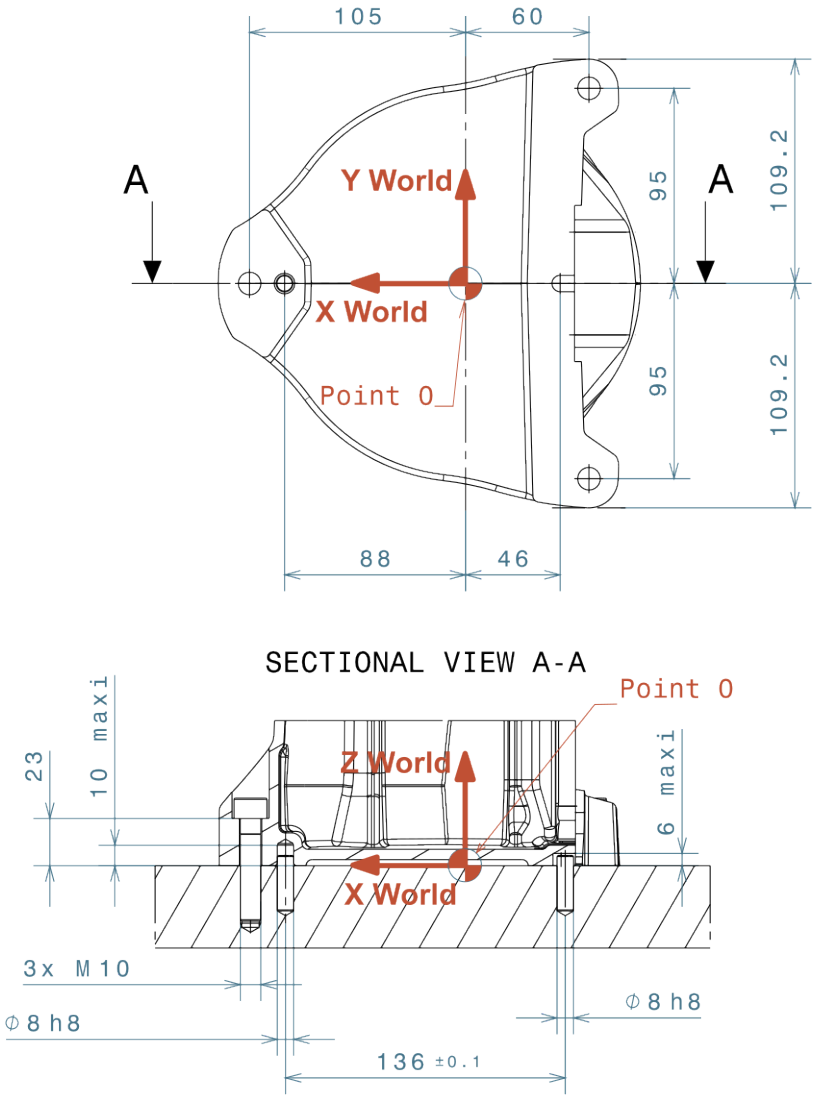
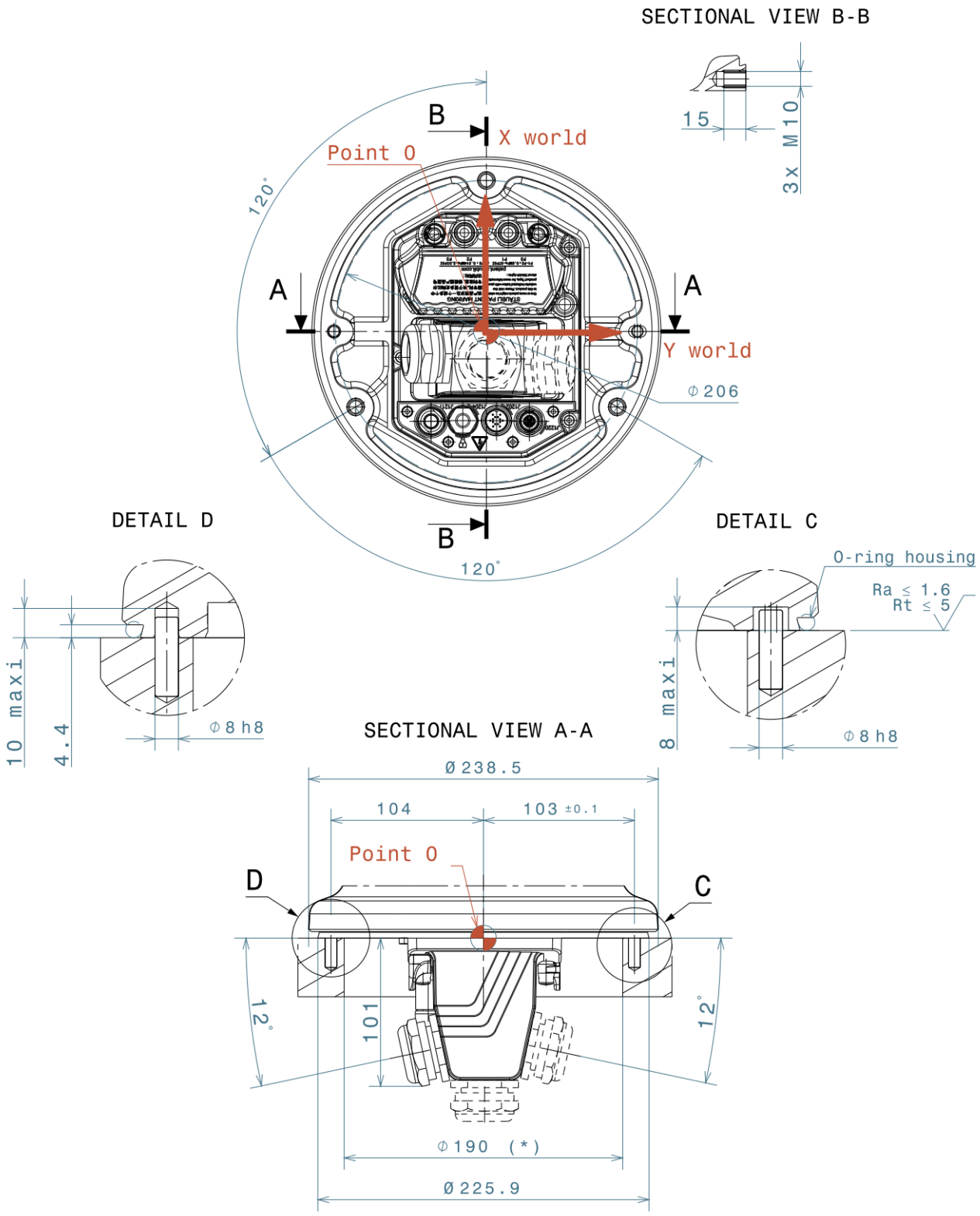


Figure 3.11: Horizontal cable outlet

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Item	Description
(*)	Opening required for the connectors and to allow removal of the interconnect plate

Figure 3.12: Vertical cable outlet

Vertical cable outlet with minimum opening size to be made or left in the attachment surface to put the cable through and access the connections. Seal on the inner face of the base on the holder using a static joint for the HE and Stericlean versions.

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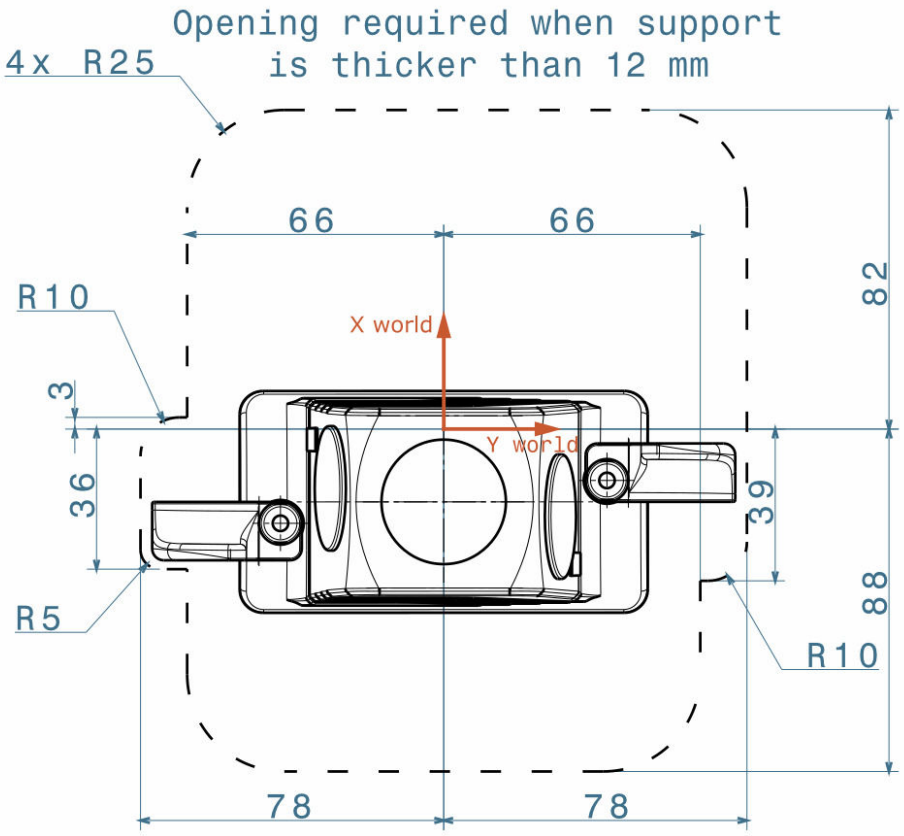


Figure 3.13:

The arm can be ordered for floor mount or ceiling mount. In all cases, it must be securely attached by 3 screws M10 class 12.9 (standard or DeltaProtect, according to ISO ISO 4762).
Tightening torque: 59 Nm (with a tool of minimum ±15% accuracy).
For the vertical cable outlet, add washer and ensure that the minimum length engagement of the screw (in the foot threaded hole) is 12 mm and the maximum is 16 mm.

DANGER	
	Specific fastening The arm will fall by breaking its attachment if the screws do not meet the requirements. The use of special treated screws is prohibited. In case of specific fastening need please contact us.

Humid environment (HE) or bio-contamination controlled environment (Stericlean):
The threaded fasteners on the robot, the underside of the base, the interconnect plate and the equipment on it, are not qualified for a humid (HE) or a bio-contamination controlled environment (Stericlean), so they must be protected from that environment.
Bio-contamination controlled environment (Stericlean):
The threaded fasteners on the wrist's mechanical interface, the threaded fasteners on the robot, the underside of the base, the electrical connection on the forearm (CW robots) or on the flange of the robot (HW robots), the interconnect plate and the equipment on it, are not qualified for a bio-contamination controlled environment (Stericlean), so they must be protected from that environment.
This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.

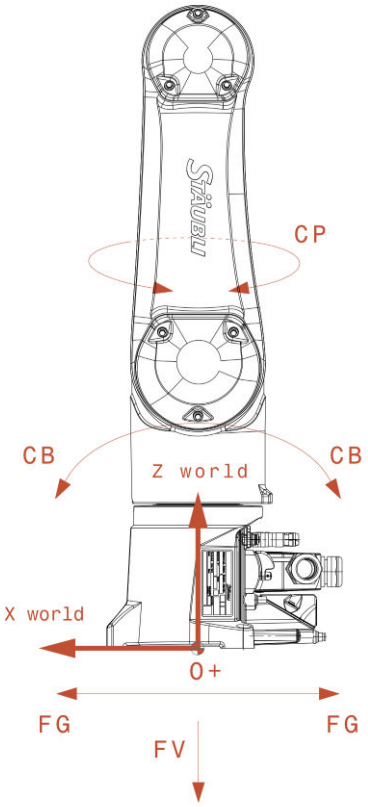
NOTICE

Environmental damage
The warranty does not cover any environmental damage to these parts.
Protection must be provided by and is the responsibility of the customer.

- Base / holder interface made waterproof with the exterior seal supplied:
  - The surface of the holder on which the robot base is fixed must have surface characteristics that provide static sealing.
  - Lubricate and fit the exterior seal.
  - The underside of the base must be protected from the humid or bio-contamination controlled environment.
- Protection of the power interconnect plate and the equipment fitted to it:
  - Protect the complete element from the humid or bio-contamination controlled environment.

To facilitate access to the connections during installation and maintenance, it is important to leave as large of an opening as possible in the robot base for the "Vertical cable outlet" version, following the dimensions shown in the figure 3.12, and to leave a safe distance in front of the base interface for both versions (figures 3.11 and 3.12).

Attachment surface shall be flat and metallic. A deformable support will greatly reduce robot performance in speed and accuracy.
When calculating the size of the support, it is necessary to take into account the maximum forces transmitted by the arm in movement at point 0, which are as follows:



Floor, ceiling or wall mounted arm:

	F_G (N)	F_V (N)	C_B (Nm)	C_P (Nm)
TX2-60 (Floor)	1930	1310	1090	360
TX2-60 (Ceiling)	1920	1430	960	330
TX2-60 (Wall)	2250	1010	1150	440
TX2-60 HW (Floor)	1030	1050	550	270
TX2-60 HW (Ceiling)	920	1200	560	280
TX2-60 HW (Wall)	1130	1290	640	310
TX2-60L (Floor)	1970	1280	1410	470
TX2-60L (Ceiling)	1950	1440	1200	410
TX2-60L (Wall)	2290	1010	1490	570
TX2-60L HW (Floor)	1050	1130	720	320
TX2-60L HW (Ceiling)	950	1210	510	330
TX2-60L HW (Wall)	1170	1400	530	350

During motion with nominal velocity and acceleration parameters and under following load conditions:

			Load position			
	Load		Axis 5		Axis 6	
	kg	lb	mm	inch	mm	inch
TX2-60	3.5	7.71	100	3.93	50	1.96
TX2-60 HW	3.5	7.71	125	4.92	50	1.96

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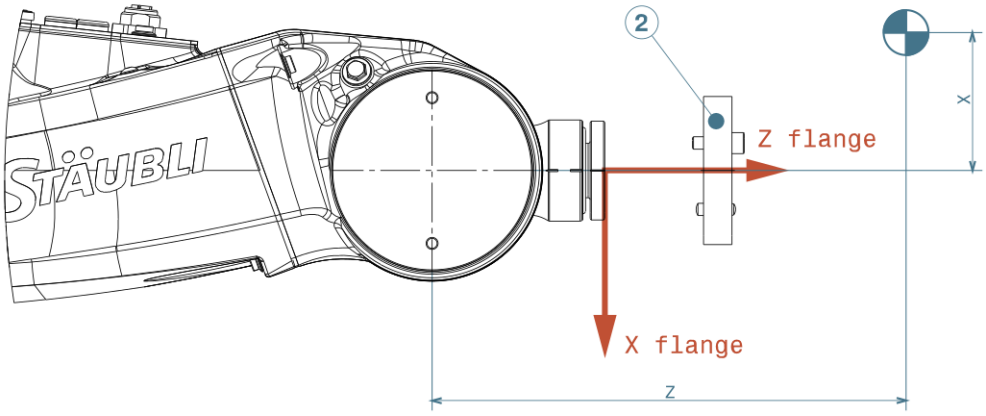
			Load position			
	Load		Axis 5		Axis 6	
	kg	lb	mm	inch	mm	inch
TX2-60L	2	4.4	100	3.93	50	1.96
TX2-60L HW	2	4.4	125	4.92	50	1.96

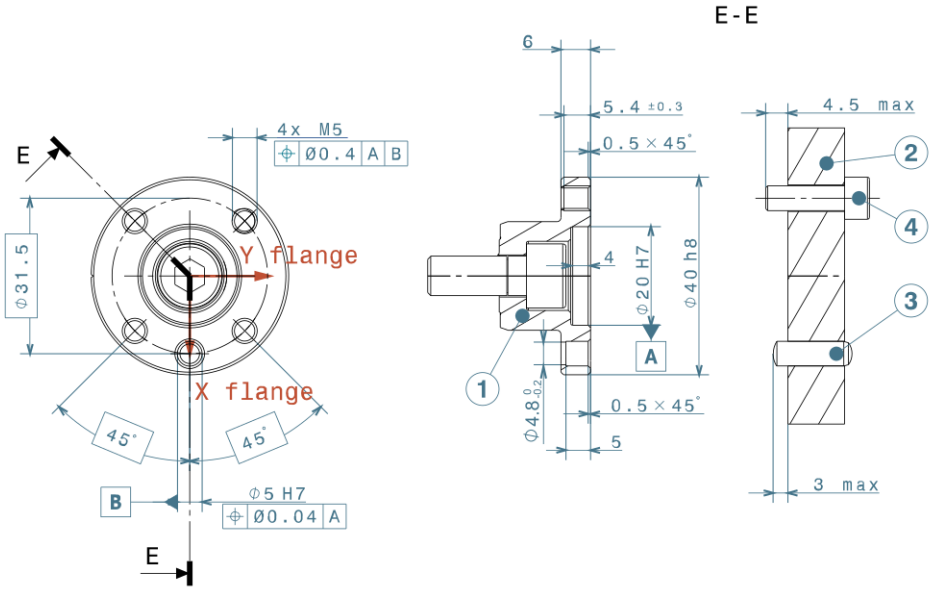
The user can accurately position the robot by means of two 8h8 diameter centering pins (not supplied).

3.3.4 - END-EFFECTOR

M0007567.1

TX2-60 Compact wrist





Item	Description
1	Mechanical interface
2	End-effector
3	Pin Ø5
4	4 M5 screws, class 12-9
(*)	M8 screw : To protect from the humid or bio-contamination controlled environment

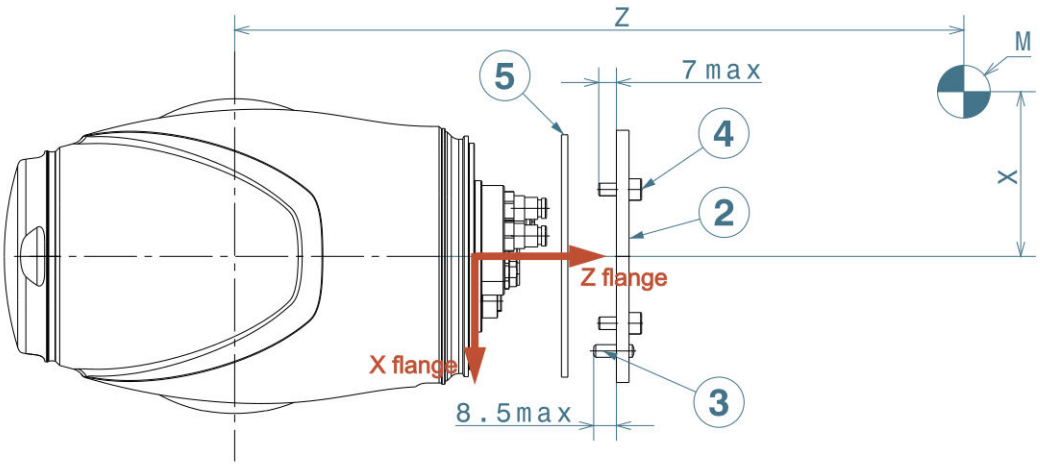
Item	Description
M	Payload center of gravity

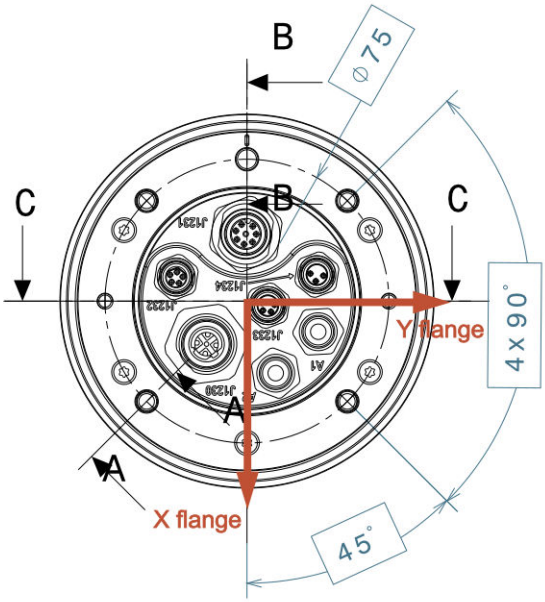
Figure 3.14: TX2-60 Compact wrist

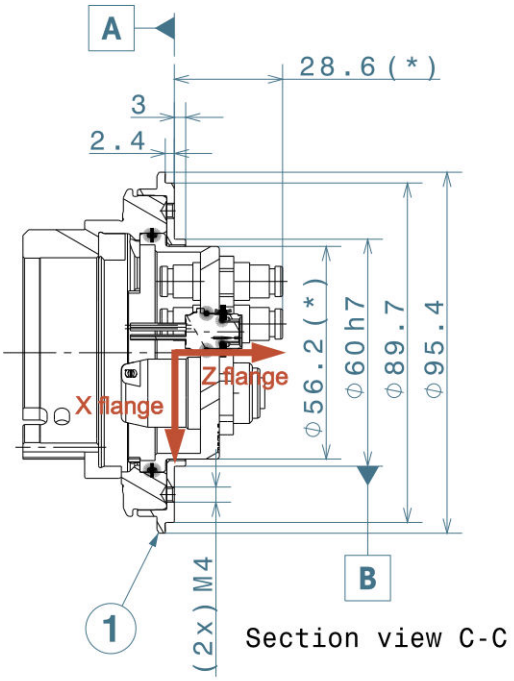
The end-effector **(2)** is mounted on the wrist's mechanical interface **(1)** (dimensions given in figure 3.14). Indexing by the Ø5 pin **(3)**.

- Tool fastening on the sensor with 4 class 12-9 screws **(M5)**. Tightening torque 8.5 N.m +/- 15%. Only valid for an aluminum tool with Re > 400 MPa or a steel tool with Re > 400 MPa
- Length of end-effector attaching screws is limited to avoid all interference with the wrist (see figure 3.14).

TX2-60 Hollow wrist

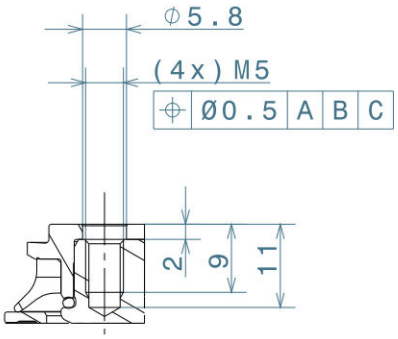




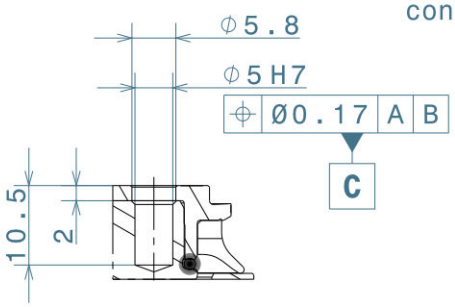


Section view C-C

(*): Only for flange-mounted connectors version



Section view A-A



Section view B-B

Item	Description
1	Mechanical interface
2	End-effector
3	Pin Ø5
4	4 M5 screws, class 12-9

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Item	Description
5	Flat seal
M	Payload center of gravity

Figure 3.15: TX2-60 Hollow wrist

The end-effector **(2)** is mounted on the wrist's mechanical interface **(1)** (dimensions given in figure 3.15). Indexing by the Ø5 pin **(3)**.

- Tool fastening on the sensor with 4 class 12-9 screws **(M5)**. Tightening torque 5.5 N.m +/- 15%. Only valid for an aluminum tool with Re > 250 MPa or a steel tool with Re > 300 MPa. Screws implantation length ≥ 5 mm
- Length of end-effector attaching screws is limited to avoid all interference with the wrist (see figure 3.15).

To guarantee the sealed design of the tool, Stäubli can provide two types of caps as an option:

- Flat caps (6), see figure 3.16.
- EHEDG caps (8) with their seal (7) (provided by Stäubli), see figure 3.17.

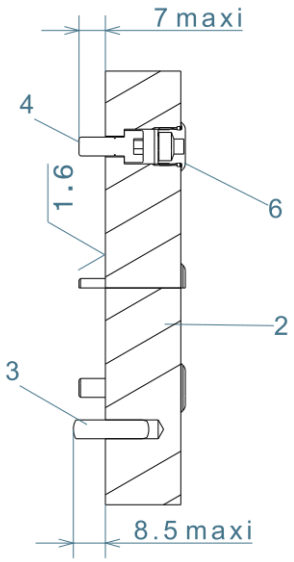


Figure 3.16: Flat caps

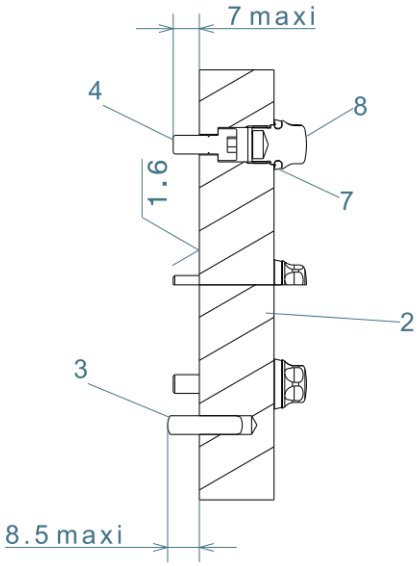


Figure 3.17: EHEDG caps

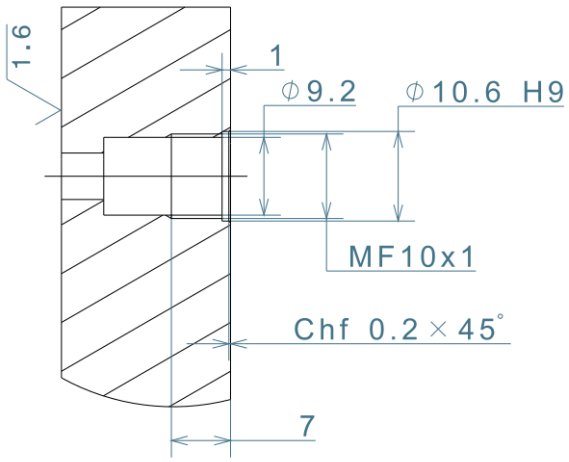


Figure 3.18: Caps fitting

To ensure the tightness, it is recommended to use a teroson sealant under the flat caps.

The static seal **(D145 289 01)**, delivered with the HW arm, can be installed between the tool (2) and the robot's wrist mechanical interface (1). The main dimensions of the seal housing are given in the figure below :

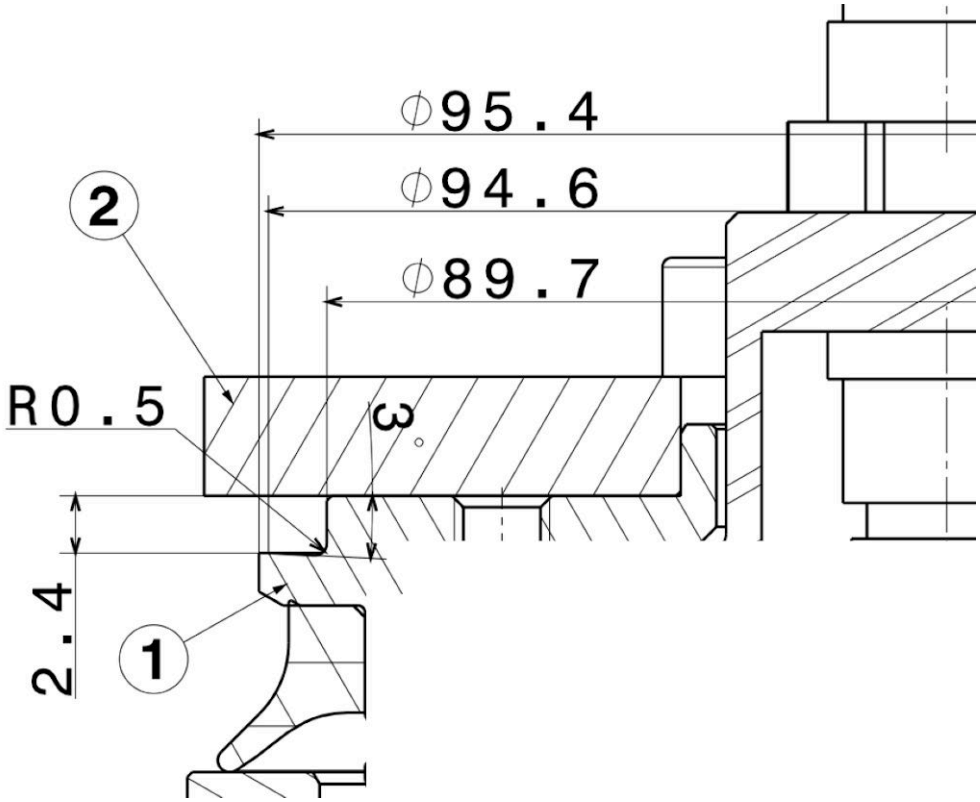


Figure 3.19: Static seal

The end-effector **(2)** is not supplied with the arm assembly; its design depends on the robot specific applications.

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NOTICE



End effector rigidity

An insufficiently rigid end effector will have a negative effect on the robot's dynamic performance and accuracy.

All studies can be carried out in collaboration with Stäubli to achieve optimum performance without exceeding the load limits of the robot arm assembly.

⚠ WARNING



User equipment grounding

The tool and any other equipment on the arm must be grounded if required by the integrator risk analysis or local standards. I/O cables can be used. (see chapter 3.3.8)

⚠ DANGER



Effector or load ejection

Ejection of the end effector or the load it carries will cause serious injury or death.

The end-effector must be correctly sized and attached in accordance with specifications.

The tool must be designed to hold the load with the programed accelerations and in the event of a cut-off in the electrical or pneumatic power supply.

At high speed, the possibility of load ejection usually requires the installation of protection fences around the robot.

Tool flange typical accelerations (with CS9LP controller version)		X, Y	Z
Translation ($m.s^2$)	TX2-60	90	63
	TX2-60L	136	96
	TX2-60 HW	198	272
	TX2-60L HW	255	329
Rotation ($rad.s^2$)	TX2-60	448	211
	TX2-60L	453	211
	TX2-60 HW	1100	223
	TX2-60L HW	1094	223

Tool flange typical accelerations (with CS9EP controller version)		X, Y	Z
Translation ($m.s^2$)	TX2-60	Not available[1]	Not available[1]
	TX2-60L	Not available[1]	Not available[1]
	TX2-60 HW	Not available[1]	Not available[1]
	TX2-60L HW	Not available[1]	Not available[1]

1 Values from CS9LP can be used (worst case)

Tool flange typical accelerations (with CS9EP controller version)		X, Y	Z
Rotation (rad.s^2)	TX2-60	Not available[1]	Not available[1]
	TX2-60L	Not available[1]	Not available[1]
	TX2-60 HW	Not available[1]	Not available[1]
	TX2-60L HW	Not available[1]	Not available[1]

[1] Values from CS9LP can be used (worst case)

NOTICE

Cable environmental damage
The warranty does not cover any environmental damage to these parts.
In humid (HE) or bio-contamination controlled environment (Stericlean or Accesspharma), the cables must be protected from the environment. This protection is to be provided by the customer, and are their responsibility.

3.3.4.1 - Load capacity

M0008500.1

Load characteristics (see figure 3.14):

Load capacity	X	Z	TX2-60 & TX2-60 HW		TX2-60L & TX2-60L HW	
	mm	mm	kg	lb	kg	lb
At nominal speed	50	100	3.5	7.71	2	4.4
At slow speed[1]	50	100	4.5	9.92	3.7	8.15
Maximum load capacity at low speed (see figure 3.20)[2]	50	100	9	19.8	5	11

[1] In all configurations with heavy payload speed and acceleration descriptors (Speed = 60%, Acceleration = 60%, Deceleration = 60%) and taking maximum inertia into account. See the table below.

[2] Limiting load for which the maximum arm extension cannot exceed that shown in figure 3.20.

	Nominal inertias (kg.m^2)				Maximal inertias (kg.m^2)[3]			
	TX2-60	TX2-60 HW	TX2-60L	TX2-60L HW	TX2-60	TX2-60 HW	TX2-60L	TX2-60L HW
For axis 5	0.045	0.045	0.025	0.025	0.325	0.45	0.125	0.2
For axis 6	0.009	0.009	0.005	0.005	0.1	0.2	0.032	0.05

[3] Under slow speed and acceleration conditions.
Generally: VEL = 60%, ACC = 60%, DEC = 60% (consult us).

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NOTICE

Higher inertias
The nominal values can be exceeded to a certain extent but imply a limitation to the speed and the acceleration of the arm.
If these limits are to be exceeded, please consult Stäubli.

For TX2-60 CW and TX2-60L CW only:
The maximum levels of inertia cannot be used simultaneously on axes 5 and 6. In all cases, the following two conditions must be complied with:

- I5 load + I6 load < I5 max,
- I6 load < I6 max.

The joint velocities and accelerations of the arm can be reduced to safely prevent the exceeding of the maximum limits on the flange. The tuning of these safety parameters is described in the "Integration" chapter of the controller instruction manual.

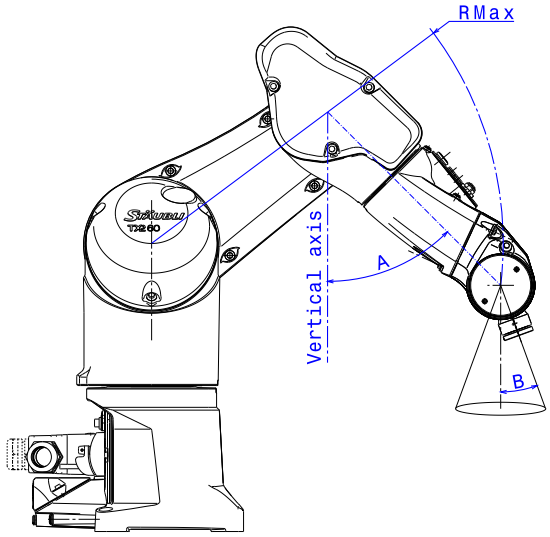


Figure 3.20: Axis 4 has to stay between -20° and +20° range

	A (°)	B (°)	Rmax (mm)
TX2-60 CW & TX2-60 HW	45	20	460
TX2-60L CW & TX2-60L HW	60	20	735

3.3.4.2 - Maximum torque

M0008045.1

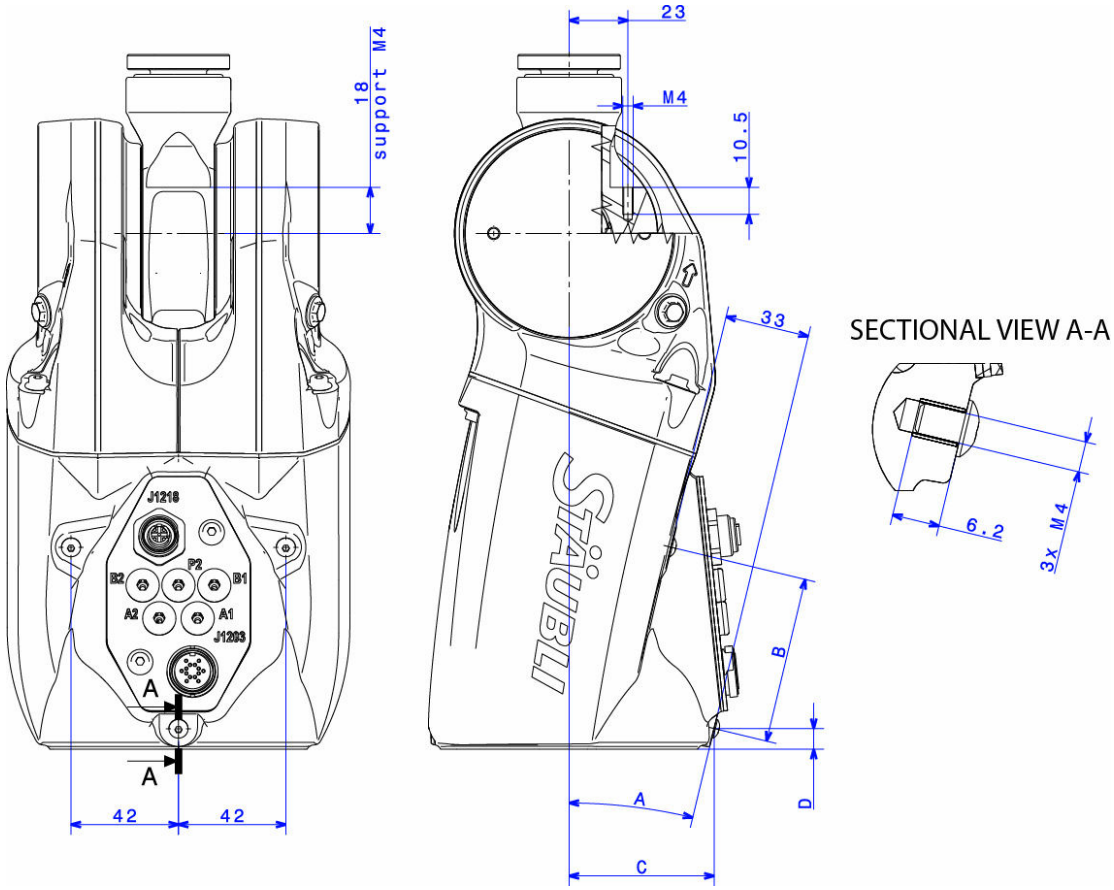
To limit the temperature of the robot, the following torques must not be exceeded.

	Reference axis					
Static torque (Nm)	**Axis 1**	**Axis 2**	**Axis 3**	**Axis 4**	**Axis 5 (Z6)**	**Axis 6 (Z7)**
TX2-60 CW TX2-60L CW	126	126	43	7	10	2
TX2-60 HW TX2-60L HW	126	126	43	7	10	2.4

These torques are global static joint torques, which include all the transmission efforts supported by the joints. These torques include forces related to gravity applied by the arm's own mass, by the tool's mass, or by any other external force that can be applied to the arm or to the tool. It is recommended to use the CHECK PAYLOAD feature on SRS to get a first analysis of the inertia and static torques, in relation to this specification, during robot integration.

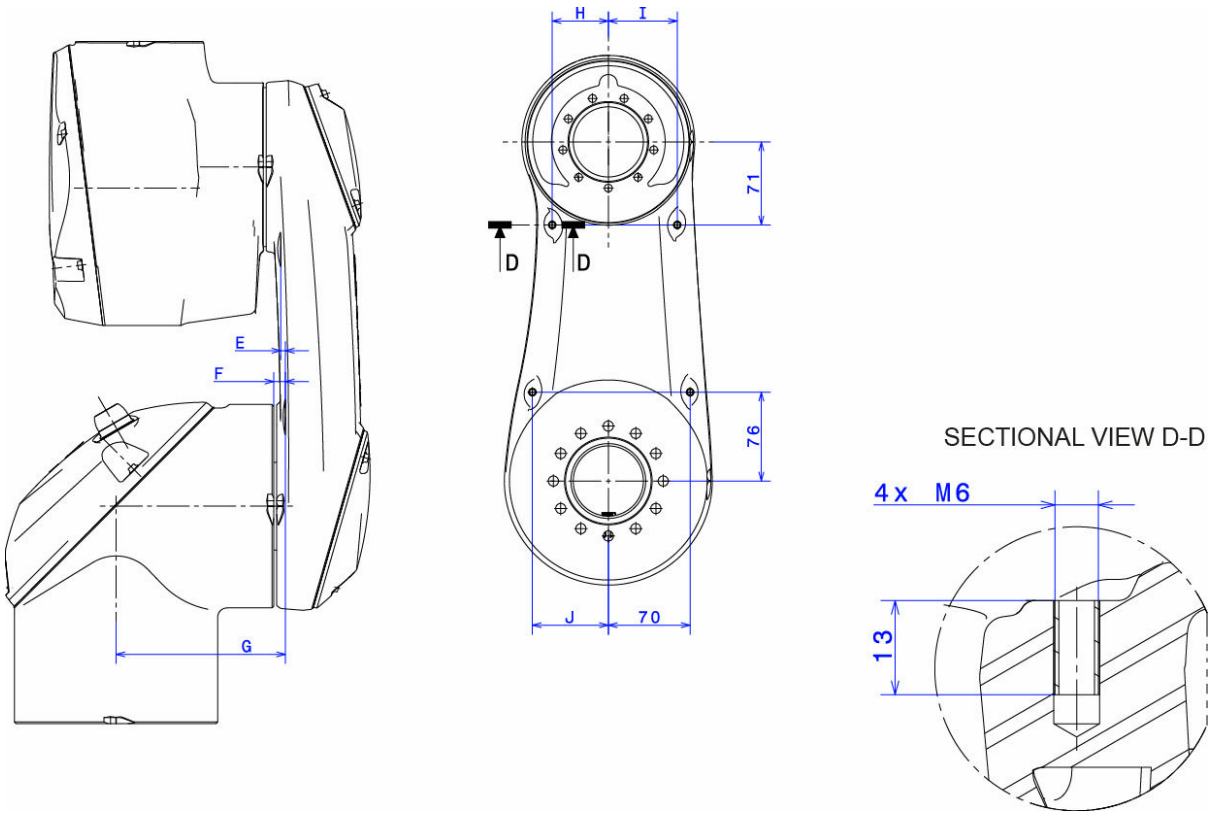
3.3.5 - USER INTERFACE (ADDITIONAL LOAD, HARNESS...) : VERSIONS CW

M0009380.1



Value	L arm
A	11°
B	100
C	55.6
D	117

Figure 3.21:



Value	L arm
E	5
F	11.5
G	146.5
H	47
I	58
J	63

Figure 3.22:

As user interface, tapped holes are available:

- 4x M6 holes on the inner link (maximum torque is 7 Nm),
- 3x M4 holes on the forearm (maximum torque is 3 Nm),
- One M4 hole on the wrist (maximum torque is 3 Nm).

NOTICE

Additional load validation

Not validating the additional load repartitioning on the arm will negatively impact its performance.

Screws for additional load are free to be placed by the client.

To validate your load configuration with additional load on segments please refer to chapter 3.3.4.1.

⚠ WARNING



Unfastened user harness or equipement

Unfastened user equipement may shock or whip the user or nearby installation.
Always secure anything you attach to the outside of the arm.

⚠ WARNING



User equipment grounding

The tool and any other equipment on the arm must be grounded if required by the integrator risk analysis or local standards. I/O cables can be used. (see chapter 3.3.8)

NOTICE



User interface protection:

For humid environment (HE) or bio-contamination controlled environment (Stericlean), in case of use of the user interface, it must be protected with mastic (Loctite Terrostat 92) or equivalent.
This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.

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3.3.6 - USER INTERFACE (ADDITIONAL LOAD, HARNESS...) : VERSIONS HW

M0009680.1

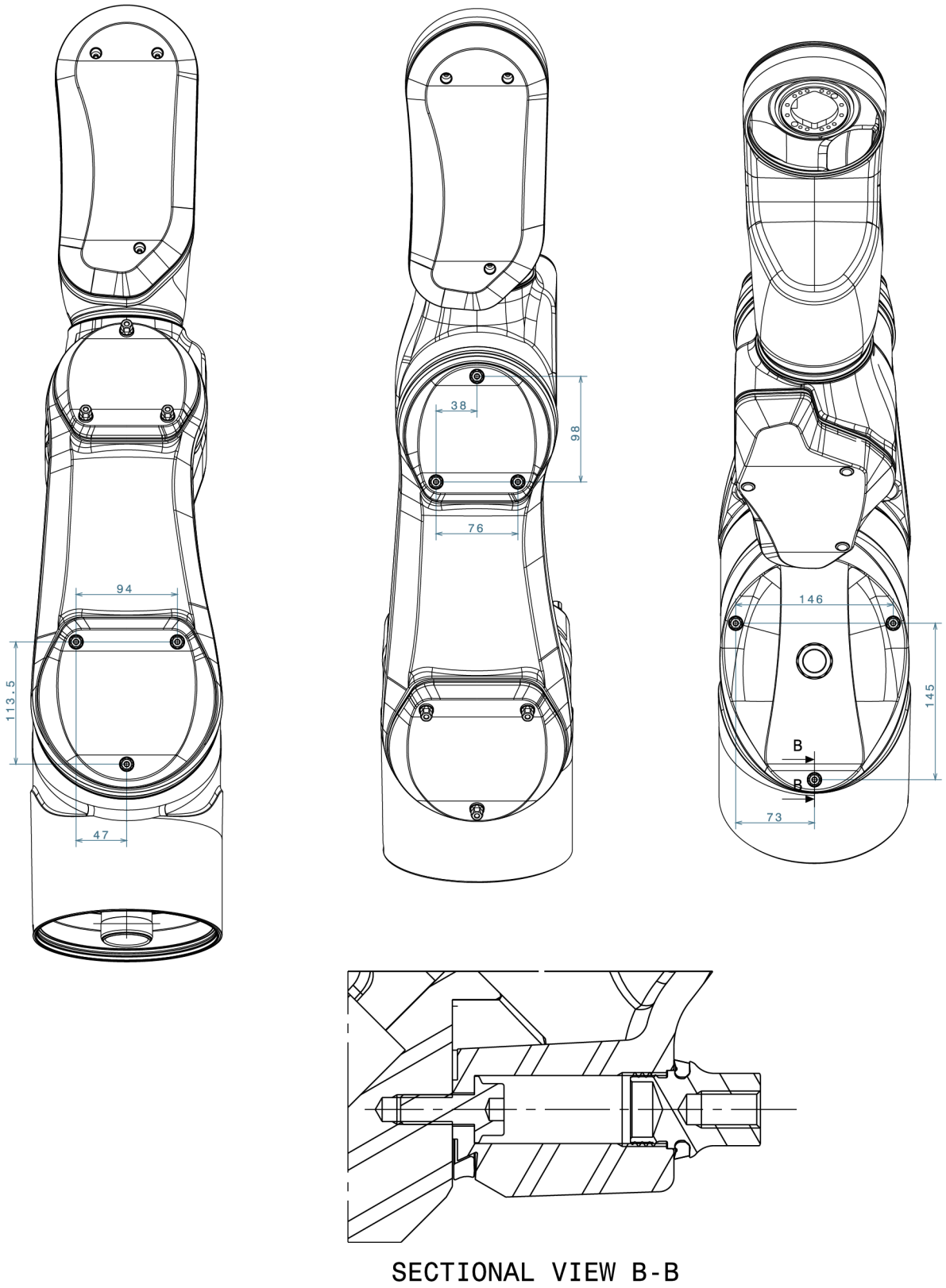


Figure 3.23: Arm cover

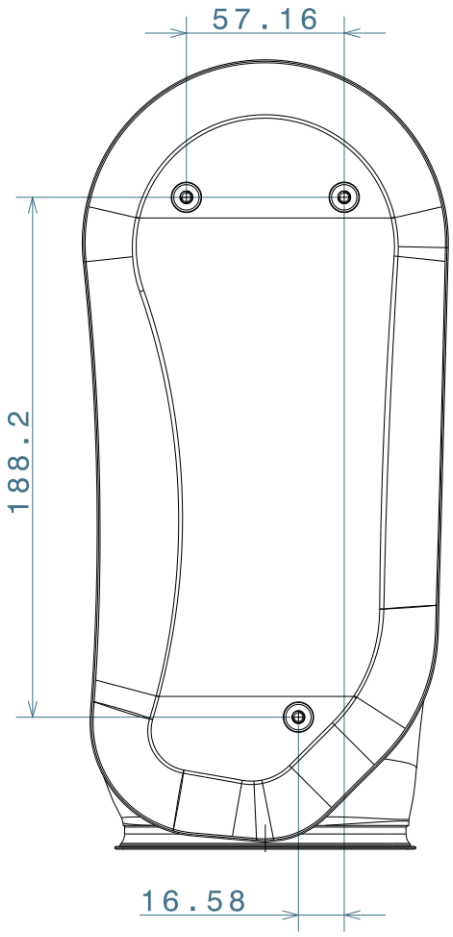


Figure 3.24: Forearm cover

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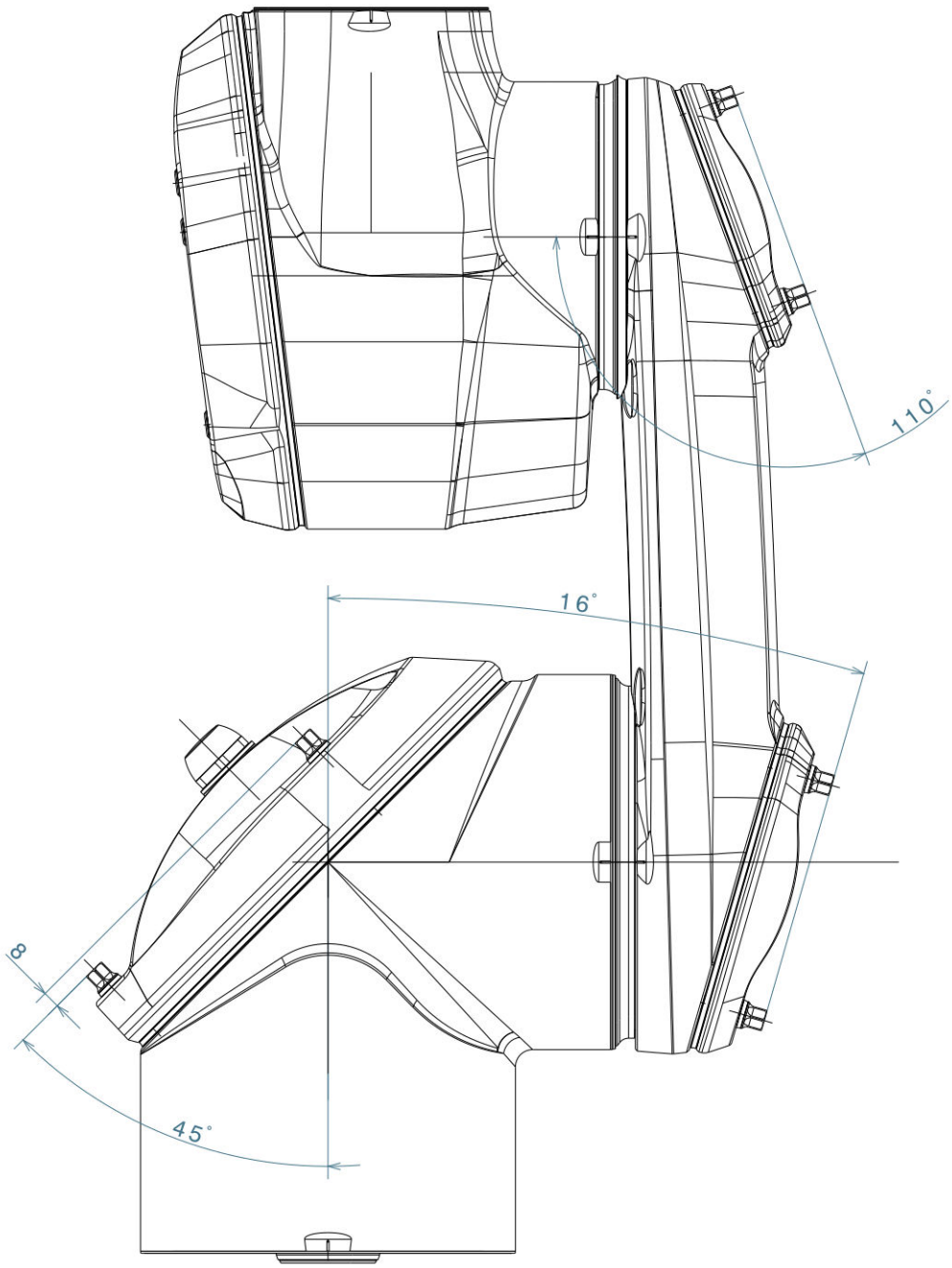


Figure 3.25: Inner link and shoulder covers

As user interface, tapped holes are available:

- 6x M6 holes on the inner link (maximum torque is 7 Nm),
- 3x M4 holes on the shoulder (maximum torque is 3 Nm),
- 3x M4 holes on the forearm (maximum torque is 3 Nm),

NOTICE



Additional load validation

Not validating the additional load repartitioning on the arm will negatively impact its performance.

Screws for additional load are free to be placed by the client.

To validate your load configuration with additional load on segments please refer to chapter 3.3.4.1.

NOTICE



User interface protection:

For humid environment (HE) or bio-contamination controlled environment (Stericlean), in case of use of the user interface, it must be protected with mastic (Loctite Terrostat 92) or equivalent.

This protection is to be provided by the customer, and are their responsibility. The guarantee does not cover any damage occurring there.

3.3.7 - PNEUMATIC CIRCUIT

M0007377.1

General information

Please note that the pneumatic systems and their components integrated in this robot comply with ISO-4414, it is the responsibility of the end user and integrator to comply with this ISO standard.

On receipt, all pneumatic ports on the arm are sealed with either a plug or screw cap.

⚠ CAUTION



Contamination

Leaving pneumatic ports open can cause contamination of the pneumatic circuit in the arm.

Only remove the caps from the ports you are using and plug them in immediately.

⚠ CAUTION



Residual pressure

Disconnecting fittings when there is residual air pressure may cause them to be ejected.

Before disconnecting any fittings, make sure that there is no residual air pressure in the circuit.

⚠ WARNING



Carried load fall

Disconnecting fittings when a load is being carried by a pneumatic actuator can cause the load to fall.

Before disconnecting any fittings, make sure that no pneumatic actuator is carrying any load.

It is the responsibility of the end user and integrator to protect the arm from overpressure.

3.3.7.1 - Compact wrist pneumatic circuit

M0009533.1

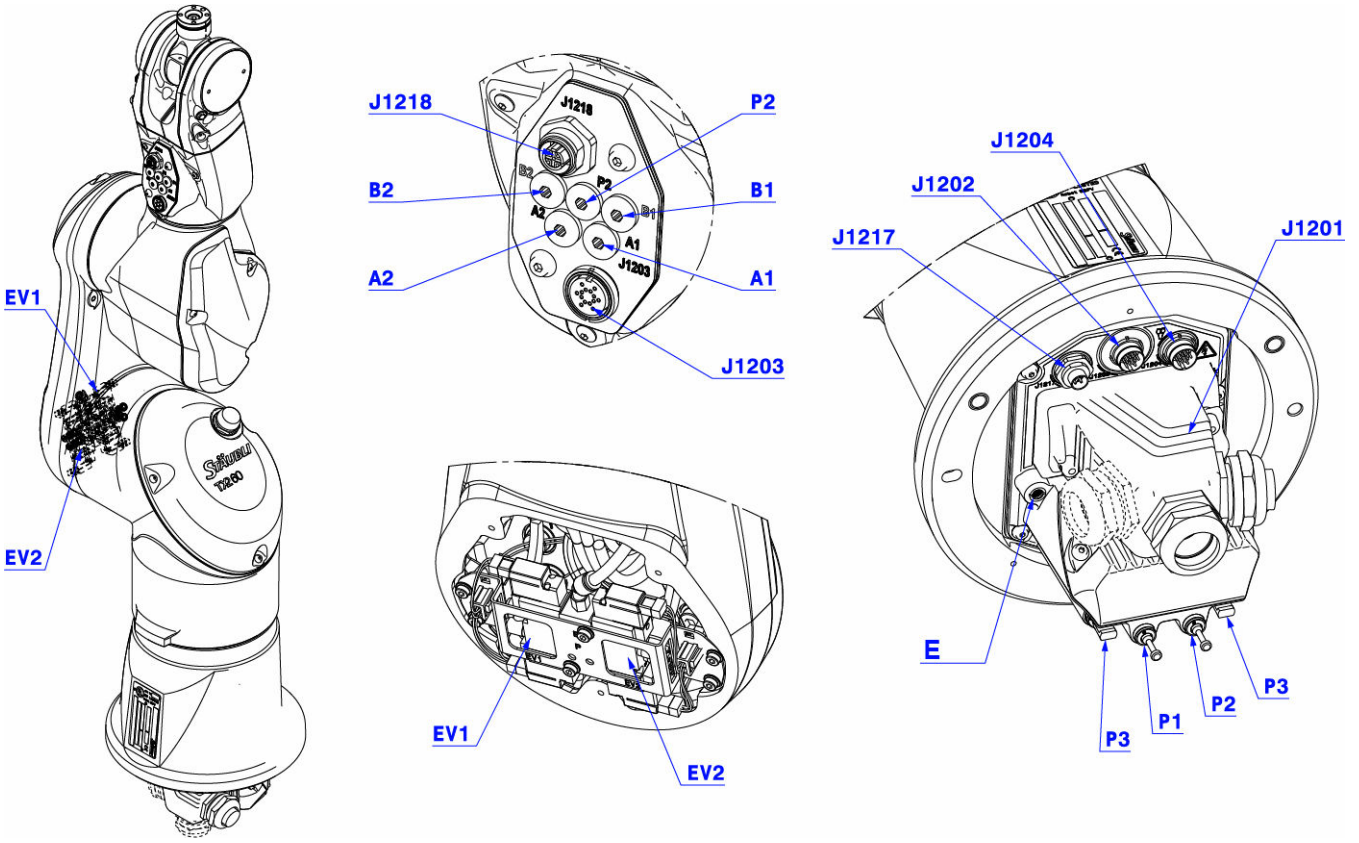


Figure 3.26: Compact wrist pneumatic circuit

The standard compact wrist arm cabling system integrates the pneumatic lines that provide 2 pressure supplies **(A1)** and **(P2)** near the tool clamp, and a hole **(A2)** for the return to the exhaust.
As an option, the harness includes pneumatic hoses which supply air to the solenoid valves (**(EV1)** and **(EV2)**).

- The robot also has a pressure source **(P2)** close to the tool clamp.
- The outlets of the solenoid valves **(EV1)** and **(EV2)** are on the forearm:
  - **(A1)** and **(B1)** for solenoid valve **(EV1)**,
  - **(A2)** and **(B2)** for solenoid valve **(EV2)**.

⚠ WARNING

Arm internal wiring modification
Adding wires or cables to internal arm wiring can cause premature wear to arm electrical wiring and affect arm safety functions.
Do not add wires or cables to internal arm wiring.

⚠ WARNING

Pneumatic circuit failure:
Even if preventive maintenance recommendations are followed perfectly, the risk of pneumatic circuit failure cannot be excluded.
This risk must be taken into account by the integrator when designing the application.

3.3.7.2 - Hollow wrist pneumatic circuit

M0009534.1

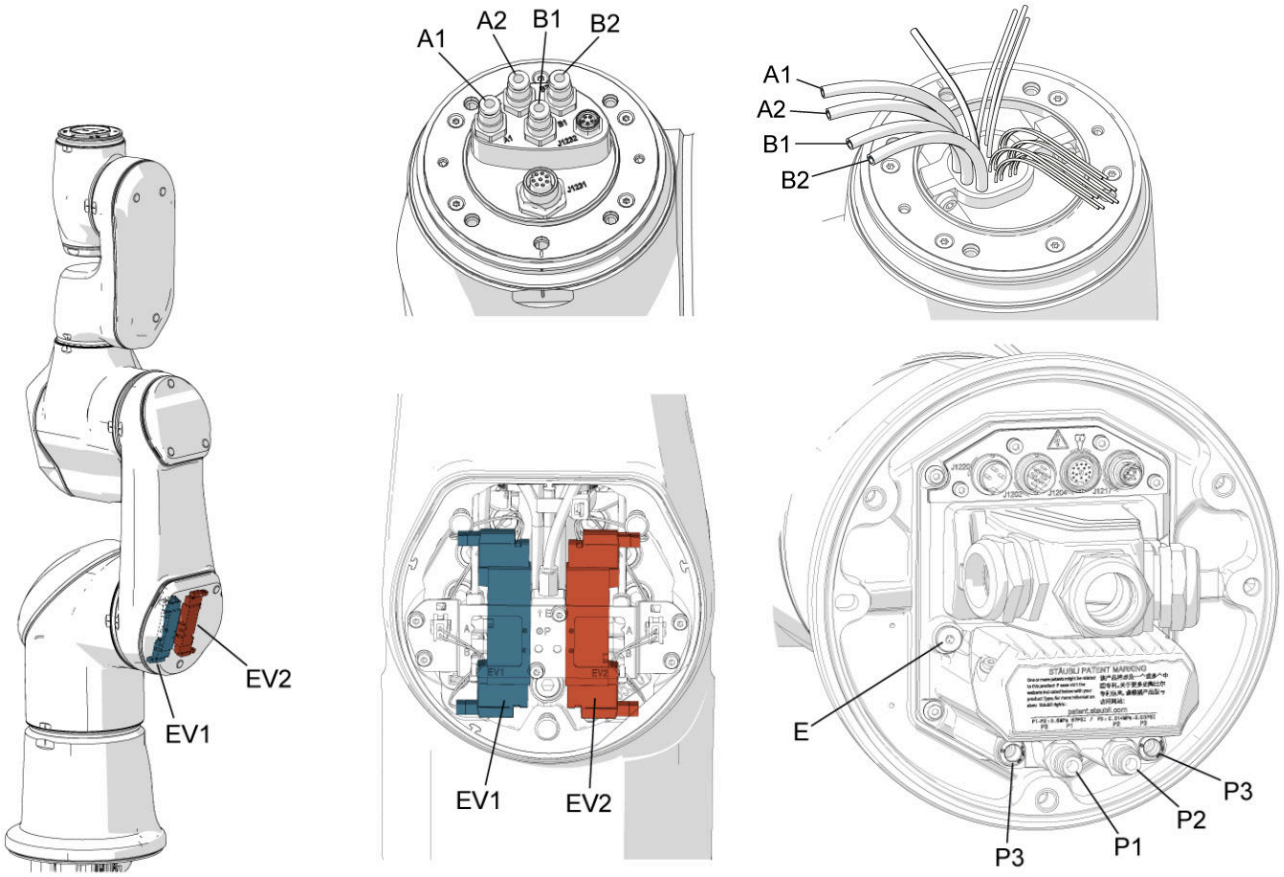


Figure 3.27: Hollow wrist pneumatic circuit

The standard wiring system of the hollow wrist arm incorporates pneumatic lines that supply 2 pressure supply hoses **(A1)** and **(B1)** on the flange, and 2 hoses **(A2)** and **(B2)** for exhaust return **(E)**.
As an option, the harness includes two solenoid valves **(EV1)** and **(EV2)**, available in 5/2, 5/3 or 3/2 for vacuum applications.
When chosen, the outlets of the solenoid valves **(EV1)** and **(EV2)** are:

- **(A1)** and **(B1)** for solenoid valve **(EV1)**.
- **(A2)** and **(B2)** for solenoid valve **(EV2)**.

SMC Pneumatic tubes are SMC® PTFE Soft Tubing TD-0425 for Ø2.5x4 and TD-0604 for Ø4x6.

⚠ WARNING	
⚠	**Arm internal wiring modification** Adding wires or cables to internal arm wiring can cause premature wear to arm electrical wiring and affect arm safety functions. Do not add wires or cables to internal arm wiring.

⚠ WARNING	
⚠	**Pneumatic circuit failure:** Even if preventive maintenance recommendations are followed perfectly, the risk of pneumatic circuit failure cannot be excluded. This risk must be taken into account by the integrator when designing the application.

3.3.7.3 - Foot pneumatic interface

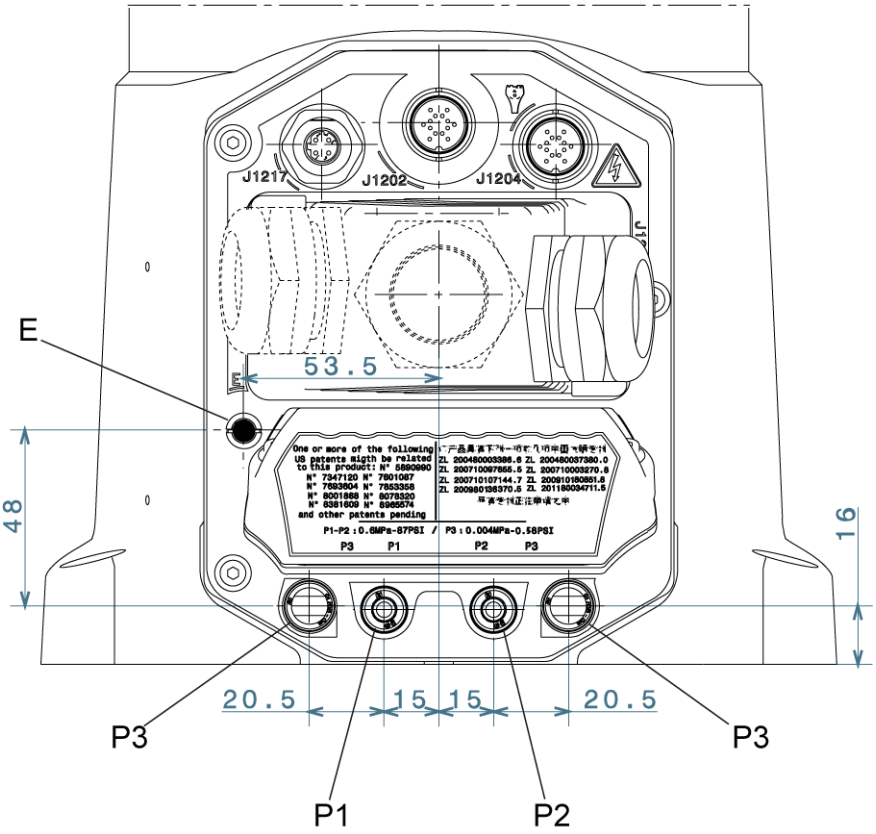


Figure 3.28: Horizontal foot pneumatic interface (CW arm)

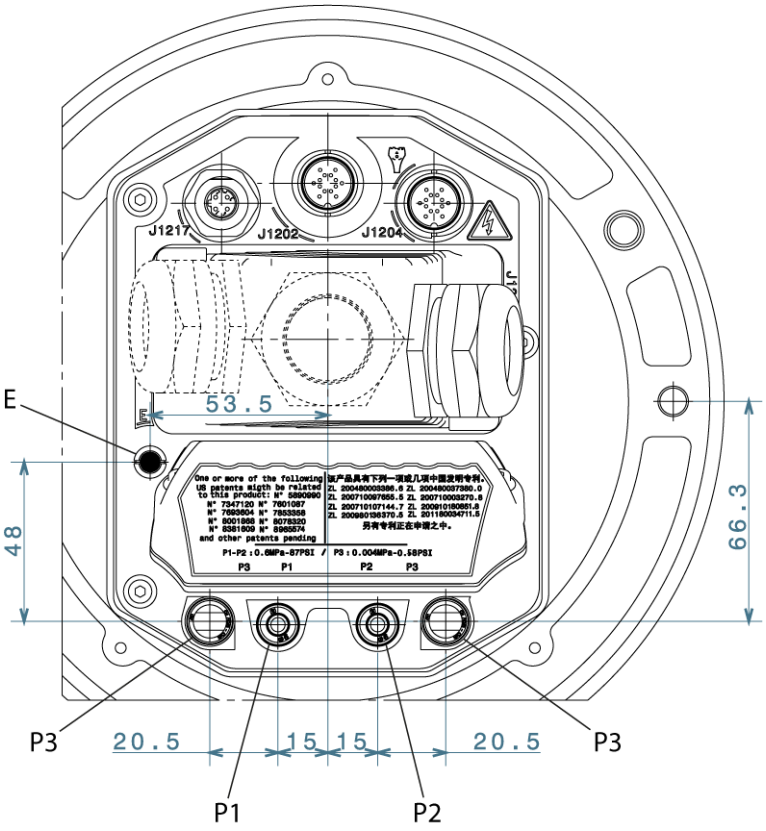


Figure 3.29: Vertical foot pneumatic interface (CW arm)

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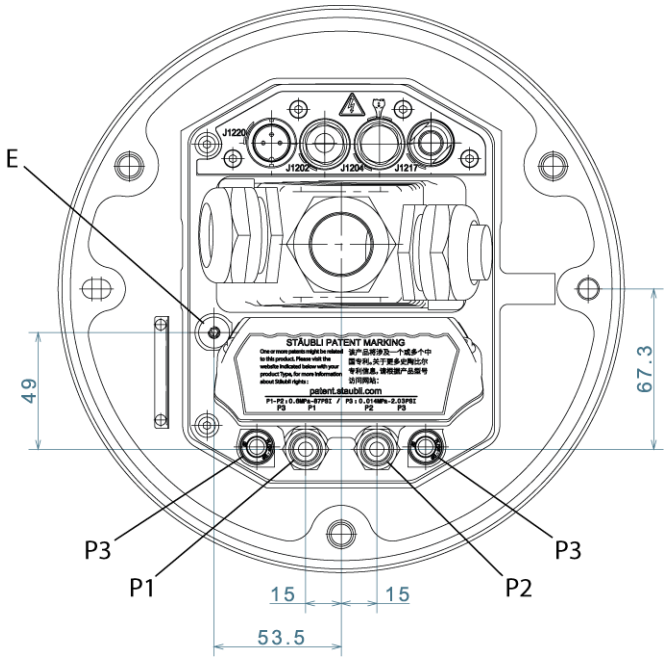
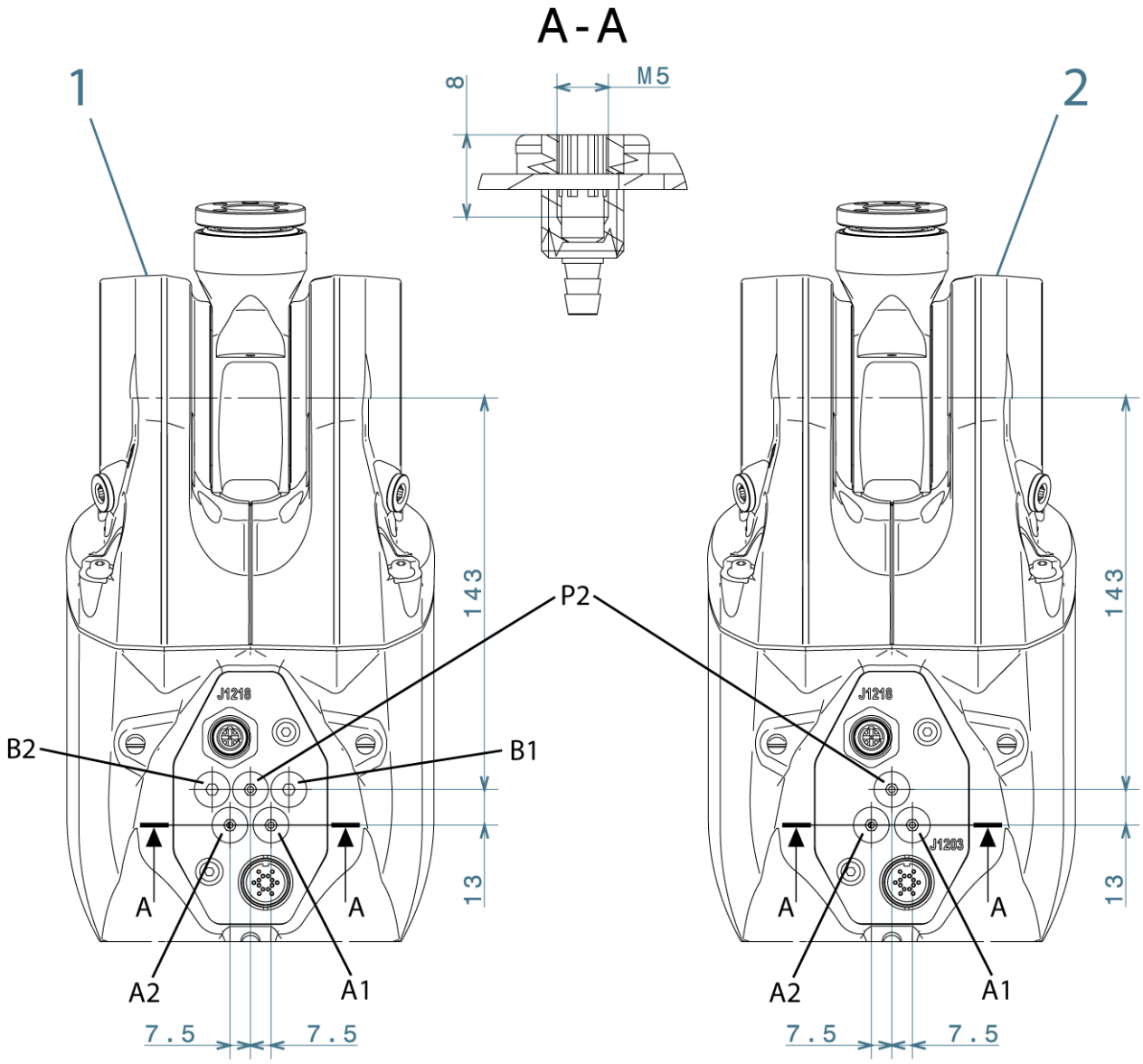


Figure 3.30: Vertical foot pneumatic interface (HW arm)

Item	Description
P1, P2, P3	Pneumatic connections
E	Exhaust muffler

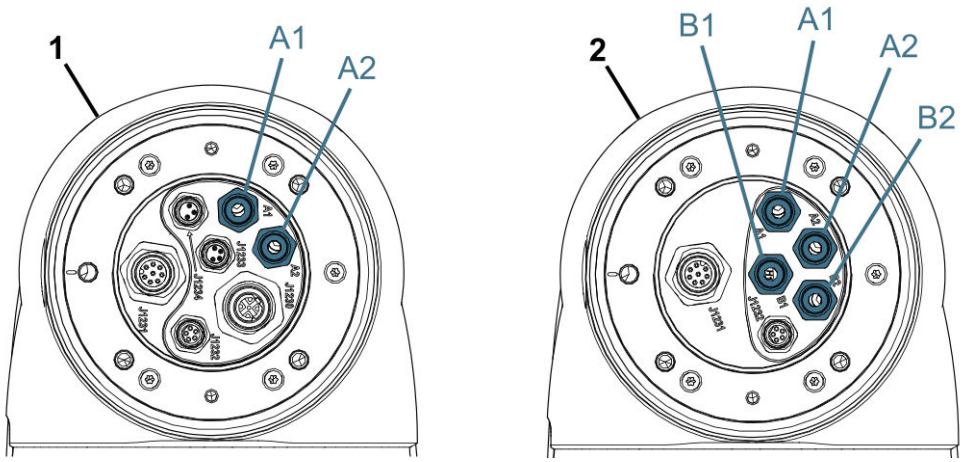
3.3.7.4 - Robot end pneumatic interface

M0009536.1



Item	Description
1	5 connections version (for 5/2 and 5/3 valves options)
2	3 connections version (for standard and 3/2 valves option)
P2, A1, A2, B1, B2	Pneumatic connections (Ø4)

Figure 3.31: Forearm pneumatic connections



Item	Description
1	2 hoses version (for harness version A1)
2	4 hoses version (for harness versions B, C and D)
A1, A2, B1, B2	Pneumatic connections (Ø4)

Figure 3.32: Flange pneumatic connections (Hollow wrist only)

3.3.7.5 - Pneumatic systems for compact wrist arms

M0009548.1

Standard pneumatic system

This harness exist with H_2O_2 resistant option in case of direct contact between piping and H_2O_2 during VHP decontamination cycle.

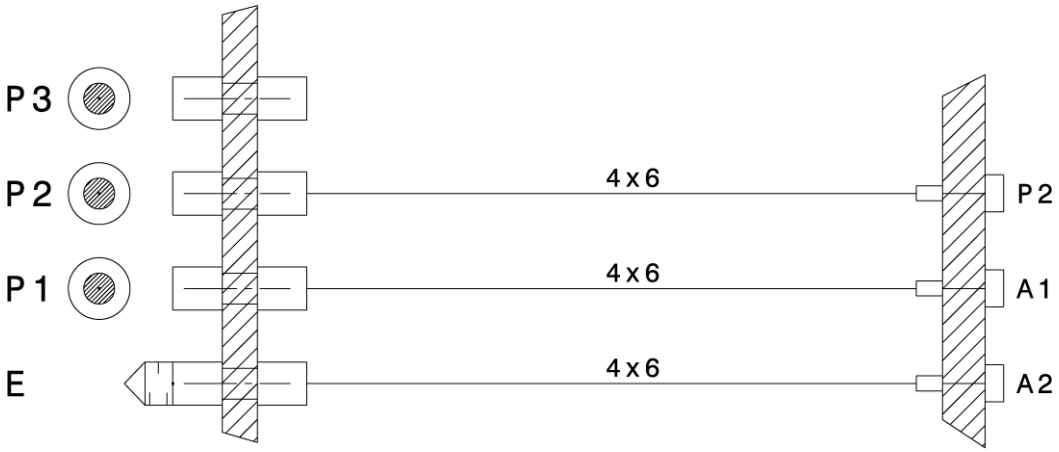


Figure 3.33: Standard pneumatic system

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- The arm is connected to the compressed air network (7 bar (100 psi) maximum and -1 bar (-14 psi) minimum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).
- There are two direct lines between the base and the forearm.
- The centralized exhaust **(A2)** is directed towards the base and its outlet is through a muffler **(E)**.

⚠ WARNING



Overpressure:
Pressure over 7 bar (100 psi) may entail risks for the user and the machine.
The air source pressure shall not overpass 7 bar (100 psi).

NOTICE



Air source
Supplying air that does not meet the following requirements will damage the arm's pneumatic system:
• Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
• The electric valves are not compatible with H_2O_2.
• Cleanliness of air must be equivalent to 5 µm filtered air.
• An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
• If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
• Ambient and fluid temperature: Do not exceed 50°C.

Pneumatic system with 5/2 bistable electric valves for use with compressed air (option)

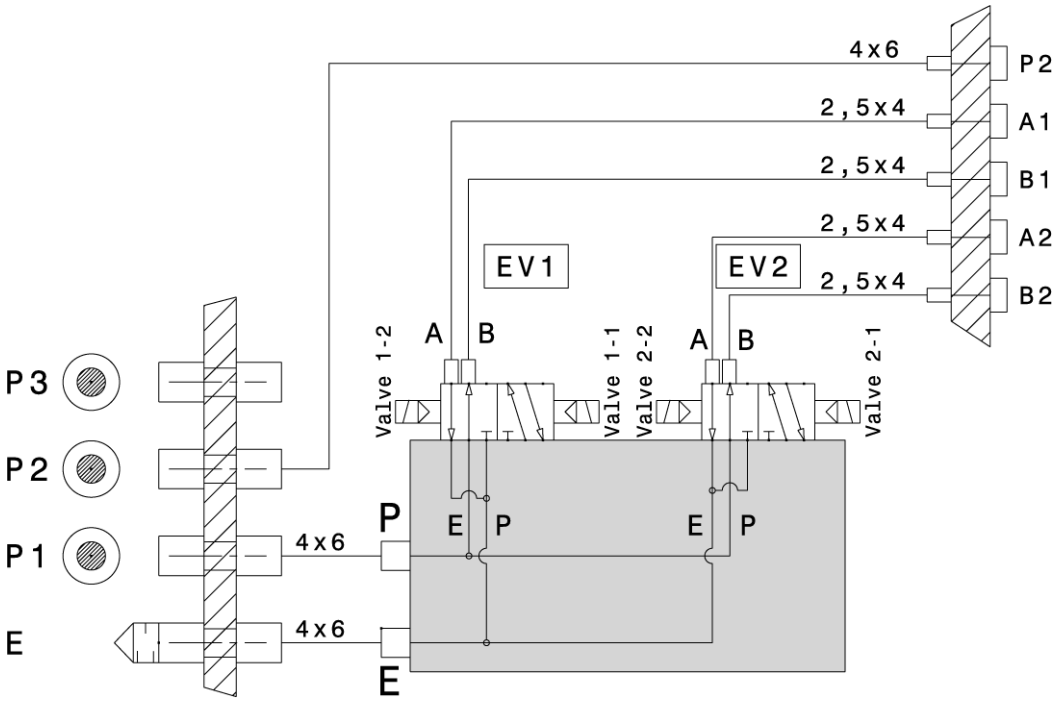


Figure 3.34: Optional 5/2 solenoid valves pneumatic system

In the event of a power failure, the solenoid valve remains in its last position.
Electric distributors (EV1 and EV2):

- 5/2-way bistable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Working pressure: 1.5 to 7 bar (21.75 to 100 psi).
- Flow coefficient: Kv = 8.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

Description:

- The arm is connected to the compressed air network (7 bar (100 psi) maximum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).

NOTICE	
	Air source Supplying air that does not meet the following requirements will damage the arm's pneumatic system: • Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction. • The electric valves are not compatible with H_2O_2. • Cleanliness of air must be equivalent to 5 µm filtered air. • An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage. • If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves. • Ambient and fluid temperature: Do not exceed 50°C.

- There is a direct line between the base (press in fittings for Ø6 hoses) and the forearm **(P2)**.
- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

Pneumatic system with 5/3 monostable electric valves for use with compressed air (option)

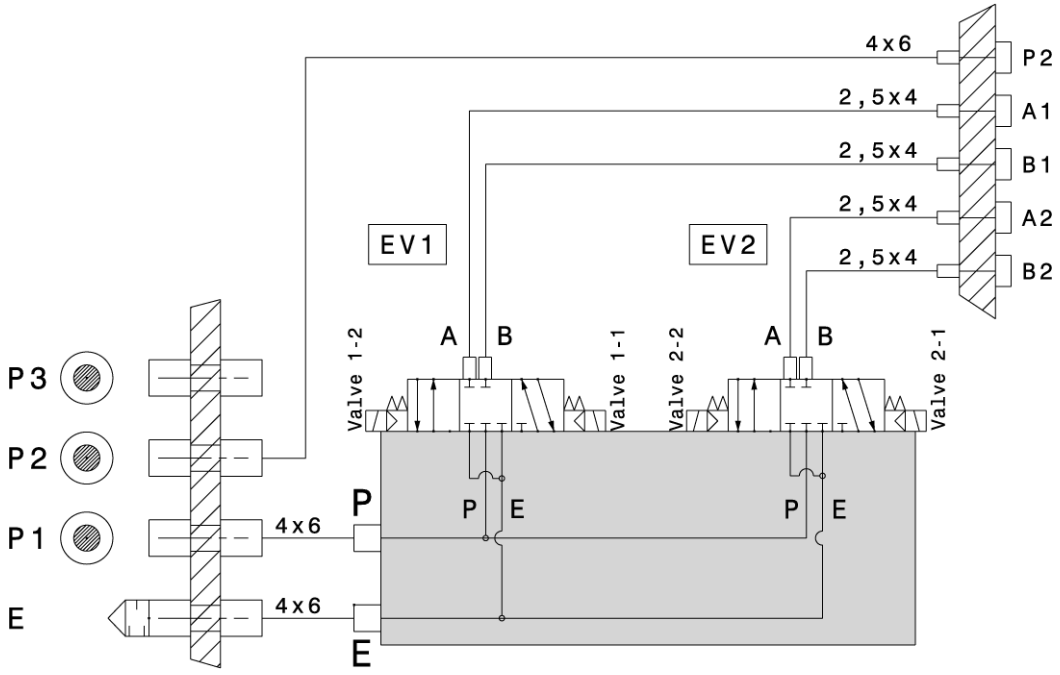


Figure 3.35: Optional 5/3 solenoid valves pneumatic system

In the event of a power failure, the solenoid valve remains in its last position.
Electric distributors (EV1 and EV2):

- 5/3-way monostable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Working pressure: 1.5 to 7 bar (21.75 to 100 psi).
- Flow coefficient: Kv = 8.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

Description:

- The arm is connected to the compressed air network (7 bar (100 psi) maximum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).

NOTICE

Air source
Supplying air that does not meet the following requirements will damage the arm's pneumatic system:
• Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
• The electric valves are not compatible with H_2O_2.
• Cleanliness of air must be equivalent to 5 µm filtered air.
• An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
• If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
• Ambient and fluid temperature: Do not exceed 50°C.

- There is a direct line between the base (press in fittings for Ø6 hoses) and the forearm **(P2)**.
- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

Pneumatic system with 3/2 monostable electric valve for use with vacuum (option)

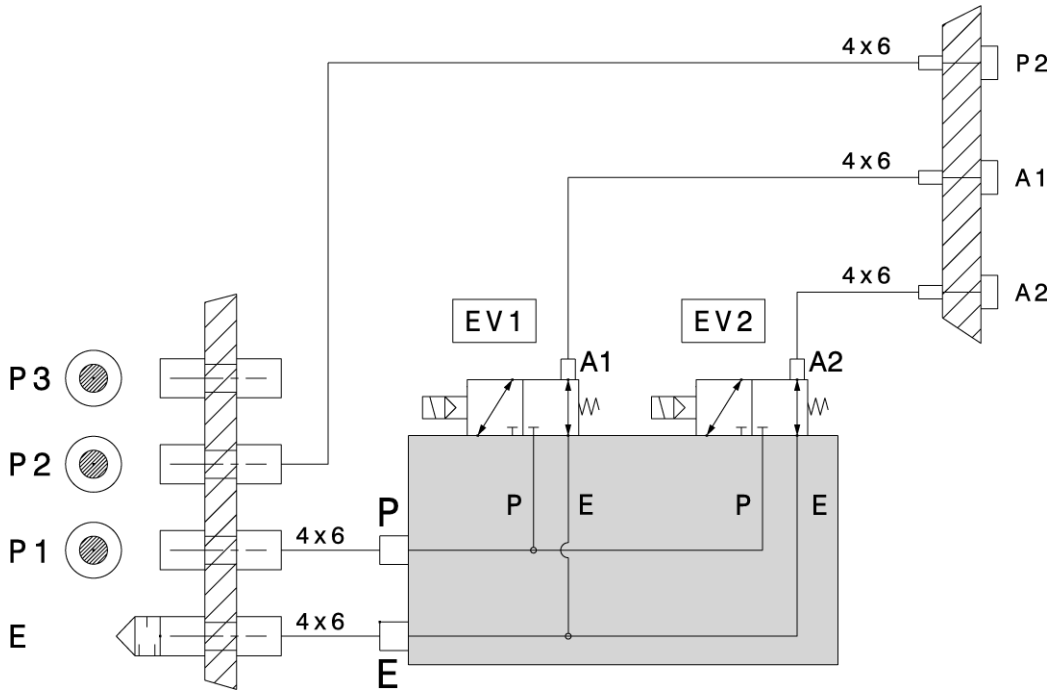


Figure 3.36: Optional 3/2 solenoid valves pneumatic system

In the event of a power failure, the solenoid valve remains in its last position.
Electric distributors (EV1 and EV2):

- 3/2-way monostable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Max. working pressure: vacuum only ≈ -0.8 bar (-11.6 psi).
- Flow coefficient: Kv = 12.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

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Description:

- The arm is connected to the vacuum network via the base **(P1)** (2 one-touch fittings for Ø6 hoses).

NOTICE	
	**Air source** Supplying air that does not meet the following requirements will damage the arm's pneumatic system: • Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction. • The electric valves are not compatible with H_2O_2. • Cleanliness of air must be equivalent to 5 µm filtered air. • An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage. • If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves. • Ambient and fluid temperature: Do not exceed 50°C.

- There is a direct line between the base (press in fittings for Ø6 hoses) and the forearm **(P2)**.
- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

NOTICE	
	**Vacuum valves:** A single-acting actuator (with a weak return spring) may be triggered in an untimely way by the air exhaust from the second valve manifold. For vacuum valves, it is preferable to use double-acting actuators.

3.3.7.6 - Pneumatic systems for hollow wrist arms

M0009549.1

2-hoses pneumatic system (version No valve 2 lines + Fieldbus)

This harness exist with H_2O_2 resistant option in case of direct contact between piping and H_2O_2 during VHP decontamination cycle.

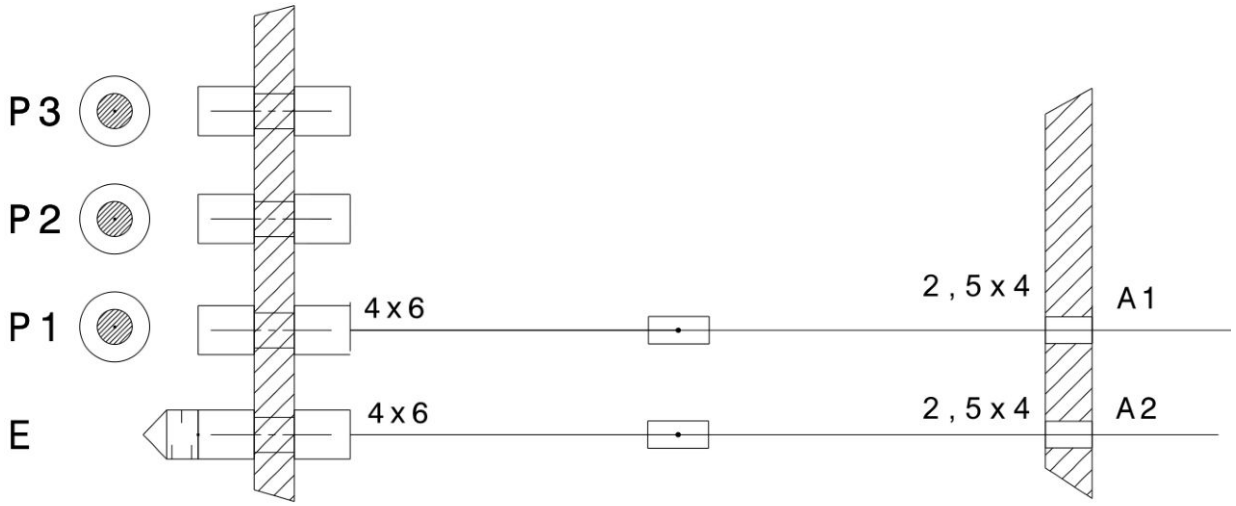


Figure 3.37: Pneumatic system version : No valve 2 lines + Fieldbus

- The arm is connected to the compressed air network (maximum 7 bar (100 psi) and minimum -1 bar (-14 psi), with or without lubrication) by means of a push-in fitting **(P1)** (for Ø6 hoses) on its base.
- There is a single direct line between the base and the flange.
- The centralized exhaust **(A2)** is directed towards the base and its outlet is through a muffler **(E)**.

⚠ WARNING

Overpressure:
Pressure over 7 bar (100 psi) may entail risks for the user and the machine.
The air source pressure shall not overpass 7 bar (100 psi).

NOTICE

Air source
Supplying air that does not meet the following requirements will damage the arm's pneumatic system:
• Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
• The electric valves are not compatible with H_2O_2.
• Cleanliness of air must be equivalent to 5 µm filtered air.
• An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
• If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
• Ambient and fluid temperature: Do not exceed 50°C.

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4-hoses pneumatic system (version No valve 2 lines)

This harness exist with H_2O_2 resistant option in case of direct contact between piping and H_2O_2 during VHP decontamination cycle.

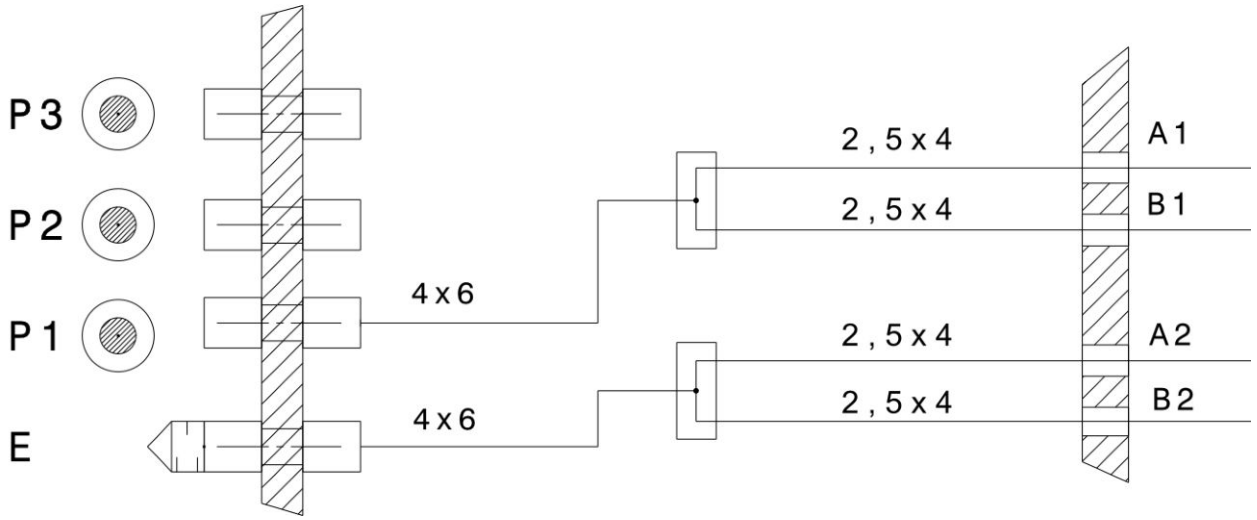


Figure 3.38: Pneumatic system version : No valve 2 lines

- The arm is connected to the compressed air network (7 bar (100 psi) maximum and -1 bar (-14 psi) minimum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).
- There are two direct lines between the base and the forearm.
- The centralized exhaust **(A2)** is directed towards the base and its outlet is through a muffler **(E)**.

⚠ WARNING

Overpressure:
Pressure over 7 bar (100 psi) may entail risks for the user and the machine.
The air source pressure shall not overpass 7 bar (100 psi).

NOTICE

Air source
Supplying air that does not meet the following requirements will damage the arm's pneumatic system:
• Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
• The electric valves are not compatible with H_2O_2.
• Cleanliness of air must be equivalent to 5 µm filtered air.
• An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
• If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
• Ambient and fluid temperature: Do not exceed 50°C.

Pneumatic system with 5/2 bistable electric valves for use with compressed air (option)

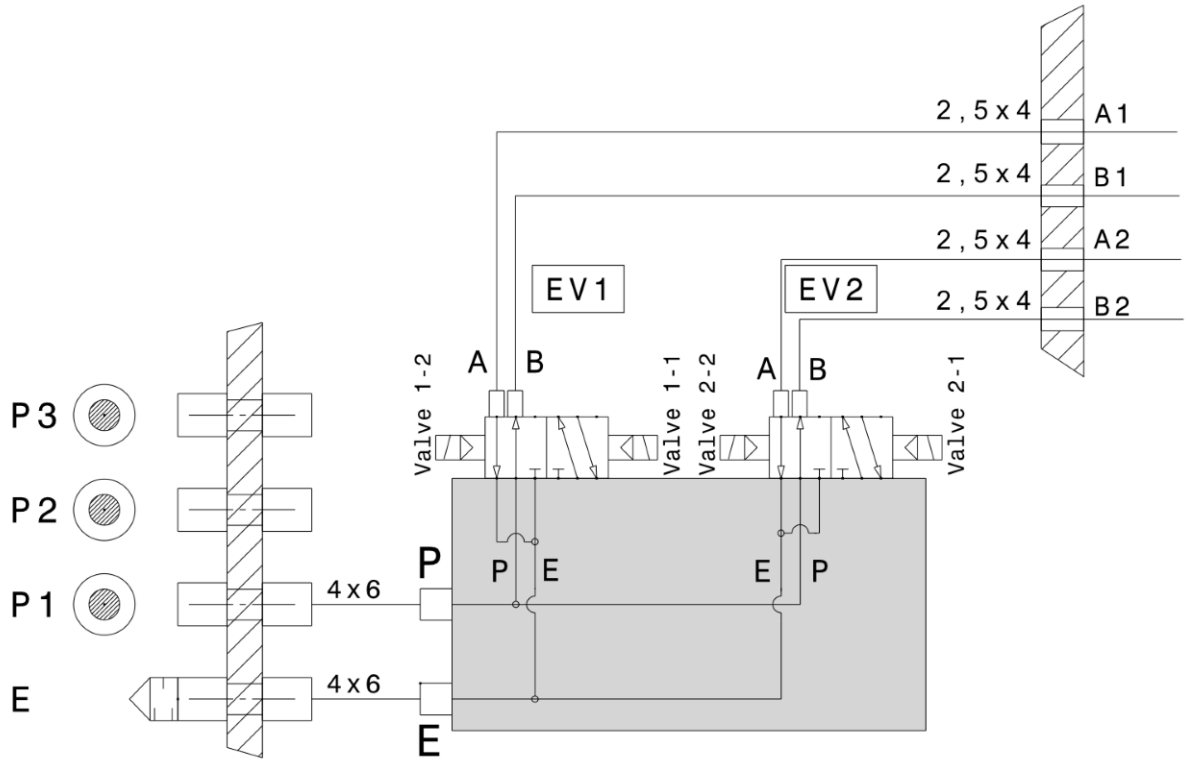


Figure 3.39: Optional 5/2 solenoid valves pneumatic system

Electric distributors (EV1 and EV2):

- 5/2-way bistable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Working pressure: 1.5 to 7 bar (21.75 to 100 psi).
- Flow coefficient: Kv = 8.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

Description:

- The arm is connected to the compressed air network (7 bar (100 psi) maximum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).

NOTICE

Air source

Supplying air that does not meet the following requirements will damage the arm's pneumatic system:

- Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
- The electric valves are not compatible with H_2O_2.
- Cleanliness of air must be equivalent to 5 µm filtered air.
- An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
- If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
- Ambient and fluid temperature: Do not exceed 50°C.

- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

Pneumatic system with 5/3 monostable electric valves for use with compressed air (option)

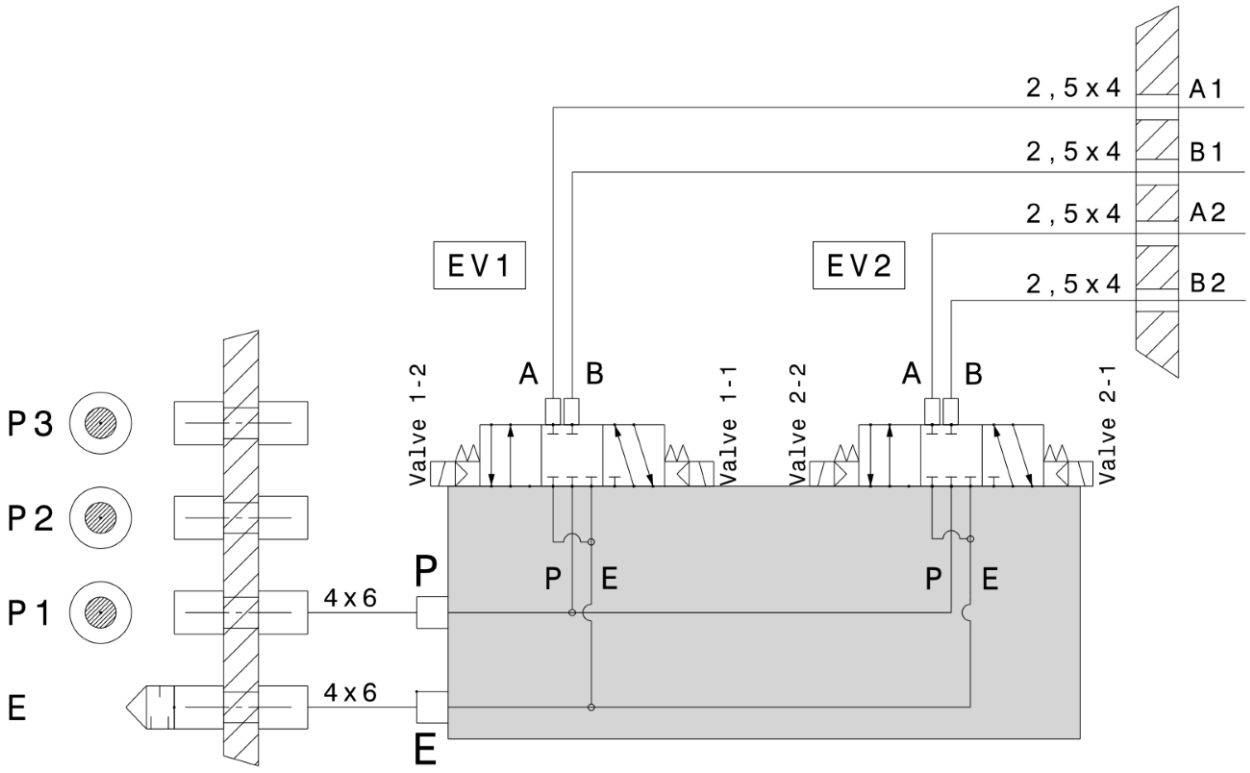


Figure 3.40: Optional 5/3 solenoid valves pneumatic system

Electric distributors (EV1 and EV2):

- 5/3-way monostable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Working pressure: 1.5 to 7 bar (21.75 to 100 psi).
- Flow coefficient: Kv = 8.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

Description:

- The arm is connected to the compressed air network (7 bar (100 psi) maximum, with or without lubrication) through 2 press in fittings (for Ø6 hoses) on its base (**(P1)** and **(P2)**).

NOTICE	
	**Air source** Supplying air that does not meet the following requirements will damage the arm's pneumatic system: • Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction. • The electric valves are not compatible with H_2O_2. • Cleanliness of air must be equivalent to 5 µm filtered air. • An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage. • If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves. • Ambient and fluid temperature: Do not exceed 50°C.

- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

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Pneumatic system with 3/2 monostable electric valve for use with vacuum (option)

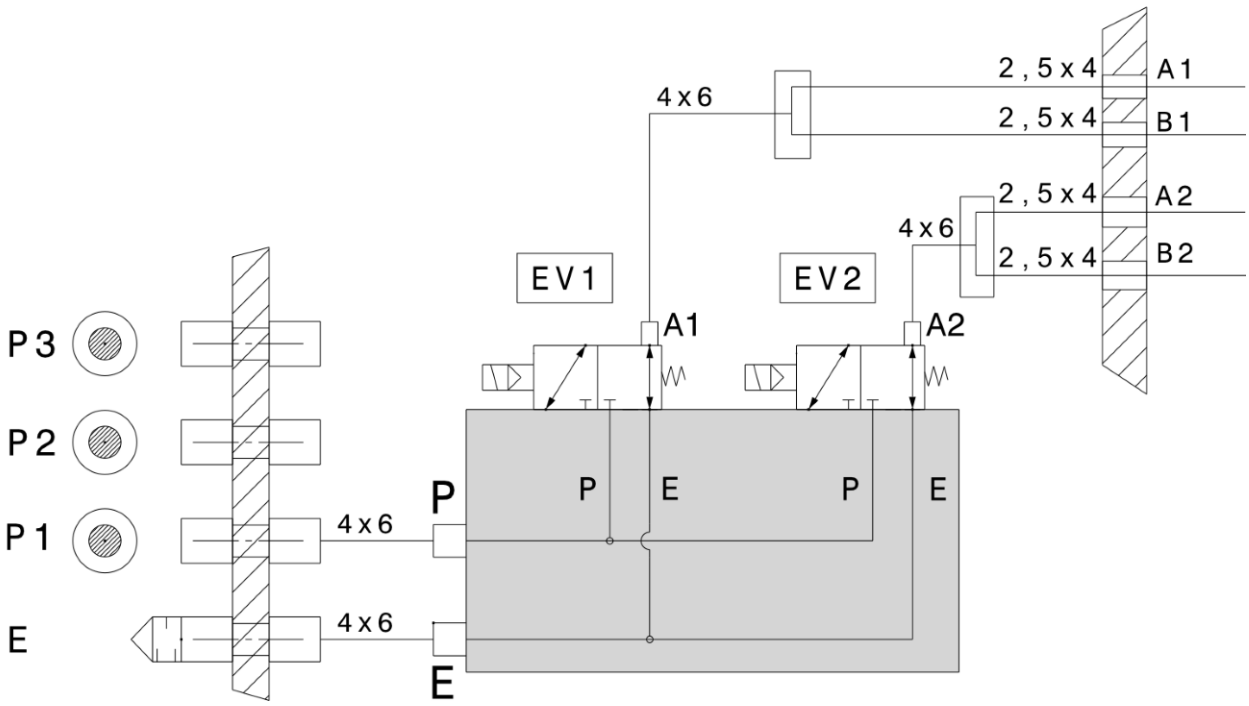


Figure 3.41: Optional 3/2 solenoid valves pneumatic system

Electric distributors (EV1 and EV2):

- 3/2-way monostable.
- Electrically controlled (24 VDC) by a safety device. See safety manual for more information.
- Max. working pressure: vacuum only ≈ -0.8 bar (-11.6 psi).
- Flow coefficient: Kv = 12.6 [m^3. h^{-1}. $bar^{-1/2}$], S = 3.6 mm^2.
- Clip-on connector.
- Overvoltage protective circuit and indicator diode.

Description:

- The arm is connected to the vacuum network via the base **(P1)** (2 one-touch fittings for Ø6 hoses).

NOTICE

Air source
Supplying air that does not meet the following requirements will damage the arm's pneumatic system:
• Do not use compressed air that contains chemicals, synthetic oils including organic solvents, salt or corrosive gases, etc. It can cause damage or malfunction.
• The electric valves are not compatible with H_2O_2.
• Cleanliness of air must be equivalent to 5 µm filtered air.
• An air dryer or water separator should be installed upstream of the filters as the compressed air may contain drainage.
• If excessive carbon dust is produced by the compressor, remove it by installing a mist separator on the upstream side of the valves.
• Ambient and fluid temperature: Do not exceed 50°C.

- The exhaust from the valve manifold is directed to the base and through a muffler **(E)**.

NOTICE

Vacuum valves:
A single-acting actuator (with a weak return spring) may be triggered in an untimely way by the air exhaust from the second valve manifold.
For vacuum valves, it is preferable to use double-acting actuators.

3.3.8 - ELECTRICAL CIRCUIT

M0009675.1

NOTICE

Permissible current
• 3-wire conductor AWG20: 3 A max. per contact.
• Shielded pairs AWG26: 1 A max. per contact.
• Pairs AWG24: 2 A max. per contact.

NOTICE

Supply voltage
60 VDC max. (14-contact connector) and 50 VDC max. (Ethernet socket).
The power supplies must be PELV according to EN 60204-1 6.4.2.

Minimum bending radius:
- 3-wire conductor AWG20: 7.5mm static, 15mm dynamic.
- Twisted pairs AWG24: 5.5mm static, 11mm dynamic.
- FIELDBUS: 24mm static, 48mm dynamic.

Stäubli recommends using a CO2 extinguisher in case of fire.

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⚠ WARNING



Electrical circuit failure

Even if preventive maintenance recommendations are followed perfectly, the risk of electrical circuit failure (short circuit or open circuit) cannot be excluded.

This risk must be taken into account by the integrator when designing the application.

⚠ CAUTION



Shields usage

Do not use the shields as a conductive cable.

⚠ WARNING



Arm internal wiring modification

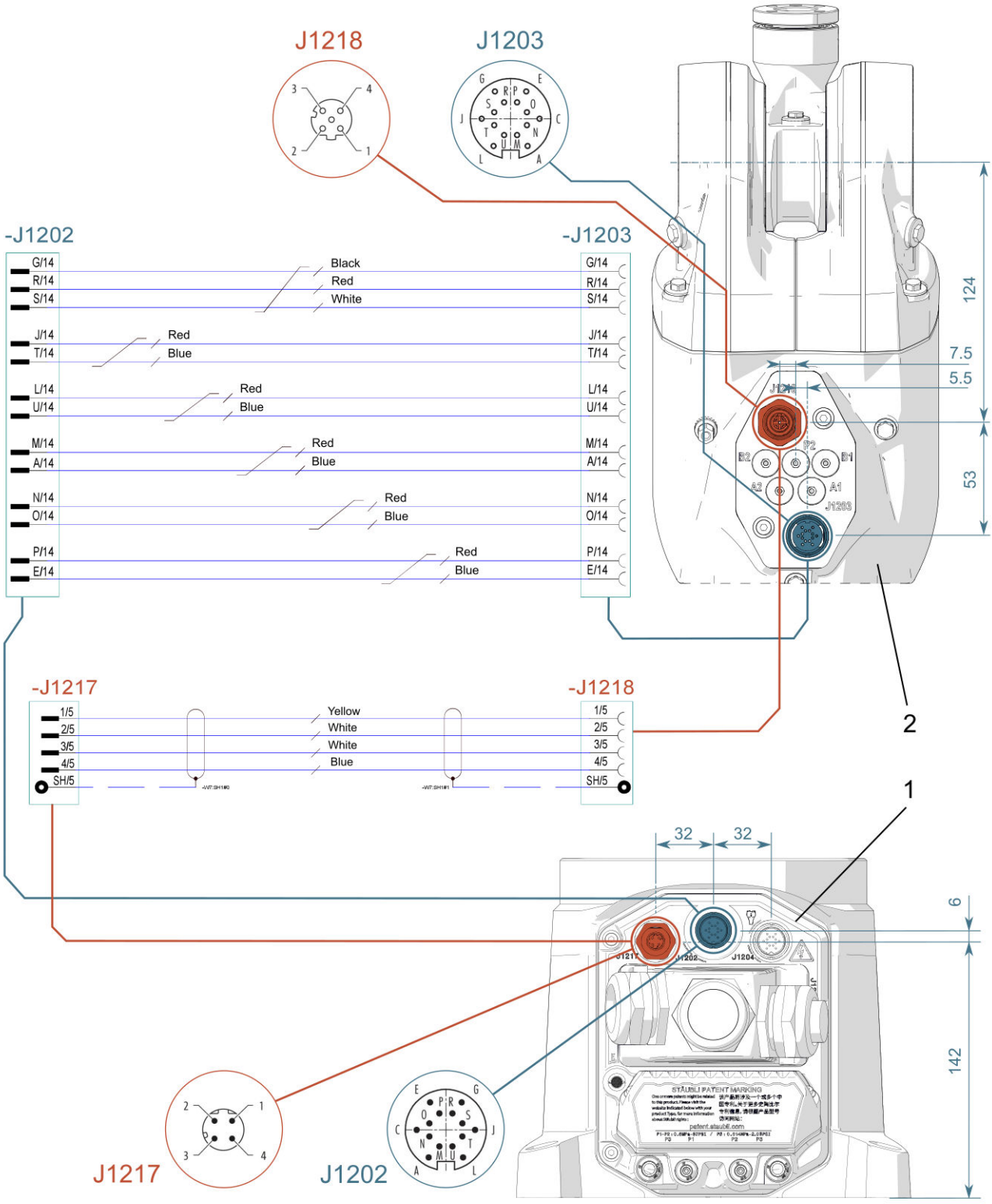
Adding wires or cables to internal arm wiring can cause premature wear to arm electrical wiring and affect arm safety functions.

Do not add wires or cables to internal arm wiring.

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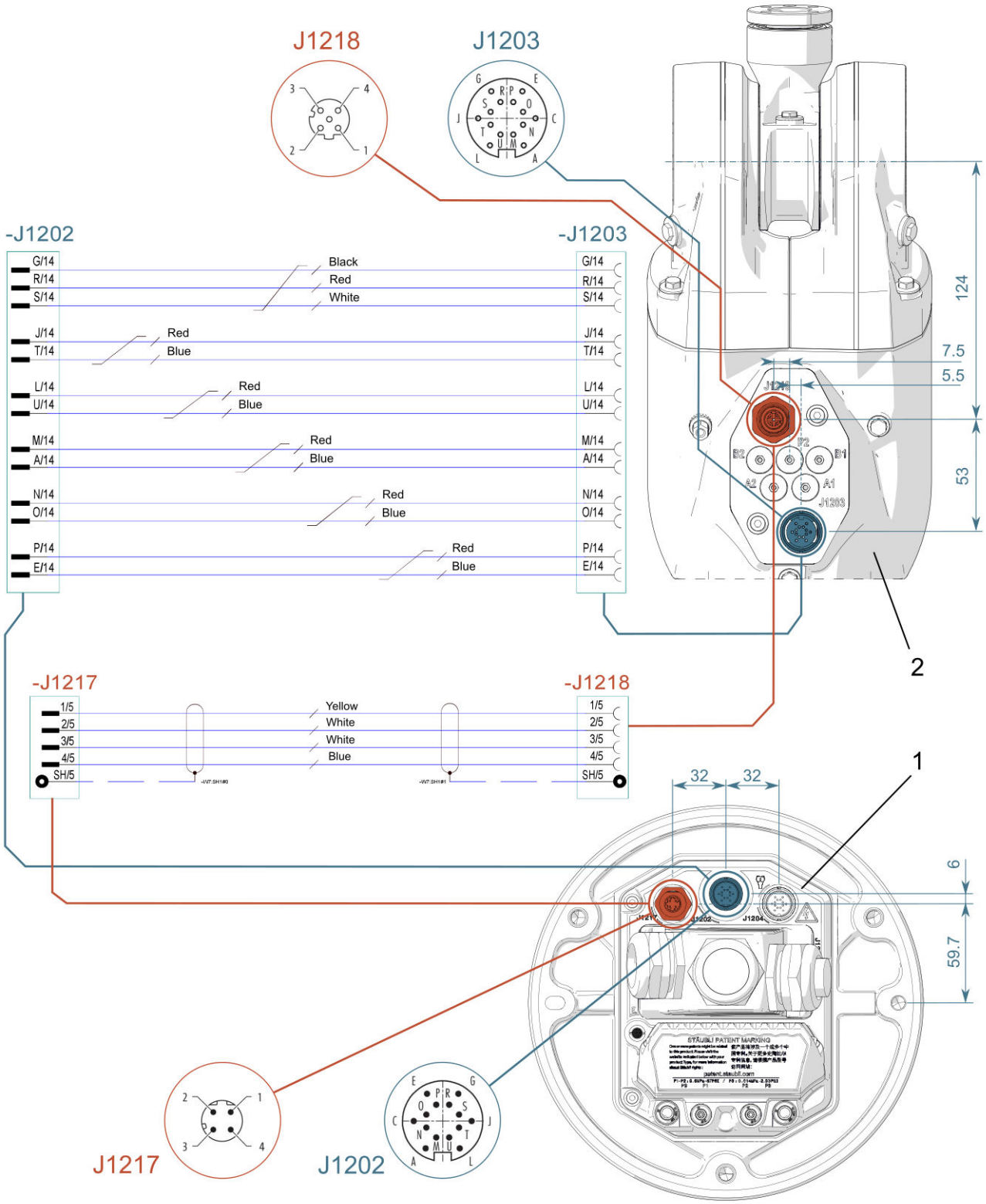
3.3.8.1 - Electrical circuit for compact wrist version

M0008497.1



Item	Description
1	Plate attached to base
2	Forearm

Figure 3.42: Compact wrist - Horizontal cable outlet



Item	Description
1	Plate attached to base
2	Forearm

Figure 3.43: Compact wrist - Vertical cable outlet

The electrical circuit consists of:

- A male 14-contact socket (J1202), M16, with only 13 connected cables, and a male 4-contact socket (J1217) (M12 Ethernet code D) at the base,
- A female 14-contact socket (J1203), M16, with only 13 connected cables, and a female 4-contact socket (J1218) (M12 Ethernet code D) on the forearm.

The 13 signal contacts in each socket are connected in the following way:

- 3 power contacts in each socket, connected by a 3-wire conductor with cross-section AWG20 (contacts G-R-S),
- 5 twisted pairs, cross-section AWG24 for the other contacts.

The 4 contacts are connected via a twin shielded pair AWG26 Ethernet cable. Cat. 5e compliant with a guaranteed transfer rate of 100 Mb/s.

NOTICE

Supply voltage
60 VDC max. (14-contact connector) and 50 VDC max. (Ethernet socket).
The power supplies must be PELV according to EN 60204-1 6.4.2.

NOTICE

Permissible current
• 3-wire conductor AWG20: 3 A max. per contact.
• Shielded pairs AWG26: 1 A max. per contact.
• Pairs AWG24: 2 A max. per contact.

In option:

Connection	Type of connection
Forearm (J1203)	Elbow male M16 (BINDER 99 0451 75 14), (Stäubli reference: D610 474 00)
Forearm (J1218)	Elbow male M12 code D (PHOENIX CONTACT SACC-M12MRD-4Q SH - 1553624), (Stäubli reference: D610 422 00)
Base (J1202)	Female cylindrical M16 (BINDER 09 0452 70 14), (Stäubli reference: D610 473 00)
Base (J1217)	Female cylindrical M12 code D (PHOENIX CONTACT SACC-M12FSD-4Q SH – 1553611), (Stäubli reference: D610 421 00)

NOTICE

Impact on the working envelope
Using these optional connectors (on the forearm) may reduce the working envelope.
Consider their volume when programming.

Stäubli recommends using a CO2 extinguisher in case of fire.

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⚠ WARNING

Electrical circuit failure
Even if preventive maintenance recommendations are followed perfectly, the risk of electrical circuit failure (short circuit or open circuit) cannot be excluded.
This risk must be taken into account by the integrator when designing the application.

⚠ CAUTION

Shields usage
Do not use the shields as a conductive cable.

⚠ WARNING

Arm internal wiring modification
Adding wires or cables to internal arm wiring can cause premature wear to arm electrical wiring and affect arm safety functions.
Do not add wires or cables to internal arm wiring.

3.3.8.2 - Electrical circuit for hollow wrist, pneumatic version (no Fieldbus)

M0009505.1

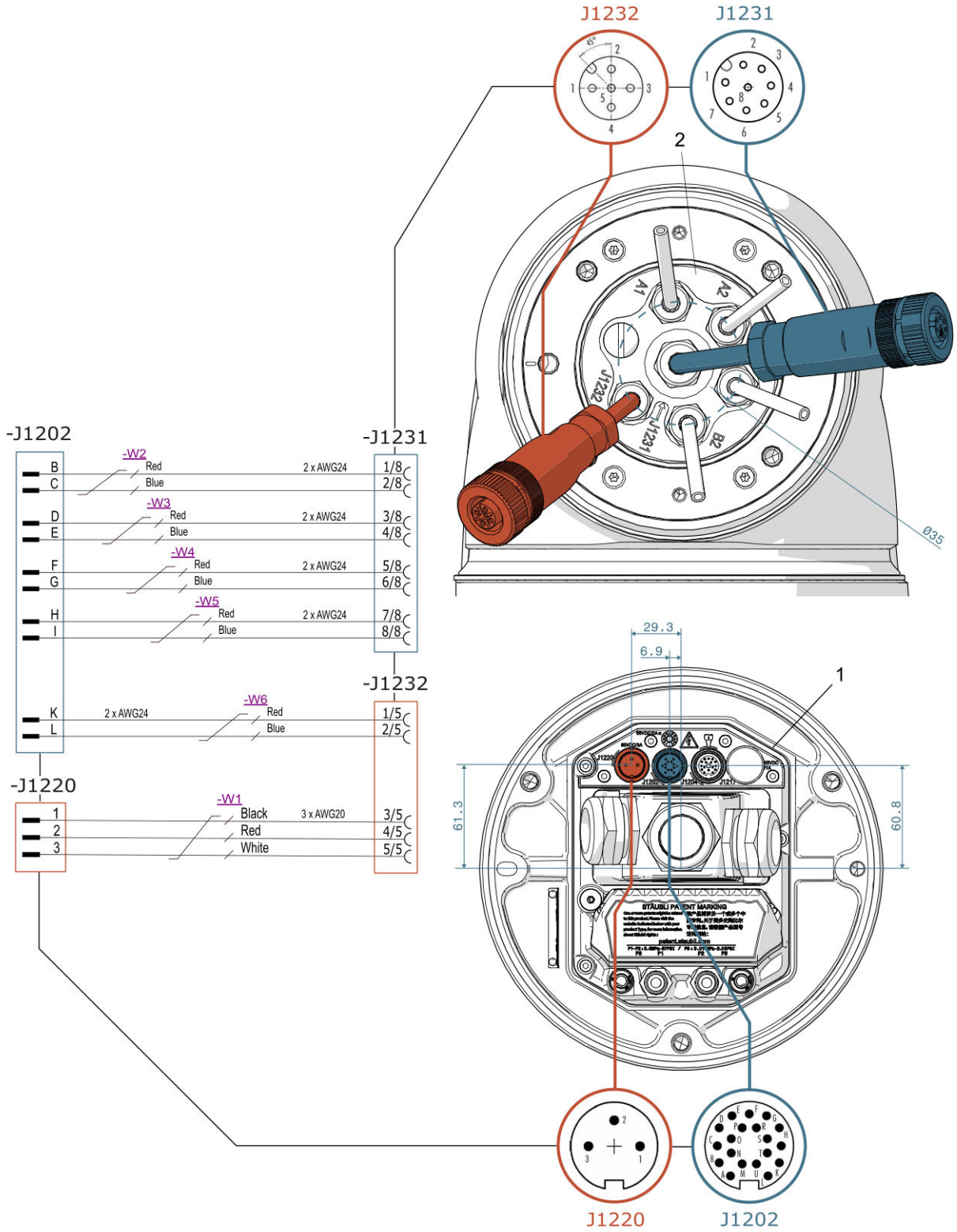
These electrical circuits apply to the following version of the wrist:

- No valve 2 lines .
- 3/2 valves 2 lines.
- 5/2 valves 2 lines.
- 5/3 valves 2 lines.

For flying connectors option, any modifications to cables or connectors made by the robot integrator will:

- Result in the loss of robots certifications
- Must ensure that the EOAT, its connectors and contacts, and its pneumatic connections can be dismantled and reassembled without the need for special tools, so that the robot can be maintained. Otherwise, additional maintenance costs may apply.

The cables and connectors of the flying connectors option must be contained inside the customer tool.



Item	Description
1	Plate attached to base
2	Cable gland on flange

Figure 3.44: Hollow wrist - Flying connectors - No valve 2 lines

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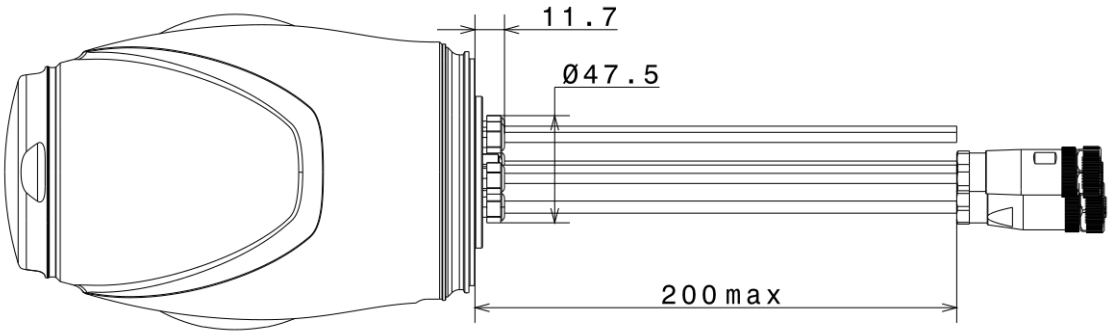


Figure 3.45: Flying connectors

The cables coming out of the flange cable gland are approximately 200 mm long.
The flying connectors located at the ends of the cables are as follows :

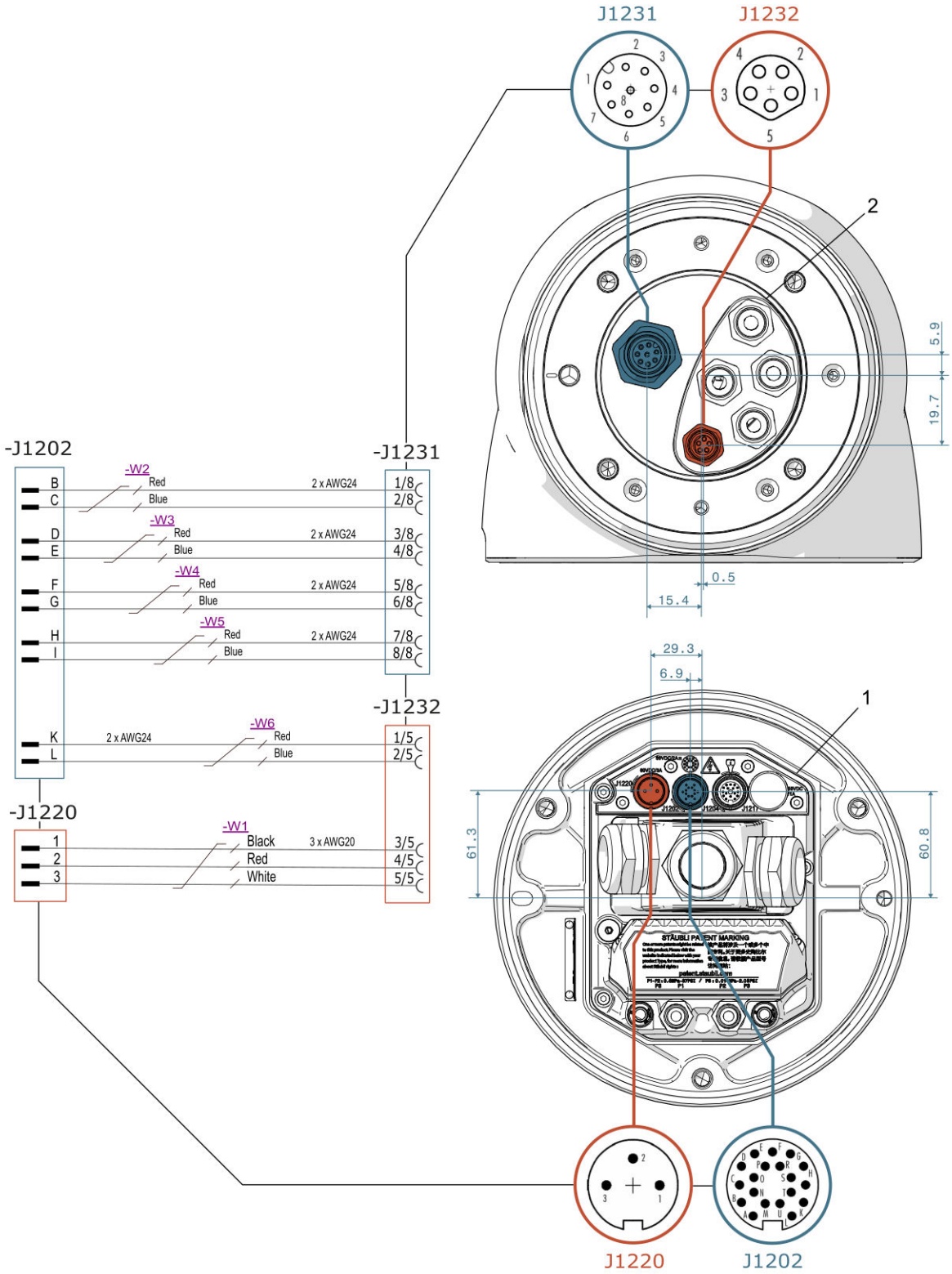
Connection	Type of connection
Hollow wrist (J1232)	Female cylindrical M12 (5-pole Binder 99 0436 14 05)
Hollow wrist (J1231)	Female cylindrical M12 (8-pole Binder 99 0486 12 08)

The connectors located on the connection plate at the base of the robot are as follows :

Connection	Type of connection
Base (J1202)	Female cylindrical M16 (19-pole Binder 09 0462 70 19), (Stäubli reference: C650 200 77)
Base (J1220)	Female cylindrical M16 (3-pole Binder 09 0106 25 03), (Stäubli reference: D610 743 00)

The electrical circuit consists of:

- A male 3-contact socket (J1220), M16, at the base, connected to a 3-wire conductor (cross-section AWG20) running up to the flange cable gland, without an output connector, labelled (W1).
- A male 19-contact socket (J1202), M16, at the base, connected to 5 twisted pairs (cross-section AWG24) running up to the flange cable gland, without an output connector , labelled (W2) to (W6).



Item	Description
1	Plate attached to base
2	Cable gland on flange

Figure 3.46: Hollow wrist - Flange-mounted connectors - No valve 2 lines

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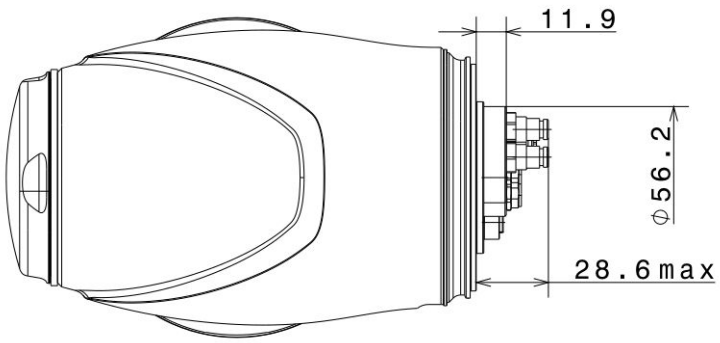


Figure 3.47: Flange-mounted connectors

The connectors located on the flange are as follows :

Connection	Type of connection
Hollow wrist (J1231)	Female cylindrical M12 (8-pole Binder 76 2632 1111 00008-0200), (Stäubli reference: D610 749 00)
Hollow wrist (J1232)	Female cylindrical M8 (5-pole Binder 76 6618 1111 00005-0200), (Stäubli reference: D610 748 00)

The connectors located on the connection plate at the base of the robot are as follows :

Connection	Type of connection
Base (J1202)	Female cylindrical M16 (19-pole Binder 09 0462 70 19), (Stäubli reference: C650 200 77)
Base (J1220)	Female cylindrical M16 (3-pole Binder 09 0106 25 03), (Stäubli reference: D610 743 00)

The electrical circuit consists of:

- A male 3-contact socket (J1220), M16, at the base, connected to a 3-wire conductor (cross-section AWG20) running up to the flange cable gland, without an output connector, labelled (W1).
- A male 19-contact socket (J1202), M16, at the base, connected to 5 twisted pairs (cross-section AWG24) running up to the flange cable gland, without an output connector , labelled (W2) to (W6).

The list below suggests connectors possible for the connection by the EOAT side:

- M12 5-pole Binder 77 3429 0000 50005-0200 (molded on cable).
- M12 8-pole Binder 77 3429 0000 50708-0200 (molded on cable).

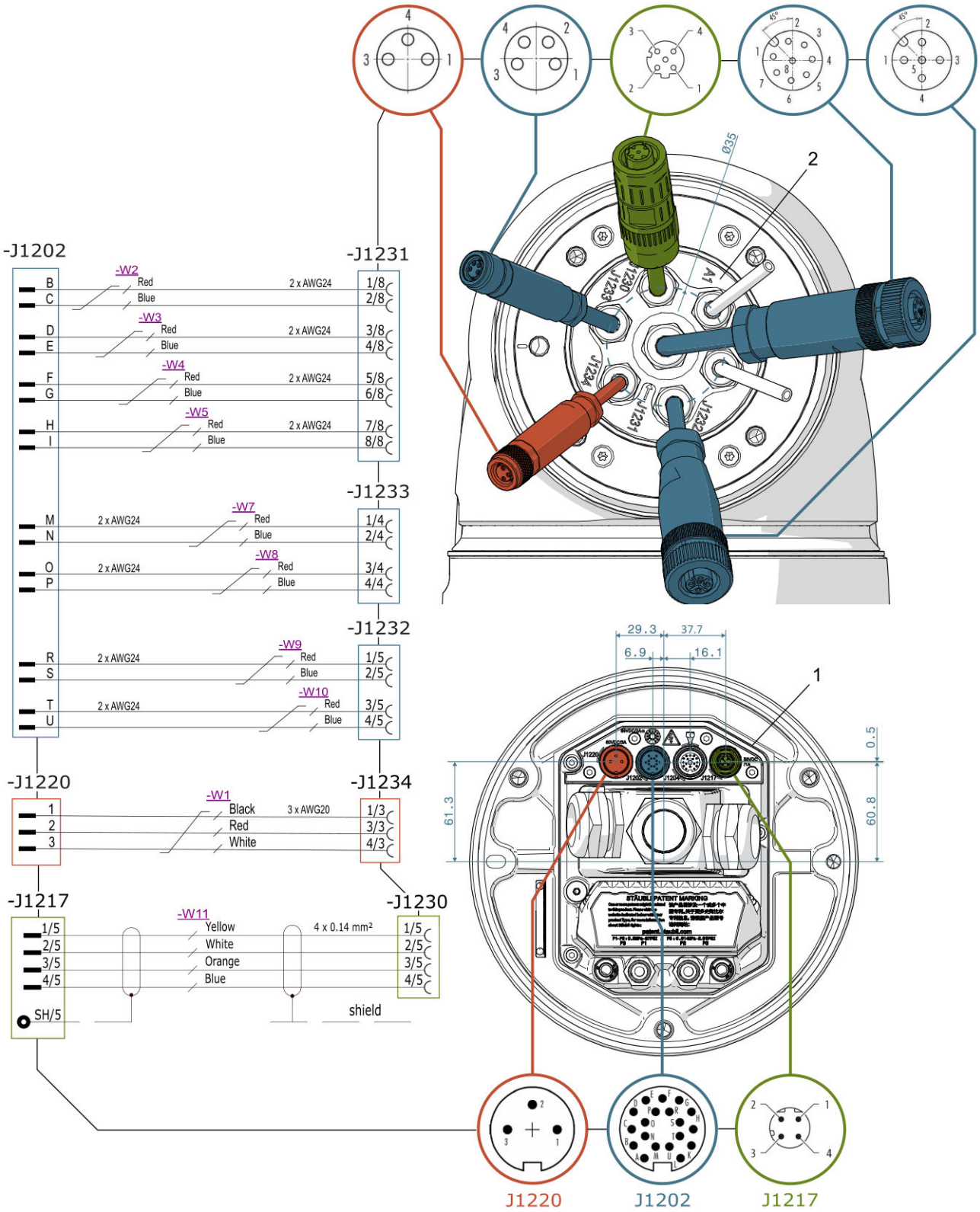
3.3.8.3 - Electrical circuit for hollow wrist, no valve 2 lines + Fieldbus

M0009671.1

For flying connectors option, any modifications to cables or connectors made by the robot integrator will:

- Result in the loss of robots certifications
- Must ensure that the EOAT, its connectors and contacts, and its pneumatic connections can be dismantled and reassembled without the need for special tools, so that the robot can be maintained. Otherwise, additional maintenance costs may apply.

The cables and connectors of the flying connectors option must be contained inside the customer tool.



Item	Description
1	Plate attached to base
2	Cable gland on flange

Figure 3.48: Hollow wrist - Flying connectors - No valve 2 lines + Fieldbus

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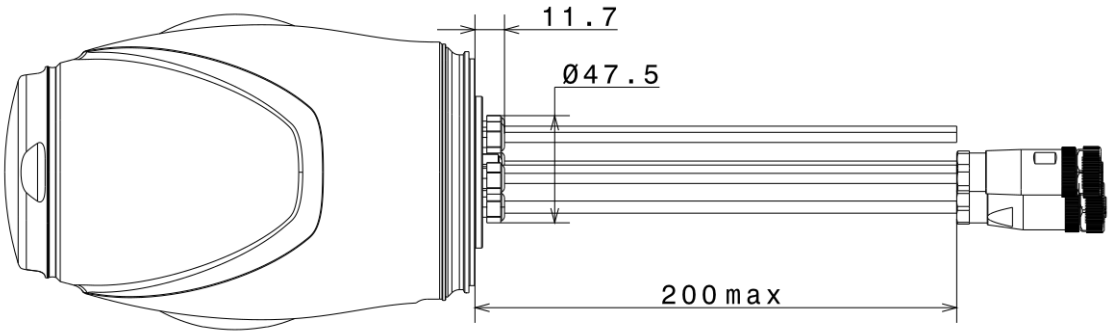


Figure 3.49: Flying connectors

The cables coming out of the flange cable gland are approximately 200 mm long.
The flying connectors located at the ends of the cables are as follows :

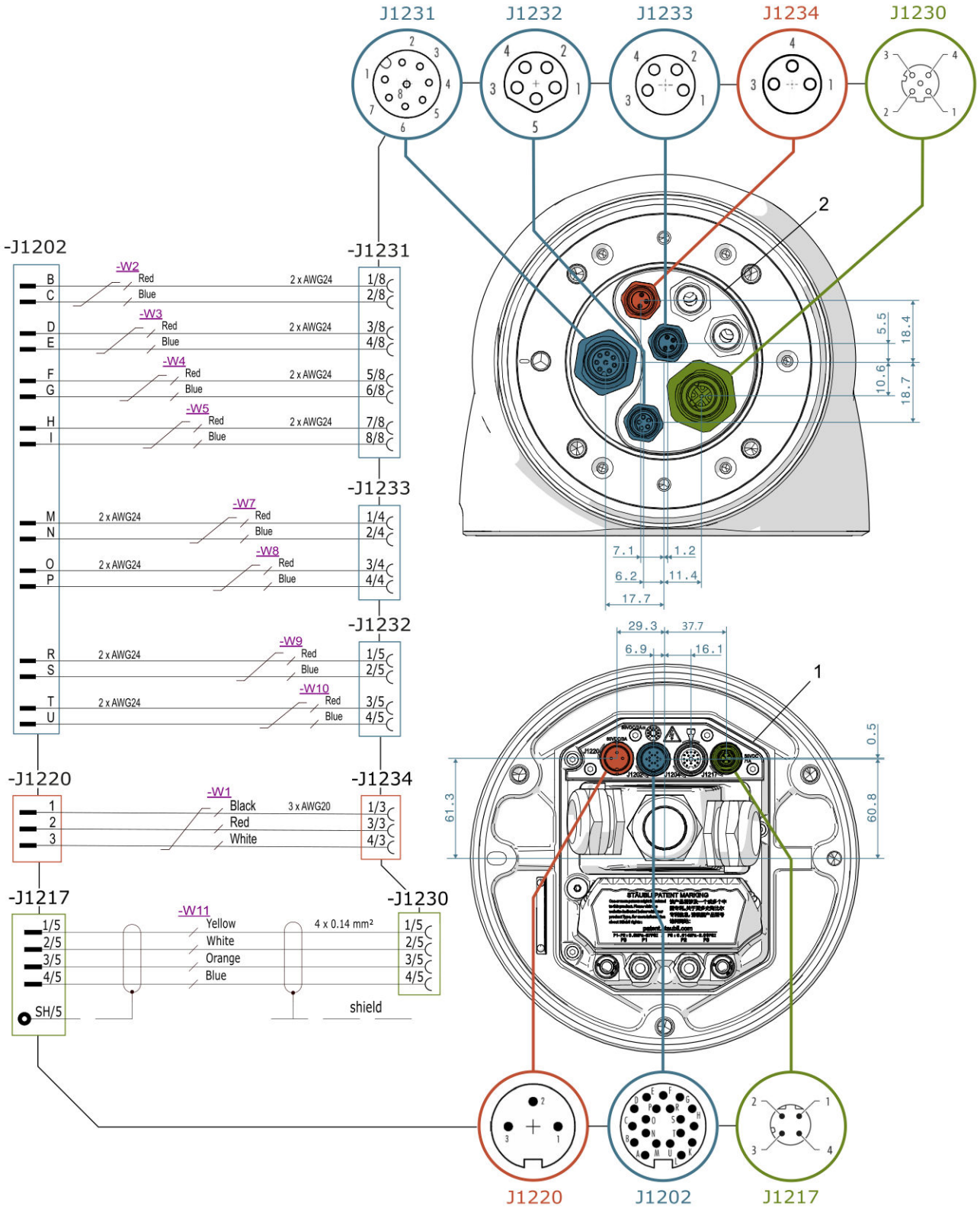
Connection	Type of connection
Hollow wrist (J1234)	Female cylindrical M8 (3-pole Binder 99 3400 100 03)
Hollow wrist (J1233)	Female cylindrical M8 (4-pole Binder 99 3376 100 04)
Hollow wrist (J1232)	Female cylindrical M12 (5-pole Binder 99 0436 14 05)
Hollow wrist (J1231)	Female cylindrical M12 (8-pole Binder 99 0486 12 08)
Hollow wrist (J1230)	Female cylindrical M12 (D-coding 4 poles Harting 21038812405)

The connectors located on the connection plate at the base of the robot are as follows :

Connection	Type of connection
Base (J1202)	Female cylindrical M16 (19-pole Binder 09 0462 70 19), (Stäubli reference: C650 200 77)
Base (J1220)	Female cylindrical M16 (3-pole Binder 09 0106 25 03), (Stäubli reference: D610 743 00)

The electrical circuit consists of:

- A male 3-contact socket (J1220), M16, at the base, connected to a 3-wire conductor (cross-section AWG20) running up to the flange cable gland, without an output connector, labelled (W1).
- A male 19-contact socket (J1202), M16, at the base, connected to 9 twisted pairs (cross-section AWG24) running up to the flange cable gland, without an output connector , labelled (W2) to (W10).



Item	Description
1	Plate attached to base
2	Cable gland on flange

Figure 3.50: Hollow wrist - Flange-mounted connectors - No valve 2 lines + Fieldbus

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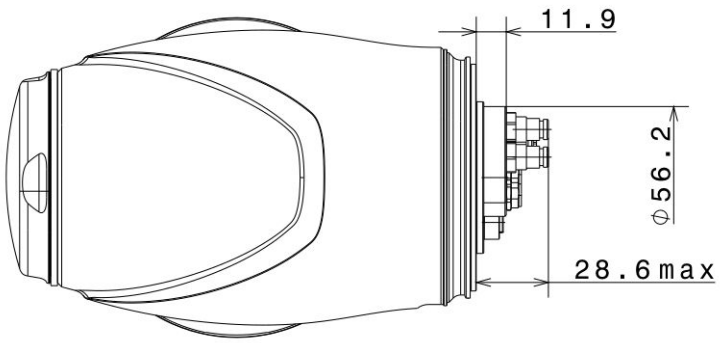


Figure 3.51: Flange-mounted connectors

The connectors located on the flange are as follows :

Connection	Type of connection
Hollow wrist (J1234)	Female cylindrical M8 (3-pole Binder 76 6618 1111 00003-0200), (Stäubli reference: D610 746 00)
Hollow wrist (J1233)	Female cylindrical M8 (4-pole Binder 76 6618 1111 00004-0200), (Stäubli reference: D610 747 00)
Hollow wrist (J1232)	Female cylindrical M8 (5-pole Binder 76 6618 1111 00005-0200), (Stäubli reference: D610 748 00)
Hollow wrist (J1231)	Female cylindrical M12 (8-pole Binder 76 2632 1111 00008-0200), (Stäubli reference: D610 749 00)
Hollow wrist (J1230)	Female cylindrical M12 (D-coding 4 poles Harting 21038812425), (Stäubli reference: D610 744 00)

The connectors located on the connection plate at the base of the robot are as follows :

Connection	Type of connection
Base (J1202)	Female cylindrical M16 (19-pole Binder 09 0462 70 19), (Stäubli reference: C650 200 77)
Base (J1220)	Female cylindrical M16 (3-pole Binder 09 0106 25 03), (Stäubli reference: D610 743 00)

The electrical circuit consists of:

- A male 3-contact socket (J1220), M16, at the base, connected to a 3-wire conductor (cross-section AWG20) running up to the flange cable gland, without an output connector, labelled (W1).
- A male 19-contact socket (J1202), M16, at the base, connected to 9 twisted pairs (cross-section AWG24) running up to the flange cable gland, without an output connector , labelled (W2) to (W10).

The list below suggests connectors possible for the connection by the EOAT side:

- M12 D-coding 4 poles Harting 21 03 881 1406 and associated contacts 09 67 000 5576.
- M8 3-pole Binder 77 3405 0000 50003-0200 (molded on cable).
- M8 4-pole Binder 77 3405 0000 50004-0200 (molded on cable).
- M8 5-pole Binder 77 3405 0000 50005-0200 (molded on cable).
- M12 5-pole Binder 77 3429 0000 50005-0200 (molded on cable).

- M12 8-pole Binder 77 3429 0000 50708-0200 (molded on cable).

3.3.9 - PRESSURIZATION UNIT

M0008849.1

For relative information about REACH 1907/2006 regulation (CE) consult the following internet address: https://www.staubli.com/en/robotics-reach/

3.3.9.1 - Purpose

M0009351.1

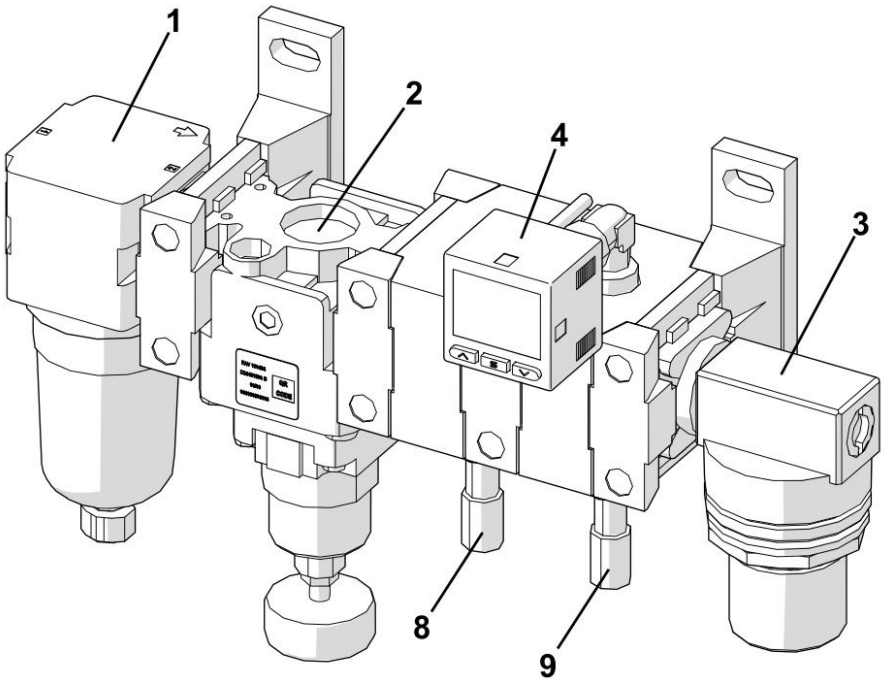


Figure 3.52:

Pressurization avoids entry of dust or liquids in the robot arm by maintaining an internal pressure that is greater than the external pressure.
The unit is made up of an air filter **(1)** (filtration 5 microns), a pressure regulator **(2)**, a pressure control valve **(3)** and a high-precision digital pressure switch **(4)**.

NOTICE	
	**Unsafe pressure control valve** Overpressure in the arm wont trigger a safety stop. The pressure control valve is not a safety valve, a safety valve must be added.

NOTICE	
	**Excessive pressurization** Excessive pressurization might damage the arm. Never apply an overpressure greater than 0.02 Mpa (2.9 psi) inside the arm.

NOTICE	
	**Inadequate pressurization** Inadequate pressurization may damage the arm and void Stäubli's warranty. Use a separate pressurization unit for each robot. Use only the Stäubli pressurization unit or one with the same characteristics.

⚠ CAUTION

Unsuitable environment for the pressurization unit

Using the Stäubli pressurization unit inside an area exposed to dust or liquids spattering might result in property damage.

Using the product in an environment that is exposed to organic solvents, chemicals, cutting fluids, synthetic oils, bases or thread locking solutions will damage the regulator filter and lubricator bowl, together with the lubricator sight dome, wich are made of polycarbonate.

Failure of the pressure unit may cause additional damage, such as rust from water intrusion or contamination of the arm with other substances.

Avoid the environments decribed above.

- Temperature of use: 0 to 50°C
- Air consumption: 1 l/mn max
- Electricity consumption: (see chapter 3.3.9.2.3)
- Ingress protection: Only the digital pressure switch has an IP65 protection index; the other pneumatic components feature exhaust openings and they must not be exposed to dust or spattering with liquids.

3.3.9.2 - Installation

M0007786.1

The installation requires the additional supply by Stäubli of pneumatic tubes and a 24 VDC power supply.

⚠ WARNING

Missinstallation

Failure to install the pressurisation kit in accordance with the following instructions may result in serious injury or death to the user.

Follow the instructions of this manual.

3.3.9.2.1 - Installation of the unit

M0009094.1

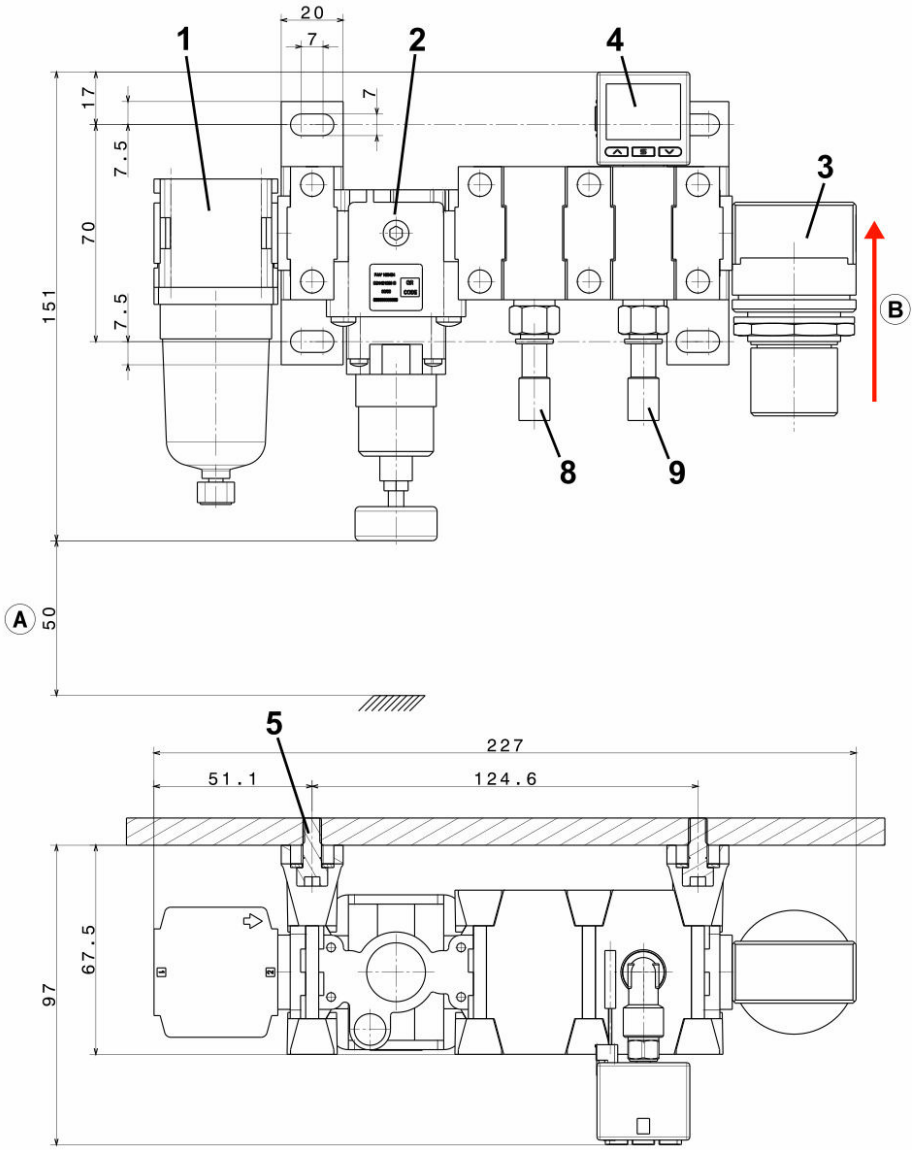


Figure 3.53:

- Fix the unit using 4 screws (Ø6) reference **(5)** (screws not supplied) on a rigid vertical surface; it is necessary to comply with the direction of the arrow **(B)** (with the air supply on the left of the unit on the air filter **(1)**).
- Leave a space **(A)** to access the lower part of the air filter bowl **(1)** for drainage operations and the pressure regulator adjustment knob **(2)**.

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3.3.9.2.2 - Pneumatic connections

M0009459.1

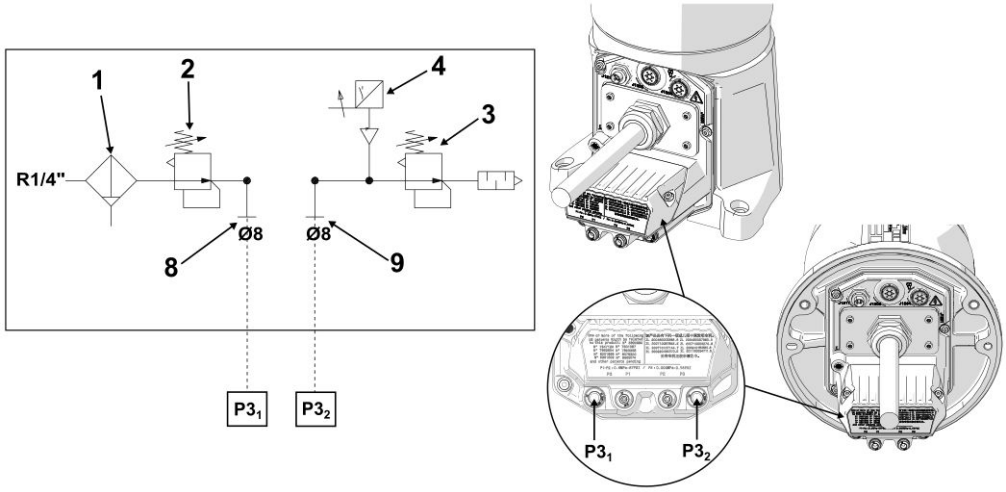


Figure 3.54: TX2

The arms must be connected to the pressurization unit via the output or outputs marked **(P3)** on the connection plate at the base of the arm by 2 hoses (feed and return).
Characteristics of the hoses (polyurethane hose, Ø 5.5 mm interior, Ø 8 mm exterior, maximum length 50 m). The hoses must be protected from risks of crushing throughout the full distance of their routing.

- The hoses must be blown clean before they are installed. The presence of pollution or swarf in the hoses can cause malfunctions. Fit a hose with an exterior diameter of Ø 8 between the unit (outlet **(8)**) and One of the **(P3)** connectors mounted on the arm.
- Fit the other hose with an exterior diameter of Ø 8 between the unit (outlet **(9)**) and The other **(P3)** connector mounted on the arm.
- Check visually that the arm is fully airtight (covers closed, plugs in the tapped holes on the arm, etc.).

3.3.9.2.3 - Electrical connections

M0007344.1

Pressure switch characteristics **(4)**:

- The pressure switch with its display requires a direct current power supply. It has 2 PNP digital outputs that can be used to monitor the arm pressurization.

Model		
Supply voltage		12 to 24 VDC ± 10%, maximum ripple (p-p) 10% (with protection from polarity reversal)
Electricity consumption		35 mA max
Detector outputs	Type	PNP open manifold
	Maximum load current	80 mA
	Protection against short-circuits	Yes
Display appearance		Screen with triple display
Indicator lights		Comes on orange when outputs OUT1, OUT2 are activated.
Ingress protection		IP65

Model	
Cable	Conductor surface area: 0.15 mm^2 (AWG26). Exterior diameter of the insulation: 1 mm. Total exterior diameter of the sleeve: 3.5 mm
Standards	CE, UL, RoHs

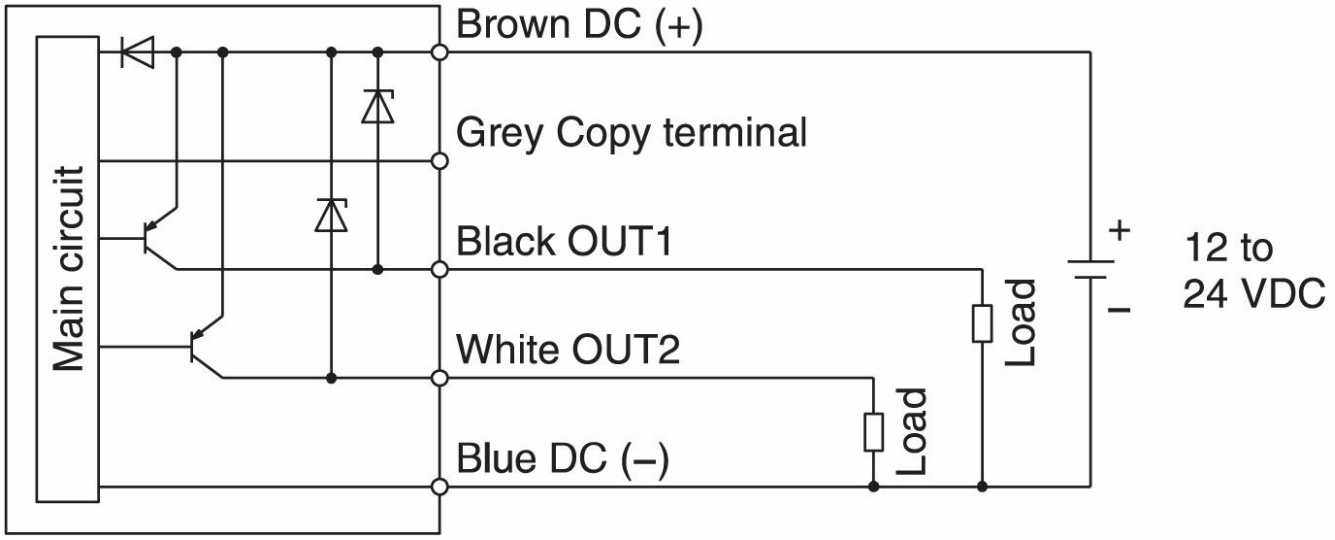


Figure 3.55:

3.3.9.3 - Settings

M0009255.1

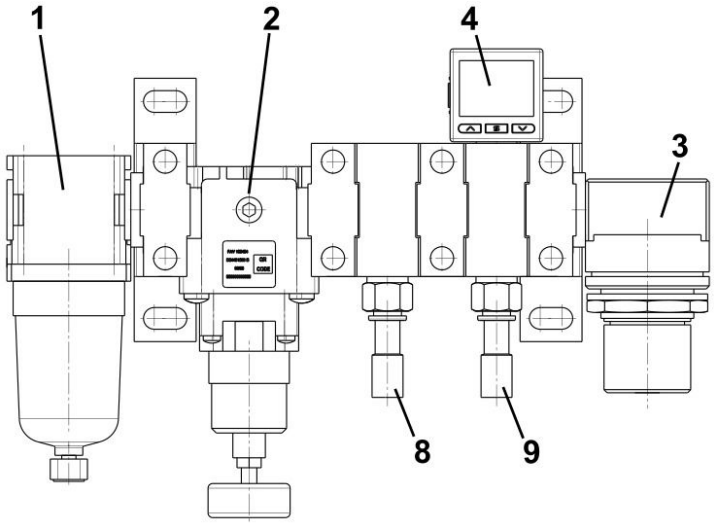


Figure 3.56:

Arm pressurization:

- The pressurization unit is factory adjusted for an air network pressure of 0,6 MPa (87 psi). Connect **(1)** to the compressed air network.
- Minimum pressure: 0,1 MPa (14,5 psi)
- Maximum pressure: 1 MPa (145 psi)
- Opening G1/4.
- Tightening torque: 12-14 Nm
- Hygrometry: The unit is fitted with a filter; the lower the hygrometry level of the air in the network, the longer the intervals between bowl draining procedures
- Use clean air. Compressed air containing chemicals, organic solvents, synthetic oil or corrosive gas can lead to damage or malfunctions.

- The pressure switch display unit **(4)** must show a pressure reading:
  - 4,0 kPa ± 2,0 kPa

  If the thresholds are exceeded, the colour of the digits on the display unit switches from green to red and the status of the PNP output changes.
  During installation, it may be necessary to adjust the setting of the pressure regulator **(2)** to move it closer to the nominal value, to within ± 0,5 kPa.
  To do so, loosen the locknut on the control knob of the pressure regulator **(2)**. Turn the knob slightly until the expected pressure level is reached on the display unit of the pressure switch **(4)**.
- If the pressure reading stays below the minimum whatever the setting, check that the various tubes are correctly connected and that the arm covers are correctly fixed.
- Set the pressure control valve **(3)** to its minimum value (0.02 MPa) by turning the knob.

3.3.9.4 - Servicing

M0007980.1

DANGER

Overflowing air filter bowl
Oveflowing condensate will pollute the interior of the arm directly; this can lead to malfunctions of the unit and the arm.
Drain the air filter bowl whenever necessary (capacity 8 cm^3).

- Check the state of the unit regularly, and remove any traces of fouling (dust and water).

CAUTION

Pneumatic or electrical failure
If electrical or pneumatic power to the pressurization unit is lost, the arm's ingress protection is reduced to its unpressurized level (see chapter 3.3.1).
Before restarting, inspect the inside of the arm for external contamination. For example, the presence of water inside the arm is an electrical hazard.

3.3.10 - CONTROLLED SPACE, RESTRICTED SPACE AND FENCES

M0007168.1

In classical robotics applications, safety is assured by the separation of the people and the moving devices; the separation itself is assured by fences and/or safety sensors to stop the robot on operator approach. They define the controlled space around the robot.

- The safety sensors must be positioned adequately with a correct separation distance.
- If the fences are used to reduce the maximum space of the robot, they must be designed adequately to stop the robot in worse-case situations.

3.3.10.1 - Separation distance

M0008680.1

The rules to compute the separation distance are given in ISO 13855:2010. This standard specifies that this computation does not need to take into account faulty conditions, as it is unlikely that the robot fails just as the operator enters the controlled space.

The separation distance must be based on the time for the robot to stop, and, possibly, on the robot distance traveled during the stop.

Different stopping times may be defined for different stop signals (see Safety manual).

The safety parameter for maximum stopping time may be further reduced, depending on the robot movement and payload. Please contact the Stäubli Customer Support for an optimization of the maximum stopping time.

DANGER

Collision due to underestimated separation distance

Using unconsolidated data to calculate safety parameters for maximum stopping time will lead to collisions and safety area exits.

A safe separation distance shall use safe parameters for robot stopping time.

CAUTION

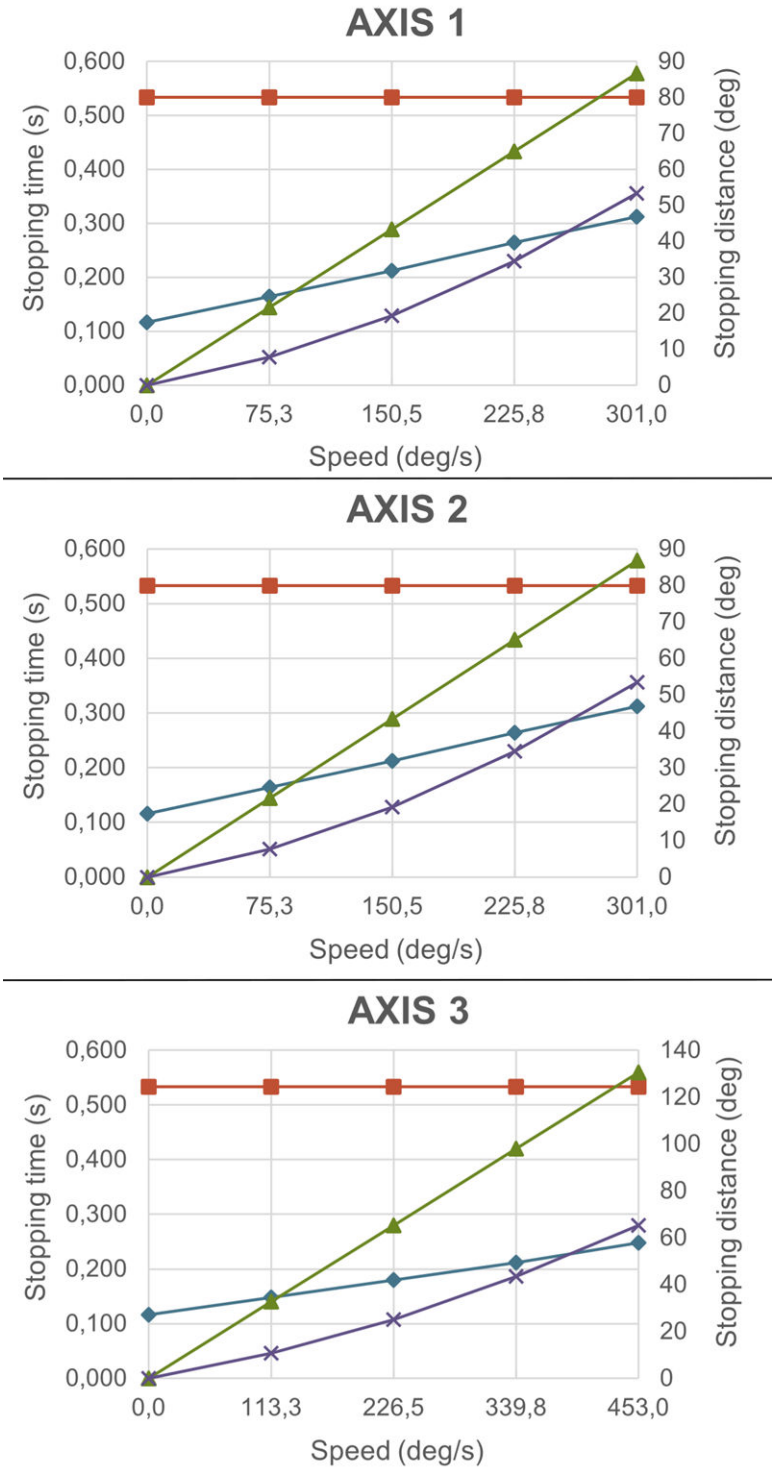
Outside nominal speed or payload specification:

When the arm is used outside its nominal speed or payload specification, it must be verified that the robot is able to stop without error in the worst-case conditions. If it's not, it will damage the robot and injure the operator.

Be aware that the abilities of the robot have been verified.

A controlled stop always stops the axes within their limits: the specified maximum stopping distance is in practice bounded by the axis limits.

TX2-60 CW



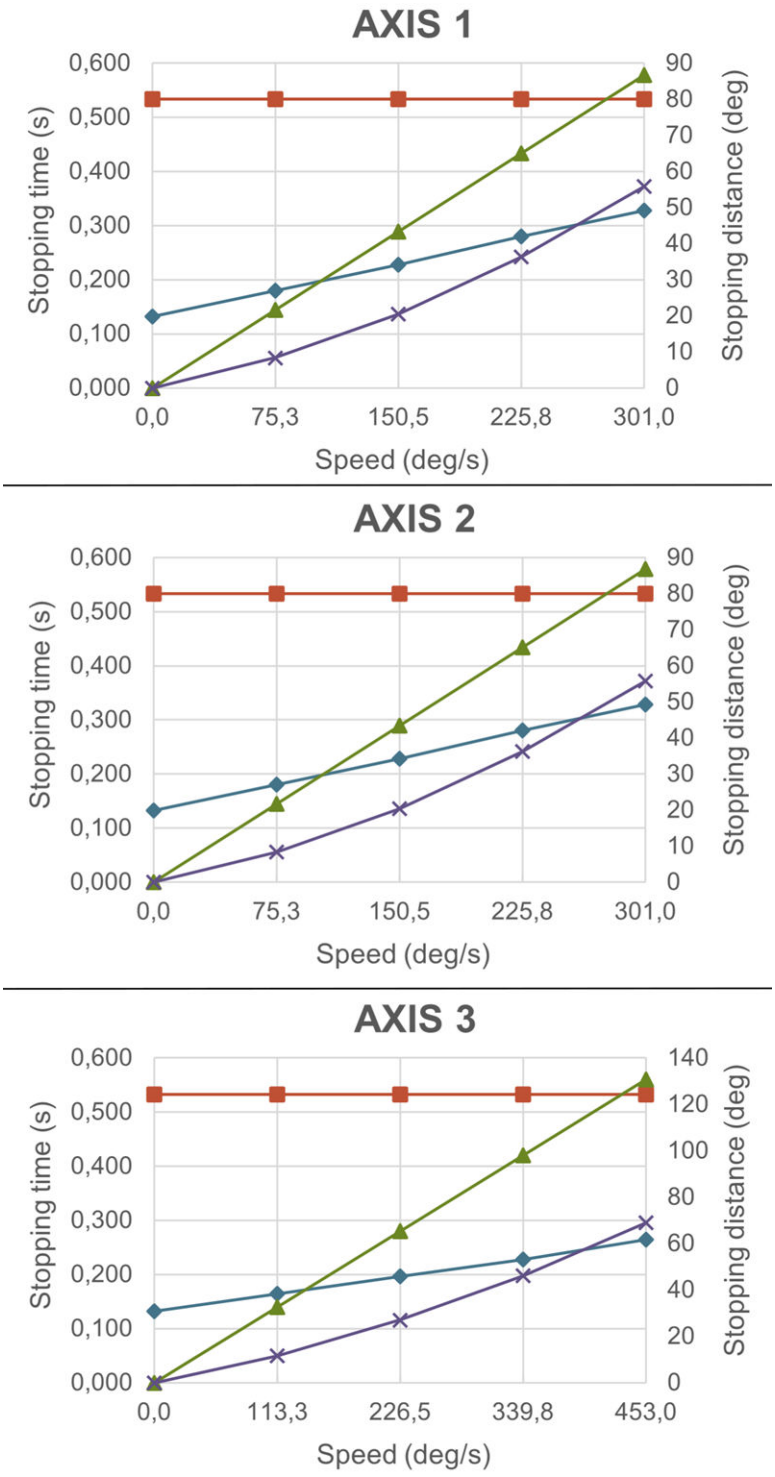
Item	Description
:	Default max. stopping time
:	Stopping time decel 100%
:	Default max. stopping distance
:	Stopping distance decel 100%

Figure 3.57:

Below you will find the typical values for the Stopping Time and Stopping Distance at speed 100 % and decel 100 %:

Test conditions	Stopping time (sec)	Stopping distance (°)
TX2-60 axis 1	0.31	53.4
TX2-60 axis 2	0.31	53.4
TX2-60 axis 3	0.24	66.0

TX2-60L CW



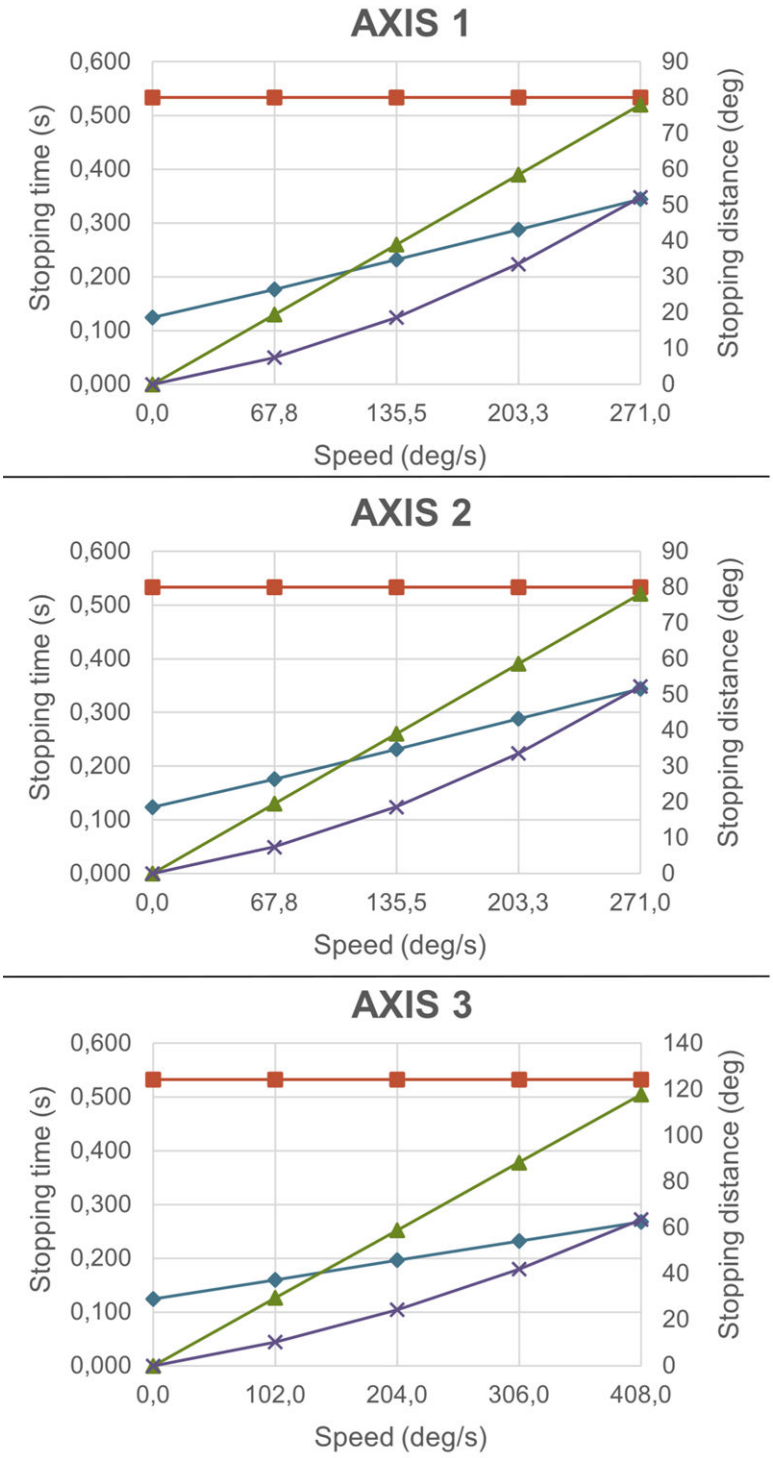
Item	Description
:	Default max. stopping time
:	Stopping time decel 100%
:	Default max. stopping distance
:	Stopping distance decel 100%

Figure 3.58:

Below you will find the typical values for the Stopping Time and Stopping Distance at speed 100 % and decel 100 %:

Test conditions	Stopping time (sec)	Stopping distance (°)
TX2-60L axis 1	0.32	55.8
TX2-60L axis 2	0.32	55.8
TX2-60L axis 3	0.26	69.0

TX2-60 HW and TX2-60 L HW



Item	Description
:	Default max. stopping time
:	Stopping time decel 100%
:	Default max. stopping distance
:	Stopping distance decel 100%

Figure 3.59:

Below you will find the typical values for the Stopping Time and Stopping Distance at speed 100 % and decel 100 %:

Test conditions	Stopping time (sec)	Stopping distance (°)
TX2-60 HW / TX2-60 L HW axis 1	0.344	52.2
TX2-60 HW / TX2-60 L HW axis 2	0.344	52.2
TX2-60 HW / TX2-60 L HW axis 3	0.268	63.46

3.3.10.2 - Restricted space and fences strength

M0007038.1

The maximum space reachable by the robot can be reduced by different means, such as:

- Fences.

The configuration of safe joint limits and safe zones is described in the safety manual. Refer to the Safety manual, Chapter "Restricted space: stopping distances and remaining energy".

The limits of the restricted space shall however take into account the possible faults of the robot, to guarantee the safety of the personal after failures such as a defective motor, amplifier, or power supply. In such worse-case situations, the robot is stopped on its brakes.

ISO 10218-2 (5.4.3) specifies some requirements to define the restricted space, and the mechanical strength of the fences when they are used to limit it.

- The safe restricted space shall be based on safe parameters for robot maximum velocity, maximum joint limits, or cartesian zones. The robot motion parameters defined in the CPU can be used to protect the equipment, but not for functional safety.
- The internal mechanical range limiters on axis 1 are not mechanical hard stops. They prevent misuse during maintenance operation, but cannot stop the robot at high speed. If a mechanical range limiter is hit, the fixed and mobile parts must be replaced (consult the Stäubli Customer service).

3.3.10.3 - Fence design

M0006963.1

The strength of the fencing shall support the maximum energy of the robot with its payload at the collision point.
The tables below give the stop distance on the brakes, and the energy of the arm at nominal conditions of use.
The maximal stopping distance on the brake is measured with the arm outstretched, at nominal load and nominal speed.

	Stopping distance (axis degrees)		
	Axis 1	Axis 2	Axis 3
TX2-60 CW	45	46	55
TX2-60L CW	58	82	60
TX2-60 HW	38	45	52
TX2-60L HW	48	68	84

The CS9 safety controller allows a significant reduction of the stopping distance on the brakes and of the remaining energy on the fences by using a monitoring of safety joint position and speed limits. A test procedure, and a stopping distance estimation tool is provided as part of the safety solution. Refer to the Safety manual, Chapter "Restricted space: stopping distances and remaining energy".

DANGER

Hazardous deformation of the fences or perimeter guards
If the arm can cause dangerous deformation of the cell fences or perimeter guards, the user is at risk of serious injury or death.
The robot shall not be able to cause a hazardous deformation of the fences or perimeter guards (ISO 10218-2, 5.4.3).

Maximum energy at nominal speed and payload at the flange

	Energy
TX2-60 CW	310 J
TX2-60L CW	480 J
TX2-60 HW	338 J
TX2-60L HW	444 J

3.3.10.4 - Mechanical hard stops (Not available on STER+)

M0008732.1

The mechanical hard stops option is not available with Stericlean +, because replaced by safety electronic options.
The workspace of the robot can be restricted with a mechanical hard stop on axis 1. The mechanical hard stop is delivered as an option to be fitted on site.
The restricted space must take the stopping distances into account:

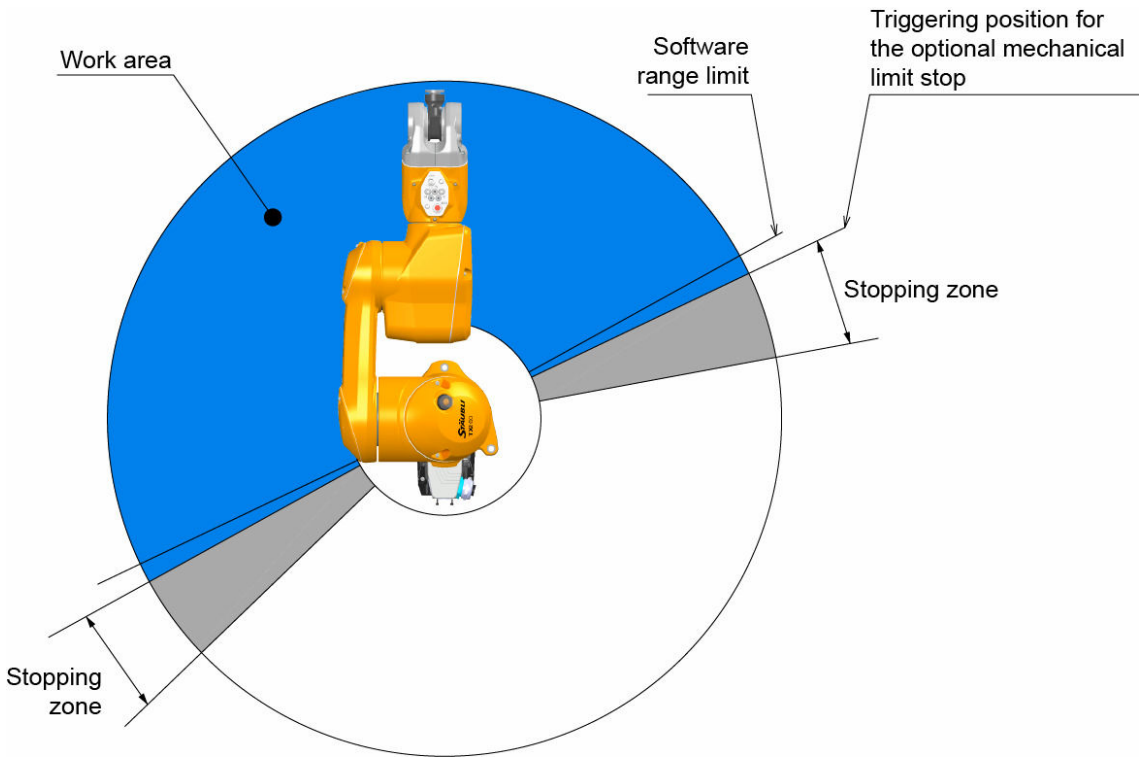


Figure 3.60:

- Work area: Angular range within which the robot moves during normal operation.
- Stopping zone (abnormal operating zone): Maximum angle necessary to stop the robot. In the event of use of the optional adjustable mechanical hard stop system, the stopping angle depends on the energy of the impact.

⚠ WARNING	
⚠	**Stopping distance on the mechanical hard stop** The arm does not stop immediately on the mechanical hard stop. In your integration study, take into account that the mechanical hard stop is designed to stop the arm with a nominal payload at nominal speed in less than 15°.

Description of the optional mechanical hard stop (see figure 3.61)

The optional mechanical hard stop system is able to stop the robot at nominal load and nominal speed. It is sized to absorb the kinetic energy of the axis.

- The optional mechanical hard stop system is made up of a two-part ring **(1)** on which one or two adjustable hard stops **(2)** can be placed to limit the working range of axis 1.
- The ring is fitted on the robot base **(3)**.
- The working range is changed by placing 1 or 2 adjustable hard stops as shown on figure 3.61.

Example of restricted workspace:

- One hard stop in position "1a":
  - Work area is limited at 178° in the positive direction and 117° in the negative direction.
- One hard stop in position "1a" and another in position "1b":
  - Work area is limited at 117° in the positive and negative directions.
- Further reduction in steps of 25° when positions 2 to 5 are used.

The installation of the mechanical hard stop is described in chapter 4.5.

NOTICE

Mechanical hard sop compatibility.
The mechanical hard stop is not compatible with humid (HE) or bio-contaminated (Sericlean) environments.
However, depending on the environment, the system may show limited chemical resistance. Regularly inspect the system to check that it is in good condition. In the event of deterioration, the system can be replaced. Because of their complex shape, the mechanical hard stops make the arm more difficult to clean and disinfect.

⚠ WARNING

Inspection after a stop on the mechanical hard stop
After the robot is stopped by the mechanical hard stop, both could be damaged.
If the robot is stopped by the mechanical hard stop, an inspection procedure must be performed (see chapters 7.14 and 7.20).

⚠ CAUTION

Check points
Check the following points at least once a year:
- No visible damage.
- Tightening torque for the 2 screws holding the ring **(1)** in place.
- No slippage for the ring **(1)**. The verification is made by checking the alignment of the ring slot with the axis marker.

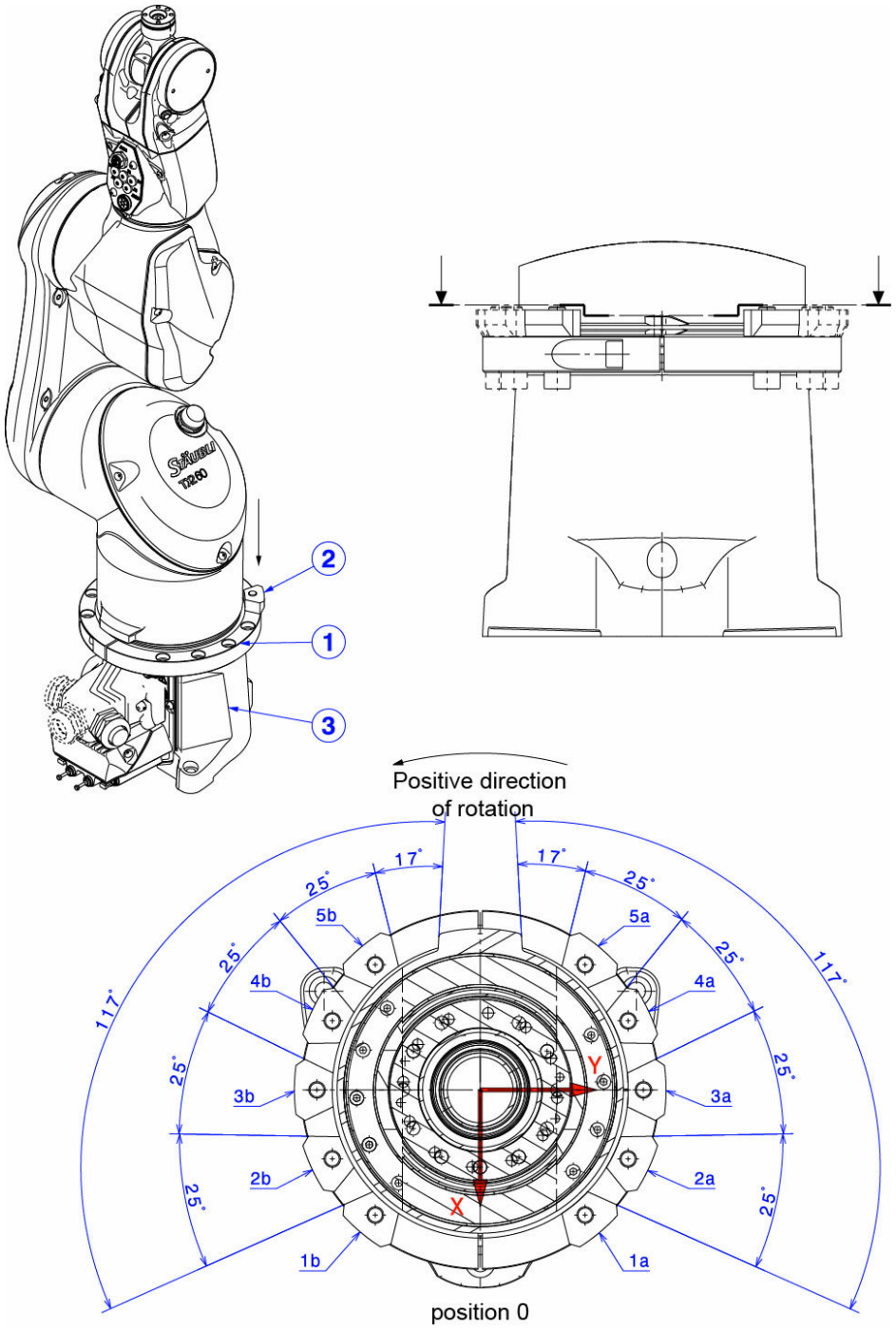


Figure 3.61:

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3.3.11 - CABLING

M0007180.1

The cable layout must comply with a minimum bend radius for each type of cable. See the table below:

Internal bend radius	
Standard interconnect cable	100 mm
RBR cable	50 mm

NOTICE

Cable environmental damage
The warranty does not cover any environmental damage to these parts.
In humid (HE) or bio-contamination controlled environment (Stericlean or Accesspharma), the cables must be protected from the environment. This protection is to be provided by the customer, and are their responsibility.

3.3.12 - SINGULARITIES

M0009589.1

A singularity is a position inside its work envelope where the robotic arm loses one or more degrees of freedom in Cartesian displacement. The singularities are the limits between the arm configurations (righty/lefty, elbow positive/negative, wrist positive/negative) and it is impossible to change the arm configuration without going through a singularity.
For a detailed description of this phenomenon, refer to the “Movement control” and “Config type” chapters of the VAL3 instruction manual.
When the robot follows a Cartesian path near a singularity, a short Cartesian displacement leads to a large joint movement. In this case, the motion generator saturates the joint velocity to avoid joint overspeed. This can significantly reduce the Cartesian speed of the robot’s tool center point or even cause it to stop.
Therefore, singularities must be avoided in applications where:

- Cycle time is critical.
- The process requires control of the Cartesian velocity of the tool center point.
- The process requires very smooth trajectory execution (the tool center point can move slightly outside the path during axis reconfiguration).

⚠ CAUTION

Shock during manual move
In Cartesian motion and near a singularity, the large joint movement may hit the nearby operator.
Do not stand too close to the robot.

⚠ DANGER

Joint overspeed limit innoperant
Using specific licences listed below the joint overspeed risk is not mitigated and is under integrator responsibility.
Applicable software licences: Alter, ValTrack (conveyor tracking), uniVal drive.

3.4 - STAFF REQUIREMENTS

M0007092.1

Stäubli robots work with computer controlled mechanisms, capable of moving at high speed and exerting considerable force. Like all robots and most industrial equipment, they must be operated with great care by the user of the machine. All staff using Stäubli robots must be familiar with the warnings and recommendations given in this manual.

DANGER

Mandatory qualifications and training

Unqualified and untrained individuals working on the robot or designing its cell may create hazardous conditions for all personnel in the area.

- The person designing and installing the robot cell shall have the mechanical, electrical and software qualifications required for the accomplishment of their assigned tasks.
- The persons configuring and validating the safety functions using the SafeCell or SafeCell+ license shall have the safety qualification required for their tasks.
- The person operating the robot shall be trained for its safe manipulation in normal conditions of use and in abnormal or emergency situations, such as movements without power using the manual brake release mode.
- The person maintaining the robot shall have the mechanical and electrical qualifications required for their interventions (see chapter 7.1).

Stäubli offers various training courses to meet these qualification requirements. These trainings also provide a better knowledge of our products allowing to fully and efficiently use their capacities.

3.5 - ROBOT SIMULATION

M0006697.1

Simulation is a required step during the design of the machine around the robot.

Our PC software suite Stäubli Robotics Suite includes the tools that are needed to test and optimize the placement of the elements of the machine, and to check that arm constraints on base attachment, end-effector and payload, are satisfied:

- Import of CAD data (STEP, IGES, STL, VRML),
- Possibility to build models based on standard primitives (cubes, spheres, etc...),
- Direct link between 3D scene and program editor (for geometric data creation/edition, etc...),
- Full multi-robot simulation capabilities (working space, realistic cycle times through the individual clocking of each controller in the cell),
- High performance collision detection (highlighting of colliding objects, minimum distance verification).

4 - STORAGE, TRANSPORT AND INSTALLATION

M0009628.1

NOTICE



Directive Machine

When lifting machinery or lifting accessories are placed on the market or put into service for the first time, the manufacturer or his authorized representative must ensure, by means of appropriate measures which he takes or arranges to be taken, that the machinery and lifting accessories ready for use, whether powered by man or motor, can perform their intended functions safely.

Tests must be performed on all lifting machinery ready to be put into service with 25% margin for static tests and 10% margin for dynamic tests.

Where the machinery cannot be assembled on the premises of the manufacturer or his authorized representative, the appropriate measures must be taken at the place of use. Failing this, the measurements may be taken either on the manufacturer's premises or at the place of use.

4.1 - ARM AND CONTROLLER PACKAGING

M0009670.1

NOTICE



Robot damage

Temperature for storage and transport: -25°C (-13°F) to 55°C (131°F)

Short period (less than 24 hours): 70°C (158°F) maximum

Maximum humidity for storage and transport: Up to 90% (non-condensing)

⚠ WARNING



Stacked package fall

Compact packaging only: do not stack over 3 high (1+2).

Incorrectly stacked packages can collapse or fall, resulting in serious injury or death.

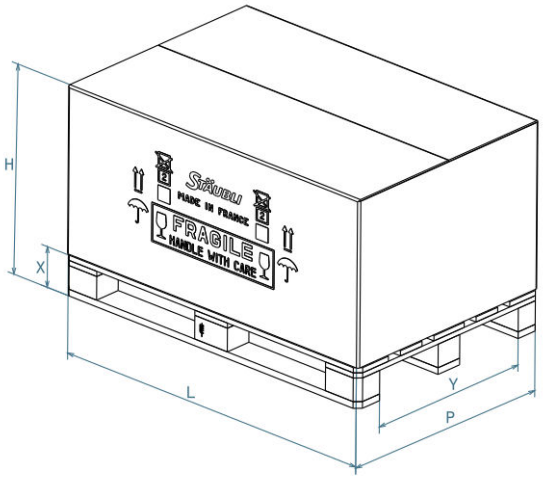
Type of packaging according to product ordered:

Product ordered	Type of packaging
Arm + Controller	Compact packaging

Robot weight:

	Weight
TX2-60 CW	52.2 kg (115.1 lb)
TX2-60 HW	52 kg (114.6 lb)
TX2-60L CW	52.9 kg (116.6 lb)
TX2-60L HW	53 kg (116.8 lb)

Compact packaging:



		TX2-60 CW	TX2-60 HW	TX2-60L CW	TX2-60L HW
L x H x P (mm) [in]		1200 x 690 x 800 [47.2 x 27.1 x 31.5]			
Gross weight (kg) [lb] (approximately)	Horizontal cable outlet	126 [278]	Not available	129 [284]	Not available
	Vertical cable outlet	126 [278]	126 [278]	129 [284]	129 [284]

Compact packaging position of the arm:

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
TX2-60 CW TX2-60L CW	+90°	–125°	+140°	0°	+90°	0°
TX2-60 HW	+90°	–125°	+125°	+90°	+90°	0°
TX2-60L HW	+90°	–125°	+140°	+90°	+90°	0°

4.2 - PACKAGE HANDLING

M0007153.1

⚠ WARNING

Stacked package fall
Compact packaging only: do not stack over 3 high (1+2).
Incorrectly stacked packages can collapse or fall, resulting in serious injury or death.

⚠ WARNING

Heavy package
Displacing the package in an inappropriate way could result in serious injury or death.
Handle the packaging with a pallet truck wherever possible, or with a forklift if necessary.

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NOTICE

Arm displacement
A wrong packaging will result in material damage.
If the arm has to be moved to another site, it shall be packed again in a similar way and protected against mechanical shocks and dust.

4.3 - UNPACKING OF THE ROBOT

M0009673.1

Before opening, inspect the packaging to detect any:
- Impact trace.
- Liquid spill.
- Pallet fracture.

NOTICE

Damaged package
Do not open a damaged package.
Contact the Stäubli support team.

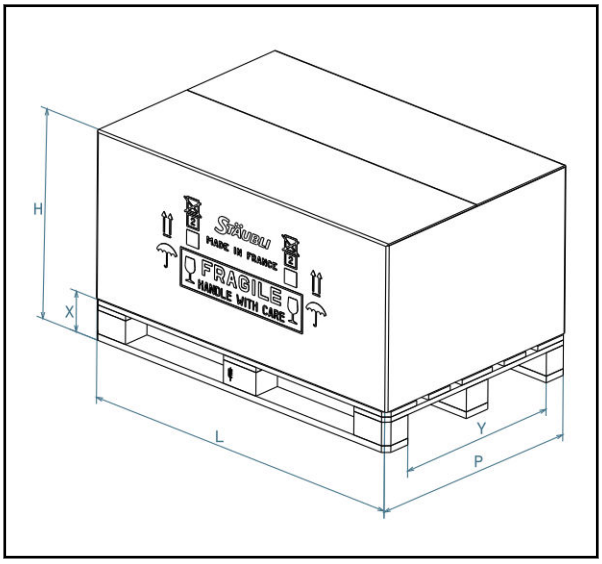
NOTICE

Connector damage
During all the operations to handle and install the arm, keep the main robot connector cover in place to avoid damaging or soiling the electrical and optical contacts.

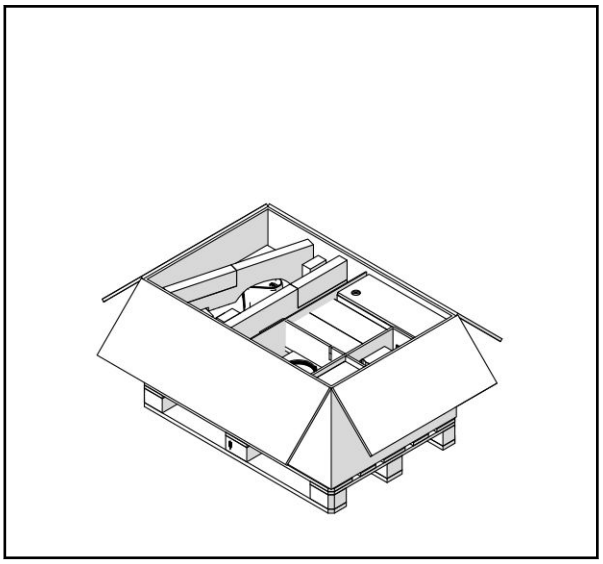
4.3.1 - COMPACT PACKAGING

M0009672.1

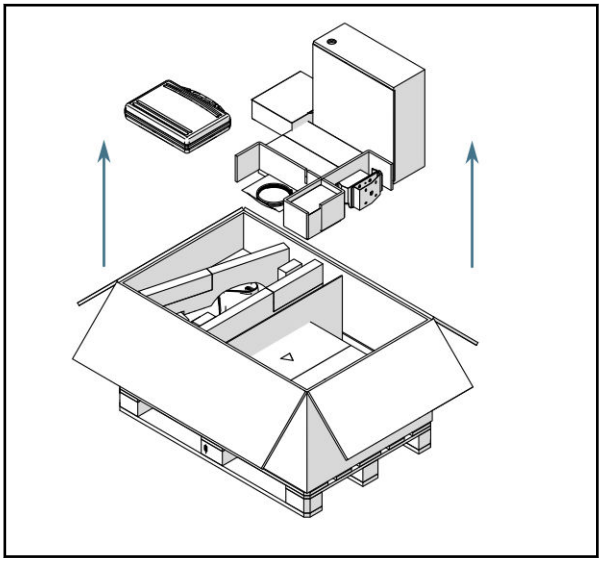
The procedure below to unpack a robot is the same for CW and HW



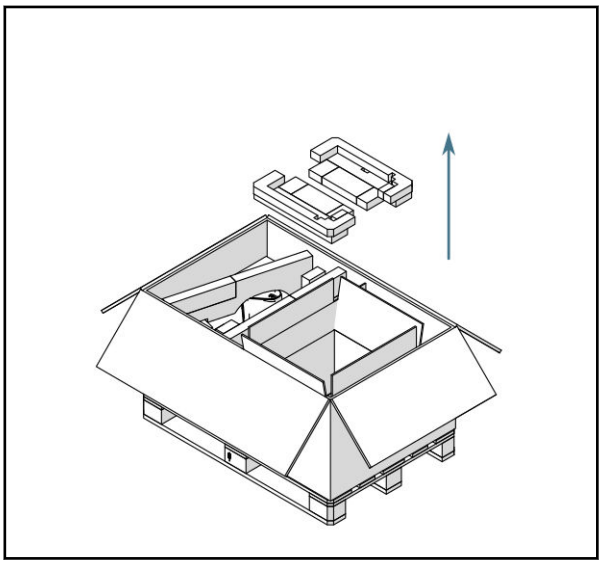
- Move the packing case as near as possible to the installation site.



- Cut the straps.
- Open the box.



- Take out the accessories.
- Take out the power supply cable.



- Open the controller box and remove the 2 upper wedges.

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⚠ WARNING

Lifting hazard
Lifting the controller in an inappropriate way could result in serious injury or death.
Only use the lifting rings to attach your lifting gear. Crane and related gears drivers must be duly trained and qualified according to local regulations.

⚠ WARNING

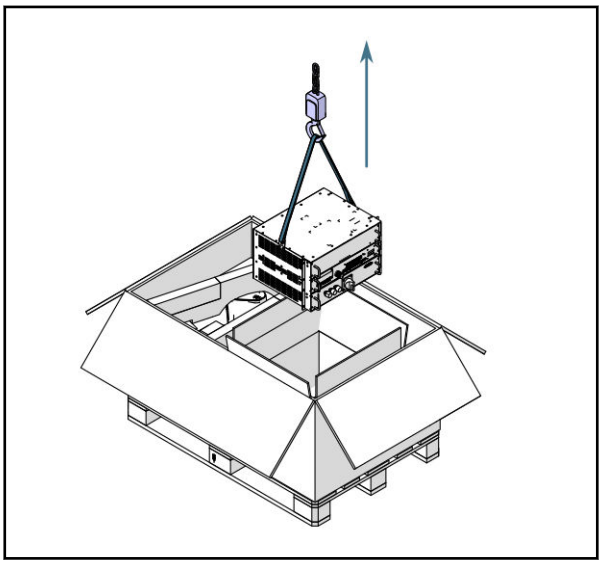
Using the sling
Improper use of slings can lead to serious injury.
Sling users must be trained and knowledgeable concerning safe and proper use of slings.

⚠ CAUTION

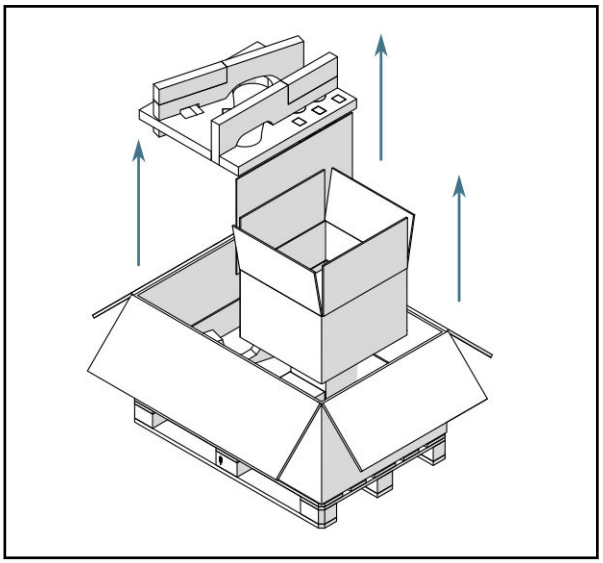
Sling regulations
A sling that does not comply with the correct regulations does not guarantee that it will function properly.
The slings used to lift the robot must comply with the Machinery Directive in Europe or with the rules in force for other countries.

⚠ WARNING

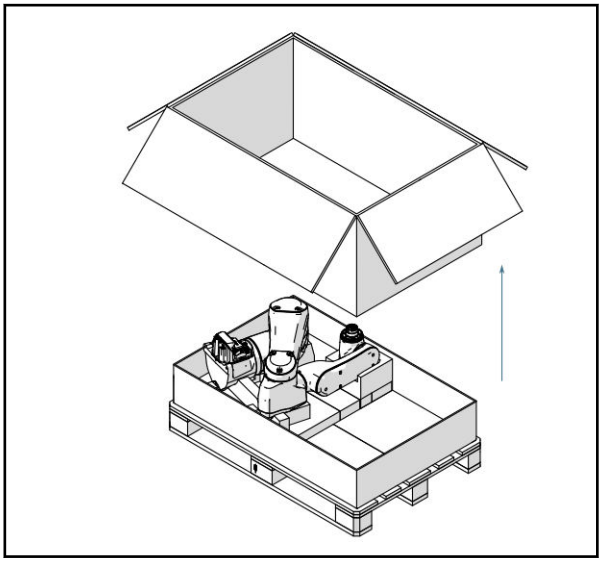
Stacked controller fall
Stacked controllers can collapse or fall, resulting in material damage, and serious injury or death.
Do not stack.



- Fit a sling (fabric sling > 80 kg / 177 lb) using hooks between the lifting rings on the controller and the hook on the lifting tackle. The hooks must also stand up to 80 kg / 177 lb.
- Slowly lift the controller and set it down beside the pallet. Each lifting ring has a maximum authorized load (30 kg / 66 lb) and shall not be used for another purpose.
- The lifting rings on the controller are removable.



- Take out the controller packing.
- Take out the foam packing pieces.



- Take out the remaining packing for an easier access to the arm.

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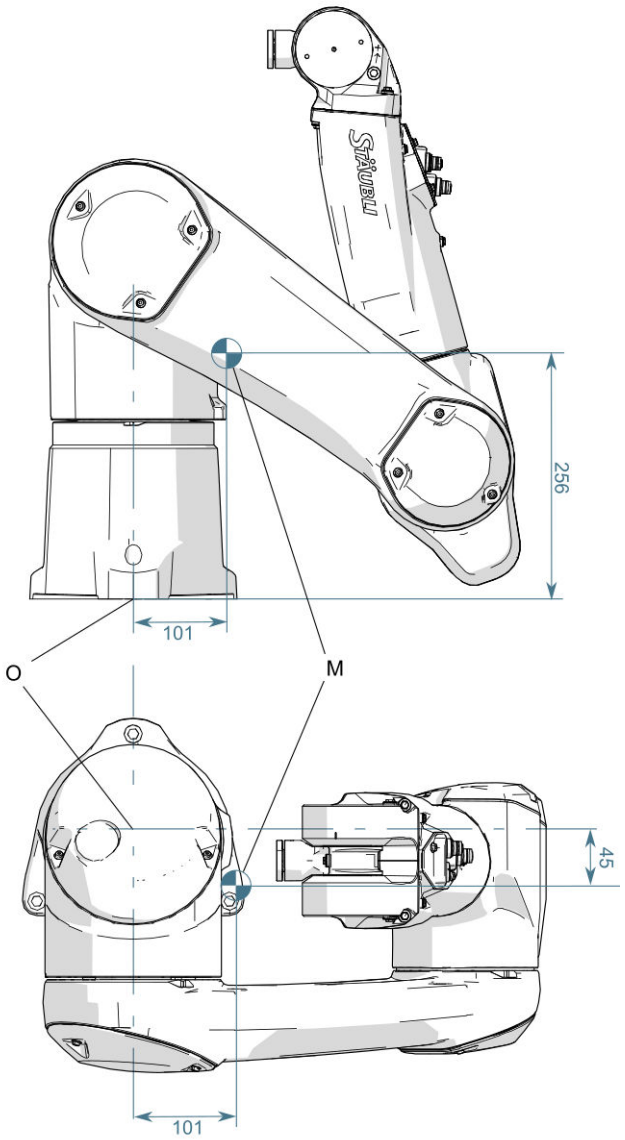


Figure 4.1: Arm center of gravity

DANGER

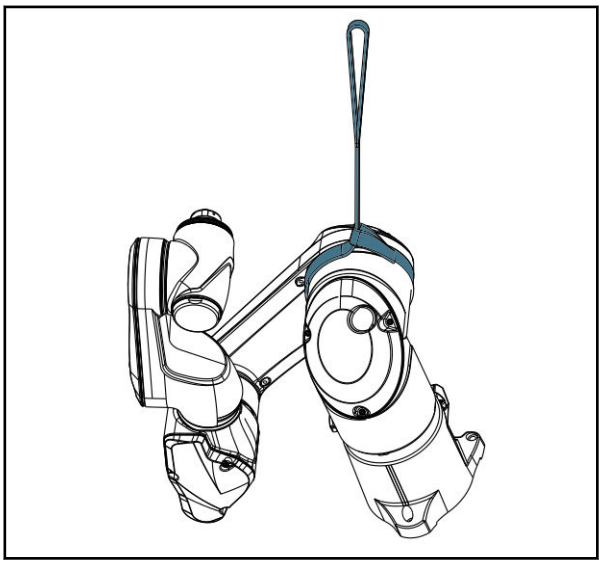
Lifting hazard
Lifting the arm in an inappropriate way will result in serious injury or death.
• Choosing an inadequate lifting mean and any deviation from the lifting instructions could lead to serious injury or death, for which the user will be held responsible.
• The arm must be in the packing position with no load attached during all lifting operations. The packing position is specified in chapter 4.1.
• Crane and related gears drivers must be duly trained and qualified according to local regulations.

WARNING

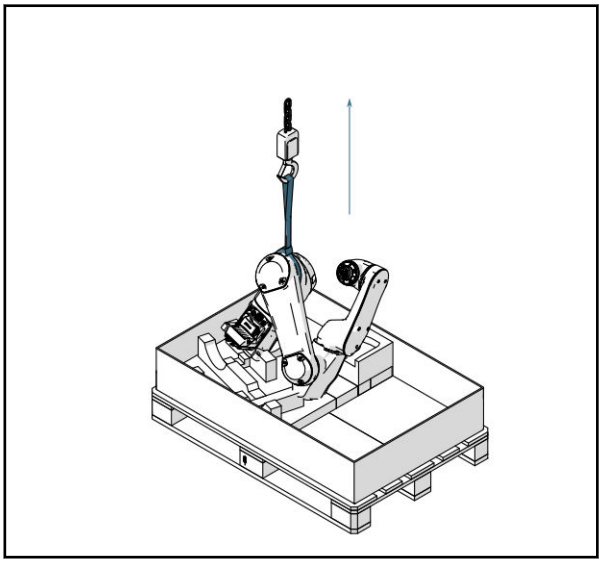
Using the sling
Improper use of slings can lead to serious injury.
Sling users must be trained and knowledgeable concerning safe and proper use of slings.

CAUTION

Sling regulations
A sling that does not comply with the correct regulations does not guarantee that it will function properly.
The slings used to lift the robot must comply with the Machinery Directive in Europe or with the rules in force for other countries.



- Place the fabric slings as shown around the arm before to move it.



WARNING

Robot swing
The robot will swing when raised.
Be aware of this risk.

- Take out the arm slowly. As a safety precaution, keep the sling fairly taut to make sure that the arm does not topple over.
- Put it on the floor.

NOTICE

Arm damage during relocation
When relocating the arm, it shall be packed in a similar way.

NOTICE

Controller damage during relocation
When relocating the controller, it shall be packed in a similar way.

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4.4 - INSTALLATION OF THE ARM

M0006501.1

NOTICE

Installation of the arm according to the controller:
A wrong installation of the arm might result in damages.
The arm can be fixed with its base facing downwards (floor-mounted version), upwards (ceiling-mounted version), or against a wall, without requiring any mechanical modifications. However, it is necessary to configure the controller accordingly. To do so, see the "Software configuration" chapter in the controller instruction manual.

NOTICE

Protection on the main connector:
A wrong installation of the arm might result in damages.
Throughout all handling and installation operations concerning the arm, it is essential to keep the protection on the main connector at the foot of the robot in place at all times to avoid damaging and soiling the electrical and optical contacts.

⚠ CAUTION

Risk of tipping or falling of the arm when relocated
During arm relocation, worn or faulty brakes may cause the arm to tip or fall due to changes in the robot's angular positions and its balance.
Before relocating the arm, Stäubli recommends to perform a brake test (Refer to the "Brake testing" chapter of the Safety manual).

- For vertical cable outlet, an O-Ring is delivered and needed between arm foot and customer interface (see figure 3.12).
- For humid (HE) or bio-contamination controlled environment (Stericlean), please refer to specifications described in chapter 3.3.3.

4.4.1 - INSTALLATION OF ARM ON FLOOR

M0008332.1

⚠ CAUTION

Sling regulations
A sling that does not comply with the correct regulations does not guarantee that it will function properly.
The slings used to lift the robot must comply with the Machinery Directive in Europe or with the rules in force for other countries.

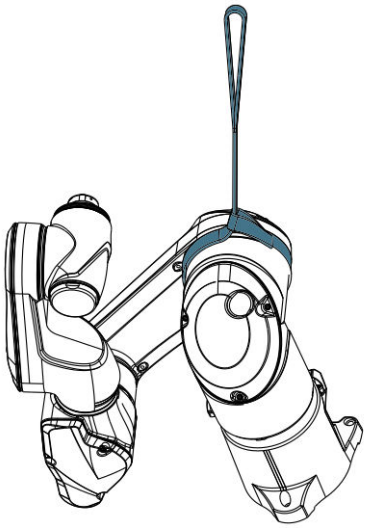


Figure 4.2:

The position of the sling is identical on the compact wrist (CW) and the hollow wrist (HW).

- Place the fabric sling around joint as shown on fig 4.2.
- Position the arm on the support at its final attachment points.
- Attach the arm with 3 class 12.9 M10 hex. socket head screws, following the recommendations previously mentioned.
- Remove the sling.

DANGER

Arm tilting
Tilting of the arm will result in serious injury or death.
Keep the lifting gear slightly tensioned until the arm is securely attached to the floor.

4.4.2 - INSTALLATION OF ARM ON CEILING OR WALL

M0008809.1

NOTICE

Arm position
Do not rotate joint 2.

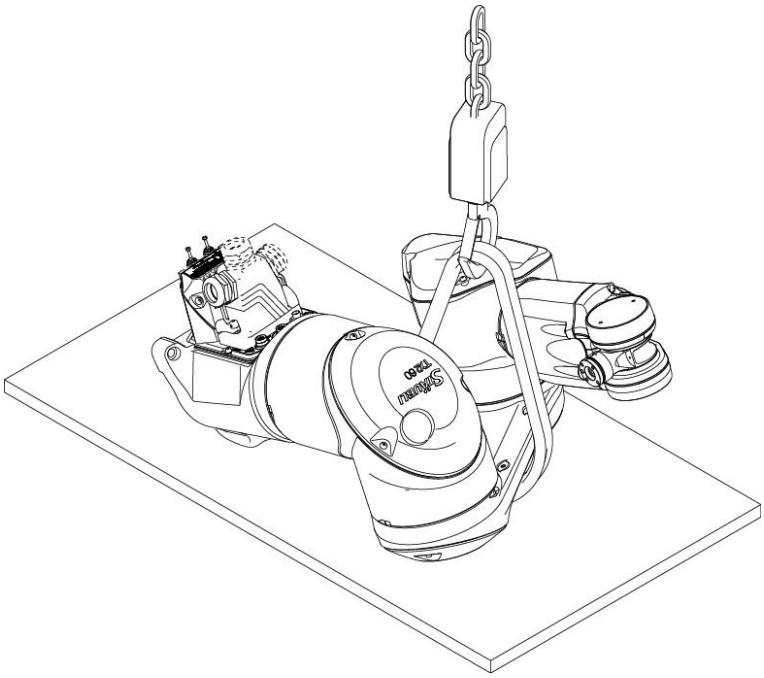


Figure 4.3:

- Carefully lay down the arm on a flexible support as shown on figure 4.3.

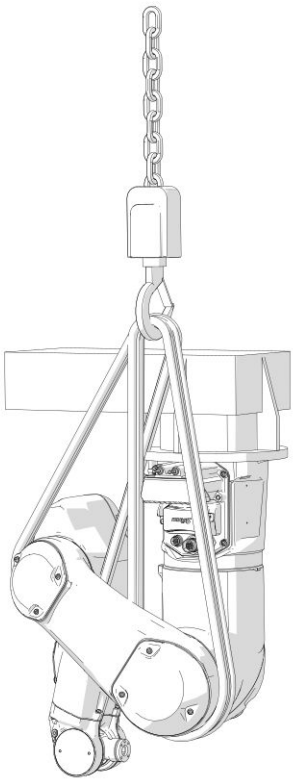


Figure 4.4:

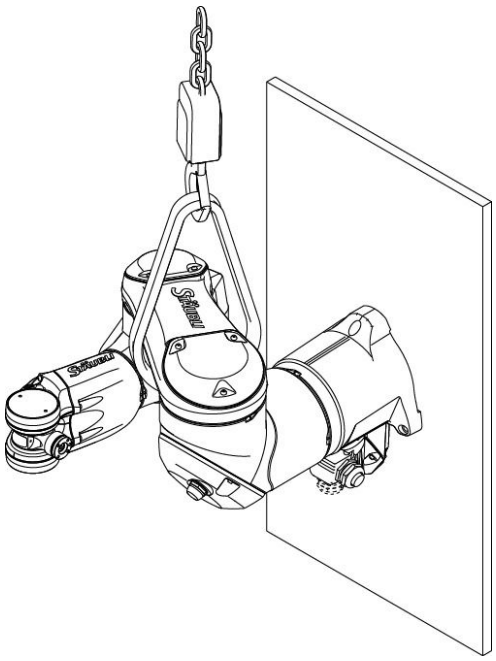


Figure 4.5:

See chapter 4.1 to select the appropriate lifting mean according to the arm weight.

The position of the sling is identical on the compact wrist (CW) and the hollow wrist (HW).

- Place the fabric sling around joint 2 as shown on figure 4.4 or 4.5.
- Position by hand the arm on the support at its final attachment points.

- Attach the arm with 3 class 12.9 M10 hex. socket head screws, following the recommendations previously mentioned.
- Remove the sling.

⚠ CAUTION

Using the sling
Improper use of slings can lead to serious injury.
For safety reasons, hold the sling slightly tensioned until arm is securely attached to the ceiling.

4.5 - INSTALLATION OF THE MECHANICAL HARD STOP (OPTION)

M0007918.1

(see figure 3.61)

1. Fit the ring **(1)** on the robot base **(3)**, resting on the chamfer.
2. Line the ring slot up with the axis marker.
3. Tighten the 2 screws supplied (CHc M10 x 45, class 12-9) holding the ring **(1)** in place, to a torque of 50 Nm ± 3.5 Nm.
4. Position the 1 or 2 adjustable hard stops **(2)**, as shown in figure 3.61.
5. Fix the one or 2 hard stops **(2)** using the screws supplied (CHc M10 x 25, class 12-9) tightened to a torque of 57 Nm ± 3.5 Nm.
6. Adjust the joint limits in the software configuration of the arm characteristics (see the controller instruction manual).

⚠ WARNING

Improper installation
The system performance levels are only guaranteed if it is correctly assembled (observing the required torques, etc.), with clean, grease-free parts. Failure to comply with these instructions may result in incorrect operation of the mechanical hard stop.
Comply with the instructions.

⚠ WARNING

Inspection after a stop on the mechanical hard stop
After the robot is stopped by the mechanical hard stop, both could be damaged.
If the robot is stopped by the mechanical hard stop, an inspection procedure must be performed (see chapters 7.14 and 7.20).

4.6 - INSTALLATION OF THE UL DRIVE POWER INDICATOR

M0007166.1

The UL drive power indicator option is not available on Stericlean +.
Standard UL1740, chapter 35.1.1 requires the presence of an indicator in the robotised cell.
If the UL indicator is not supplied on the robot arm (Stericlean + option for example), the integrator has to install an indicator in the cell, under his own responsibility, in compliance with the recommendations set out in UL1740 35.1.1.b and 35.1.1d.
There are 2 ways to connect the drive power indicator on the robot:

- Use the USOC safety outputs J102-9 or J102-10 (see Electrical diagram manual).
  The USOC safety configuration should be set to "USOC_power": This is the case by default, and this is always the case using the Initial safety version (without safety license).
- Use the controller IOMAP configuration to have the "power" robot status delivered on the digital output of your choice, on which is wired the drive power indicator.
  This is described in the CS9 manual (chapter system inputs/outputs, "power" keyword)
  The IOMAP configuration is easily accessible using the SRS IO Map Editor, that is part of the SRS Physical IOs editing tool.

5 - COMMISSIONING

M0006623.1

Before using the robot for the first time and putting it into production, verification tests must be carried out to ensure a reliable and safe use of the robot over time.

5.1 - VERIFICATION OF THE MECHANICAL INTEGRITY

M0006504.1

- Arm mounting and attachment.
- End effector attachment and load capacity.
- Additional load.
- Hard stop.
- Environmental conditions and specificities for CR,SCR, HE or Stericlean applications.
- Access to covers and preventive maintenance positions (oil level check).

⚠ WARNING

Periodic visual inspection

Failure to perform a periodic visual inspection could result in unpredictable wear, failure and/or environmental contamination.

Based on the customer's risk analysis, a visual inspection schedule for the robot must be established. If abnormal rust, paint cracks, missing components, etc. are found at this time, the robot must be repaired prior to use.

5.2 - VERIFICATION OF THE PNEUMATIC CIRCUITS

M0006685.1

- Base plate connections.
- Forearm connections.
- Air pressure and cleanliness.
- Pressurization unit.

5.3 - VERIFICATION OF THE ELECTRIC CIRCUITS

M0006968.1

- Base plate connections.
- Forearm connections.

5.4 - VERIFICATION OF THE ROBOT APPLICATION

M0006594.1

⚠ DANGER

Testing in automatic mode

Testing the application in automatic mode significantly extends the stopping distance required if a programming error is detected by the operator. This situation heightens the risk of severe bodily injury to the user and extensive material damage to the cell.

First robot moves and application tests must be done using the manual or the maintenance working mode.

- Torque limits.
- Temperature limits of motors and DSI9 board (at startup, stabilized temperature after continuous operation).

- Check that the application does not create collisions.

5.5 - VALIDATION OF THE SAFETY FUNCTIONS

M0006686.1

The validation of the safety functions and safety measures is detailed in the Safety Programming manual:

- Stopping time and separation distances.
- Position limits and restricted workspace.
- Speed limits.
- Working mode and restart control.
- Diagnostic functions on brakes and referencing.
- Safety organizational measures.

5.6 - VALIDATION OF ORGANIZATIONAL MEASURES

M0007149.1

- Testing of the emergency stops (regularly, after a shock).
- Spare parts and maintenance equipment.
- Software and configuration backup.
- Training and procedures for emergency rescue, preventive maintenance.

6 - WARRANTY

M0007111.1

The Stäubli warranty for the robot applies during the warranty period as defined in applicable contractual documents for defects which occur under a normal use of the robot in accordance with the robot's intended use, best use and integration practices, excluding counter-indications, as detailed in chapter 3.1. The Stäubli warranty is only delivered to robots that have undergone preventive maintenance defined in chapter 7.1.
The same conditions apply to the application of the extended warranty contract if the case may be.

⚠ WARNING

Robot modification
Modifying the robot voids the warranty and exposes the user to unpredictable risks.
Do not modify or alter the Stäubli robot without Stäubli's prior indication.

Register your robots via MyStaubli portal, your single-entry point to manage your robot information:
https://login.staubli.com/

7 - MAINTENANCE

7.1 - RULES TO BE APPLIED FOR MAINTENANCE

M0006412.1

To ensure high performance of your robot, help prevent equipment failure, and keep production moving, a Preventive Maintenance must be performed on a yearly basis or 7000 working hours whichever is the sooner (please refer to chapter 7.4 for more details).
Preventive maintenance must be carried out solely by Stäubli technicians or a certified maintenance technician who has undergone specific Stäubli training. The technician must also have an electrical accreditation. Failure to comply with this precaution can cause unexpected risks undermining the robot performance and the component lifetime and may result in loss of Stäubli's liability and warranty for any subsequent default related to the robot. A routine inspection must be done by the operator and defined on the chapter 7.4.

Customer support information:
Whenever requesting information, ordering spare parts, or requesting intervention, please contact the customer support by email or by phone.
The up-to-date contact information is available via the link: https://www.staubli.com/global/en/contacts.html

Please have the following information ready before you contact our Customer Support:

- Robot type
- Serial number of the machine (as shown on the identification plate)
- Software version: (Access this information on the teach pendant’s display screen by selecting ROBOT / INFORMATION)
- Error logger of the robot: (Access this information on the teach pendant’s display screen by selecting ROBOT / LOGGER)

DANGER

Movements with a removed cover
When a cover is removed, all movements of the arm are prohibited.

WARNING

Poor maintenance
Poor maintenance can lead to premature wear, or even cause safety hazards for production or maintenance operators.
Follow maintenance instructions.

NOTICE



Seal replacement

Not replacing the flat seal after a cover is removed may cause premature wear of the flat seal and reduce the ingress protection of the arm.

To ensure a correct seal, it is essential to replace the seal each time a cover is removed (see chapter 7.10).

NOTICE



Equipement pollution

When a cover is removed, the system loses its ingress protection. It is then exposed to pollution.

During maintance, be aware of the ingress of solids and liquids and check the cleanliness of the system before closing the covers again.

Arm: In the event of contamination, remove it with a dry wipe or a vacuum cleaner.

If the contamination can only be removed with a cleaning product or detergent, please contact Stäubli.

NOTICE



Particular case of HE and Stericlean robots

In the event of damage, any failure to comply with these instructions can affect the product's technical characteristics, and hence bring into question Stäubli liability as to guarantee.

In the event of damage to paintwork (HE or Stericlean arm), it is essential to touch up the paintwork as soon as possible, to protect the part concerned from corrosion (see chapter 7.6).

We strongly recommend contacting the Stäubli after-sales services to ensure that the repairs are carried out correctly.

Stäubli recommends an additional visit 6 months after the yearly Preventive Maintenance.

DANGER



Non-original parts

Not using original Stäubli parts when making repairs can cause unexpected risks resulting in severe injury and may result in loss of Stäubli's liability and warranty for the subsequent defaults related to the robot.

To ensure correct operation of the robot, it is essential to use original Stäubli parts when making repairs.

DANGER



Older or incompatible part versions

Stäubli's declaration of conformity with the requirements set out in the applicable machinery regulations and other national legislation is lost if some safety-related parts are replaced with older or incompatible versions.

An updated list of compatible spare parts can be obtained at your Stäubli Customer Support contact.

DANGER

Electrical hazard (power cut-off before maintenance)
Hazardous voltages may remain inside the controller, its connectors or the arm after the switch off. Some externally powered cables inside the controller can remain live after the switch off. The non-respect of the following points will result in injuries or death.
Wait 5 minutes after switching off the controller before carrying out any work on the controller or the arm. Disconnect all the externally powered cables before carrying out any work on the controller or the arm (Main power supply, external 24V (J213), digital inputs comming from externally powered devices (J212)).
Do not disconnect the X1 plug **(2)** under load.
Do not disconnect the J1200 plug under load.

DANGER

Residual voltage:
Residual voltages may remain present between motor phases even in "disable power" status.
Before carrying out any work on the motor connections, the controller must be switched off.

DANGER

Mandatory checks after a crash
If a crash of the arm occurs, safety components might have been damaged.
If a crash of the arm occurs, all safety components involved in the safety have to be checked carefully to verify they are still operating and not damaged: tools and grippers, hard stop devices on the arm, electrical limit switches, calibration and structure of the robot. Don't hesitate to call Stäubli service for any doubt.

DANGER

Calibration control
Each time a calibration, adjustment or recovery procedure is done, the calibration of the arm has to be controlled carefully to verify that the robot is able to move in its expected angular range and not more. This verification has to be done at slow speed.

CAUTION

Pressurized pneumatic circuit (cut-off before maintenance)
Pressure remaining inside the pneumatic circuit of the arm can cause minor to moderate injury.
Cut off the pneumatic supply to the arm before any maintenance operation.

DANGER

Lockout
Lockout and label the equipment before any work is carried out. See the controller instruction manual for the lockout procedure.

DANGER

People in the working area
Certain robot working modes such as the "brake release" mode can lead to unforeseeable arm movements.
All people are prohibited from remaining in the working area in which the robot arm operates.

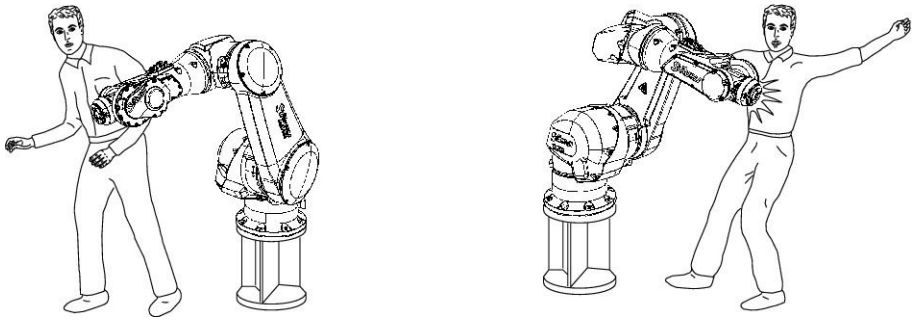


Figure 7.1:

Following maintenance work, whether it involves mechanical, electrical, pneumatic or software operations, it is advisable to make sure that the robot functions correctly, first at low speed while the person stays outside the cell, and then under the normal conditions of use. In particular, make sure that all the protective and safety systems are correctly in place, and that calibration of the robot is correct.

7.1.1 - MANDATORY LUBRICANTS

M0006841.1

Only Stäubli-approved lubricants must be used. Type of lubricants and approximate quantities (the quantities are given for information purposes only, oil is not fully drainable, please refer to the oil level for accurate estimation):

Compact wrist:

	Standard lubricant	H1 option	Quantity
Axis 1	ROBOLUB 68-S	ROBOLUB 100-H1	430 ml
Axis 2	ROBOLUB 68-S	ROBOLUB 100-H1	240 ml
Axis 3	ROBOLUB 68-S	ROBOLUB 100-H1	195 ml
Axis 4	ROBOLUB 1000-H1	ROBOLUB 1000-H1	90 ml
Axis 5/6	ROBOLUB 1000-H1	ROBOLUB 1000-H1	173 ml

Hollow wrist:

	Standard lubricant	H1 option	Quantity
Axis 1	ROBOLUB 68-S	ROBOLUB 100-H1	430 ml
Axis 2	ROBOLUB 68-S	ROBOLUB 100-H1	240 ml
Axis 3	ROBOLUB 68-S	ROBOLUB 100-H1	195 ml
Axis 4	ROBOLUB 1000-H1	ROBOLUB 1000-H1	90 ml
Axis 5	ROBOLUB 100-H1	ROBOLUB 100-H1	42 ml
Axis 6	ROBOLUB 100-H1	ROBOLUB 100-H1	24 ml

7.1.2 - SPARE PARTS

M0006572.1

Failure to comply with these two precautions can cause unexpected risks resulting in severe injury and may result in loss of Stäubli's liability and warranty for the modified the subsequent defaults related to the robot.
For a detailed list of spare parts, in accordance with the risk analysis for the application, please connect yourself to MyStaubli portal and access to our Spare parts Catalogue: https://login.staubli.com/

7.1.3 - TOOLING

M0006665.1

- Maintenance plug.
- SP2/SP2+ (if not used in production phase).
- RBR interface (if required).
- External 24 V power supply for CPT (RSI9) (if required).
- External 24 V power supply for RBR (if required).

7.2 - SAFETY

7.2.1 - MAINTENANCE MODE

M0006704.1

Use of the Maintenance Mode is not required for all maintenance operations. It must only be used when it is necessary to move the robot or activate SDO and SDI independently of the safety program.
The maintenance plug resets the safety configuration to a default configuration so that the controller and the arm can be tested. An external safety PLC can force the maintenance mode by activating the manual slow, automatic local and automatic remote working modes together, either on the J101 connector, or using the slave profile FS bus interface. The robot maintenance mode can thus be integrated as part of a global cell-defined maintenance mode.

- User safety stops are disabled; personnel safety is assured by keeping the robot motion at slow speed and under the control of the enabling device, even in automatic modes. Fast speed in automatic modes can be activated and controlled in the same way as for the manual fast speed working mode.
- Working mode selection and safety restart acknowledge are made from the pendant interface.

The maintenance plug is meant to prevent the electrical short-circuiting of the user safety stops during service. After such a short-circuit, the safety of the personnel cannot be assured when an automatic mode is used (voluntarily or by mistake).

DANGER

Restriction of use cases
The maintenance plug shall be used only when a temporary change in the safety configuration is needed for maintenance reasons.

DANGER

Untrained personnel
Service personnel shall be trained on the risks near the robot and related to the use of the maintenance plug.

DANGER

Maintenance plug accessibility
Access to the maintenance plug shall be limited to trained people.

7.2.2 - ABILITY

M0009052.1

Service personnel have to have the experience to be able to perform electrical and mechanical interventions. We differentiate 3 levels of intervention. Refer to the regulations currently in force in the country concerned.
Level 1: Operations that can be carried out by a maintenance technician without specific Stäubli training.
Level 2: Operations that can be carried out by a maintenance technician who has undergone specific Stäubli training. The documentation corresponding to these operations will be provided during the training.
Level 3: Operations that must be carried out by the Stäubli Customer Support.

WARNING

Failure to comply
Failure to comply with the levels of intervention can lead to incorrect operation of the robot and entail risks for the user and the machine environment.
Comply with the levels of intervention.

7.2.3 - RISKS

M0007177.1

WARNING

Residual risks
The maintenance technician must be informed of the hazards linked to the robot environment, and especially the risks of concurrent activity on the machine during maintenance.
Refer to the identified residual risk in chapter 3.2.5.

DANGER

Axis fall
The arm brakes alone shall not be considered as a safe protection against axis fall.
Make sure that the arm and the load are suitably supported during all work on the robot arm.

DANGER

Residual voltage:
Residual voltages may remain present between motor phases even in "disable power" status.
Before carrying out any work on the motor connections, the controller must be switched off.

DANGER

Disconnect power supplies:
Carrying out any work when electrical and pneumatic power supplies are activated can result in material damages and human injuries or death.
Before carrying out any work on the controller or the arm, it is highly recommended to disconnect all the electrical and pneumatic power supplies and to wait 5 minutes before starting to work. If this is not the case, the appropriate protection equipment must be worn, as required by the electrical accreditation (see the controller instruction manual).

7.2.4 - RE-COMMISSIONING

M0006485.1

After a change of a safety-related component, the corresponding safety functions must be validated:

Changed component	Function/component to test
DSI9	Power enabling; solenoid valves control.
Brake	Brake test; manual brake release.
Brake encoder module	Brake test; manual brake release; referencing.

For restarting after any maintenance operation:
- Make sure that the maintenance procedure was followed until its end.
- Check that the connections of the arm have been restored.
- Stay away from the arm during the functional tests.
- Perform first tests at slow speed, using the manual or the maintenance mode.

The DSI9 firmware version shall be consistent with the safety controller (RSI9) firmware version. A new DSI9 board can only replace a DSI9 board with a compatible firmware.

DSI9 firmware	RSI9 compatibility
0.1.1.36	1.0.0.10F
1.0.0.4	2.0.0.3
1.1.0.0	2.0.0.3

7.3 - PROTECTION AGAINST ELECTROSTATIC DISCHARGES

M0007069.1

7.3.1 - INFORMATION ON ELECTROSTATIC DISCHARGES

M0007008.1

What is an electrostatic discharge?
Everyone has felt the effects of static electricity on their clothes or when they touch a metal object, without being aware of the damage that can be done to electronic components by static electricity.
Our desire to integrate the notions of quality and reliability in our products makes it necessary to prevent electrostatic discharges from causing damage to them. This means that all the staff and login users must be informed.

Storage of a charge

An electric charge is created simply by combining a conductor, a dielectric and the ground (lowest reference potential, usually the ground in the case of an electrostatic charge).
Example: People, printed circuits, integrated circuits, components, conducting mats when separated from the ground by a dielectric.

Electrostatic discharges or ESD

Most people have experienced ESD by receiving an electric shock when walking on a carpet and touching a doorknob or when getting out of a car.

In most cases, the following is true:

- To feel an ESD, a charge of at least 3 500 V is required.
- To hear one, a charge of at least 5 000 V is required.
- To see a spark, a charge of at least 10 000 V is required.

This shows that it is possible to develop charges of over 10 000 V before noticing an electrostatic discharge!

Risks created by an electrostatic discharge

A high ESD voltage (several thousand volts) creates danger for electronic components. A semi-conductor must be handled carefully to prevent destruction by ESD. It is estimated that ESD destroy only 10% of the components that they effect. The other 90% of components fall into the "deteriorated" category. A component may be damaged with simply 25% of the voltage required to destroy it.

These hidden faults can lead to problems that appear several days, weeks or even months after the incident. Components may also undergo a change in their operating characteristics. Initial tests are successfully passed but an intermittent error occurs under vibration or temperature constraints. The same components will pass the "on/off" test successfully, as carried out during repairs, but the problem will reappear again once on site.

Typical ESD voltages

Source	Low relative humidity 10 - 20%	Average relative humidity 40%	High relative humidity 65 - 90%
Walking on carpet	35 kV	15 kV	1.5 kV
Walking on vinyl	12 kV	5 kV	0.3 kV
Working at the workstation	6 kV	2.5 kV	0.1 kV
Plastified instructions	7 kV	2.6 kV	0.6 kV
Polyethylene bags	20 kV	2 kV	1.2 kV
Cellular polyurethane	18 kV	11 kV	1.5 kV

Charge sources

- Work surfaces
- Packaging
- Floors
- Handling
- Chairs
- Assembly
- Carriages
- Cleaning
- Clothes
- Repairing

Parts sensitive to static charges

- Electronic boards
- Power supplies
- Encoders
- Etc

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7.3.2 - PREVENTION OF DAMAGE DUE TO ELECTROSTATIC DISCHARGES

M0006861.1

It is essential to guard against electrostatic discharges during an intervention concerning electronic components, subassemblies and complete systems.
Elimination of the danger due to ESD requires a combined team effort. By complying with the following instructions, the user can substantially reduce the potential damage caused by ESD and ensure long-term reliability for the robot:

- Inform the staff of the risks stemming from ESD.
- Know the critical zones sensitive to ESD.
- Know the rules and procedures to deal with ESD.
- Always carry components and boards in a tray to protect them from electrostatic charges.
- Always ground yourself before working on a workstation.
- Keep non-conducting equipment (static charge generators) away from components and boards.
- Use tools providing protection from ESD.

Stäubli workstation

To handle electronic cards, Stäubli workstations are given a grounded coating that dissipates static electricity.

Work zones

An antistatic bracelet is required to handle boards or electronic components.
Remove objects that generate static electrical charges from the work area.
Printed circuits, boards and electronic components must be kept in antistatic bags.

Antistatic wrist strap

The wrist straps are supplied as part of the standard equipment for the robot.

NOTICE	
	**Electrostatic discharges** Boards and electrical components are sensible to electrostatic discharges. Use an antistatic wrist strap and an antistatic mat connected to the controller for all work involving handling of boards or electrical components.

7.4 - MAINTENANCE FREQUENCY

M0006718.1

		Routine inspection	Check up	Preventive maintenance		
				Level A	Level B	Level C
		Monthly	Yearly	7000 h or 1 year	20 000 h or 5 years	40 000 h or 8 years
Safety[1]	Mechanical hard stops (option): Visual check, change if necessary	◊	●	●	●	●
	Remote brake release (option): Check operation, replace if necessary	◊	●	●	●	●
	Fastenings (base, tool interface flange, etc.): Check tightness and for corrosion	◊	●	●	●	●
	Power light (option): Check, change if necessary	◊	●	●	●	●
	Check that the motors and brakes are working properly	◊	●	●	●	●
General state of the arm	External: Visual check for corrosion and collision damage	●	●	●	●	●
	Painted parts: Visual check, touch up the paintwork for small areas (<2cm^2) without oxidation	Δ		●	●	●
	Checking of the factory marks		●	●	●	●
	Checking all moving parts (MPS010, external harness, bellow, …)	●	●	●	●	●
	Lubrication of moving parts			●	●	●
	Execution of a robot test program lasting from 15 minutes to 1 hour maximum			●	●	●
	Control of the general tightening of the screws			●	●	●
	Checking the known reference position			●	●	●
Gear boxes	Checking the joints for backlash and proper operation		●	●	●	●
	Oil level check of all joints		●	●	●	●
	Oil level adjustment of all the joints			●	●	●
	Adjustment of backlash			●	●	●
	Draining of all axes				●	●

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		Routine inspection	Check up	Preventive maintenance		
				Level A	Level B	Level C
		Monthly	Yearly	7000 h or 1 year	20 000 h or 5 years	40 000 h or 8 years
Sealing	Visual check of external leakage	●	●	●	●	●
	Visual check of internal leakage		●	●	●	●
	Covers, interconnect plate, forearm I/O plate and forearm/wrist interface flat seals: Visual check	Δ	●			
	Replacement of flat seals			●	●	●
	Pressurization unit: Check operating pressure on dial	●	●	●	●	●
Belt	Checking the condition of the belts		●	●	●	●
	Adjustment of belt tension			●	●	●
	Replacement of belts				●	●
	Replacement of pulleys and rollers				●	●
Electrical and pneumatic harness	Checking the electrical and pneumatic harnesses and connections		●	●	●	●
	Solenoid valves: Check operation(2)	◊	●	●	●	●
	Replacement of solenoid valves					●
	Checking the ground wiring of the robot arm segments		●			

(1): Visual check only - if the part needs to be changed, it will be charged extra

(2): The valves are factory lubricated for life and do not require additional lubrication

◊: Recommended operations: To be adapted in the light of the risk analysis for the application.

●: Recommended standard operations.

Δ: Recommended operations for applications in harsh environments (such as damp environments) or clean environments (such as CRs).

Recommended periodicity:

	Level A	Level B	Level C
Standard robot	Every 7 000 h or 1 year	Every 20 000 h or 5 years	Every 40 000 h or 8 years
Robot using H1 oil OR robot with heavy duty applications	Every 7 000 h or 1 year	Every 10 000 h or 5 years	Every 40 000 h or 8 years
Robot using H1 oil AND with heavy duty applications	-	Every 7 000 h or 1 year	Every 40 000 h or 8 years

The interval depends on the operation cycle of the robot, its working environment and movement pattern. In case of doubt, please refer to your Stäubli customer support to define the proper maintenance plan for your robot.

Dynamic parts such as gearbox, wrist, worm/gear, lip seals, brakes, balancing system, bellow, ball screw and harness are not replaced during the preventive maintenance. Nevertheless, these items will be checked during the preventive maintenance visit and a recommendation will be made on the identified parts to be replaced.

Those dynamic parts have been designed to achieve the statistic lifetimes ISO 281* listed below under optimal robot operating conditions. Lifetime can be calculated for a dedicated application with Optimize Lab software.

Spare parts	Lifetime estimated for average applications (h)	Life time estimated for Robot with heavy duty applications (h)*	Details
Gearbox	60 000	20 000	
Worm gear	40 000	15 000	
Wrist	40 000	15 000	
Lip seals	40 000	20 000	
Harness	60 000	20 000	
Motor (without brake)	80 000	40 000	
Motor (with brakes)	40 000	15 000	
Brake encoder module (BEM)	40 000	15 000	

* ISO 281 - L10 = Life span (statistical average) based on 90% reliability, i.e. 90% survival after testing on the same batch.

For H1 robots, the estimated lifetime of the gearboxes is divided by two.

7.4.1 - REGULAR MAINTENANCE

M0008281.1

The air filter on the ventilation system must be cleaned and/or replaced as and when necessary, depending on its level of fouling. A visual check should be done at least once a year.

If unusual sounds or vibrations are noted on the robot arm, it is necessary to inspect the tool and gripper fastenings carefully and make diagnoses at low speed.

DANGER

Mandatory checks after a crash

If a crash of the arm occurs, safety components might have been damaged.

If a crash of the arm occurs, all safety components involved in the safety have to be checked carefully to verify they are still operating and not damaged: tools and grippers, hard stop devices on the arm, electrical limit switches, calibration and structure of the robot. Don't hesitate to call Stäubli service for any doubt.

DANGER

Calibration control

Each time a calibration, adjustment or recovery procedure is done, the calibration of the arm has to be controlled carefully to verify that the robot is able to move in its expected angular range and not more. This verification has to be done at slow speed.

7.5 - MAINTENANCE PROCEDURES

M0009590.1

The only maintenance procedures described in this manual are the ones that require no specific training from Stäubli, classified as Level 1 (refer to Chapter 7.2.2).
All the available procedures are described in documents titled "Field Service Procedure (FSP)". These are listed in the table below, sorted by ability level:

Level	Name	Number	Chapter
1	Paint touch-up	FSP-000300	7.6
1	Flat seal exchange	FSP-000302	7.10
1	Covers	FSP-000303	7.8
1	Oil level check	FSP-000308	7.11.2
1	Pneumatic circuit test	FSP-000326	7.15
2	Encoder disconnection and connection	FSP-000600	
2	BEM exchange	FSP-000607	
2	Calibration	FSP-000611	
2	DSI exchange	FSP-000614	
2	Oil draining	FSP-000620	
2	Belt check	FSP-000706	
2	Power light exchange	FSP-000707	

To obtain these documents or schedule the required training, please contact the Stäubli support team.

7.6 - PAINT TOUCH-UP PROCEDURE FOR HUMID, ACCESSPHARMA, STERICLEAN AND STERICLEAN+ ROBOTS

M0006401.1

To protect the surfaces of painted parts from the environment, it is essential to make all necessary touch-ups in the event of scratches.
A paint touch-up kit is available. To order one, contact the Stäubli Customer Support.
The kit includes the paint and the procedure to be followed to carry out the touch-up work.

NOTICE

Immediate maintenance operation
In the event of damage to the paintwork, failure to comply with these instructions may lead to deterioration in the product's technical characteristics and bring into question Stäubli liability concerning warranty.
In the event of damage to the paintwork, immediatly make the paint touch-ups using the dedicated Stäubli touch-up kit.

7.7 - CLEANING AND DISINFECTING RECOMMENDATIONS

M0006993.1

Cleaning process and products which are listed below are compatible with the robot (performances, lifetime, ..). It does not replace an integrator analysis to confirm that the cleaning process (agent, periodicity, ...) is compatible with the application (especially in food, pharmaceutical and cosmetics applications).

Some versions permit a horizontal outlet connection. In any event, these components (including electrical/pneumatic connectors and interface plates) are not suitable for cleaning with chemical products.

Stäubli standard, CR, SCR external robot surfaces

- Unpainted metal parts:
  - Use wipes wetted with product recommended in the following compatibility chart.
- Painted metal parts or plastic parts:
  - Use wipes soaked with a neutral pH aqueous soap solution (For example, use a product listed in the following compatibility chart).
  - Rinse with wipe wetted with clean water.
  - Dry with wipes.

The interval between two cleaning or disinfecting actions depends on robot dirtiness (Typical periodicity: 1 week).

Stäubli HE external robot surfaces (Humid environment application):

- Use an aqueous product having a pH between 4.5 to 8.5 (For example, use a product listed in the following compatibility chart).
- Rinse with clean water.
- Dry with wipes.

Stäubli Accesspharma external robot surfaces (Bio-contamination controlled environment application):

- Unpainted metal parts:
  - Use wipes wetted with product recommended in the following compatibility chart.
- Painted metal parts or plastic parts:
  - Use wipes soaked with a neutral pH aqueous soap solution (For example, use a product listed in the following compatibility chart).
  - Rinse with wipe wetted with clean water.
  - Dry with wipes.

The interval between two cleaning or disinfecting actions depends on robot dirtiness (Typical periodicity: 1 week).

Stäubli Stericlean and Stericlean + external robot surfaces (Bio-contamination controlled environment application):

- Use an aqueous product having a pH between 4.5 to 8.5 STER or STER + (pH 2 to 12 under condition, please consult us). For example, use a product listed in the following compatibility chart
- Rinse with clean water.
- Dry with wipes.

Or

- Use other products recommended in the following compatibility chart (e.g. 70% Isopropyl alcohol (IPA 70), hydrogen peroxide (H2O2) 50% on a wipe or in vapor phase) without any water rinse.

The interval between two cleaning or disinfecting actions depends on robot dirtiness (Typical periodicity: End of every production).

NOTICE	
	**Use of alcohol:** Alcohol could damage the coder. Do not use alcohol to clean the coder. Use only dry cloth.

Information given herein shall not relieve you from checking the cleanliness of the robot after the cleaning and disinfecting process. Stäubli shall not be liable for any damage arising because of persistence of residue after the cleaning process described herein.

NOTICE

Other products
Use of chlorinated products for cleaning or disinfecting is prohibited. Use of any other cleaning or disinfecting product than those specified may damage robot external surfaces.
Use a protective covering for the robot when additional products are required for your application.

⚠ CAUTION

Mandatory pressurization
In case of cleaning or disinfecting process with liquid splash or low pressure liquid jet, the interior of the arm will be contaminated if not pressurized.
In case of cleaning or disinfecting process with liquid splash or low pressure liquid jet, the robot arm has to be pressurized (see chapter 3.3.9).

To avoid scaling down the paint brightness during cleaning, we recommend to gently wiping the surfaces with a soft non-woven cloth.

In case of cleaning or disinfecting agents being out of specification, please consult Stäubli.

NOTICE

Indicator light (optional):
The chemical compatibility table below doesn't includes the indicator light (optional).
It should be noted that due to its transparent plastic design, the indicator light is not optimal in harsh environments, and makes the arm more difficult to clean and disinfect.

Type of product			Disinfectant				Cleaner		
pH			Neutral	Neutral	2 to 4	-	8.5	Neutral	-
Cleaner or disinfecting product name			P3 alcodes	P3 topax 990	Hydrogen peroxide H2O2	70% Isopropyl alcohol (IPA 70)	P3 aquanta tin	ELPON	70% Isopropyl alcohol (IPA 70)
Manufacturer			ECOLAB http://www.ecolab.com				ECOLAB http://www.ecolab.com		
Concentration			100%	3%	35% on a wipe or vapor phase	100%	0.2%	3%	100%
Robot version	Standard	Uncoated parts	√*	No	No	√*	No	No	√*
		Coated parts or plastic parts	No**	√	No	No**	√	√	No**
	CR	Uncoated parts	√*	No	No	√*	No	No	√*
		Coated parts or plastic parts	No**	√	No	No**	√	√	No**
	SCR	Uncoated parts	√*	No	No	√*	No	No	√*
		Coated parts	√	√	No	√	√	√	√
	HE	Uncoated parts	√	√	√*	√	√	√	√
		Coated parts	No**	√	No	No**	√	√	No**
	Stericlean	Uncoated parts	√	√	√	√	√	√	√
		Coated parts	√	√	√	√	√	√	√
Usable for food industry			√	√	√	√	√	√	√

Please contact the product manufacturer to define cleaning or disinfected process.

NOTICE

*: Use only wipes wetted with the product.
**: For painted part, it is also possible to use this product with a soft action of wiping and without too long application. To avoid scaling down the paint brightness during cleaning, we recommend to gently wiping the surfaces with a soft non-woven cloth.

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Chemical compatibility table[1]

Chemical Compound	Concentration	Accesspharma	Stericlean	Stericlean +
Hydrogen peroxide	30%	N/A	++	+++
Hydrogen peroxide	50%	N/A	++	+++
Isopropanol	70%	++[2]	++[2]	+++
Ethanol	>99%	N/A	++[2]	+++
Hydrochloric acid[3]	5%	N/A	- -	++
Sulfuric acid	5%	N/A	+	++
Peracetic acid	15%	N/A	-	+
Sodium Hydroxyde	5%	N/A	-	+++
Sodium hypochlorite	5%	N/A	-	+
Ammonium	25%	N/A	+	+++
Cyclohexane	Pure	N/A	++	+++

2 Use only wipes wetted with the product. The repetitive mechanical action of wiping may damage the brightness of the plastic parts (indicator light).

3 Chlorine-containing chemicals must be rinsed so as not to damage stainless steel parts.

7.8 - COVER LOCATIONS AND REMOVAL

M0006423.1

The cover screws on Stericlean+ versions are protected by caps, either hexagonal or flat-headed.
Removal and installation procedures for Stericlean+ caps are described in chapter 7.9.

Corresponding FSP: FSP-000303 (see chapter 7.5)

DANGER

Movements with a removed cover
When a cover is removed, all movements of the arm are prohibited.

DANGER

Axis fall
The arm brakes alone shall not be considered as a safe protection against axis fall.
Make sure that the arm and the load are suitably supported during all work on the robot arm.

1 High temperatures will weaken chemical resistance: Maximum temperature recommended 40°C.

⚠ WARNING



Incorrectly tightened covers

Incorrectly tightened covers compromise the arm's ingress protection and could detach during operation, posing a severe injury or fatality risk to personnel nearby.

All cover screws must be present and tightened to the required torque. If it is necessary to replace any of them, use only Stäubli references (see the spare parts catalogue in chapter 7.1.2).



Tools required:

- 3 mm Allen key.



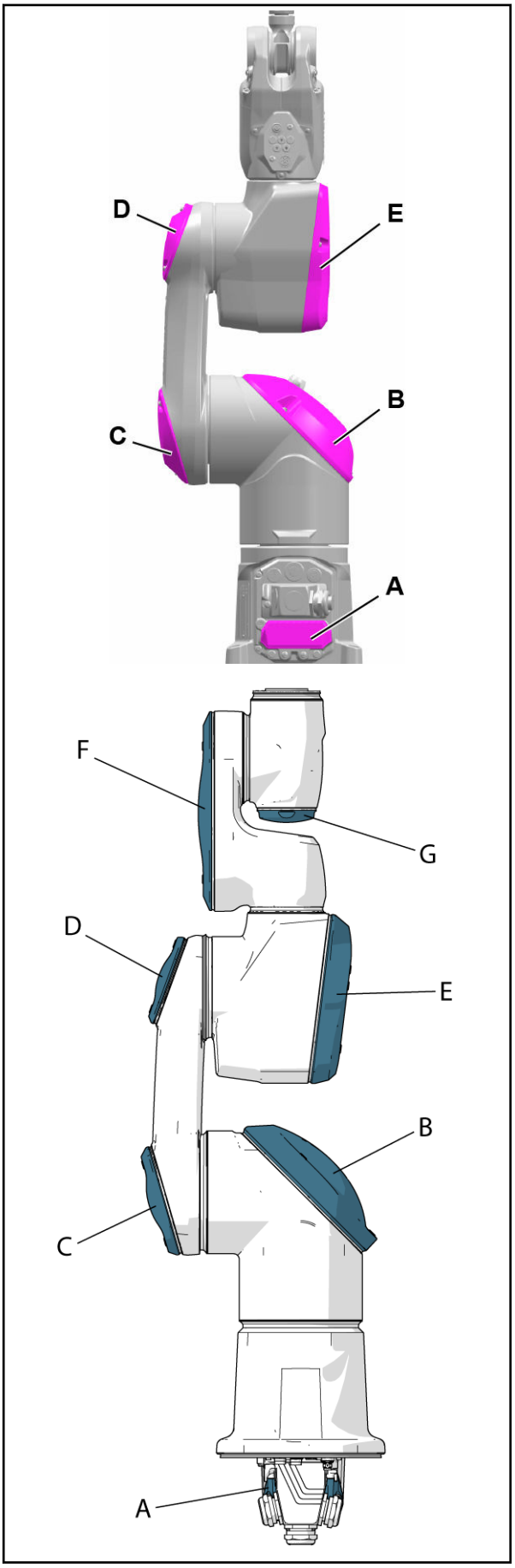
Recommended tools:

- Torque wrench.

Initial position:

- Not applicable.

Procedure:



- **(A) DSI cover:** 2 M4x20 screws (tightening torque 3.6 Nm).
- **(B) Shoulder cover:** 3 M4x20 screws (tightening torque 3.6 Nm).
- **(C) Inner link bottom cover:** 3 M4x20 screws (tightening torque 3.6 Nm).
- **(D) Inner link top cover:** 3 M4x20 screws (tightening torque 3.6 Nm).
- **(E) Elbow cover:** 3 M4x20 screws (tightening torque 3.6 Nm).
- **(F) Forearm cover (Hollow wrist only):** 3 M4x10 screws (tightening torque 3.6 Nm).
- **(G) Wrist cover (Hollow wrist only):** Screwable cover (tightening torque 25 Nm).

- The indicated tightening torques must be within ±7%.
- To ensure correct compression of the flat seal, it is preferable to apply progressive tightening.

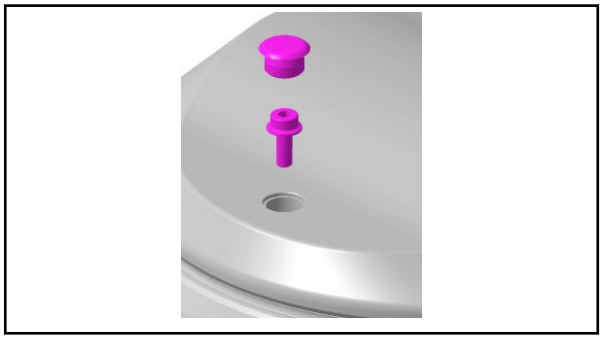
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- Sealed screws for specific versions (tightening torque 3.6 Nm).
- Sealed screws for Stericlean and Accesspharma versions (tightening torque 3.6 Nm).

DANGER

Covers grounding
Threadlocker glue can compromise the cover's grounding through its screws, creating an electrical hazard.
Do not use threadlocker glue on screws for covers grounding.



- Sealed screws with caps for Stericlean+ version (tightening torque 3.6 Nm).

DANGER

Covers grounding
Threadlocker glue can compromise the cover's grounding through its screws, creating an electrical hazard.
Do not use threadlocker glue on screws for covers grounding.

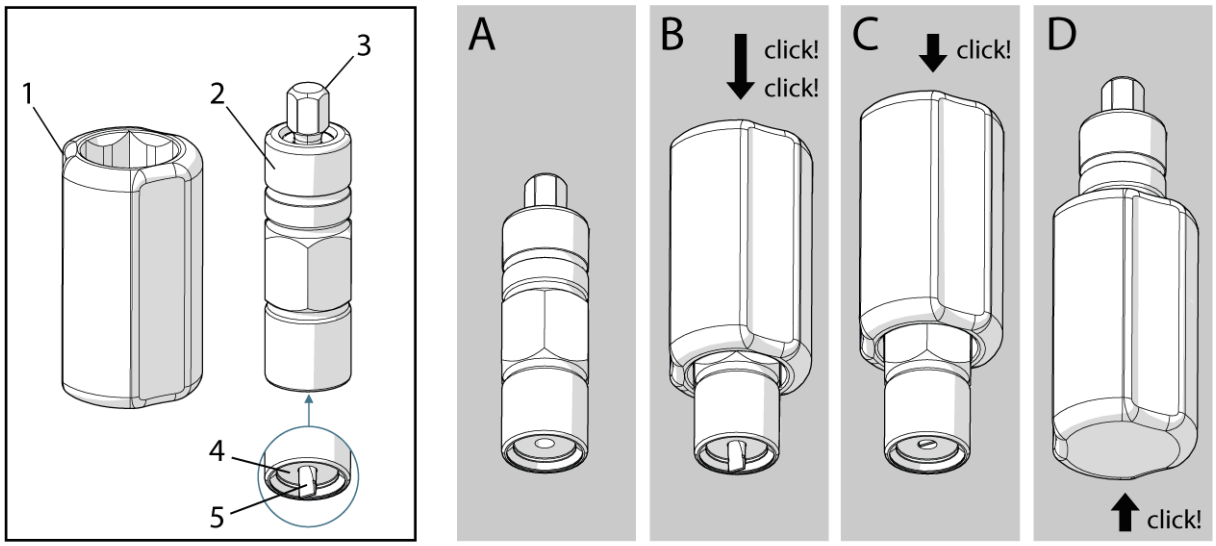
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7.9 - STER+ SCREW CAPS

M0009558.1

7.9.1 - SPECIFIC TOOL FOR FLAT HEAD STER+ CAPS

M0009556.1



English	Translation
click!	Click!

Item	Description
1	Outer part (cap or handle)
2	Inner part (main body)
3	Rear end (chisel head)
4	Front end (cap support face)
5	Retractable flat screwdriver tip/Retractable chisel cutting edge
A	Punching configuration, the outer part is removed
B	Unscrewing configuration, the outer part is slid from the back until the second click is heard
C	Screwing configuration, the outer part is slid from the back until the first click is heard
D	Storage configuration, the outer part is slid from the front until a click is heard

Figure 7.2: Specific tool

7.9.2 - FLAT HEAD STER+ CAPS REMOVAL

M0009552.1

Tools required:

- Hammer.
- Specific tool D 231 959 XX. (see tool description in chapter 7.9.1)
- Maintenance kit D 244 857 XX. (includes the double-sided stickers and the caps)

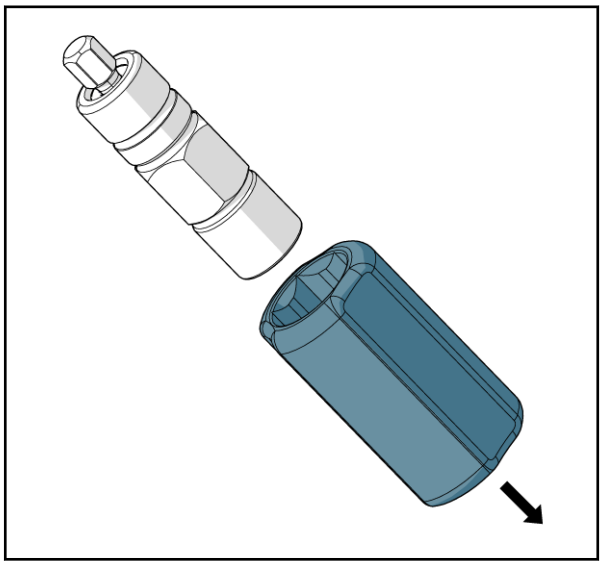
Recommended tools:

- Not applicable.

Initial position:

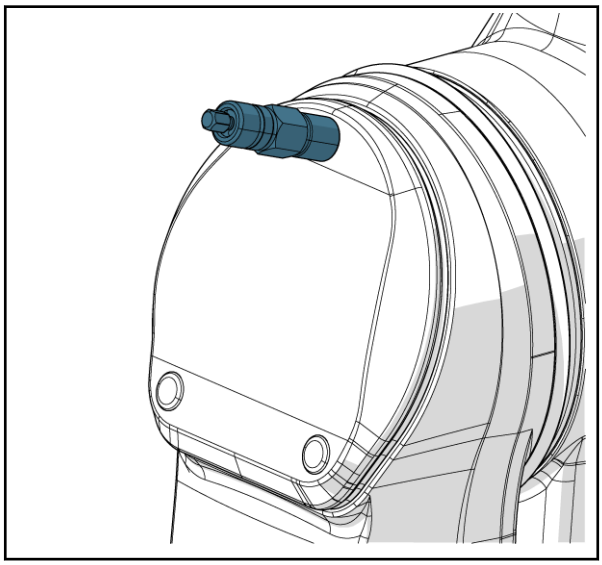
- Not applicable.

Procedure:



Tool preparation:

- Remove the outer part of the specific tool **(1)**.

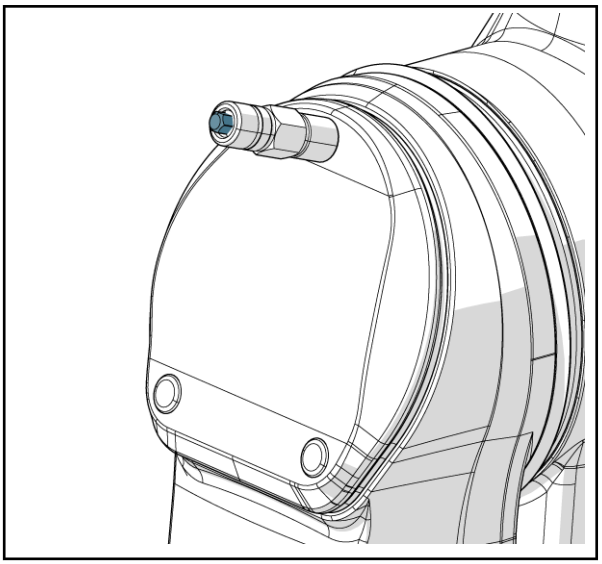


Placing the tool:

- Place the flat screwdriver tip side of the specific tool onto the cap.
- Perfectly align the specific tool on the cap.

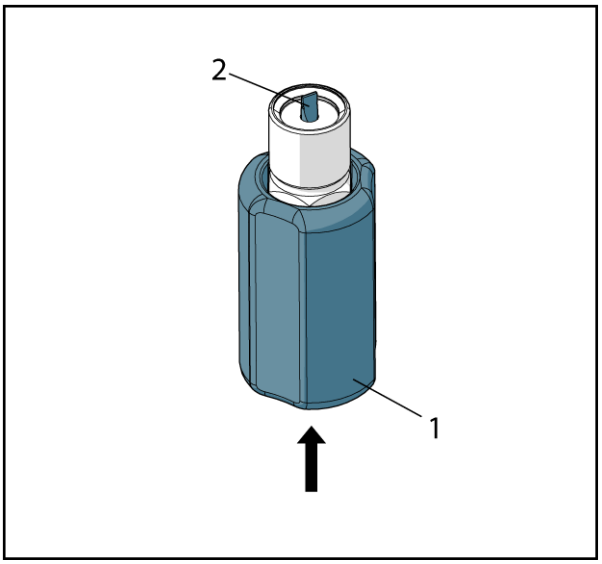
A misaligned tool could slip on impact and damage the arm surface.

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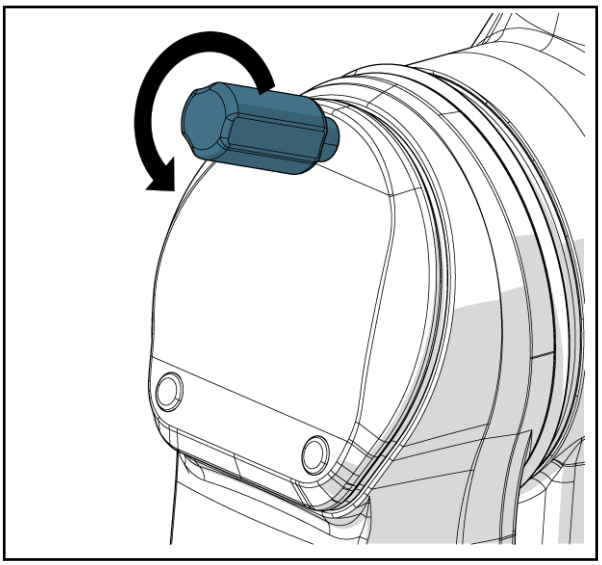
Punching holes in the caps:

- Using a hammer, strike the back of the specific tool until the cap is successfully punched.
- Repeat this operation for all the caps you want to remove.



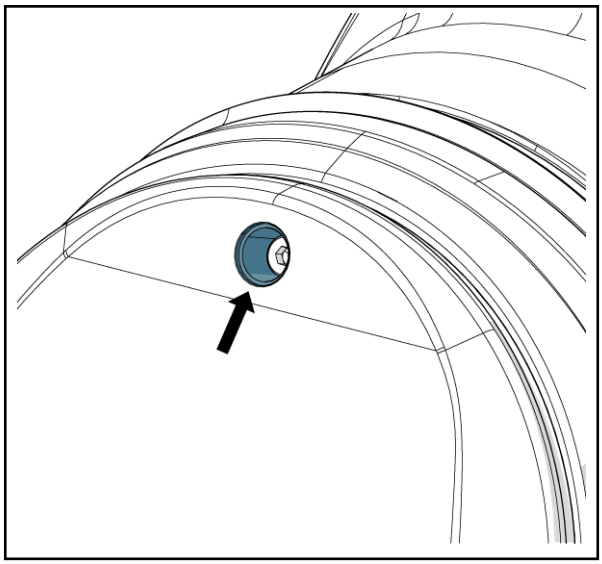
Tool preparation:

- Replace the outer part **(1)** of the specific tool on its back (see tool configuration **(B)** in chapter 7.9.1).
- Push it until you hear a second click.
- The flat screwdriver tip **(2)** should be out.



Unscrewing the caps:

- Place the flat screwdriver tip of the specific tool into the hole you've made earlier.
- Unscrew the cap by turning the specific tool counterclockwise.
- Discard the used cap.
- Repeat this operation for all the caps you want to remove.



Cleaning:

- Using a dry cloth, remove as much of the sealant residue as possible from the cover threads.

  Use only a dry cloth for cleaning to avoid contamination

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7.9.3 - FITTING THE FLAT HEAD STER+ CAPS

M0009553.1

Tools required:

- Sealant cartridge D 600 069 XX.
- Specific tool D 231 959 XX. (see tool description in chapter 7.9.1)
- Maintenance kit D 244 857 XX. (includes the double-sided stickers and the caps)

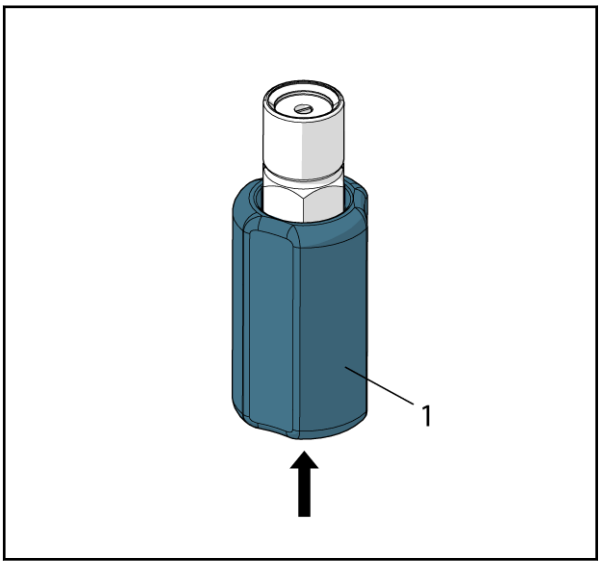
Recommended tools:

- Not applicable.

Initial position:

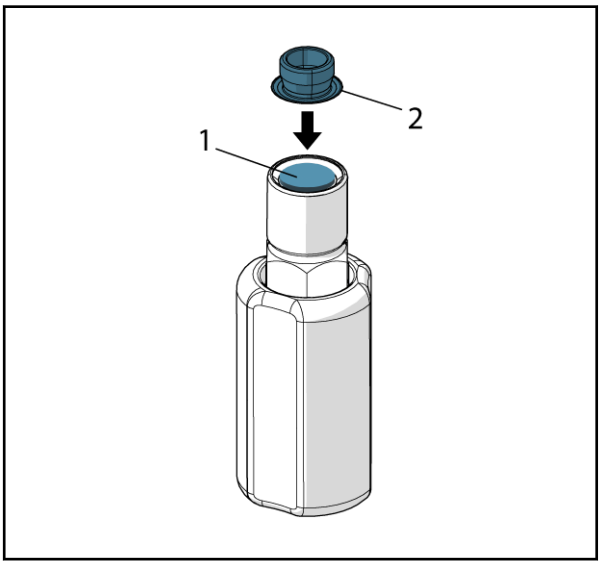
- Not applicable.

Procedure:



Tool preparation:

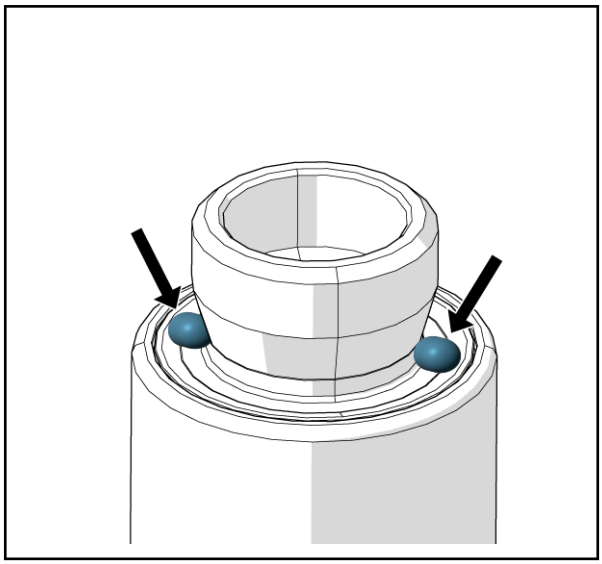
- Slide the outer part **(1)** of the specific tool at the back end, until you hear the first click (see tool configuration **(C)** in chapter 7.9.1).
- The flat screwdriver tip should be fully retracted.



Placing the new cap on the specific tool:

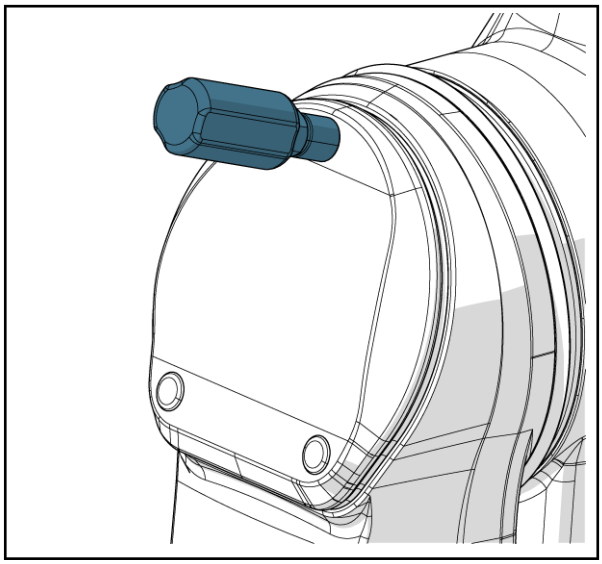
- Apply a double-sided sticker **(1)** to the end of the specific tool.
- Place a new cap **(2)** on the specific tool.
- Press it against the sticker.

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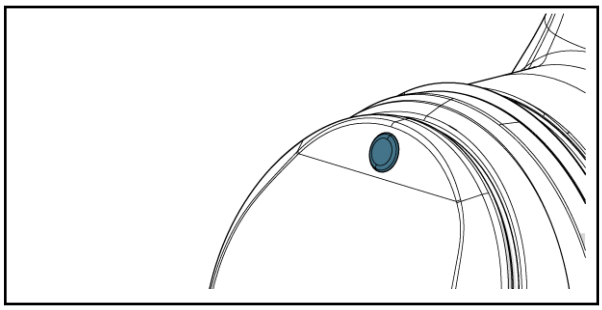
Sealant application:

- Place 2 small dots of sealant on the new cap, opposite each other.



Fitting the new cap:

- Tighten the new cap until it releases from the tool.

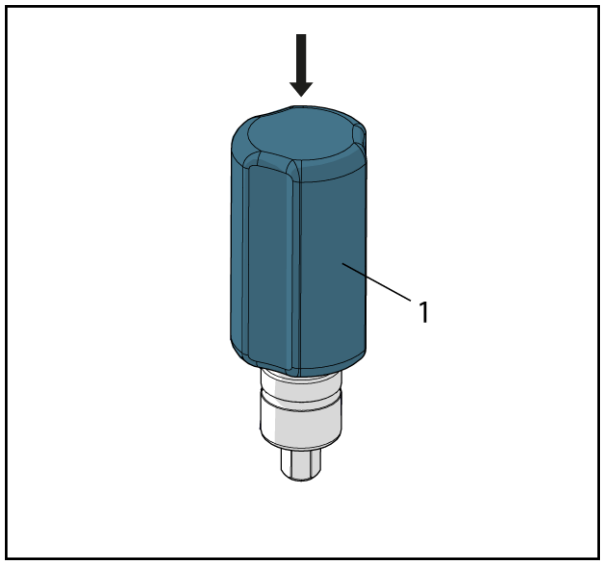


Cleaning:

- Remove and discard the double-sided sticker.
- Use a dry cloth to clean any sealant that has spread around the cap.

To avoid unscrewing the cap, do not perform a counterclockwise circular motion when cleaning

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Tool storage:

- Clean your specific tool.
- Slide the outer part **(1)** of the specific tool at the front end to cover the screwdriver tip.

7.9.4 - HEXAGONAL HEAD STER+ CAPS REMOVAL

M0009560.1

Tools required:

- 10 mm socket wrench.

Recommended tools:

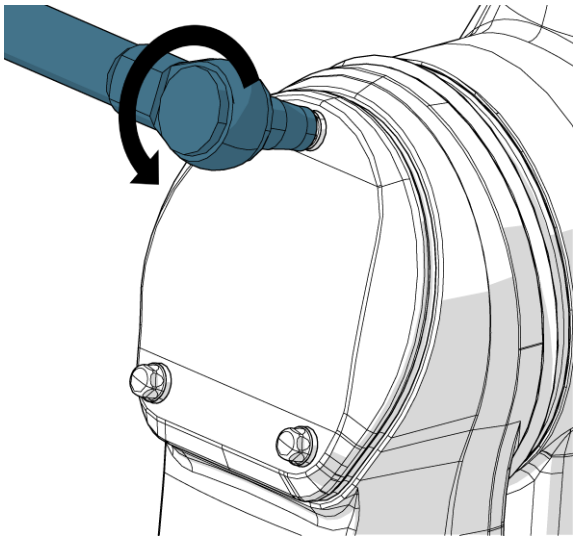
- Not applicable.

Initial position:

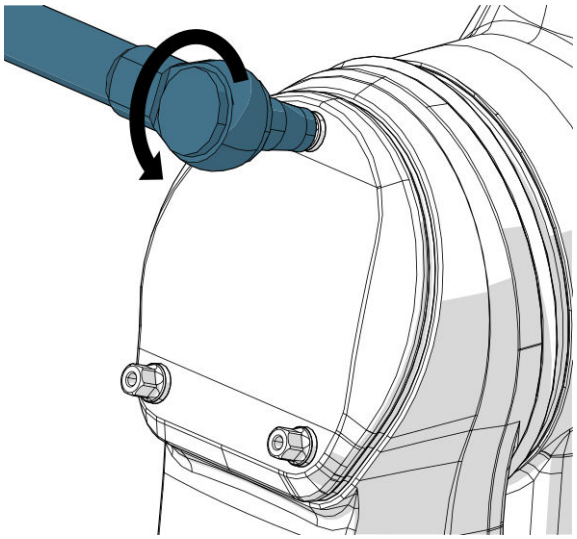
- Not applicable.

Procedure:

Hexagonal caps:

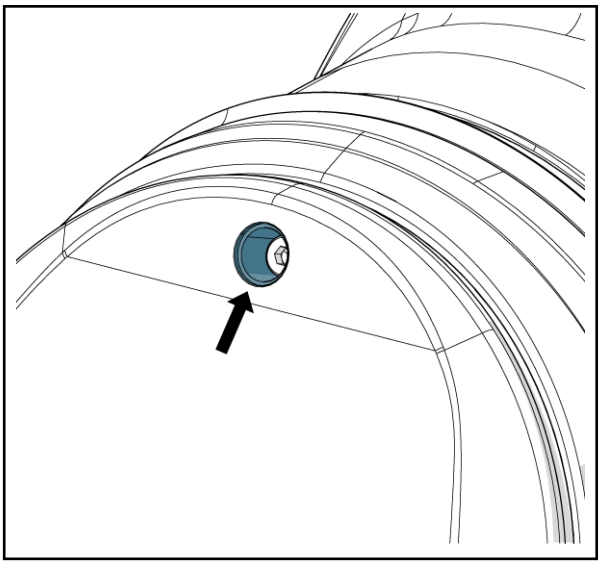


Spacer caps (option):



- Unscrew the caps.

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Cleaning:

- Using a dry cloth, remove as much of the sealant residue as possible from the cover threads.

Use only a dry cloth for cleaning to avoid contamination

7.9.5 - FITTING THE HEXAGONAL HEAD STER+ CAPS

M0009559.1

Tools required:

- Sealant cartridge D 600 069 XX.
- Torque wrench with 10 mm socket.

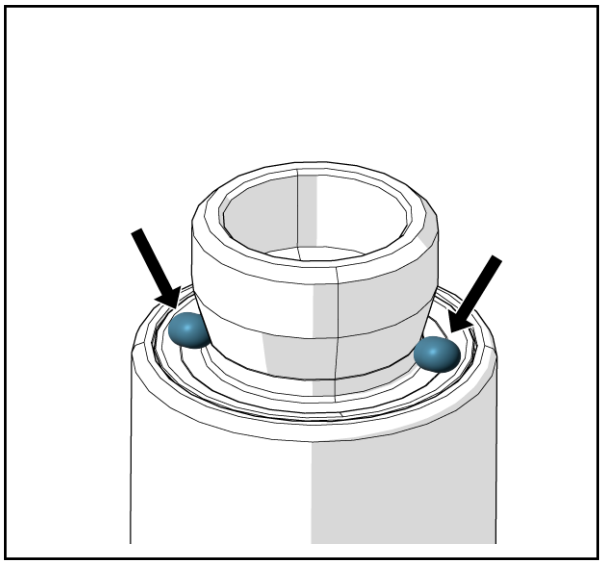
Recommended tools:

- Not applicable.

Initial position:

- Not applicable.

Procedure:

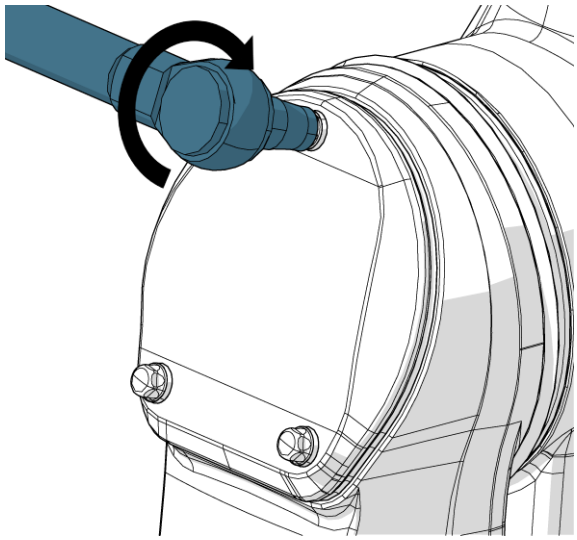


Sealant application:

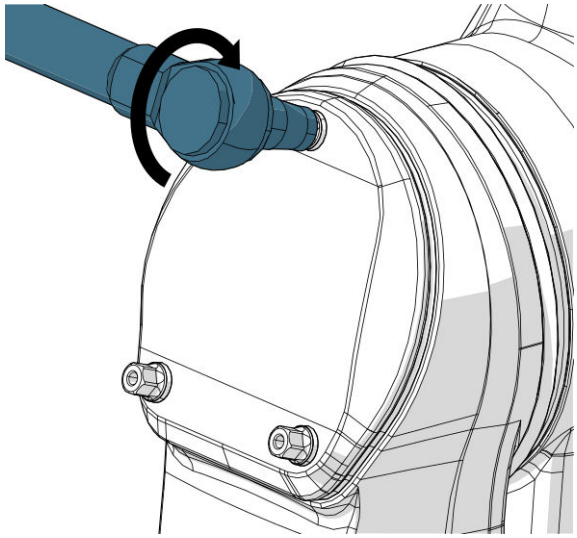
- Place 2 small dots of sealant on the new cap, opposite each other.

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Hexagonal caps:



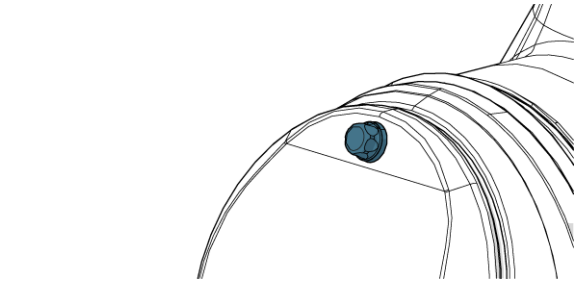
Spacer caps (option):



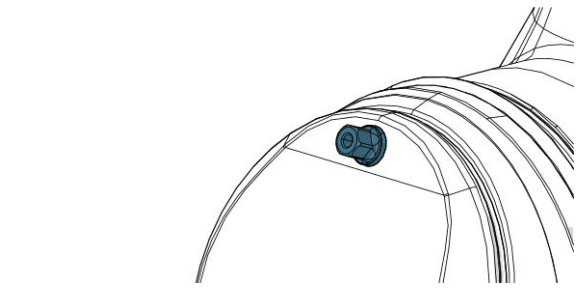
Fitting the new cap:

- Tighten the cap to a torque of 3.6 Nm.

Hexagonal caps:



Spacer caps (option):



Cleaning:

- Use a dry cloth to clean any sealant that has spread around the cap.

To avoid unscrewing the cap, do not perform a counterclockwise circular motion when cleaning

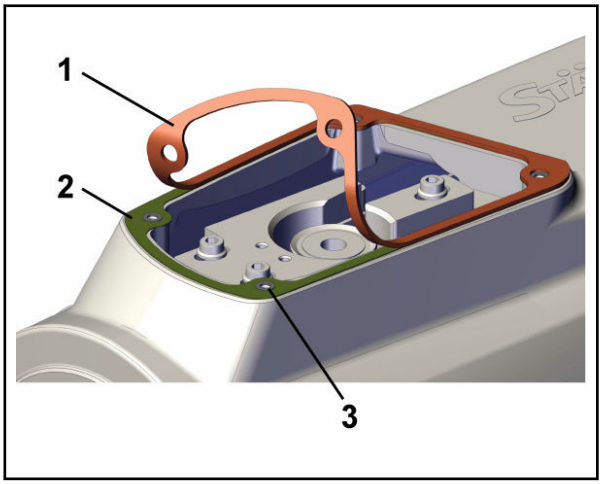
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7.10 - PROCEDURE FOR REPLACING THE FLAT SEAL

M0006902.1

7.10.1 - REMOVING THE SEAL

M0006415.1

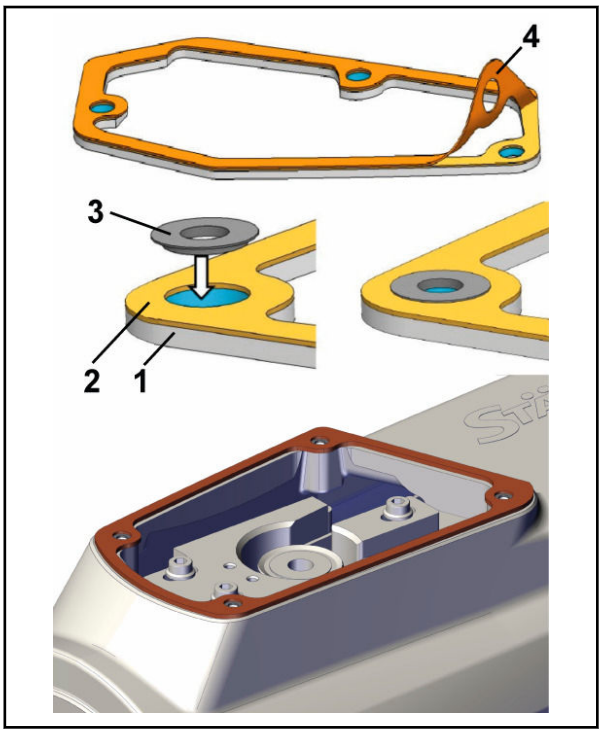


The flat seal is made up of 3 sections:

- A section made of foam **(1)**.
- An adhesive section **(2)**.
- Metal inserts **(3)**.
- To remove the flat seal, peel off a corner of the seal and pull it towards you.
  If the adhesive section **(2)** remains stuck to the cover, use "C" fluid (ethyl acetate) to remove it.
- Remove the metal inserts **(3)** if they are still stuck in place.
- Clean the surface, and remove any particles present on the surface, without scratching it.
- Clean the whole surface using "C" fluid (ethyl acetate).

7.10.2 - FITTING A NEW SEAL

M0006413.1



The new seal is made up of 4 sections:

- A section made of foam **(1)**.
- An adhesive section **(2)**.
- Metal inserts **(3)**.
- A protective paper covering the adhesive **(4)**.
- Remove the cut-out sections such as screw holes.
- Present the seal.
- Remove the protective paper **(4)** and place the metal inserts **(3)** in the screw hole recesses.
- Apply the new seal to the surface. Line the contour of the seal up with that of the surface.

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7.11 - LUBRICATION

M0006376.1

The greasing and lubrication work is carried out by Stäubli in the factory. At the time of delivery, the robot is ready for use.

7.11.1 - GENERAL INFORMATION

M0006373.1

The robot requires very little maintenance, but lubrication and the inspections listed below must be carried out at regular intervals to ensure its efficiency over time.

DANGER

Arm left under power between each maintenance position
Performing maintenance on an energized arm is hazardous.
Read the general safety precautions before carrying out any maintenance operation (see chapter 3.2).
Despite the need to move the arm between each maintenance position, power must be disconnected each time.

NOTICE

Preventive maintenance:
A failure of maintenance will result in material damages.
It is advisable to carry out regular, precise preventive maintenance to ensure that the robot keeps its original operating characteristics over long periods.

7.11.1.1 - Lubricate

M0006377.1

The type and quantity of lubricant for each lubrication sequence are listed in the table in chapter 7.1.1.

NOTICE

Inadequate oil levels
Too much or too little oil can prevent the robot from operating correctly.
The oil levels for joints 1 to 4 must be with the marks on the sight glasses.
For joints 5 and 6, the oil level should reach the bottom of the filling hole.

7.11.1.2 - Precautions for use of lubricants

M0006374.1

CAUTION

Health risks
During assembly in the Stäubli factory, the robot is filled up with lubricants that are free of substances harmful to health; however, in some cases, repeated and prolonged exposure to the product can cause skin irritation, or sickness if swallowed.
Avoid prolonged and repeated contact with the product and do not ingest it.
In the event of contact with the eyes or the skin, wash the affected areas with plenty of water; if irritation persists, consult a doctor.
In the event of swallowing, do not provoke vomiting or administer any products orally; consult a doctor as soon as possible.

7.11.2 - CHECKING OIL LEVELS

M0006445.1

Tools required:

- Not applicable.

Recommended tools:

- Not applicable.

Initial position:

- Not applicable.

Procedure:

NOTICE

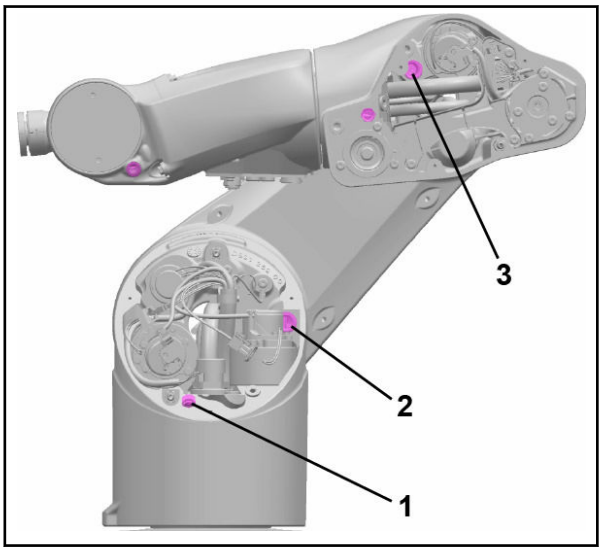
Inadequate oil levels
Too much or too little oil can prevent the robot from operating correctly.
The oil levels for joints 1 to 4 must be with the marks on the sight glasses.
For joints 5 and 6, the oil level should reach the bottom of the filling hole.

7.11.2.1 - Arm in floor-mounted position

M0006424.1

Checking the levels for axes 1, 2 and 3 with the arm on the floor

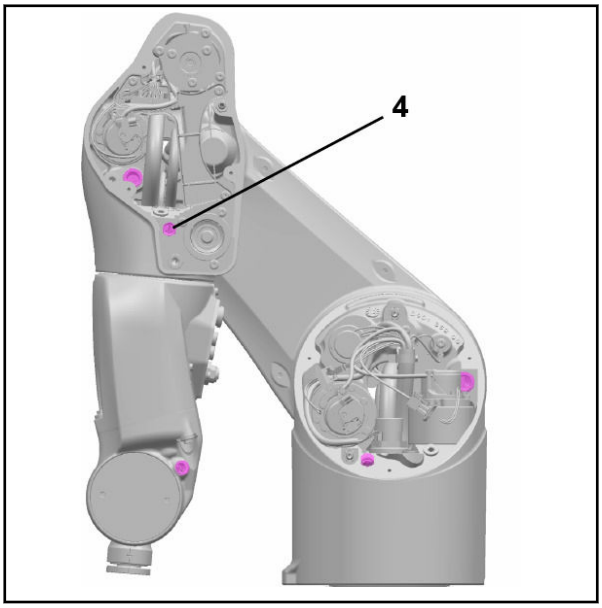
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position (Compact wrist only)	-	+45°	-127°	-	-	-
Axis position (Hollow wrist only)	-	+45°	-126°	-	-	-



- **Axis 1:** Oil level has to be in the middle of sight glass **(1)** ± 3 mm.
- **Axis 2:** Oil level has to be in the middle of sight glass **(2)** ± 3 mm.
- **Axis 3:** Oil level has to be in the middle of sight glass **(3)** ± 3 mm.

Checking the level for axis 4 with the arm on the floor

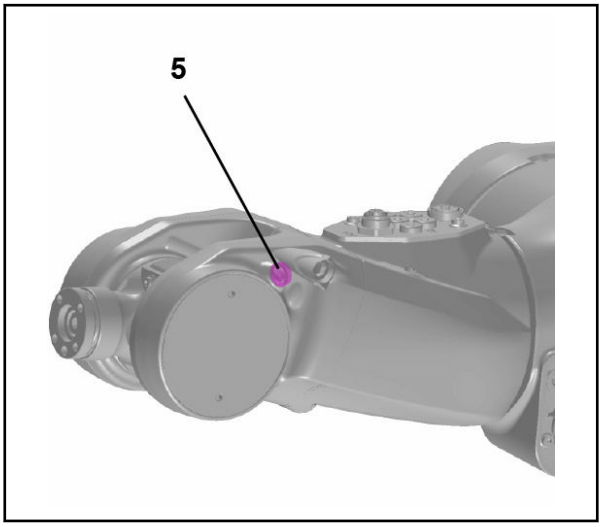
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position (Compact wrist only)	-	-45°	-135°	-	-	-
Axis position (Hollow wrist only)	+90°	-54°	-126°	-180°	-	-



Axis 4: The oil level is flushed with the upper section of the wheel after removing cap **(4)**.

Checking the levels for axes 5 and 6 with the arm on the floor (Compact wrist only)

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	+45°	-127°	-180°	-	-

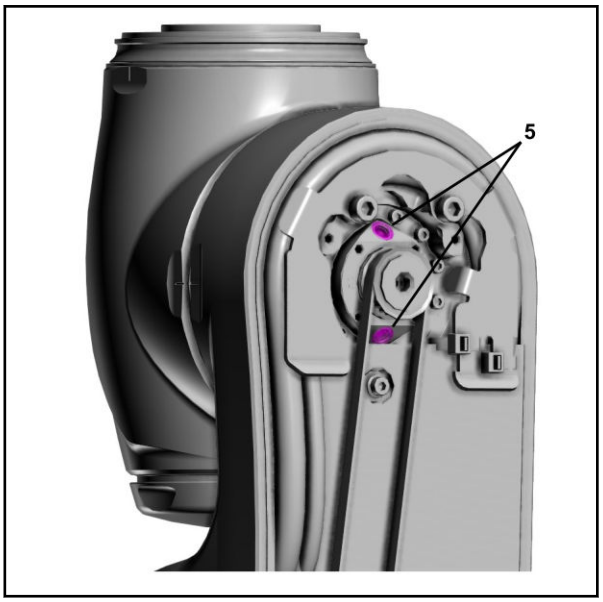


Axis 5-6:

- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove both caps **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Re-install both caps **(5)** (3 Nm).

Checking the level for axis 5 with the arm on the floor (Hollow wrist only)

The forearm must be positioned vertically, as in the picture below.



Axis 5:

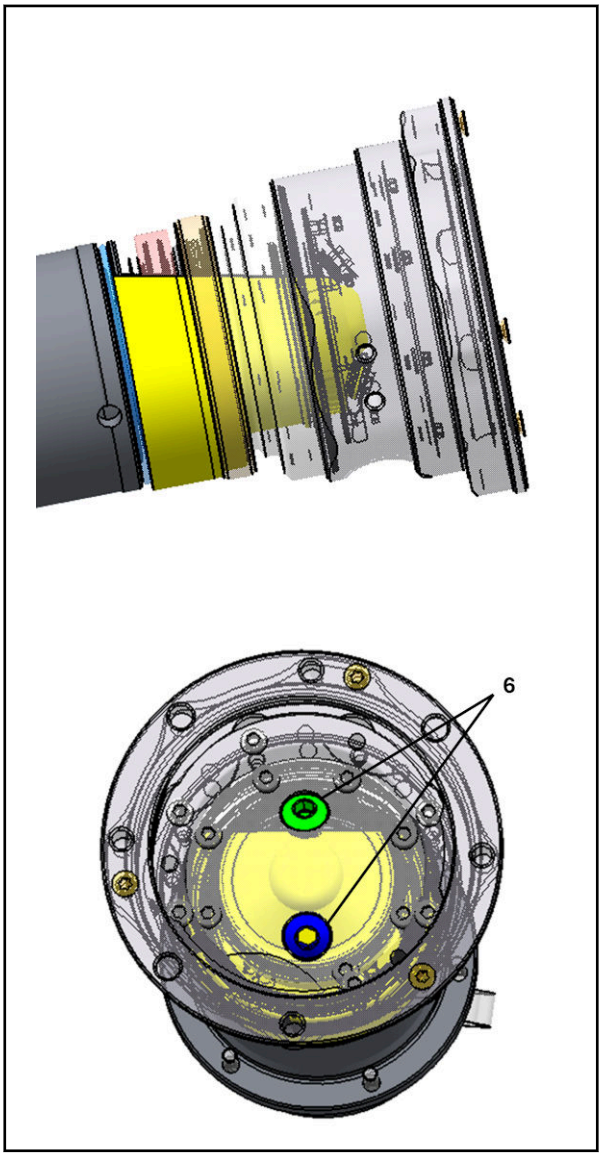
- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the upper cap **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(5)** (3 Nm).

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Checking the level for axis 6 with the arm on the floor (Hollow wrist only)

The following position is recommended to check the level for axis 6. The axis 6 must be inclined at 11° to the horizontal with the oil drain plugs vertically aligned.

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	+45°	-124°	-	-	+45°



Axis 6:

- Run test program at SP60 on the largest joint 6 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the stainless steel flange sleeve (non visible on the picture).
- Place the cable gland support plate (non visible on the picture) on the side (regardless of flying or flange-mounted connectors).
- Remove the upper cap **(6)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(6)** (3 Nm).

7.11.2.2 - Arm in vertical wall-mounted position

M0006425.1

The positions for checking depend on arm orientation

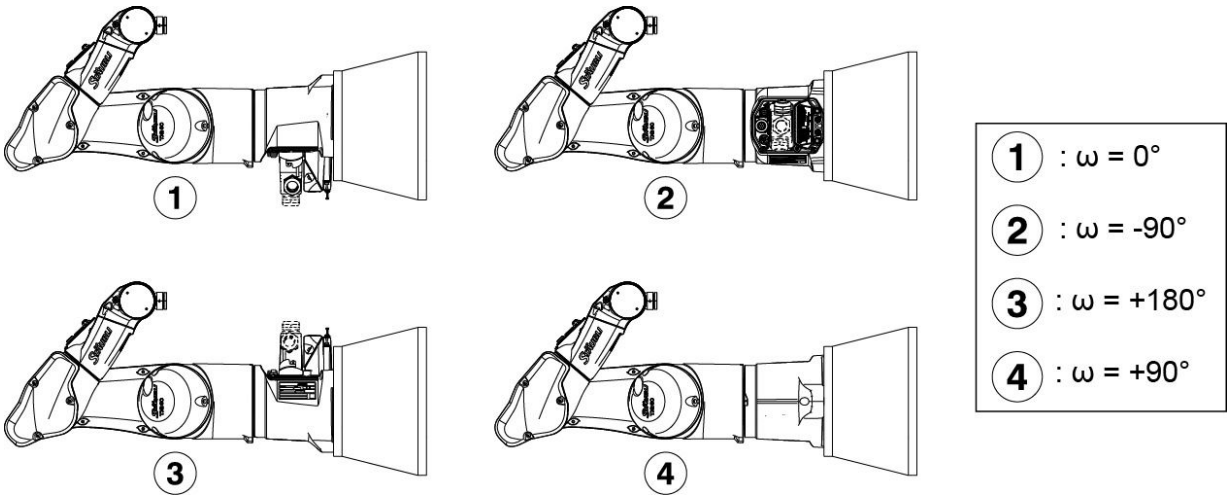
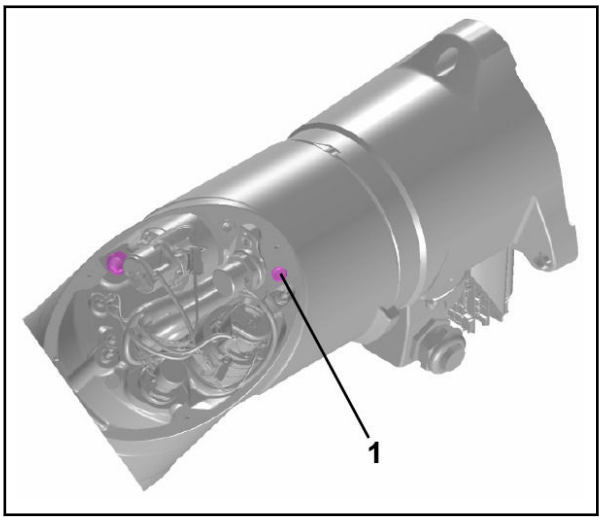


Figure 7.3:

Checking the level for axis 1 with the arm on a wall

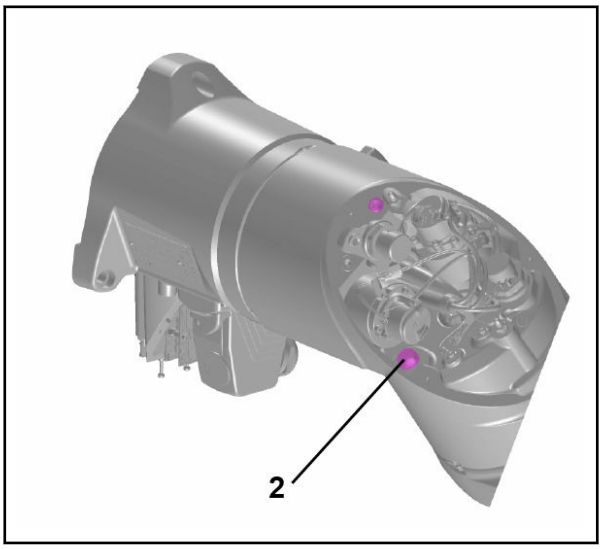
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	+53° + ω	-	-	-	-	-



Axis 1: Oil level has to be in the middle of sight glass **(1)** ± 3 mm.

Checking the level for axis 2 with the arm on a wall

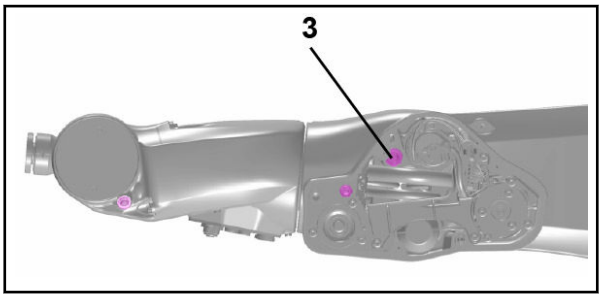
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	+126° + ω	-	-	-	-	-



Axis 2: Oil level has to be in the middle of sight glass **(2)** ± 3 mm.

Checking the level for axis 3 with the arm on a wall

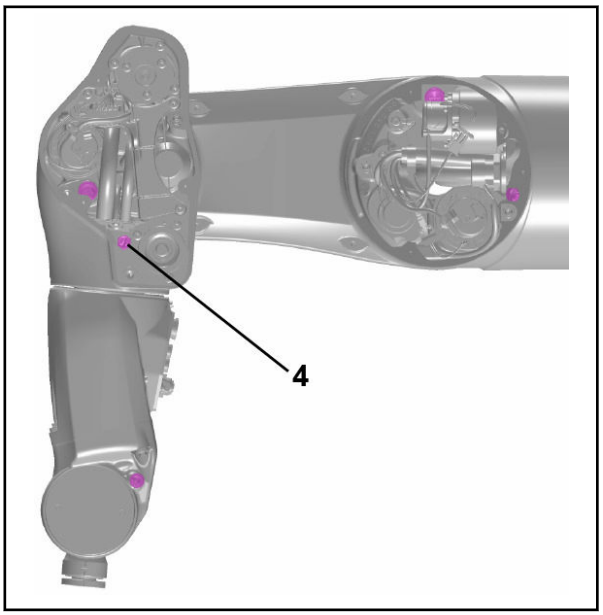
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	ω	0°	-8°	-	-	-



Axis 3: Oil level has to be in the middle of sight glass **(3)** ± 3 mm.

Checking the level for axis 4 with the arm on a wall

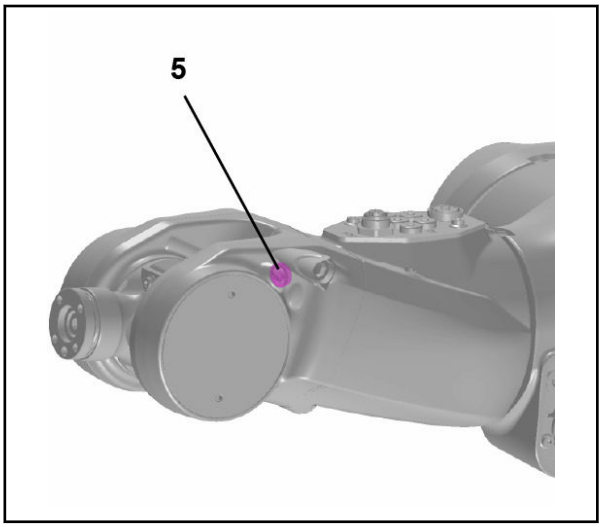
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	ω	0°	-90°	-	-	-



Axis 4: The oil level is flushed with the upper section of the wheel after removing cap **(4)**.

Checking the levels for axes 5 and 6 with the arm on a wall (Compact wrist only)

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	ω	0°	-8°	180 °	-	-

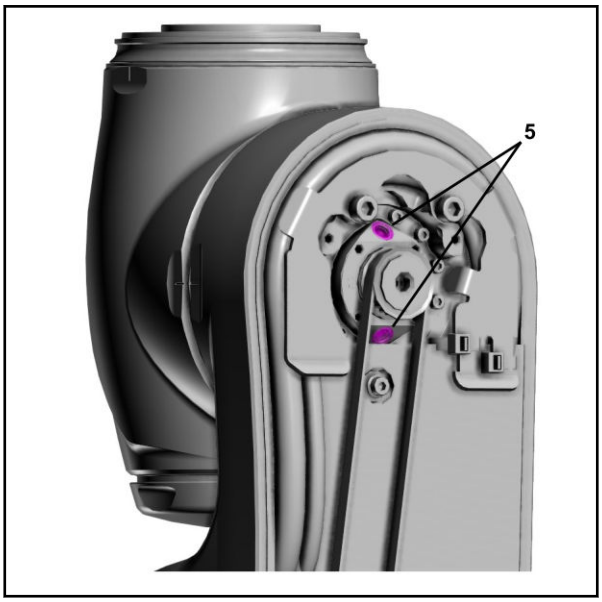


Axis 5-6:

- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove both caps **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
  - Re-install both caps **(5)** (3 Nm).

Checking the level for axis 5 with the arm on a wall (Hollow wrist only)

The forearm must be positioned vertically, as in the picture below.



Axis 5:

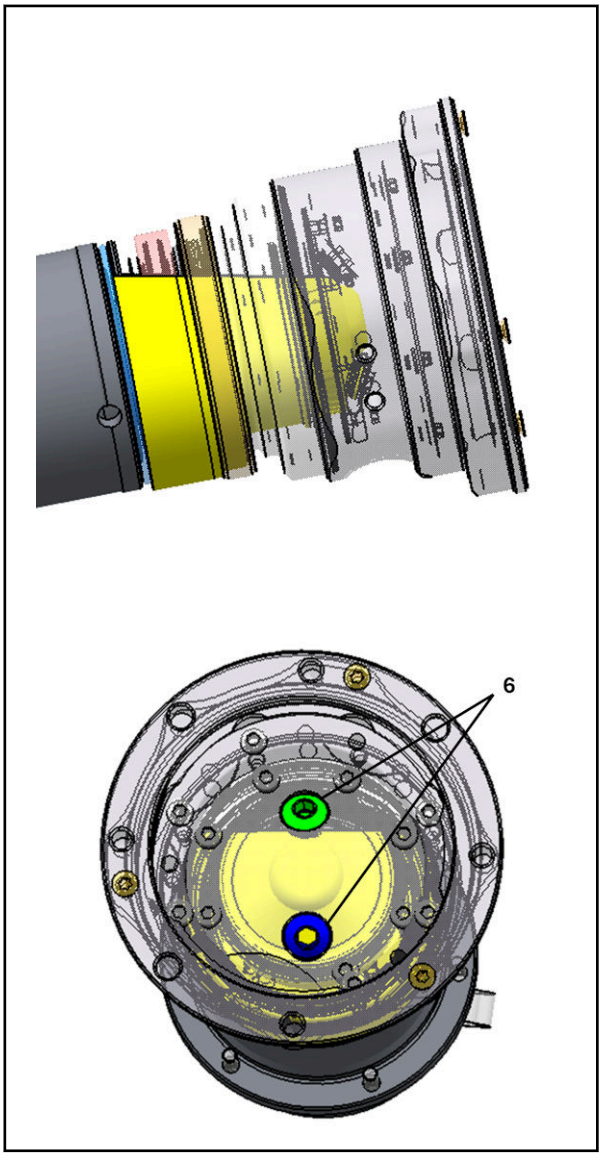
- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the upper cap **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(5)** (3 Nm).

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Checking the level for axis 6 with the arm on a wall (Hollow wrist only)

The following position is recommended to check the level for axis 6. The axis 6 must be inclined at 11° to the horizontal with the oil drain plugs vertically aligned.

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	ω	-	-	-	+11°	+45°



Axis 6:

- Run test program at SP60 on the largest joint 6 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the stainless steel flange sleeve (non visible on the picture).
- Place the cable gland support plate (non visible on the picture) on the side (regardless of flying or flange-mounted connectors).
- Remove the upper cap **(6)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(6)** (3 Nm).

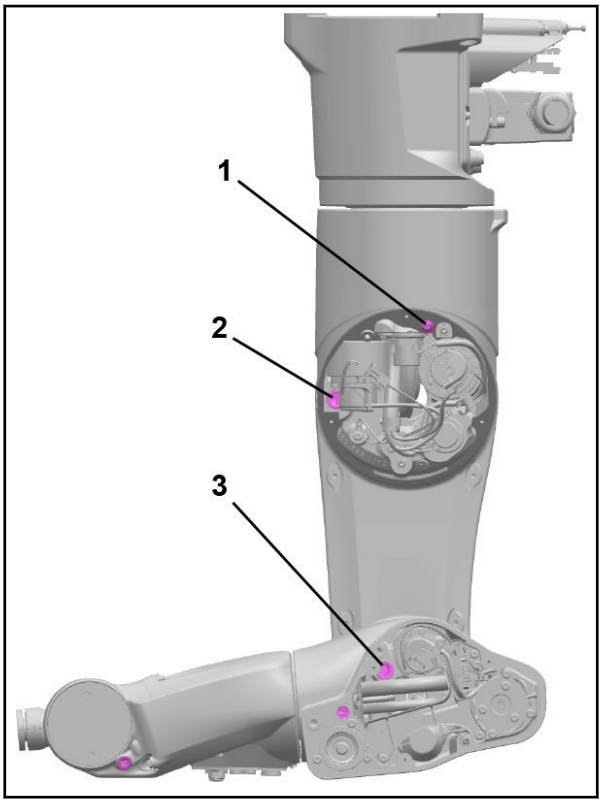
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7.11.2.3 - Arm in ceiling-mounted position

M0006419.1

Checking the levels for axes 1, 2 and 3 with the arm on the ceiling

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	0°	+85°	-	-	-



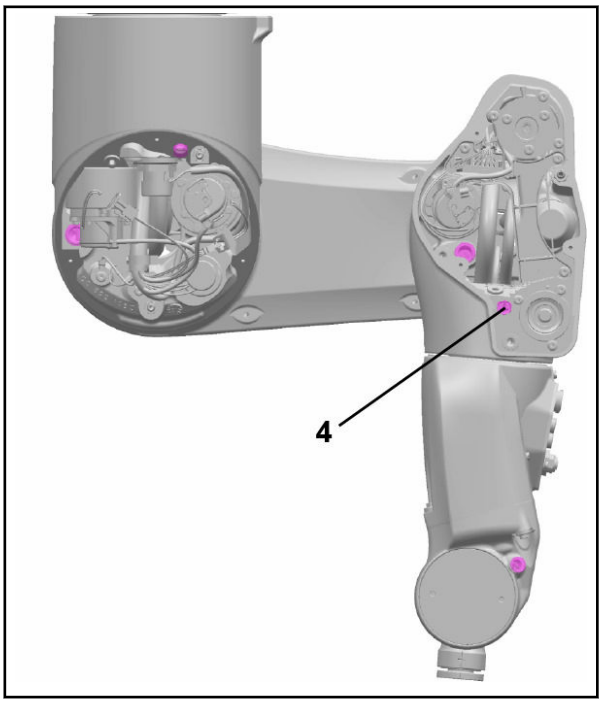
Axis 1: No checks.

Axis 2: Oil level has to be in the middle of sight glass **(2)** ± 3 mm.

Axis 3: Oil level has to be in the middle of sight glass **(3)** ± 3 mm.

Checking the level for axis 4 with the arm on the ceiling

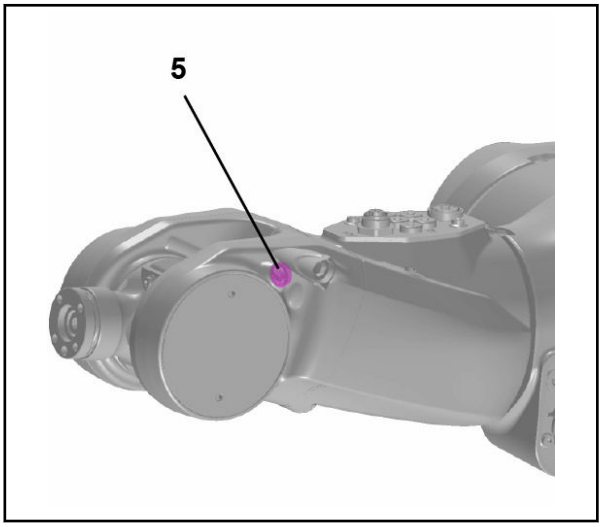
	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	-90°	+90°	-	-	-



- **Axis 4:** Oil level has to be in the middle of sight glass **(4)** ± 3 mm.

Checking the levels for axes 5 and 6 with the arm on the ceiling (Compact wrist only)

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	0°	-98°	-	-	-



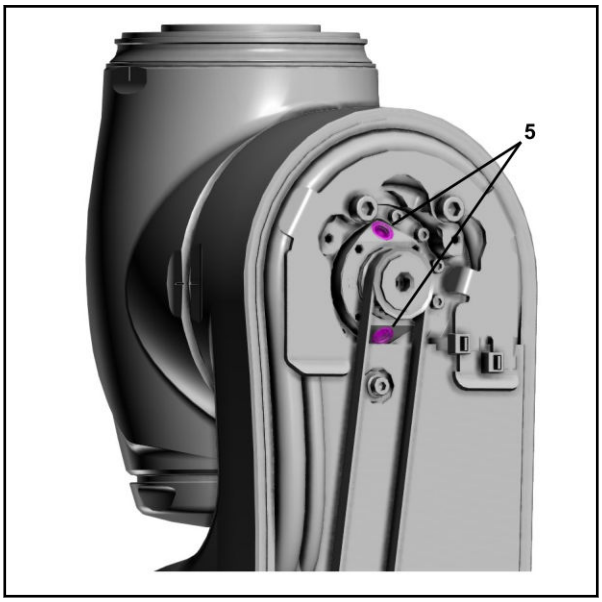
Axis 5-6:

- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove both caps **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Re-install both caps **(5)** (3 Nm).

Checking the level for axis 5 with the arm on the ceiling (Hollow wrist only)

The forearm must be positioned vertically, as in the picture below.

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Axis 5:

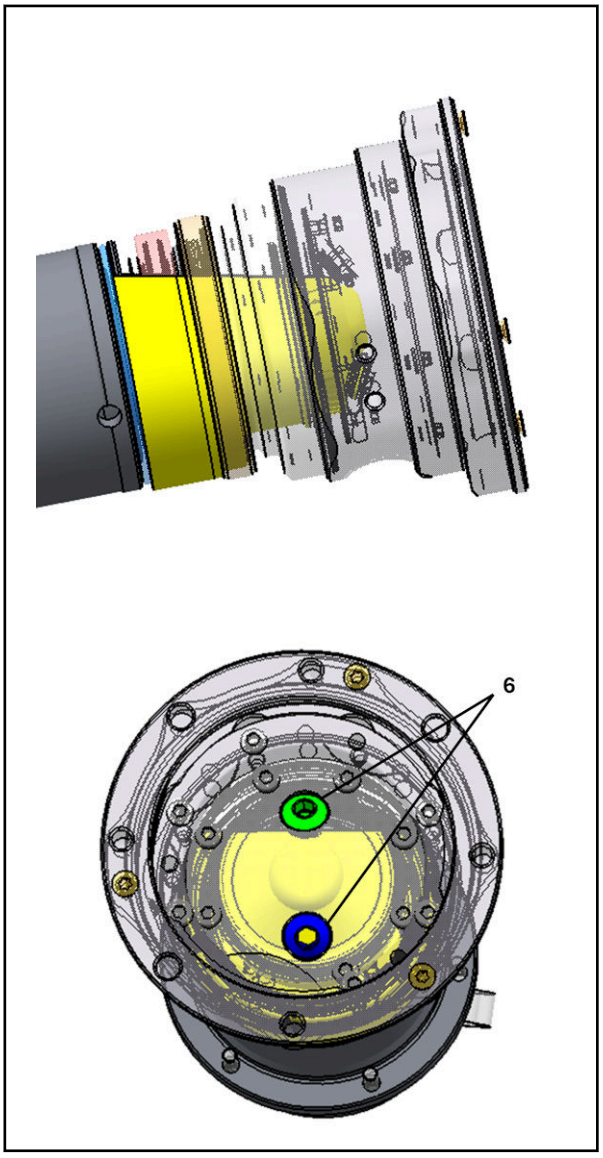
- Run test program at SP50 on the largest joint 5 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the upper cap **(5)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(5)** (3 Nm).

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Checking the level for axis 6 with the arm on the ceiling (Hollow wrist only)

The following position is recommended to check the level for axis 6. The axis 6 must be inclined at 11° to the horizontal with the oil drain plugs vertically aligned.

	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Axis position	-	+23°	-124°	-	-	+45°



Axis 6:

- Run test program at SP60 on the largest joint 6 envelop possible during 5 minutes.
- Wait for 10 minutes (because of the viscosity and the air bubbles).
- Remove the stainless steel flange sleeve (non visible on the picture).
- Place the cable gland support plate (non visible on the picture) on the side (regardless of flying or flange-mounted connectors).
- Remove the upper cap **(6)**.
- The oil level must flush the edge of the filling hole:
  - If too much oil is in the gearbox, let oil overflow.
  - If oil is not flushing the edge of the hole, use a syringe to adjust this level of oil.
- Replace the cap **(6)** (3 Nm).

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7.12 - CHECKING THE STATE OF THE JOINT BELT 4

M0006446.1

Tools required:

- Not applicable.

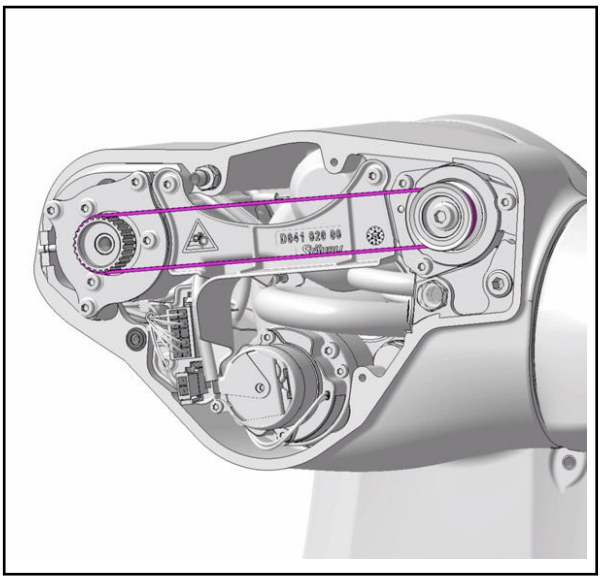
Recommended tools:

- Not applicable.

Initial position:

- Elbow cover removed.

Procedure:



- Check the condition of the belt without removing it (do not force or twist the belt).
- Replace the belt if necessary.

7.13 - CHECKING THE STATE OF THE JOINT BELT 5

M0007623.1

Tools required:

- Not applicable.

Recommended tools:

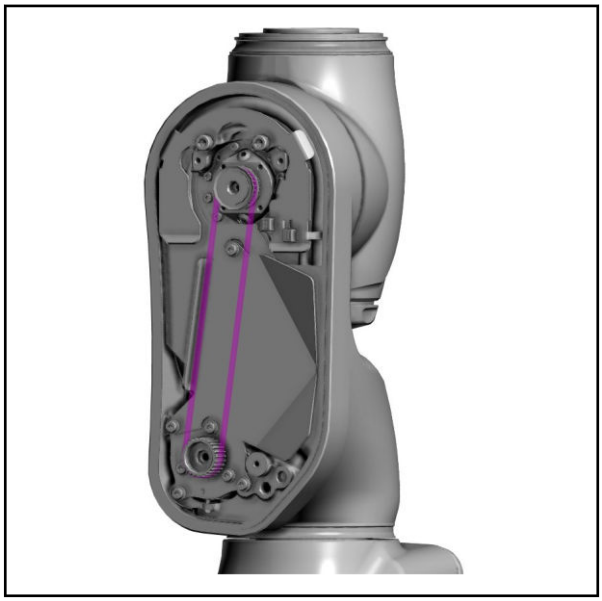
- Not applicable.

Initial position:

- Elbow cover removed.
- Axis 5 at 0° position. The wrist must be in vertical position.
- Axis 6 at 11° position.

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Procedure:

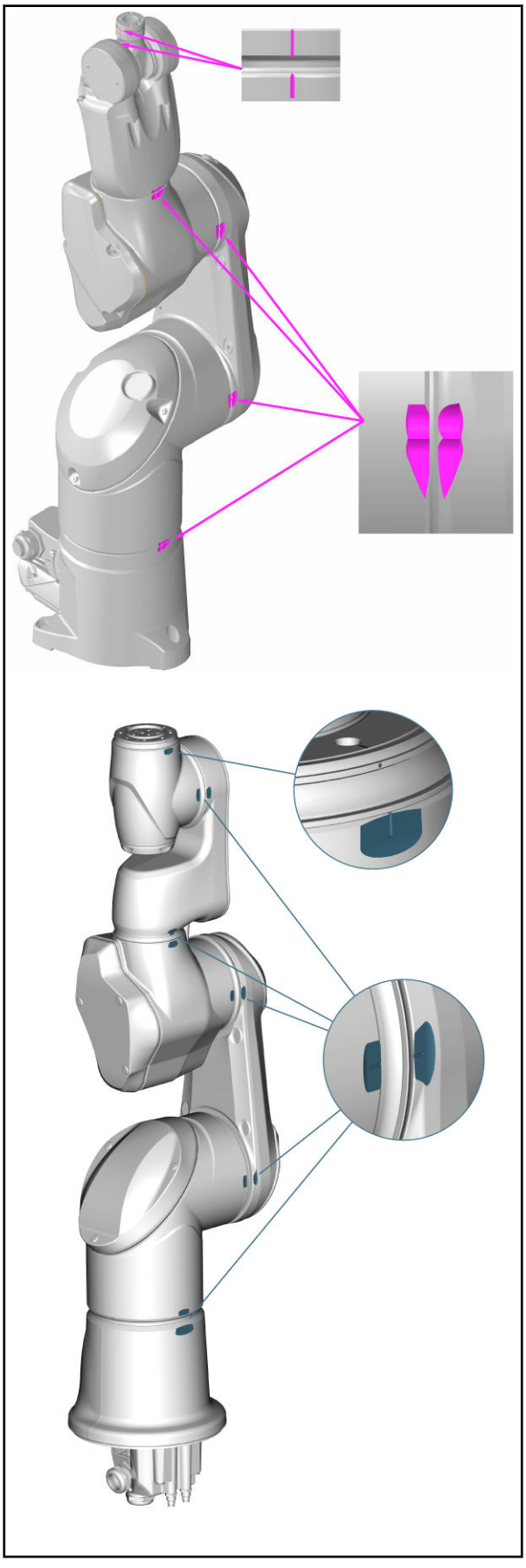


- Check the condition of the belt without removing it (do not force or twist the belt).
- Replace the belt if necessary.

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7.14 - RECOMMENDED CHECKS AFTER A ROBOT COLLISION, OVERCURRENT, OR IGBT DESATURATION

M0006441.1



In the event of a handling or programming error, a robot movement may result into a crash. Also, in other cases, error messages concerning an overcurrent on a motor phase or an IGBT desaturation may occur and affect internal mechanical elements. To make sure the robot's internal mechanical elements were not affected by those events, check the alignment of the marks calibrated during assembly at Stäubli prior to dispatch of the arm or the references determined by the customer (UserMarks, RefPos).

Before making any such checks, make sure you can do so in full safety by ensuring that the movement speed is set to a suitable level (we recommend 25% of the nominal speed).

In the event of an offset, carry out a calibration. Consult the Stäubli Customer Support.

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7.15 - RECOMMENDED CHECKS AFTER AN OVERPRESSURE IN THE AIR SUPPLY

M0009580.1

Corresponding FSP: FSP-000326 (see chapter 7.5)

If for any reason overpressure occurs in the pneumatic circuit:

- Standard pneumatic circuit: check for air leakage before restarting the circuit (sealings might be at fault), we recommend that you contact Stäubli after-sales service to schedule a diagnostic.
- Versions 5/2 and 5/3: contact Stäubli after-sales service to schedule a replacement of the pneumatic circuit.

7.16 - VISUAL CHECK OF THE HARNESS CONDITION

7.16.1 - VISUAL CHECK OF THE HARNESS CONDITION

M0006422.1

Tools required:

- Not applicable.

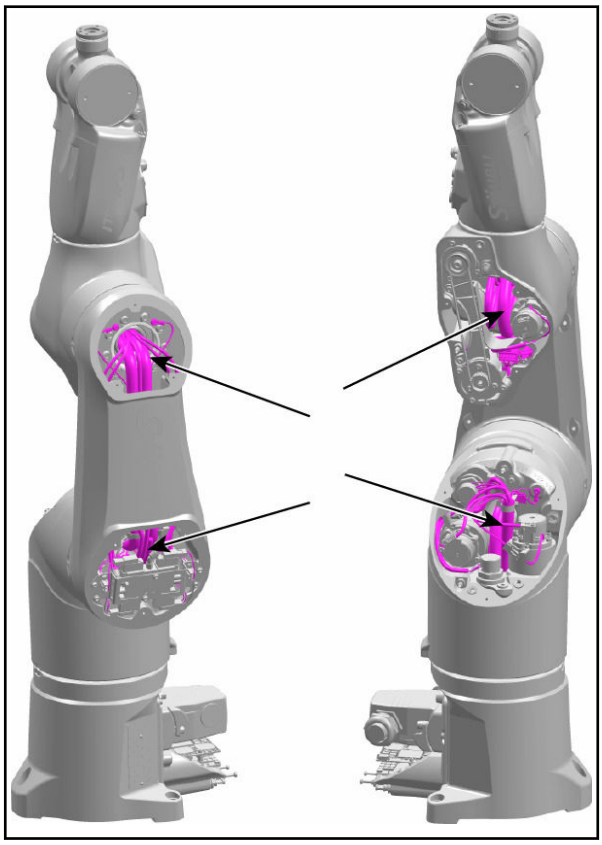
Recommended tools:

- Not applicable.

Initial position:

- Shoulder cover removed.
- Inner link covers removed.
- Elbow cover removed.

Procedure:



- Check the general condition of the harness over its full length and especially at the points shown.

NOTICE

Exceeding the arm's joint position limits

Attention, axis 4 does not feature an internal joint limitation system. This means that during calibration, it is possible to mistakenly offset the axis by 360°. This type of calibration error can lead to harness damage.

To check that axis 4 is correctly oriented, proceed as described in chapter 7.17.

7.16.2 - VISUAL CHECK OF THE HARNESS CONDITION (HOLLOW WRIST ONLY)

M0009561.1

Tools required:

- Not applicable.

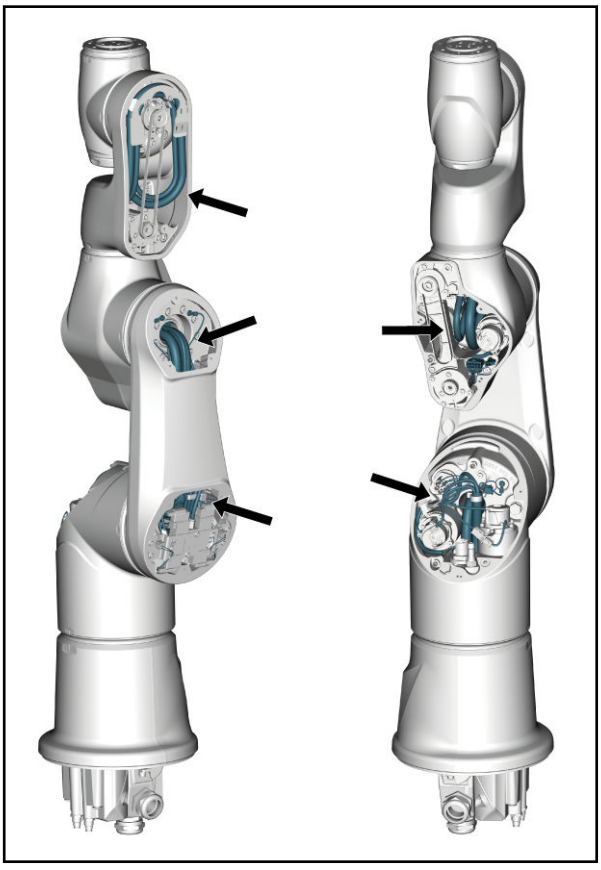
Recommended tools:

- Not applicable.

Initial position:

- Wrist cover removed.
- Forearm cover removed.
- Shoulder cover removed.
- Inner link covers removed.
- Elbow cover removed.

Procedure:



- Check the general condition of the harness over its full length and especially at the points shown.

NOTICE

Exceeding the arm's joint position limits

Attention, axis 4 does not feature an internal joint limitation system. This means that during calibration, it is possible to mistakenly offset the axis by 360°. This type of calibration error can lead to harness damage.

To check that axis 4 is correctly oriented, proceed as described in chapter 7.17.

7.17 - ORIENTATION OF AXIS 4 AFTER AN ADJUSTMENT

M0006386.1

Tools required:

- Not applicable.

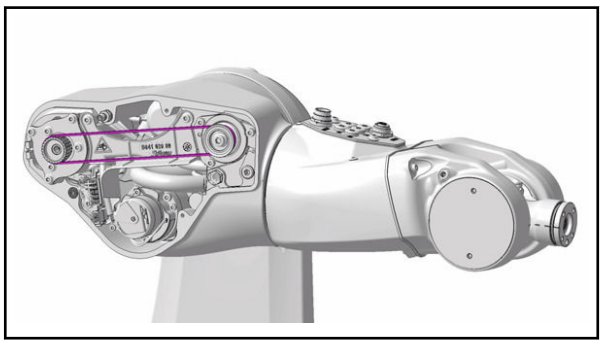
Recommended tools:

- Not applicable.

Initial position:

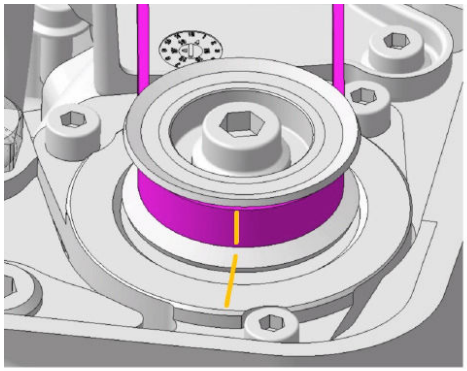
- Elbow cover removed.
- Axis 4 at 0° position.

Procedure:

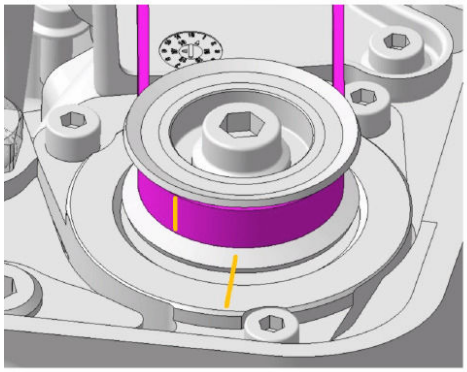


- When axis 4 is at 0° position, the 2 lines (one on the belt and the other on the base of joint 4) have to be aligned.

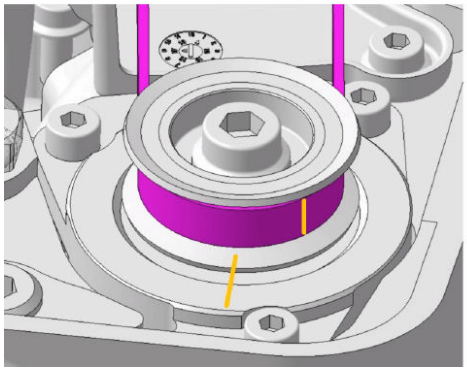
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1



2



2

- **1:** OK.
- **2:** Not OK.

7.18 - ORIENTATION OF AXIS 5 AFTER AN ADJUSTMENT

M0009683.1

Tools required:

- Not applicable.

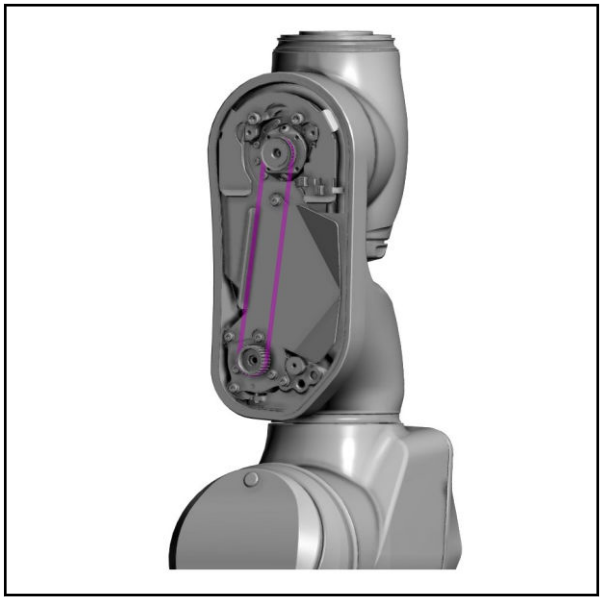
Recommended tools:

- Not applicable.

Initial position:

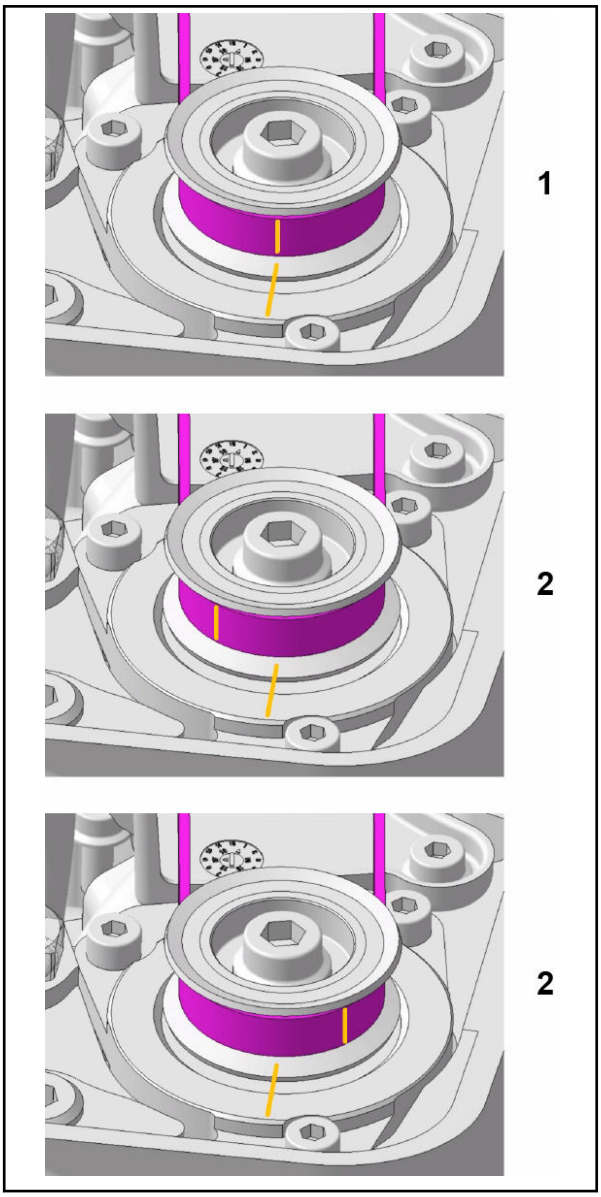
- Elbow cover removed.
- Axis 5 at 0° position. The wrist must be in vertical position.
- Axis 6 at 11° position.

Procedure:



- When axis 5 is at 0° position, the 2 lines (one on the belt and the other on the base of joint 5) have to be aligned.

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- **1:** OK.
- **2:** Not OK.

7.19 - PROCEDURE FOR REPLACING PNEUMATIC VALVES AND POWER LIGHT (OPTION)

7.19.1 - ELECTRIC VALVE FOR USE WITH COMPRESSED AIR (OPTION)

M0006447.1

Tools required:

- 3 mm Allen key.

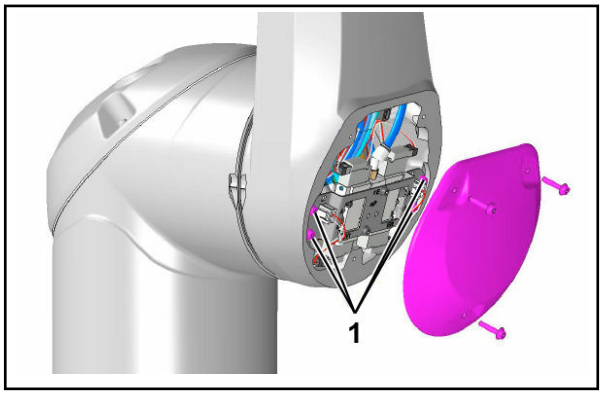
Recommended tools:

- Torque wrench.

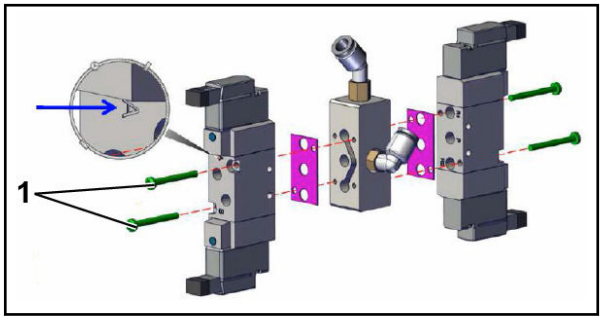
Initial position:

- Not applicable.

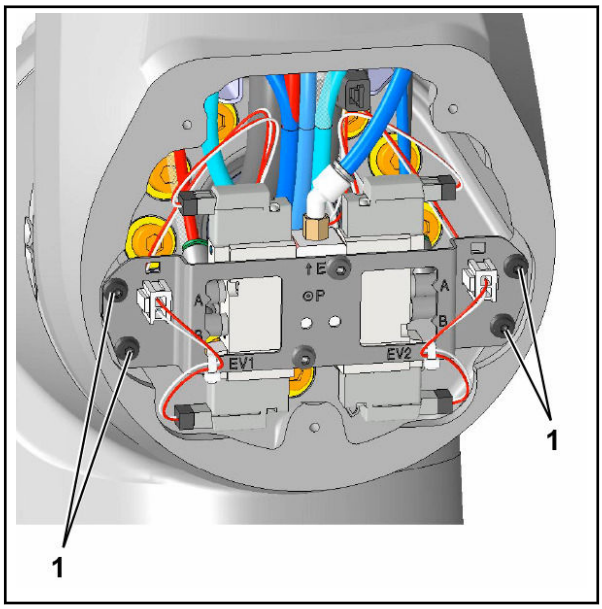
Procedure:



- Remove the cover.
- Disconnect lines and minifit connectors.
- Remove the 4 M4x10 screws **(1)** and the valve assembly.



- Replace one or both valves if needed, using the seal and the M2.5x25 screws **(1)** (0.45 Nm) supplied (contact Stäubli to get the reference of your electric valves).
- Pay attention to the A and B markings on the valves in order to install it correctly.



- Reassemble the assembly in the arm, using 4 M4x10 screws **(1)** tightened to 4.8 Nm.
- Connect the lines and minifit connectors, according to the labels on them.
- Re-install the cover.

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7.19.2 - ELECTRIC VALVES FOR USE WITH VACUUM (OPTION)

M0006402.1

Tools required:

- 3 mm Allen key.

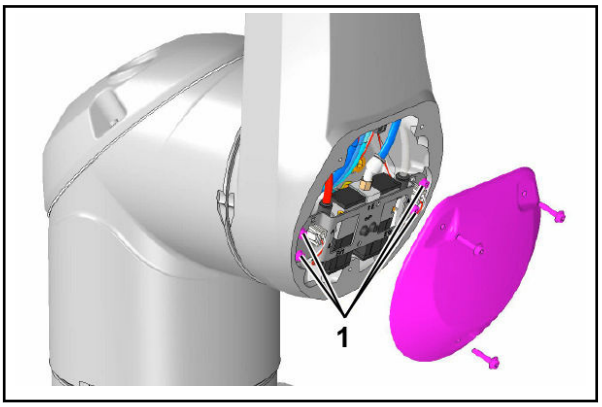
Recommended tools:

- Torque wrench.

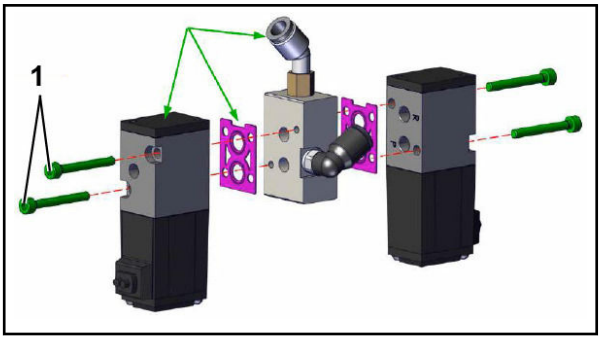
Initial position:

- Not applicable.

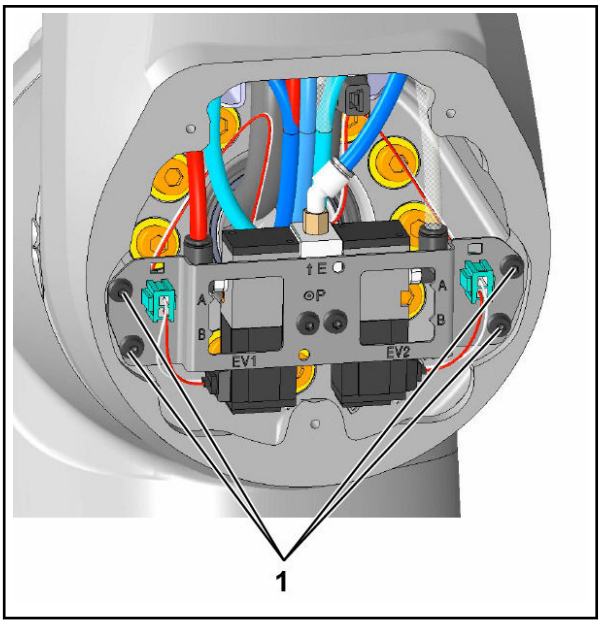
Procedure:



- Remove the cover.
- Disconnect lines and minifit connectors.
- Remove the 4 M4x10 screws **(1)** and the valve assembly.



- Replace one or both valves if needed, using M3x26 **(1)** screws tightened to 0.6 Nm (contact Stäubli to get the reference of your electric valves).
- Pay attention to the orientation of the valves, see the arrows on the picture.



- Reassemble the assembly in the arm, using 4 M4x10 screws **(1)** tightened to 4.8 Nm.
- Connect the lines and minifit connectors, according to the labels on them.
- Re-install the cover.

7.19.3 - REPLACEMENT OF THE POWER LIGHT (OPTION)

M0006430.1

Tools required:

- 3 mm Allen key.

Recommended tools:

- Torque wrench.

Initial position:

- Not applicable.

Procedure:



- Remove the elbow cover.
- Disconnect the power light minifit connector.



- Remove the screws M20x1.5 **(1)**.
- Replace the power light and the seal.
- Tighten the M20x1.5 screw **(1)** to 3 Nm.

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7.20 - MAINTENANCE OF THE MECHANICAL HARD STOP (OPTION) M0006380.1

What should be done if the robot is stopped by an optional mechanical hard stop ?

1. Check that the robot has not suffered any visible damage.
2. Check transmission. Place arm on reference mark. If there is an offset, contact Stäubli.
3. Check that there is no unusual noise or backlash on the various axes during brake release and in manual mode.
4. Put the ring back in place if it has slipped. Follow the points concerned in the installation chapter (see chapter 4.5).
5. Check operation at low speed.

DANGER



Damaged or slipped ring

Operating the robot with a damaged or slipped mechanical hard stop is hazardous.

If any ring damage or slippage is noted (or in the event of doubt following an impact or a suspected impact), consult the Stäubli Customer Support for an arm diagnostic.

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8 - RECYCLING

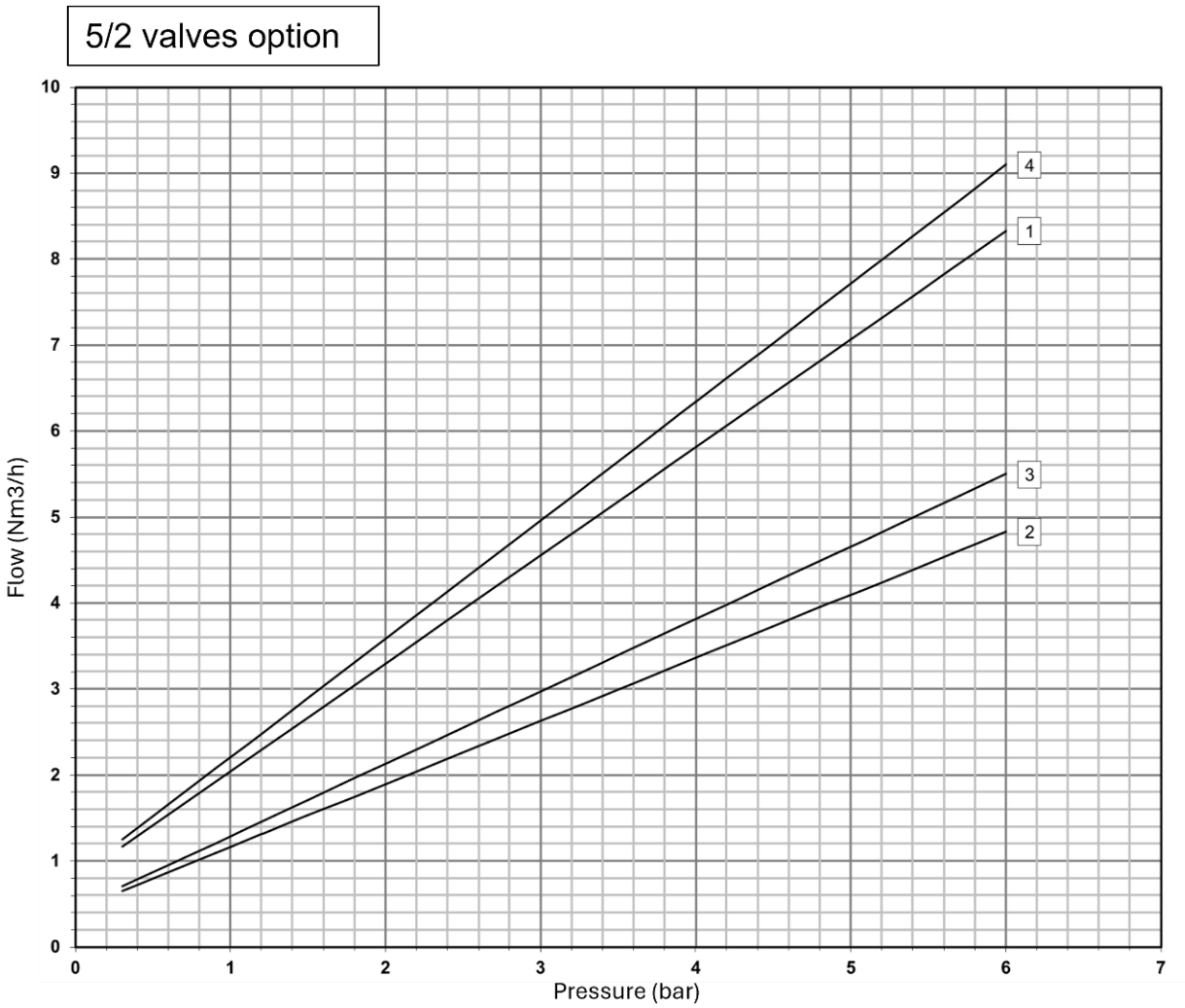
M0007049.1

Disposal and recycling of robot components shall be made in accordance to national rules and directives (DEEE 2012/19/UE - July 2012 for Europe).
All critical components shall not be disposed in household garbage and shall be collected and recycled as defined in directives.
Critical components categories included in the robot:

- Electronic boards, identified with  ,
- Battery inside CPT,
- SP2/SP2+ and CPT displays,
- Electrical cables:
  - Arm and controller wiring,
  - Arm to controller cable,
- Oil in gear boxes of the arm.

9 - APPENDIX -CHARTS OF OUTLET FLOW VERSUS INLET PRESSURE

M0009579.1

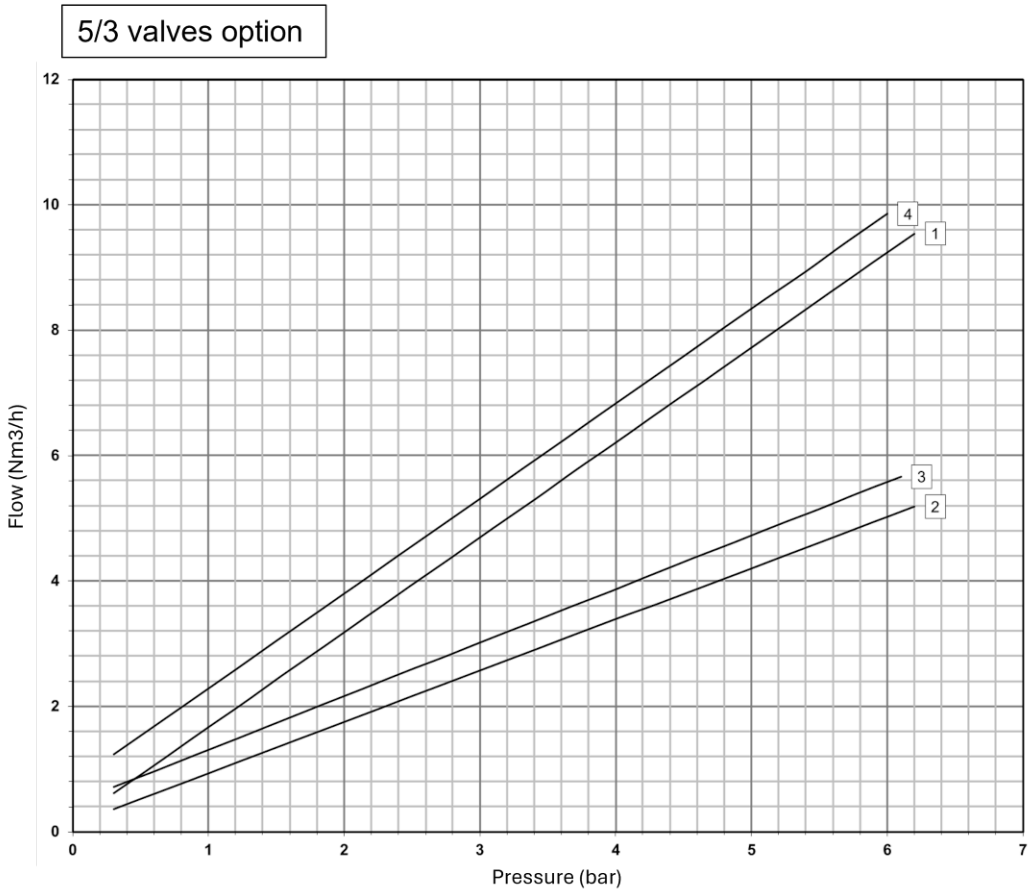


Curve	Product	Circuit	Flow at 1 bar (Nm3/h)	Flow at 3 bar (Nm3/h)	Flow at 6 bar (Nm3/h)
1	EV 5/2	P1->B1+B2	2,04	4,55	8,32
2	EV 5/2	P1->B1	1,16	2,63	4,83
3	EV 5/2	A1->E (1)	1,29	2,97	5,5
4	EV 5/2	A1+A2->E (1)	2,21	4,96	9,09

(1) with silencer

Figure 9.1: Outlet flow versus inlet pressure for the 5/2 valves option

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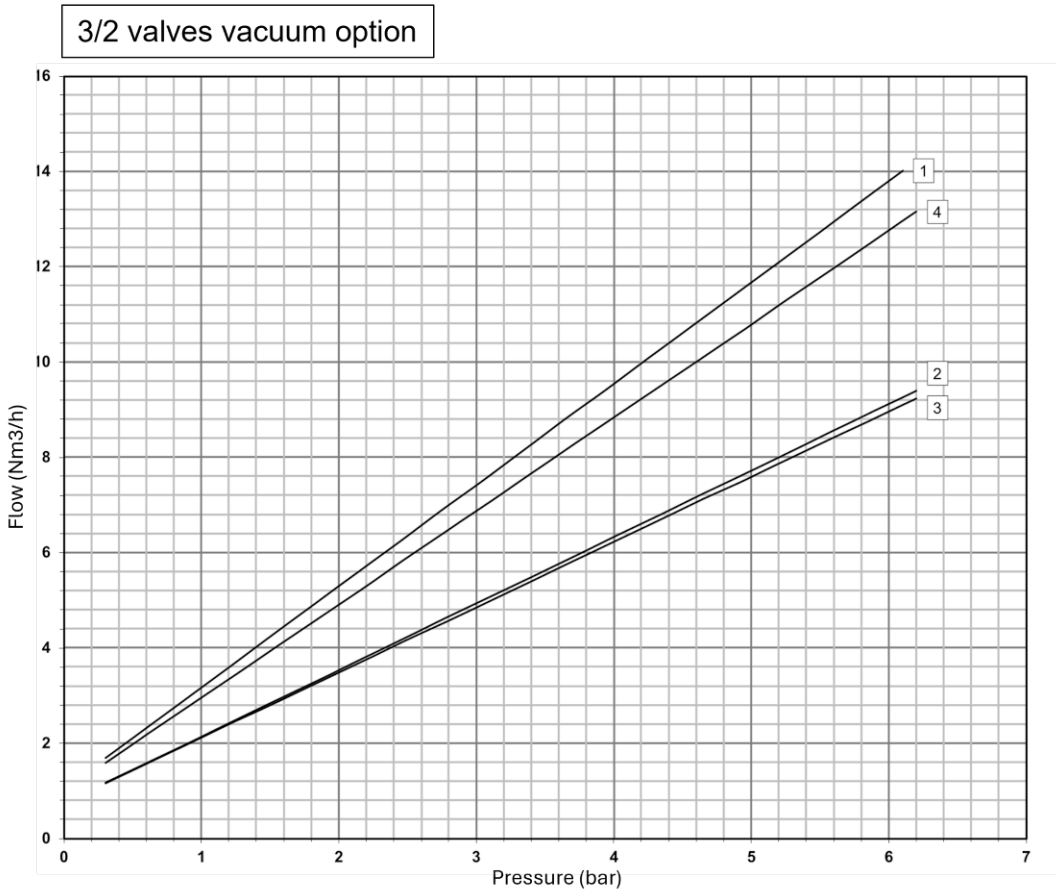
Curve	Product	Circuit	Flow at 1 bar (Nm3/h)	Flow at 3 bar (Nm3/h)	Flow at 6 bar (Nm3/h)
1	EV 5/3	P1->B1+B2	1,67	4,7	9,24
2	EV 5/3	P1->B1	0,93	2,57	5,02
3	EV 5/3	A1->E (1) (2)	1,31	3,02	5,58
4	EV 5/3	A1+A2->E (1) (2)	2,29	5,31	9,85

(1) with silencer

(2) with 6 bar A.C (network) in P1

Figure 9.2: Outlet flow versus inlet pressure for the 5/3 valves option

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Curve	Product	Circuit	Flow at 1 bar (Nm3/h)	Flow at 3 bar (Nm3/h)	Flow at 6 bar (Nm3/h)
1	EV 3/2 vaccum option	P1->A1B1 + A2B2	3,17	7,42	13,8
2	EV 3/2 vaccum option	P1->A1B1	2,14	4,93	9,12
3	EV 3/2 vaccum option	A1B1->E (1)	2,12	4,86	8,96
4	EV 3/2 vaccum option	A1B1+A2B2->E (1)	2,95	6,87	12,76

(1) with silencer

Figure 9.3: Outlet flow versus inlet pressure for the 3/2 vacuum valves option



Curve	Product	Circuit	Flow at 1 bar (Nm3/h)	Flow at 3 bar (Nm3/h)	Flow at 6 bar (Nm3/h)
1	Direct lines	P1->A2	1,49	3,53	6,59
2	Direct lines	P1->A1+B1	2,57	6,25	11,78
3	Direct lines	A2->E	1,43	3,42	6,4
4	Direct lines	A1+B2->E	2,42	5,81	10,9

Figure 9.4: Outlet flow versus inlet pressure for the direct lines valves option

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